Geopolitical Risks and International Business

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Foreword

Every multinational now says that geopolitics matters to it. Far fewer can say *how much*, *through which channel*, or *what they would do differently if the number changed*. The gap is not one of attention — boards discuss sanctions, export controls, wars and elections every quarter — but of measurement. Geopolitical risk reaches a firm through its market access, its inputs, its finance, its people and its data, and each of those channels can be counted. Most firms do not count them.

This book is the lecture notes of a twelve-session graduate course that tries to close that gap. Its method is simple and a little stubborn: **every big idea is read, then rebuilt from its data.** A landmark article or book — Caldara and Iacoviello's geopolitical risk index, Baldwin's *Great Convergence*, Kuziemko and Werker on the price of a Security Council seat, the Global Sanctions Data Base — is paired with a practice round in which the data are collected at source and its central figure is replicated in Tableau. A famous chart is a set of decisions: which countries, which years, which deflator, which coding rule. You only see the decisions when you make them yourself.

The data will disagree with the paper. Sources are revised, codebooks change, countries are renamed and recoded. Every practice round records at least one place where the replication did not match — and asks what that disagreement means for a firm that would have relied on the original.

Each chapter ends on the same question: *what does this figure not license a multinational to conclude?* That is the habit the course is built to leave behind.

How to use it

If you want to	Go to
Prepare for next week	the session's pre-session page, linked at the top of each chapter
Re-read the argument after class	the chapter itself
Know exactly what the practice round asks	the chapter's last section, **In the practice**
Find where something was covered	the **search box**, top of the sidebar
Find a dataset, its URL and its traps	Data sources
Know how you are graded	Assessment

The search box is the reason this book exists. Slides are good for a room and bad for finding something three weeks later. Search covers every session at once, so a term you half-remember — *weaponised interdependence*, *ideal points*, *Herfindahl* — takes you to the session that introduced it, with its reading and its exercise.

The slides

Each chapter links to the decks it was built from. The slides remain the version delivered in class; the book sets the same argument out at length, as continuous prose — about six thousand words a session, with the definitions, the history and the literature the slides only point to. All decks are also listed on the course site.

The twelve chapters

	Chapter	Question
1	What is geopolitical risk?	Before we can manage geopolitical risk, what exactly are we counting?
2	The international economic order	Who wrote the rules multinationals play by — and what happens to the rules when the writers' share of the world shrinks?
3	Anarchy and the balance of the major powers	In a system with no government above states, who sets the terms of trade — and how would a multinational know that the balance is shifting?
4	Trade, conflict and geoeconomic fragmentation	If states trade more with friends than with rivals, how does a multinational find out which side each of its markets is on — and how fast the lines are hardening?
5	International institutions	When a multinational relies on an international institution to finance, insure or protect its business, is it relying on a rule — or on the states that buy and sell influence inside it?
6	Private authority, standards and global regulation	Most of the rules that govern a multinational's daily operations were not written by a government. Who wrote them, with whose power — and does trade spread them up or down?
7	Measuring risk: content analysis, coding, big data and composite indices	Every geopolitical risk score a multinational buys or builds is a chain of choices: which texts, which words, which coder, which weights. Can you see the chain — and defend each link?
8	Cyber geopolitics and digital threats	Is cyberspace a new battlefield for great powers — or a new way for old rivals to hurt each other, with multinationals as collateral?
9	The redefinition of economic diplomacy	When states compete for markets, investment and technology, is the multinational the object of diplomacy, its instrument — or a diplomat in its own right?
10	Economic sanctions	Sanctions have become the instrument of first resort. Are they an efficient mechanism for regulating geopolitical risk — or a source of it?
11	Space geopolitics	Satellites now carry the position, timing, connectivity and imagery a multinational runs on. When orbit is crowded, contested and dominated by one operator, who holds the risk — and what can the firm do about it?

	Chapter	Question
12	Team presentations	Can you make a board act on your assessment — and say what would change your mind?

Chapter 1

What Is Geopolitical Risk? Counting the Threat Before the War

Session 1 · Concepts, indicators, and the syllabus Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Before we can manage geopolitical risk, what exactly are we counting?**

Big work	Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225. doi:10.1257/aer.20191823
Target figure	The monthly GPR index since 1985, with its spikes labelled — and its historical counterpart since 1900
Data you collect	The GPR index, from the authors' page: matteoiacoviello.com/gpr.htm
Tableau skill	Connect to an Excel file · date fields · line chart · reference line · annotations
Course themes	Methodology (text-based indicators); the vocabulary of the whole course

1.1 A question about counting

Every risk function in every multinational firm already has a view about geopolitics. It sits in the annual report, in the paragraph on principal risks; it sits in the treasury's hedging policy, in the procurement team's list of single-source suppliers, in the country manager's sense of which visas will be hard to renew next year. What it rarely has is a *number* — and when it does, the number usually comes from somewhere else, built by someone else, for a purpose the firm never examined.

This course begins with a deliberately narrow question: before a firm can manage geopolitical risk, what exactly is it counting? The question sounds pedantic. It is not. A board that is told "geopolitical risk is at its highest level in forty years" has been told something precise about a particular index, constructed from particular newspapers, normalised to a particular base period. Whether that statement should change a capital-expenditure plan in Monterrey or a supplier contract in Penang depends entirely on what was counted, and the person presenting the slide often does not know.

The answer this chapter builds towards is that geopolitical risk is not one thing to be measured but a family of things, each visible to some instruments and invisible to others. We start with the oldest distinction in the subject, between risk and uncertainty; separate three kinds of risk that are routinely confused; sort the instruments that try to measure them into six families; and then take one instrument apart in detail — the Geopolitical Risk index of Caldara and Iacoviello (2022), the big work of this session and the target of its replication. The chapter closes on what that index cannot tell a firm, because a measure whose limits you cannot state is a measure you should not use.

1.2 Risk, uncertainty, and the ground in between

The modern vocabulary starts in 1921, with an economist trying to explain where profit comes from.

From the archive · Knight, 1921

Frank Knight's *Risk, Uncertainty and Profit* began as a doctoral dissertation at Cornell and was published in 1921. Its problem was a theoretical embarrassment: in a perfectly competitive economy with full information, competition should drive profit to zero, yet profit plainly exists. Knight's answer was that some of what businessmen face can be reduced to known odds and therefore insured, pooled or priced into costs — and some cannot.

The first he called *risk*: a situation in which the distribution of outcomes is known, either a priori, as with dice, or statistically, from the experience of many similar cases, as with fire insurance. The second he called *uncertainty*: a situation that is unique enough that no such distribution can be formed, and in which the decision-maker must act on judgement. Risk, being insurable, becomes a cost like any other. Uncertainty cannot be shifted to an insurer, and it is the bearing of uncertainty — the entrepreneur's willingness to commit resources on a judgement no one can verify in advance — that profit rewards (Knight 1921).

A century later, the distinction is quoted more often than it is read, and usually as a binary. Knight's own treatment was more graded: what matters is whether instances can be grouped into classes homogeneous enough to support an estimate. That is precisely the question a geopolitical risk analyst faces every morning.

Risk (in Knight's sense) — a situation in which the probabilities of outcomes can be estimated, from theory or from the experience of sufficiently similar cases, so that the exposure can be priced, pooled or insured.

Uncertainty (Knightian uncertainty) — a situation in which no reliable probability distribution over outcomes can be formed, because the case is too unlike any class of past cases. It cannot be insured away; it can only be judged.

Geopolitics lives in the ground between the two. A shipping insurer pricing piracy off the Horn of Africa is close to Knightian risk: there are thousands of transits, hundreds of incidents, and a market for war-risk premiums that updates as the incidents accumulate. A board asking whether a particular strait will be blockaded within five years is close to Knightian uncertainty: there have been blockades, but never this one, by this state, against this coalition, with these weapons. Wars recur, coups recur, sanctions recur — the categories are populated — but each instance is unlike the last in ways that matter for the estimate.

The honest description of this course is that it is an attempt to move questions from the second category towards the first, and to be candid about how far they move. Every index you will replicate is a device for grouping unlike events into classes so that something can be counted. Grouping is never neutral. When the Geopolitical Risk index counts an article about a nuclear test and an article about a terrorist bombing as two instances of the same thing, it has made a modelling choice about what is homogeneous. When the Correlates of War project averages steel output and population into one number for "power" (Session 03), it has made another. The skill the course tries to teach is not to avoid these choices — no measurement is possible without them — but to see them, state them, and ask what they do to the conclusion.

There is a practical corollary for the firm. Knightian risk can be transferred: political-risk insurance, war-risk cover on cargo, credit insurance on export receivables are all markets for it. Knightian uncertainty cannot be transferred, only managed — by flexibility, by diversification, by keeping options open, by being willing to pay for the ability to change one's mind. Much of the strategic response to geopolitical change that firms now describe as "resilience" is, in Knight's terms, an admission that the relevant exposures are uncertain rather than risky, and that the premium has to be paid in redundancy because there is no one to pay it to.

1.3 Three kinds of risk, and the layer that comes from outside

The second distinction is between three phrases that firms, consultancies and journalists use almost interchangeably: political risk, country risk and geopolitical risk. They answer different questions, attach to different units of analysis, and call for different instruments.

Political risk — the possibility that the government of a host country changes the rules applying to a foreign firm in a way that reduces the value of its investment: expropriation, forced renegotiation, discriminatory taxation, licence withdrawal, restrictions on repatriation. Its natural unit is the firm in a host country.

Country risk — the possibility that a country as a whole fails to pay, convert or transfer: sovereign default, capital controls, currency inconvertibility, a banking crisis. Its natural unit is the country, and its natural user is the lender.

Geopolitical risk — the possibility that relations *between* states and other political actors move against the firm: war, terrorism, sanctions, export controls, the formation of blocs, state-sponsored cyber operations. Its natural unit is the international system or a pair of states, and it frequently originates outside any country in which the firm operates.

Political risk is the oldest of the three in the international business literature, and its archetypes are dramatic. When Mexico nationalised the assets of foreign oil companies in 1938, or Iran nationalised the Anglo-Iranian Oil Company in 1951, the threat came from the host government and fell on identifiable foreign firms. The scholarly literature that grew up in the 1960s and 1970s, as decolonisation multiplied the number of host governments and expropriations became frequent, was preoccupied with this kind of event. Its classic synthesis is Kobrin (1979), who reviewed a decade and a half of work and complained that "political risk" was used to mean too many things — sometimes the political event itself, sometimes the instability of a regime, sometimes the consequences for a firm's operations — and that the literature rarely connected the first to the last.

Kobrin's lasting contribution was to insist that risk is a property of the relationship between an environment and a particular firm, not of the environment alone. Political instability in a country matters to a foreign investor only through specific *contingencies* that affect specific operations; a coup that changes nothing about the mining code may be irrelevant to the mining company, while a quiet amendment to a tax treaty may be decisive. From this followed the distinction between **macro** risk, which affects all foreign firms in a country, and **micro** risk, which falls on a sector, an industry or a single firm. Expropriation of every foreign asset after a revolution is macro; a windfall tax on telecommunications operators is micro.

Country risk grew from a different root — the bank lending of the 1970s to developing-country governments, and the debt crisis that followed in the 1980s. Its question is whether a country, as an economic and financial unit, will service its obligations and let money move. Its instruments are sovereign ratings, debt-sustainability analysis and the spreads on sovereign bonds and credit default swaps. It cares less about what the government will do to a particular investor than about whether the treasury will have the dollars.

Geopolitical risk adds a third layer that neither of the earlier traditions was designed to see: risk that originates *outside* the host country. A Canadian firm operating in Vietnam can be hurt by a decision taken in Washington about Chinese inputs, by a decision in Beijing about rare-earth exports, by a decision in Brussels about which payment systems may be used for which counterparties, or by a naval incident in a sea lane neither Canada nor Vietnam controls. The host government may be perfectly stable and perfectly friendly; the threat travels through the firm's suppliers, customers, banks and networks.

The three are not mutually exclusive, and the clearest way to see that is a single case. Take a Canadian bank with a subsidiary in Moscow in February 2022. It faces *political* risk when the host state restricts the ability of foreign firms to sell their assets and leave. It faces *country* risk when capital controls trap the subsidiary's earnings and the rouble's convertibility is suspended for foreigners. And it faces *geopolitical* risk in its purest form when its own government, acting with its allies, forbids transactions on which the subsidiary depends. All three arrive in the same quarter; they arrive through different channels; and they call for different responses — a write-down, a liquidity plan, a compliance review. A risk function that has only one of the three words in its vocabulary will see one-third of the problem.

1.4 Six ways of counting

If geopolitical risk is a family of phenomena, it is measured by a family of instruments, and the instruments disagree. It helps to sort them by *what they count*, because what an indicator counts determines what it cannot see.

Family	What it counts	Examples in this course	Strength	Blind spot
Event-based	things that happened, coded from reports	cyber incidents (S08), sanctions episodes (S10)	concrete, datable	only what was reported and coded
Text-based	how much the press or firms talk about a risk	GPR (today), EPU, firm-level risk (S07)	timely, long, high-frequency	measures attention, not exposure

Family	What it counts	Examples in this course	Strength	Blind spot
Expert-based	scores assigned by analysts or surveys	governance indicators, commercial ratings	reaches latent concepts	opaque, sticky, perception-driven
Structural	slow-moving features of the system	material capabilities (S03), trade dependence (S04, S11)	explains *why*	rarely says *when*
Market-based	prices that embed risk	sovereign CDS spreads, implied volatility	forward-looking, daily	mixes risk with risk appetite
Firm-level	risk as perceived by firms themselves	earnings-call text (S07), disclosed exposures	speaks to the decision	self-reported, listed firms only

Event-based indicators record occurrences — a war began, a sanction was imposed, an intrusion was attributed — coded by researchers from reports. They are concrete, but they see only what was reported and coded, and they are backward-looking: a threat never carried out leaves no event. *Text-based* indicators count words. The Geopolitical Risk index is one; the Economic Policy Uncertainty index of Baker, Bloom, and Davis (2016) is its older sibling; the firm-level political-risk measure of Hassan et al. (2019) applies the idea to earnings calls. Text is abundant, long and timely, but it measures *attention*: how much the writers thought a subject worth discussing, which is related to, but not the same as, how much risk any particular reader faces.

Expert-based indicators ask people to score what cannot be observed directly — the rule of law, the likelihood of political violence. The Worldwide Governance Indicators aggregate many such sources (Kaufmann, Kraay, and Mastruzzi 2011). They reach latent concepts, at the price of opacity, stickiness and shared perceptions. *Structural* indicators — the distribution of capabilities, the concentration of a country's imports — explain *why* a situation is fragile but rarely *when* the fragility will matter, which is the question a firm is usually paid to answer. *Market-based* indicators are prices, bets placed by people with money at stake; but a price combines the quantity of risk with the appetite for bearing it, and a spread can widen because danger rose or because investors grew more fearful of the same danger. *Firm-level* indicators record what the people who allocate capital say worries them; they are also self-reported, shaped by liability and investor relations, and available almost only for listed firms.

No family is enough on its own. The team project in this course requires a risk assessment that **triangulates** across at least three, and the reason is not that more is better. It is that the families fail in different ways, so their disagreements carry information. When the text is calm and the market is not, someone may be pricing something the press has not yet seen. When the structure is fragile and the events are quiet, risk is accumulating rather than absent. When the experts are alarmed and the firms are not, either the exposure is concentrated in a few firms or the experts are behind. A single index can only tell you its own story; three indices that quarrel force you to say which one you believe, and why.

1.5 How a newspaper archive becomes an index

The rest of the chapter takes one text-based instrument apart. Dario Caldara and Matteo Iacoviello, economists at the Federal Reserve Board, published the Geopolitical Risk index in the *American Economic Review* in 2022, after circulating it for several years as a working paper. It has become the default series for anyone — central bank, investment bank, consultancy or doctoral student — who needs a number for geopolitical risk. That ubiquity is the reason to understand its construction exactly.

1.5.1 The definition

The authors begin with a definition, and it is worth copying word for word, as the pre-session asked you to do. Geopolitical risk is "the threat, realization, and escalation of adverse events associated with wars, terrorism, and any tensions among

states and political actors that affect the peaceful course of international relations" (Caldara and Iacoviello 2022).

Read it for what is in and what is out. *In* are threats as well as realisations — the index is meant to rise when war becomes more likely, not only when it starts. *In* are non-state actors: "political actors", not only states, which is how terrorism enters. *In* are tensions short of war. *Out* are domestic politics, elections, regulation and economic policy, which belong to other indices. The definition is therefore closer to the third row of the typology above than to the first two: it is about relations between actors, not about what a host government does to a firm.

Geopolitical Risk (GPR) index — a monthly text-based index, constructed by Caldara and Iacoviello (2022), equal to the share of articles in a fixed set of major newspapers that discuss adverse geopolitical events and threats, rescaled so that its average over a base period equals 100.

1.5.2 Four steps

The construction has four steps, and each is a decision that could have been taken differently.

The corpus. The recent index, from 1985, is built from the electronic archives of ten leading newspapers published in the United States, the United Kingdom and Canada. The historical index, from 1900, uses three American newspapers whose archives reach that far back. Already the scope is clear: this is the anglophone press of three allied countries, and the historical series is the American press alone.

The dictionary. An article counts as a GPR article if it contains words from one of eight categories: war threats, peace threats, military build-ups, nuclear threats, terror threats, the beginning of war, the escalation of war, and terror acts. In most categories, the search requires a combination — a word for the threat and a word that anchors the article to geopolitics — so that an article about a "price war" or a "threat" to a football team's league position does not qualify. Designing such a dictionary is a trade-off between precision (only relevant articles) and recall (all relevant articles), and it is the step on which most of the paper's validation effort is spent.

The share. For each month, the number of GPR articles is divided by the total number of articles published. The division matters more than it looks. Newspapers grow and shrink; their archives are digitised unevenly; the number of articles a newspaper publishes in a month changes for reasons that have nothing to do with geopolitics. A raw count would rise whenever the archive grew. The share does not care how large the newspaper is, only what fraction of its attention went to war and terror.

The normalisation. Finally the share is rescaled so that its average over 1985–2019 equals 100. The historical index is normalised to its own average over 1900–2019. A reading of 300 therefore means that, in that month, the share of articles discussing geopolitical threats and acts was three times its average over the base period. It does not mean that risk was three times higher in any sense a probability could express; it means that attention was.

The misconception to drop now. A high GPR reading does not mean that risk is high *for your firm*. It means that the anglophone press was writing a great deal about geopolitical tension that month. The second statement is what the index measures; the first is an inference that requires everything else in this course.

1.5.3 Threats and acts

The eight categories are split into two groups. The first five — war threats, peace threats, military build-ups, nuclear threats, terror threats — form the **GPR Threats** sub-index. The last three — the beginning of war, the escalation of war, terror acts — form **GPR Acts**. The split is the most useful thing in the paper for a firm, because the two need not move together. Threats can build for months before anything happens, and they can subside without anything happening at all. Acts can arrive with almost no prior threat, as a terrorist attack usually does.

The authors report that threats, and not only realised acts, depress economic activity. For a firm, that finding has a blunt implication: you do not need the war to pay its cost. If what moves investment is the threat, then the cost is borne in the months of tension, by firms that delay, hedge and wait, whether or not the feared event ever occurs.

From the archive · The Cuban missile crisis, 1962

In mid-October 1962, photographs taken by American U-2 reconnaissance aircraft showed Soviet medium-range ballistic missile sites under construction in Cuba. On 22 October President Kennedy announced the discovery in a televised address

and a naval "quarantine" of the island. For the next six days the two nuclear powers negotiated in public and in private while ships approached the quarantine line. On 28 October Khrushchev announced that the missiles would be dismantled and returned to the Soviet Union; the United States pledged not to invade Cuba and, privately, agreed to withdraw its own Jupiter missiles from Turkey.

No war began. No shot was exchanged between the principal forces. In the categories of the GPR dictionary, the crisis is almost entirely *threats*: nuclear threats, military build-ups, war threats. It is also one of the episodes at which the historical index spikes, which is why the practice round asks you to annotate it alongside the two World Wars and Korea.

The episode makes two points at once. First, it is a case in which the index is doing exactly what its definition promises — registering the threat of adverse events, not only their realisation. Second, it is a case in which the risk was arguably greatest at a moment, not over a month: the crisis lasted thirteen days. A monthly index records it as one high observation. A firm deciding whether to load a ship that week needed a finer clock.

1.5.4 Country-specific indices

For several dozen countries, the authors also publish country-specific series: the same search, restricted to articles that also mention the country. In the monthly file these are the `GPRC_` columns, one per country identified by its three-letter ISO code. They are not normalised to 100; they are the percentage of articles that both discuss geopolitical risk and mention the country. They measure, in other words, how much the anglophone press associates a given country with geopolitical tension. That is useful, and it is also a very particular thing: a country can be the *source* of tension, its *victim*, its *mediator* or merely its *location*, and the count does not distinguish them.

1.6 Does it measure something real?

A text-based index is a hypothesis about the world: that the frequency of certain words in certain newspapers tracks a phenomenon of interest. The paper tests the hypothesis in three ways, and reading them is the best training this course offers in how to evaluate an index you did not build.

The first is **narrative**. The index should spike when, on any reasonable reading of history, the world's geopolitical tension spiked: at the outbreak of the two World Wars, the Korean War, the Cuban missile crisis, the Gulf War, the attacks of 11 September 2001, the invasion of Iraq in 2003, the Russian invasion of Ukraine in 2022. It does. This is a necessary test and a weak one: any index that failed it would be discarded, but many indices would pass it, including some that measure something other than what is claimed.

The second is **audit**. The authors drew samples of articles flagged by the dictionary and had them read by people, to estimate how often the algorithm counted an article that a human would not consider to be about geopolitical risk, and how often it missed one that a human would. A text measure whose false-positive rate is unknown is a measure whose movements cannot be interpreted; the audit is what licenses the claim that the counted articles are, for the most part, about what they are supposed to be about.

The third is **economic content**. If the index measures something real, it should be related to economic outcomes in the ways theory predicts. The paper finds that higher geopolitical risk predicts lower investment and employment, and that the effect is stronger in industries more exposed to geopolitical risk. That is the bridge to the firm, and the subject of the next section.

None of the three tests establishes that the index measures *risk* in Knight's sense — a probability of adverse outcomes. What they establish is that it measures *attention to geopolitical adversity*, that this attention is concentrated in the right months, that the counted articles are mostly about the right things, and that the attention is associated with economic decisions. That is a considerable achievement. It is also a narrower one than the word "risk" in the name suggests, and the gap between the two is where the misuse of the index lives.

1.7 Why waiting has value

Why should a firm's investment fall when the newspapers write about war, if the war has not yet begun and may never begin? The answer comes from the theory of irreversible investment under uncertainty, and it is one of the few results in economics that every manager already knows intuitively.

The real-options channel — when an investment is costly to reverse and its payoff depends on information that will arrive later, the opportunity to wait has value; an increase in uncertainty raises that value and so leads firms to postpone investment, hiring and entry, even if expected returns are unchanged.

Bernanke (1983) made the argument for cyclical investment: a firm choosing between investing now and waiting for news gives up some return by waiting, but avoids the possibility of having sunk capital into a project that the news would have shown to be a mistake. Because the investment is irreversible — a plant cannot be un-built at its cost — the bad outcomes weigh more heavily in the decision than the good ones, and more uncertainty about the bad outcomes makes waiting more attractive. Dixit and Pindyck (1994) gave the idea its general treatment, borrowing the mathematics of financial options: an unexercised investment opportunity is like a call option, and exercising it early destroys the option's time value.

Bloom (2009) turned the theory into a macroeconomic prediction and tested it: a spike in uncertainty should produce a rapid drop in hiring and investment as firms pause, followed by a rebound as the uncertainty resolves and pent-up decisions are taken. The pause-and-rebound pattern is the fingerprint of the real-options channel, and it is what distinguishes uncertainty effects from a simple loss of confidence about the level of future demand. Political uncertainty has been shown to work the same way: Julio and Yook (2012) find that corporate investment falls in the period before national elections, when the identity of the next government is unknown, and recovers after it.

Geopolitical risk is a natural candidate for the same mechanism, and it fits the threat-versus-act distinction exactly. A threat raises uncertainty about future states of the world without yet changing the present one; it is the purest form of the shock the real-options model describes. Caldara and Iacoviello's finding that threats depress investment is, read this way, an application of Bernanke's 1983 argument to a new source of uncertainty.

The firm-level implication is concrete. The decisions most exposed to geopolitical threats are those that are large, lumpy and hard to reverse: a greenfield plant, an acquisition, a long-term supply contract with take-or-pay terms, a decision to localise production for one market. The decisions least exposed are those that can be scaled up and down: inventory, temporary staff, contract manufacturing. A firm that wants to keep investing through a period of geopolitical tension is, in effect, paying to convert the first kind of decision into the second — leasing rather than building, dual-sourcing rather than committing, staging an investment in tranches with exit points. That conversion has a cost, and the measure of what the firm is willing to pay for it is a measure of how much it believes the tension is uncertainty rather than noise.

The classroom question that closes the lecture is this: in March 2022, with the GPR index several times its long-run average, you run a Canadian fertiliser exporter. What do you do differently this month? "Monitor the situation" is not an answer. A real answer names a decision — hedge the currency, delay the warehouse in Europe, reroute shipments, sign longer contracts with customers who have lost a Russian or Belarusian supplier — and names the threshold that would trigger it. The index can be an input to that threshold. It cannot be the threshold, because it says nothing about the exporter in particular.

1.8 From the system to the profit and loss account

The real-options channel explains why aggregate investment responds to geopolitical risk. A firm needs something more specific: the routes by which a change in relations between states reaches *its* costs, revenues and assets. This course uses five, and they return in every session.

Channel	How it reaches the firm	An example	Where it returns
Market access	tariffs, quotas, bloc-based trade diversion, procurement bans	a market closes to firms from a rival bloc	Session 04
Inputs	a critical input or service controlled by a foreign state or a single provider	a critical service from one satellite operator	Session 11

Channel	How it reaches the firm	An example	Where it returns
Finance and payments	sanctions on banks, asset freezes, exclusion from payment networks	a correspondent bank exits a jurisdiction	Session 10
Assets and people	conflict, expropriation, evacuation, detention	staff evacuated, a plant abandoned	Sessions 03, 05
Data and operations	state-sponsored cyber operations, data-localisation demands	operations halted by an intrusion	Session 08

The channels matter because the same geopolitical event reaches different firms through different channels, and therefore with different magnitudes and different timing. A war in a grain-exporting region reaches a food processor through *inputs*, a shipping line through *assets and people* and the insurance market, a bank through *finance and payments*, and a software firm through *data and operations* when the conflict spills into cyberspace. A global index records one event. A firm's exposure is a vector, and the index is a scalar.

The channels also tell you which data to look for. Market-access exposure is visible in the geographic breakdown of revenue that most listed firms publish. Input exposure is visible, if at all, in the supply-chain maps that firms are increasingly asked to produce by regulators and customers. Financial exposure sits in the treasury: which banks, which currencies, which payment routes. Asset exposure sits in the fixed-asset register and the list of staff posted abroad. Data exposure sits with the chief information security officer. A geopolitical risk assessment that does not ask each of these offices a question is a geopolitical risk assessment written from the newspaper.

1.9 What the index cannot see

The practice round ends every week with "the catch": one sentence saying what the figure you have just rebuilt does *not* license a multinational to conclude. This session supplies the template, and the GPR index supplies at least four limits.

The first is that it measures **attention in the anglophone press**. The newspapers are American, British and Canadian. Their editors decide what is news for their readers, and their readers are not in Abidjan or Shenzhen. A violent dispute that dominates the regional press of West Africa can leave the index untouched; a crisis involving the United States can dominate it. The index is therefore partly a measure of what is geopolitically salient *from the perspective of three allied countries* — which is a legitimate thing to measure, and which is not the same as the risk faced by a subsidiary elsewhere.

The second is that it is **global**. One number per month for the whole world cannot distinguish a risk that falls on the Baltic from one that falls on the Taiwan Strait, and for most firms that distinction is the whole question. The country indices help, but only for the countries the newspapers write about often enough to produce a stable series, and only in the sense of "mentioned together with tension".

The third is that **attention fades while risk persists**. Newspapers cover change. A frozen conflict — an unresolved territorial dispute, a ceasefire line, a sanctions regime in its tenth year — can be a constant and material risk for firms operating near it while generating ever fewer articles. On the index, a persistent risk looks like a declining one. This is the most important limit for a firm, because firms often hold assets for longer than editors hold attention.

The fourth is that the index is **not a forecast**. It rises with events and with threats as they are reported. It does not tell you what the next event will be, or when. Its predictive content for investment arises because firms *respond* to the same reported threats; that is a statement about behaviour, not about the probability of war.

A fifth, which the lecture invites you to supply, can be drawn from the construction itself. The denominator is all articles: if newspapers devote their space to an election or a pandemic, the geopolitical share can fall with no change in geopolitics. Each such limit is a reason to triangulate.

1.10 How a concept becomes data

The practice round follows a sequence that recurs in every session. A concept — geopolitical risk, defined in words — becomes an *operational definition*: the eight categories of the dictionary, the ten newspapers, the share, the base period. The operational definition becomes a *dataset*: a spreadsheet, published by the authors, with one row per month and columns for the recent index, its threats and acts components, the historical index and the country series. The dataset becomes a *figure*. And the figure becomes an *argument*, the moment someone puts it on a slide in front of a board.

Every link in that chain is a place where meaning can be lost. Between concept and operational definition, everything the dictionary does not capture is dropped. Between operational definition and dataset, the series is revised — the index is updated monthly and the authors refine it, so the values you download need not equal the values printed in the paper. Between dataset and figure, choices of frequency, smoothing and axis change what the eye sees: the same series averaged by year loses most of its spikes, because a geopolitical shock is a monthly phenomenon and a yearly average buries it. And between figure and argument, the slide's title replaces the paper's caveats.

That is why every practice round in this course has the same rule: the deliverable documents at least one place where the data disagreed with the paper. The disagreement is not a failure of the replication. It is evidence that you looked.

1.11 A short review of the literature

The distinction between measurable risk and unmeasurable uncertainty is due to Knight (1921), and it remains the vocabulary of the field even where the measurement it describes has moved on. In international business, political risk became a subject in its own right in the 1960s and 1970s, when expropriations were frequent; Kobrin (1979) is the synthesis that defined it as a property of the firm–environment relationship rather than of the environment alone, and that distinguished macro from micro risk. Later work moved from case-based assessment to measurement: Henisz (2000) built an index of political constraints on policy change from the structure of political institutions, and Jensen (2003) showed that democratic regimes attract more foreign direct investment, consistent with their greater credibility towards investors. The expert-based tradition is summarised in the methodology of the Worldwide Governance Indicators (Kaufmann, Kraay, and Mastruzzi 2011). Bremmer and Keat (2009) wrote the practitioner's case for treating political knowledge as an investment input.

The economics of why uncertainty matters for investment runs from Bernanke (1983) through the real-options synthesis of Dixit and Pindyck (1994) to the empirical work of Bloom (2009), who showed that uncertainty shocks produce a sharp fall and rebound in activity. Julio and Yook (2012) documented the same mechanism for political uncertainty around elections.

The measurement of uncertainty from text is the most recent strand. Baker, Bloom, and Davis (2016) built the Economic Policy Uncertainty index by counting newspaper articles that combine terms for the economy, policy and uncertainty, and validated it by human audit — the template Caldara and Iacoviello (2022) followed for geopolitical risk, adding the split between threats and acts, a historical series to 1900, and country-specific indices. Hassan et al. (2019) moved the method from newspapers to the firm, measuring each company's political risk from the transcripts of its own earnings calls — the measure Session 07 replicates.

The through-line is a shift of the unit of analysis: from the host country, to the system, and back to the firm. The indices have become better at counting what is said. Whether what is said is what a firm is exposed to remains the open question.

1.12 Into the practice round

The practice round rebuilds the paper's central figure: the monthly GPR index since 1985 with its spikes labelled, then its historical counterpart since 1900, then the threats and acts sub-indices on one axis. The steps are in the brief that follows. What this chapter adds is the question to hold while you build.

When you place a reference line at 100, remember that it is the base-period average of a share of newspaper articles, and that the two series you plot are normalised to different base periods. When you annotate the spikes, check the months yourself rather than trusting any list, and ask which episodes were mainly threats and which mainly acts. When you compare your figure with the paper's, expect differences — the series is revised — and record one. And when you change the monthly view

to a yearly average, watch which spikes disappear, because that is a demonstration, on your own screen, that a measurement choice can erase the phenomenon the index was built to see.

Then write the catch. Not a criticism of Caldara and Iacoviello, whose index does what it says, but a sentence about a firm: what this figure does not license a multinational to conclude. The strongest answers will name a channel and a place — a supplier, a subsidiary, a payment route — that a global count of anglophone newspaper articles cannot see. That sentence is the first entry in the risk register your team will build for its multinational over the term, and the next eleven sessions are, in a sense, a sequence of instruments designed to see what this one misses.

1.13 In the practice

> **The target.** Caldara & Iacoviello (2022), Figure 1: the monthly GPR index since 1985, its spikes labelled with the events behind them. Then its historical counterpart since 1900.

Teams are not yet formed: today you work **in pairs**. Everyone builds their own workbook.

1.13.1 1 · Collect — 15 min

If you did the pre-session, `data_gpr_export.xls` is already in `data/raw/`. If not, download it now from matteoiacoviello.com/gpr.htm. Details: `../data/README.md`.

Start your **provenance log** — copy `templates/provenance-log.csv` into `data/` and fill one line:

file	source	url	version / release	accessed	licence	notes
data_gpr_export.xls	Caldara & Iacoviello	matteoiacoviello.com/gpr.htm	*the "last updated" date on the page*	*today*	free, cite AER 2022	*anything odd*

1.13.2 2 · Prepare — 15 min

1. Tableau Desktop → **Connect** → **Microsoft Excel** → `data_gpr_export.xls`.
2. Drag the data sheet (`Sheet1`) to the canvas. Look at the grid at the bottom. The last two columns, `var_name` and `var_label`, are the authors' notes, not data.
3. Check the type icon of `month`. It must be a **date** (calendar). If it is text (`Abc`) or a date-time, click the icon and change it to *Date*.
4. Check `GPR`, `GPRT`, `GPRA`, `GPRH` are **numbers** (`#`). Empty cells in early years of the recent index are expected: it starts in 1985.
5. Write in your log what Tableau got wrong on its own, if anything.

1.13.3 3 · Replicate — 30 min

1.13.3.1 Sheet 1 · "GPR since 1985"

1. `month` → **Columns**. Right-click the pill → choose the *continuous* **Month** (green pill, the second "Month" in the menu — the one showing *May 2015*, not *May*).
2. `GPR` → **Rows**. Keep the aggregation `SUM` — there is one row per month, so the sum is the value.
3. Filter `month` to January 1985 onward.
4. **Analytics** pane → drag **Reference Line** onto the view → *Table* → **Constant 100**, label "1985–2019 average = 100".
5. Find the five highest peaks. Right-click each → **Annotate** → **Point** and write the event: Gulf War (1991), 9/11 (2001), Iraq invasion (2003), Paris attacks (2015), Russia invades Ukraine

(2022) — but **check the months yourself**; do not trust this list.

6. Title the sheet the way the paper would, and put the source in a caption: *Source: Caldara & Iacoviello (2022), matteoiacoviello.com, accessed ...*

1.13.3.2 Sheet 2 · "GPR since 1900"

Duplicate Sheet 1, replace `GPR` with `GPRH`, remove the 1985 filter. Relabel the reference line: `GPRH` is normalised to its **1900–2019** average = 100, not 1985–2019. Annotate the two World Wars, Korea and the Cuban missile crisis.

1.13.3.3 Sheet 3 · "Threats and acts"

Put `GPRT` and `GPRA` on the same axis: drag `GPRA` onto the `GPRT` axis until the double-ruler icon appears (**Measure Values**). Colour by **Measure Names**.

1.13.4 4 · Compare and break — 20 min

Put your Sheet 1 beside Figure 1 of the paper.

1. **Where do they differ?** Your series runs further: the index is updated monthly, and the authors also revise it. Are past values exactly the paper's? Write down one difference.
2. **Threats versus acts.** Find one episode where threats rose long before acts did. What would a firm watching only *acts* have missed?
3. **Break one decision.** Replace the monthly view by a **yearly average** (change the pill to continuous *Year*, aggregation `AVG`). Which spikes disappear? What does that tell you about the frequency at which a risk dashboard should run?
4. **Extend.** Add a `GPRC_` country index for a country of your choice (Sheet 4). It is a *percent of articles*, not an index to 100: use a dual axis. Is the country index correlated with the global one? When does it diverge?

1.13.5 5 · The catch — 10 min

Build **one slide** (a Tableau dashboard exported as an image is fine) with three panels:

Original	Ours	The catch
Figure 1 of the paper	Your Sheet 1	One sentence: what this figure does **not** license a multinational to conclude

A pair drawn at random presents in two minutes.

1.13.6 Hand in

Upload to ZoneCours before leaving: your `.twbx` (packaged workbook, *File → Export Packaged*) and your provenance log. Today's round is **formative**: it is commented on, not marked, and it counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

1.13.7 What loses credit, from Session 02 onwards

- A chart with no source caption
- A provenance log with no version or access date
- "The data matched exactly" — it rarely does; look harder
- A catch that is a limitation of the *paper* rather than of what the *firm* may conclude

Chapter 2

The International Economic Order: Who Wrote the Rules, and What Happens When Their Share Shrinks

Session 2 · The system in which multinational firms operate Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Who wrote the rules multinationals play by — and what happens to the rules when the writers' share of the world shrinks?**

Big work	Baldwin, R. (2016). *The Great Convergence: Information Technology and the New Globalization*. Belknap Press of Harvard University Press.
Target figure	The **G7 share of world GDP**, 1970–today — the "great convergence" that begins around 1990
Data you collect	World Bank, World Development Indicators: GDP in current US$ (`NY.GDP.MKTP.CD`), GDP at PPP (`NY.GDP.MKTP.PP.CD`), trade in % of GDP (`NE.TRD.GNFS.ZS`) — DataBank WDI
Tableau skill	**Pivot** wide years to long · filter on ISO codes, not names · area chart · **percent of total**
Course themes	The system of international economic order in which multinational firms operate

2.1 The rules a firm does not notice

A multinational firm lives inside a set of rules it rarely thinks about. When a Canadian manufacturer ships components to a plant in Mexico, assembles them, and sells the result in Germany, it assumes a great deal: that the tariff it pays in Germany is the same as its competitors pay; that a dispute over that tariff will be settled by procedure rather than by retaliation; that the dollars it earns can be converted and moved; that the patent on the component will be respected in the country where it is assembled. None of these assumptions is a law of nature. Each was written, by particular states, at particular moments, in their own interest, and each is held in place by something — power, shared interest, or institutions with momentum of their own.

This chapter is about that set of rules: the international economic order. Its question is the one the session poses. Who wrote the rules multinationals play by, and what happens to the rules when the writers' share of the world shrinks? The answer has three parts. The first is historical: how the post-1945 order was built, in layers, by the United States and its allies. The second is economic, and comes from the session's big work, Richard Baldwin's *The Great Convergence*: how a fall in the cost of moving ideas after about 1990 moved industry and income from the countries that wrote the rules to a handful that did not. The third is political: why that convergence produced a backlash, and what the backlash means for a firm whose supply chain was built on the old assumptions.

2.2 What an order is

The word "order" is used loosely, so it is worth fixing.

International order — a set of rules, institutions and shared expectations governing relations between states, sustained by some distribution of power among them.

International economic order — the part of the international order that governs trade, money and investment: tariff commitments and their exceptions, exchange-rate and capital-account arrangements, the treatment of foreign investors and their property, and the procedures by which disputes about all of these are settled.

Three questions describe any economic order, and they are worth keeping in view for the rest of the course. *Who wrote the rules?* — which states designed the institutions, and in whose interest. *What do the rules say?* — how open they are to trade, to capital and to investment, and how disputes are resolved. *What holds them in place?* — whether it is the power of a dominant state, a common interest among many, or the inertia of institutions that are costly to abandon. The first question is about history, the second about law, and the third about politics. A firm's exposure to geopolitical risk is largely a matter of the third: rules that are held in place by one state's power are as durable as that power and that state's willingness to use it.

The third question has a classic answer in political economy, and a classic rejoinder. Kindleberger (1973), writing about the Great Depression, argued that the interwar collapse of trade and finance happened because no state was both able and willing to stabilise the system — Britain was able no longer, and the United States was not yet willing. An open world economy, on this view, needs a stabiliser: a state that keeps its market open in a slump, lends counter-cyclically, and acts as lender of last resort. Krasner (1976) generalised the point into what became known as hegemonic stability theory: openness is most likely when a single state is economically dominant, because the dominant state gains most from openness and has the means to enforce it. The rejoinder came from Keohane (1984), whose *After Hegemony* argued that institutions, once established, can sustain cooperation even as the hegemon that created them declines, because they reduce the costs of making and monitoring agreements and states have reasons to keep them. Whether the post-1945 order outlives the relative decline of its authors is the empirical test of that debate, and it is being run now.

Hegemonic stability — the proposition that an open international economy is most likely to be created and maintained when a single dominant state has both the capacity and the interest to supply its rules and absorb the costs of enforcing them.

2.3 The post-1945 order, in four layers

The order that multinationals take for granted was assembled in layers over half a century. It is easier to understand as a stack than as a single design.

From the archive · Bretton Woods, 1944

In July 1944, while the war in Europe and the Pacific was still being fought, delegates of forty-four allied nations met at the Mount Washington Hotel in Bretton Woods, New Hampshire. The two dominant figures were John Maynard Keynes for the United Kingdom and Harry Dexter White for the United States Treasury, and the conference largely ratified a compromise their governments had negotiated in advance. The British wanted a system that would protect a debtor country's freedom to pursue full employment; the Americans, who would be the world's creditor, wanted a system in which their money would not be used to finance others' deficits indefinitely.

The conference created the International Monetary Fund, to oversee a system of fixed but adjustable exchange rates and to lend to countries facing temporary balance-of-payments difficulties, and the International Bank for Reconstruction and Development, the core of what became the World Bank. Currencies were pegged to the dollar; the dollar was convertible into gold at a fixed price for foreign official holders. Capital controls were not merely tolerated but expected: the founders had watched short-term capital flows destabilise the interwar economy and did not intend to repeat the experience (Eichengreen 2019).

What is striking, read from the twenty-first century, is how much of the design reflected the balance of power in the room. The United States held most of the world's monetary gold and was the only major economy undamaged by the war. The institution's voting shares, its headquarters in Washington, and the central role of the dollar all followed. The rules were

multilateral in form and American in their foundations — which is the observation that every later debate about the order starts from.

The second layer was trade. The planned International Trade Organization, which was to have stood beside the Fund and the Bank, was never established: its charter was negotiated but not ratified by the United States. What survived was a provisional agreement, the **General Agreement on Tariffs and Trade** of 1947, which turned out to be one of the most durable provisional arrangements in history. The GATT rested on two principles. **Most-favoured-nation** treatment required each member to extend to all members any tariff concession it gave to one; **national treatment** required that imported goods, once across the border, be treated no worse than domestic ones. On that basis, successive negotiating rounds cut tariffs on manufactured goods among the industrial countries.

The third layer is not an institution but a bargain, and the political scientist John Ruggie gave it its name.

Embedded liberalism — the post-war compromise in which states committed to multilateral openness in trade while retaining the domestic policy autonomy — macroeconomic management, social insurance, capital controls — needed to make that openness politically sustainable at home (Ruggie 1982).

Ruggie's point was that the post-war order was not the liberalism of the nineteenth century restored. It was liberalism *embedded* in a domestic social settlement: governments would open their markets, but they would also cushion their citizens from the dislocations openness caused, and they would keep the instruments to do so. Free trade was managed; capital was controlled; full employment was a government responsibility. The bargain held for roughly a quarter of a century, and its unravelling is the background to most of what follows.

The fourth layer is the neoliberal turn and the creation of the World Trade Organization. The monetary pillar of Bretton Woods broke first.

From the archive · The end of gold convertibility, 1971

On 15 August 1971, President Nixon announced that the United States would suspend the convertibility of the dollar into gold for foreign official holders, and imposed a temporary surcharge on imports. The announcement was made without consulting the other members of the IMF. A realignment of exchange rates negotiated at the Smithsonian Institution in December 1971 tried to restore a system of pegs; by 1973 the major currencies were floating against one another, and they have floated since (Eichengreen 2019).

The episode is usually told as the end of Bretton Woods, and in its monetary design it was. But it is more instructive as a demonstration of the third question — what holds the rules in place. The system had depended on the United States' willingness to subordinate its own policy to the gold parity. When that willingness ended, the rule ended, unilaterally, on a Sunday evening. The institutions survived: the IMF found new roles, and the dollar remained the centre of the system, now without the anchor. The order changed form without collapsing, which is the pattern to look for whenever someone announces the end of an order.

Through the 1980s and 1990s, capital controls in the industrial countries were dismantled, and many developing countries followed. In trade, the Uruguay Round of negotiations, launched in 1986, ended with the creation of the **World Trade Organization** in 1995. The WTO differed from the GATT in two ways that matter for firms. Its rules reached far beyond tariffs on goods — into services, into intellectual property through the TRIPS agreement, into investment measures through TRIMs — so that the order began to govern not only what crossed borders but what happened behind them. And it had a binding dispute-settlement mechanism, with an Appellate Body, so that rulings could no longer be blocked by the losing party. China joined in 2001, after fifteen years of negotiation, on terms that bound it to many of these rules.

How should the result be characterised? Ikenberry (2011) argues that the American-led order was **easy to join and hard to overturn**: open, rule-based, and institutionalised enough that rising states would gain more by joining it than by challenging it. On this reading, the order is not merely an expression of American power but a set of institutions that bind the hegemon too, and that therefore give others a reason to accept it. Session 03 sets the realist reply beside this view: that institutions reflect the distribution of power and will be revised when it shifts.

2.4 Firms as authors as well as subjects

It would be a mistake to read the order as something states wrote and firms merely obeyed. At several points, firms were among its authors, and the layers that matter most to multinationals today are the ones in which their influence was greatest.

The clearest case is intellectual property. Before the Uruguay Round, patents, copyrights and trademarks were governed by conventions administered outside the trade regime, with weak enforcement and wide latitude for national policy; many developing countries did not grant patents on pharmaceuticals at all. The TRIPS agreement changed that, by making minimum standards of intellectual-property protection a condition of WTO membership, enforceable through the dispute-settlement mechanism. Sell (2003) traces how this happened: a small coalition of chief executives from American pharmaceutical, software and entertainment firms organised in the 1980s, defined the problem as one of trade rather than of intellectual property, persuaded the United States government to adopt their framing, and worked with counterparts in Europe and Japan to build a joint position. Twelve corporations, in her account, effectively drafted the core of a public international law. The episode is the strongest evidence that the post-1945 order was not only American in its foundations but, in some of its later layers, corporate.

A second case is the protection of foreign investment, which the GATT and the WTO largely left aside. Here the instrument was the bilateral investment treaty. The first was signed between West Germany and Pakistan in 1959; the Convention establishing the International Centre for Settlement of Investment Disputes, under World Bank auspices, followed in 1965; and in the 1990s the number of such treaties multiplied. Their distinctive feature is that they allow a foreign investor to bring a claim directly against a host state before an international tribunal, without needing its own government to take up the case. For a multinational, this is the layer of the order that most directly answers Kobrin's political risk (Session 01): it is the legal response to expropriation and discriminatory treatment. It is also the layer that has attracted the most political contestation in the 2010s, from governments that came to see investor-state arbitration as a constraint on regulation — Rodrik's trilemma, applied to a single clause.

The point for this course is twofold. First, the rules a firm relies on are not all of a piece: some are multilateral (tariff bindings), some are bilateral (investment treaties), some are private or semi-private (standards, the subject of Session 06). They are held in place by different things and they will fail in different ways. Second, if firms helped write some of the rules, they have a stake — and sometimes a voice — in how they are rewritten. A firm's government affairs function is, in this sense, part of its risk function: the influence a firm exercises over the rules is one of the few geopolitical exposures it can partly manage from the inside.

2.5 Three costs, three unbundlings

Baldwin's book offers a different way into the same history, one that starts not from institutions but from costs. Globalisation, he argues, has been shaped by three successive constraints: the cost of moving **goods**, the cost of moving **ideas**, and the cost of moving **people** (R. Baldwin 2016). Each time one of these costs fell sharply, it "unbundled" something that geography had previously held together.

Unbundling — in Baldwin's account, the separation, made possible by a fall in a particular cost of distance, of activities that had previously had to take place in the same location: first production from consumption, then the stages of production from one another.

2.5.1 The first unbundling: goods move, ideas stay

From the nineteenth century, steam ships and railways made it cheap to move goods. Production no longer had to happen near consumption: a factory in Manchester could clothe customers in India. But the cost of moving ideas remained high, because coordinating complex production still required people to be close to one another. So industry clustered — in Britain, western Europe, North America, later Japan — where know-how could circulate cheaply within a local area. The result was the great divergence: the industrialising North pulled away from the rest of the world in income, because the knowledge that made factories productive stayed where it was invented. Trade grew, but it was trade in finished goods between nations, and the comparative advantage that governed it was a national property.

2.5.2 The second unbundling: ideas move, inside the firm

From about 1990, information and communication technology made it cheap to move ideas — to coordinate complex activities at a distance. The factory itself could now be unbundled. Stages of production that had to sit together in one place could be dispersed across countries, each stage located where its costs were lowest, and coordinated from headquarters by telecommunications and computers. A firm could combine its own technology, management and marketing with low-wage labour in a nearby developing country — for the United States, Mexico; for Germany, Poland and central Europe; for Japan, the rest of East Asia.

The crucial move in Baldwin's argument is what travelled with the stages of production. When a Japanese firm moved assembly to Thailand, it did not move only the assembly. It moved the know-how to perform it: the process engineering, the quality systems, the managerial practice. Know-how that had been confined within national borders now flowed across them — but inside the firm's own production network, under its control. This is why the second unbundling produced convergence rather than divergence. A handful of developing economies, those that joined the supply chains of northern firms, industrialised at a speed previously unknown, because they did not have to invent the know-how; they had it brought to them.

Global value chain — the international fragmentation of a production process into stages located in different countries and coordinated, usually by a lead firm, so that what crosses borders is increasingly intermediate goods, services and know-how rather than finished products.

Baldwin's empirical picture of the shift emphasises its regional character. Supply-chain trade, R. Baldwin and Lopez-Gonzalez (2015) show, is organised around a few hubs — "Factory North America", "Factory Europe", "Factory Asia" — each pairing high-technology headquarter economies with lower-wage neighbours. The second unbundling was less a globalisation than a set of regionalisations, linked by the headquarters economies.

2.5.3 The third unbundling: people, virtually

The third constraint is the cost of moving people, and specifically of face-to-face interaction. Baldwin's prospective third unbundling would come from telepresence and telerobotics: technologies that let a worker in one country perform services in another without travelling. If it arrives at scale, competition would reach occupations rather than sectors, and services would be traded as tasks, the way manufacturing stages are now. It is a forecast rather than a finding, and it is the least developed part of the argument.

2.5.4 The multinational as the vehicle

For this course, the key point of the book is a single sentence's worth of argument: in the second unbundling, *the multinational firm is the vehicle of convergence*. The convergence of incomes between the old industrial economies and the new ones did not happen through anonymous markets in finished goods. It happened through supply chains that multinational firms built, owned or controlled, along which they moved technology and organisational capability.

That is why geopolitics now reaches firms so directly. If know-how, jobs and capabilities travel along supply chains, then supply chains are where states will intervene when they decide that the movement of know-how threatens their security. The export controls of the 2020s on advanced semiconductors and the equipment to make them are attempts to stop exactly the flow the second unbundling made cheap. The firm's supply chain has become a geopolitical object, not because firms chose to make it one, but because it carries what states care about.

2.6 The target figure, and what it measures

Baldwin's evidence for the convergence is a picture: the share of the G7 — Canada, France, Germany, Italy, Japan, the United Kingdom and the United States — in world income, which rose for over a century and then fell sharply from around 1990. The share lost by the G7 was gained mostly by a small group of developing economies, which he calls the "Industrialising Six": China, Korea, India, Poland, Indonesia and Thailand. That figure is the session's replication target.

The replication uses the World Bank's World Development Indicators, and building it forces three decisions, each of which matters for interpretation.

The first is that **the G7 is not in the data**. The World Bank publishes the world total and many aggregates — income groups, regions, the euro area — but no G7. You must construct it by summing the seven members, and you must do so by their ISO codes rather than their names, because names change between releases and between datasets and codes are designed not to. The G7 is a political grouping, and any measure of it is a construction by the analyst.

The second is the **unit of account**. The target figure uses GDP in current US dollars at market exchange rates. That is the natural measure if the question is purchasing power on world markets — what an economy's output can buy abroad, how large it looks to an exporter selling in dollars. But it means that the share moves with the dollar. When the dollar strengthens, every other economy shrinks in dollar terms and the US share rises, without any change in what anyone produces. Part of any year-to-year movement in the series is the exchange rate.

Purchasing-power parity (PPP) — a conversion of national output into a common unit using the relative prices of a common basket of goods and services rather than market exchange rates, so that an economy's output is valued at what it would cost in a reference country.

The alternative is GDP at purchasing-power parity. Because many goods and services — especially non-traded services — are cheaper in lower-income economies, converting their output at market exchange rates understates its real volume. At PPP, the economies of China and India are much larger relative to the G7 than they are at market rates, and the G7's share is correspondingly smaller. The World Development Indicators' PPP series begins in 1990, so the comparison covers only the convergence period.

Which measure is right depends on the decision, which is the point the lecture puts to the room. If a board is deciding where the purchasing power for its *exported* product sits — how many dollars or euros a market can pay — market exchange rates are the relevant measure. If it is deciding where real production capacity and local markets for locally made goods are, PPP is closer. A firm that shows its board only one of the two has made a strategic choice without stating it. Measurement choices are strategy choices.

The third decision is about **who is in the denominator**. The world total is a World Bank aggregate that includes economies for which data are sparse or estimated, particularly in the early decades. Some members of the Industrialising Six lack values for parts of the 1970s and 1980s. A stacked chart of shares — G7, I6, rest of the world — will therefore show the "rest of the world" absorbing whatever is missing from the I6, which is an artefact of the data, not a fact about the world. The practice round asks you to find it and record it.

2.7 From convergence to backlash

The convergence was, by most measures of global welfare, a remarkable success: incomes in the economies that joined supply chains rose at a pace without historical precedent, within a single generation. It was also politically destabilising in the countries whose share was shrinking, and the most useful framework for understanding why comes from Dani Rodrik.

Rodrik's trilemma — the proposition that a country can have at most two of three things: deep economic integration with the world economy, national sovereignty in policy-making, and democratic politics (Rodrik 2011).

The logic is that deep integration requires harmonised rules — on products, on investment, on capital flows — and those rules constrain what national governments can do. If the nation-state keeps its sovereignty and pursues deep integration, democratic choice must be constrained: governments adopt whatever policies markets reward, a condition Rodrik describes with Thomas Friedman's phrase, the "golden straitjacket". If integration and democracy are both kept, the rules must be made democratically above the state — a global federalism no one has built. If sovereignty and democracy are kept, integration must be shallow: the Bretton Woods compromise, with managed openness and room for national policy.

The trilemma is a reading of the history in the previous sections. Embedded liberalism was the third option: shallow integration that left room for democracy and sovereignty. The neoliberal turn and the WTO pushed towards deep integration — Rodrik calls it hyperglobalisation — and, for a while, the countries concerned accepted the straitjacket. The backlash of the 2010s and 2020s — Brexit, the tariff wars that began in 2018, the return of industrial policy, export controls, investment screening, "de-risking" — can be read as a sequence of choices of sovereignty and democracy over deep integration.

The economics behind the politics became clearer with evidence on the distribution of the gains. Autor, Dorn, and Hanson (2013) showed that US local labour markets more exposed to rising Chinese import competition after 2000 suffered larger falls in manufacturing employment and wages, and higher take-up of public transfers, with effects that persisted. Trade theory had always predicted losers as well as winners; what the "China shock" literature showed was that the losers were geographically concentrated, that adjustment was slow, and that the compensation the embedded-liberal bargain had promised did not reach them. A bargain that is not honoured stops being a bargain.

2.7.1 Slowbalisation

The phenomenon that the press calls *slowbalisation* is visible in a single series: world trade as a percentage of world GDP. It rose steeply from the early 1990s to 2008, fell in the financial crisis, and has broadly flattened since. Whether the flattening is a retreat from globalisation or a change in its form is contested. A retreat would show up as firms reshoring and supply chains shortening. A change of form would show up as trade in goods stagnating while trade in services and cross-border data flows — much of which is poorly measured in trade statistics — continue to grow. The practice round's *extend* step asks you to plot the series and judge it; the honest answer is that the series alone cannot settle the question, because its denominator, GDP, has shifted towards services, which are traded less, in exactly the period in question.

2.7.2 The dispute mechanism

One institutional fact belongs in every firm's risk register. The WTO's Appellate Body has been unable to hear appeals since December 2019, when the terms of its members expired and the United States blocked the appointment of replacements. A panel ruling can now be appealed "into the void", where no body exists to hear it, and so never become binding. The dispute-settlement mechanism that distinguished the WTO from the GATT therefore functions only partially. For a firm, that means that a discriminatory measure by a trading partner is now more likely to be resolved by bilateral bargaining, retaliation or negotiation between governments than by a legal ruling — that is, by power rather than by procedure.

2.8 Interdependence as an instrument

The first two parts of this chapter describe an order designed to make interdependence safe. The third must add that interdependence was never only an economic fact. Albert Hirschman, writing in 1945 about Nazi Germany's trade with south-eastern Europe, showed that trade relations could be structured deliberately to create dependence: by trading with small partners for whom the trade was a large share of their total, a great power could make the threat of interrupting it a source of political influence (Hirschman 1945). The gains from trade, in his account, were also the measure of the power the trade conferred — the cost a partner would bear if it stopped.

Farrell and Newman (2019) brought the argument into the age of networks. The infrastructure of globalisation — financial messaging, dollar clearing, internet backbones, cloud platforms — is not flat; it has hubs, and the hubs sit disproportionately in a few jurisdictions. States that control the hubs can use them for surveillance, observing what flows through, and for exclusion, cutting adversaries off. Interdependence, on this reading, has been "weaponised": the very networks that the economic order built become instruments of coercion. Sessions 08, 10 and 11 develop the argument for cyber, sanctions and space. Its relevance here is that the order's openness and its vulnerability are the same thing seen from two sides.

For a multinational, the practical form of weaponised interdependence is jurisdictional reach. A firm that clears payments in dollars, stores data with a provider headquartered in one country, or uses components containing technology controlled by another, has placed part of its operations under that country's law, wherever the firm itself is incorporated and wherever the transaction takes place. The order built after 1945 made it cheap and natural to route business through American banks, American platforms and American technology; the same routing now carries American regulatory authority with it, and other states — China, the European Union — have begun to build their own instruments of reach in response. A firm's exposure to the rules of the international economic order is therefore not only a matter of where it sells and produces. It is also a matter of which networks it runs through, and whose law travels along them. That is a question few firms asked when the networks were built, because the order was designed to make it unnecessary.

2.9 What it means for the multinational

The history and the economics lead to four implications that a firm's strategy team can act on.

Your market is no longer where the rules are written. For most of the post-war period, the largest markets and the rule-makers were the same countries. The convergence separated them. A European firm now earns a growing share of its revenue in markets whose governments did not design the WTO, did not participate in Bretton Woods, and do not feel bound by the spirit — as distinct from the letter — of rules they joined late. The firm's legal environment and its commercial environment have come apart.

Your supply chain is a geopolitical object. If the second unbundling moved know-how along supply chains, then supply chains are where governments now intervene to stop it: export controls, outbound investment screening, restrictions on technology transfer, requirements to locate certain stages of production in trusted jurisdictions. The question for the firm is no longer only where a stage is cheapest, but where it is permitted, and by whom.

Multilateral protection is weaker. With the Appellate Body unable to hear appeals, the guarantee that a trading partner's discriminatory measure could be challenged and removed through procedure has weakened. Firms that relied implicitly on that guarantee — exporters into markets with a history of trade remedies, for example — face disputes settled more often by negotiation and retaliation. The firm's government's bargaining power has become part of the firm's risk profile.

Measurement choices are strategy choices. At market exchange rates, the G7 still looks large; at PPP, it looks much smaller. Which figure a board sees will shape where it builds, where it sells, and how it assesses the bargaining position of the governments it deals with. A risk analyst's first duty is to show both and say which question each answers.

Mapped onto the five channels of this course, the order's transformation runs mainly through **market access** — the tariffs, trade remedies and bloc-based diversion of Session 04 — and through **inputs**, where the know-how and components that the second unbundling dispersed are now subject to control. But it reaches **finance and payments** too, because the dollar-centred monetary system that survived 1971 is the platform on which financial sanctions operate.

2.10 What the figure does not license

The target figure is a powerful image, and its power is a reason for care. It does not license four conclusions that boards and commentators routinely draw from it.

A falling G7 share is **relative**. The G7 economies kept growing in absolute terms through the whole period; the denominator grew faster. A falling share does not mean the G7 market shrank, and a firm selling into the G7 did not lose customers because the line went down.

At current US dollars, the share **moves with the dollar**. Part of any year-to-year change is the exchange rate, not production. A strong dollar flatters the United States and with it the G7, and a reading in any single year can say as much about currency markets as about convergence.

Convergence is **concentrated**. Most of the gain went to a handful of economies, and above all to China. "The South" did not converge; specific economies that joined specific supply chains did. A firm that reads the figure as a story about emerging markets in general will over-invest in the markets that did not participate.

It shows **income**, not **rule-making power**. Whether a shrinking share of world GDP means a shrinking ability to set the rules is a question about the relation between economic size and political influence — the subject of Session 03. Hegemonic stability theory predicts that it does; institutionalists predict that rules can outlast the share. The chart cannot settle the dispute, because it does not measure the thing in dispute.

2.11 A short review of the literature

The political economy of the post-war order begins with the hegemonic-stability debate. Kindleberger (1973) attributed the interwar collapse to the absence of a stabilising power; Krasner (1976) generalised the argument to the structure of trade; Keohane (1984) replied that institutions can sustain cooperation after hegemony. Ruggie (1982) named the domestic bargain — embedded liberalism — on which post-war openness rested, and Ikenberry (2011) reinterpreted the American-led order as liberal, institutional and self-binding, and therefore "easy to join and hard to overturn". The monetary history, from Bretton Woods through the end of gold convertibility, is told by Eichengreen (2019).

The economics of the new globalisation is organised by R. Baldwin (2016) around falling costs of moving goods, ideas and people, with the multinational's supply chain as the conduit for the second unbundling; R. Baldwin and Lopez-Gonzalez (2015) map the regional structure of supply-chain trade. The distributional consequences in the rich countries are documented most influentially by Autor, Dorn, and Hanson (2013), and their political consequences are read through Rodrik (2011)'s trilemma of integration, sovereignty and democracy.

A third strand treats economic interdependence as a source of power rather than only of gains. Hirschman (1945) showed how asymmetric trade could be designed to create political dependence; Farrell and Newman (2019) extended the argument to the networked infrastructure of finance and communications, introducing "weaponised interdependence" — the strand that Sessions 08, 10 and 11 follow.

The through-line is that every account of the order answers the same question — what holds the rules in place — and that the answers differ chiefly in how much weight they give to the distribution of power. The great convergence is the event that puts that weight to the test.

2.12 Into the practice round

The practice round rebuilds Baldwin's figure from the World Bank's data: the G7's share of world GDP in current dollars, from 1970 to the latest year, followed by a stacked picture of who gained. The steps are in the brief that follows. What this chapter adds is a set of questions to carry into it.

When you construct the G7, remember that you are building a political aggregate that the statistics do not contain, and that you build it by code rather than by name. When you find the peak and the latest value, ask how much of the movement between any two years is the dollar. When you switch to PPP, notice that the series starts in 1990 — just as the convergence begins — and ask which board decision each line informs. When the Industrialising Six have missing values in the early decades, record what that does to the "rest of the world", because that is your first place where the data disagree with the book. And when you plot world trade as a share of GDP, ask whether the flattening after 2008 is a retreat or a change of form, and what evidence would tell the two apart.

Then write the catch, about your team's firm rather than about Baldwin. A falling G7 share does not tell a multinational that its G7 customers are poorer, that the emerging markets in general are richer, or that the rules it relies on have already changed. It tells it that the economies that wrote the rules now produce a smaller share of the world's income than when they wrote them — and that whether the rules follow the income is the question the next session takes up.

2.13 In the practice

> **The target.** Baldwin (2016): the **G7 share of world GDP**, rising for a century and falling sharply after about 1990. You rebuild it from 1970 with World Bank data, then ask what changes when the dollars change.

2.13.1 0 · Form your team — 10 min

Teams of three or four, for the rest of the term. Agree on **one multinational** from your shortlists and register it on ZoneCours (first come, first served). Write the team name and firm at the top of your provenance log.

2.13.2 1 · Collect — 10 min

If you did the pre-session, your DataBank extract is in `data/raw/`. If not, build it now — `../data/README.md`. One extract per team is enough; everyone builds their own workbook from it.

Copy `templates/provenance-log.csv` into `data/` and fill one line per file. For WDI, the **version** is the "Last Updated" date DataBank shows for the database.

2.13.3 2 · Prepare — 20 min

1. **Connect → Microsoft Excel →** your extract. Drag the data sheet to the canvas.
2. If there are notes rows at the bottom ("Data from database…", "Last Updated…"), tick **Use Data Interpreter**, or filter them out later. Write which you did.
3. **Pivot the years.** Click the first year column, shift-click the last → right-click → **Pivot**. Two new columns appear.
4. Rename `Pivot Field Names` → `Year`, `Pivot Field Values` → `Value`.
5. `Year` is text like `1990 [YR1990]`. Create a calculated field from the data source pane: `Year (n)` = `INT(LEFT([Year], 4))`. (If your extract has plain `1990`, just change the type to *Number (whole)*.)
6. `Value` is text — DataBank writes missing data as `..`. Change its type to **Number (decimal)**; the dots become nulls. Check the null count: is it plausible?
7. Look at `Country Code` and `Series Code` (in the Route B CSV the second is `Indicator Code`, the years are plain `1990`, and missing values are blank). **From now on you filter on codes, never on names.**

2.13.4 3 · Replicate — 25 min

2.13.4.1 Sheet 1 · "The G7 share of world GDP"

1. Filters: `Series Code` = `NY.GDP.MKTP.CD` only.
2. Create two calculated fields:

```
G7 GDP     = IF [Country Code] IN ("CAN","FRA","DEU","ITA","JPN","GBR","USA") THEN [Value] END
World GDP  = IF [Country Code] = "WLD" THEN [Value] END
```

3. And the share:

```
G7 share   = SUM([G7 GDP]) / SUM([World GDP])
```

   Format it as a percentage (right-click → *Default Properties* → *Number Format*).

4. `Year (n)` → Columns as a **continuous dimension** (green); `G7 share` → Rows.
5. Annotate the peak year and the latest year with their values. Title the sheet with the finding, not the variable. Caption: *World Bank, WDI, NY.GDP.MKTP.CD, last updated …, accessed ….*

2.13.4.2 Sheet 2 · "Who gained?" — a stacked area of world shares

1. Two more calculated fields:

```
I6 GDP              = IF [Country Code] IN ("CHN","KOR","IND","POL","IDN","THA") THEN [Value] END
Rest of world GDP   = SUM([World GDP]) - SUM([G7 GDP]) - SUM([I6 GDP])
```

2. `Year (n)` → Columns. Drag `G7 GDP`, `I6 GDP` and `Rest of world GDP` onto Rows as **Measure Values** (drop the second and third onto the first axis).
3. **Marks → Area**. `Measure Names` → Colour.
4. Right-click `Measure Values` → **Quick Table Calculation → Percent of Total**. Then **Edit Table Calculation → Compute Using → Measure Names**, so that the three shares sum to 100% **within each year** — not across years. Check one year by hand against Sheet 1.

2.13.5 4 · Compare and break — 15 min

1. **Compare.** Put Sheet 1 beside Baldwin's figure. Where does the curve peak in yours? How far has it fallen by Baldwin's last year, and by the latest year? Write down one difference and its most likely cause (dollars, years, data revision).
2. **Missing data.** Which I6 members have no GDP value in the 1970s or 1980s? What does that do to the I6 and "rest of world" areas in those years? This is your "the data disagreed" entry.
3. **Break — change the dollars.** Duplicate Sheet 1 and switch the filter to `NY.GDP.MKTP.PP.CD`. The PPP series starts in 1990. How much smaller is the G7 at PPP? Which board decision would you use each line for?
4. **Extend — slowbalisation.** New sheet: `NE.TRD.GNFS.ZS` for `WLD` — world trade in % of GDP (an index, so check the aggregation: there is one value per year; `AVG` or `SUM` give the same only because you filtered to one country — say so). Annotate 2008. Is the line after 2008 a retreat?

2.13.6 5 · The catch — 10 min

One slide (a dashboard exported as an image):

Original	Ours	The catch
Baldwin's G7 share figure	Your Sheet 1, plus the PPP line	One sentence: what this figure does **not** license your team's multinational to conclude

A member drawn at random presents in two minutes.

2.13.7 Hand in

On ZoneCours before leaving, one per team: the `.twbx` (*File → Export Packaged Workbook*), the provenance log, and the slide. Formative: commented, not marked; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the book.

2.13.8 What loses credit

- Filtering on country *names* ("Korea, Rep." will bite you in Session 04)
- A percent of total that sums to 100% across years instead of within each year
- A share chart in current dollars with no mention of exchange rates
- A catch about Baldwin's book rather than about what your firm may conclude

Chapter 3

Anarchy and the Balance of Power: Who Sets the Terms When No One Governs

Session 3 · The geostrategic balance that dictates international trade relations Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **In a system with no government above states, who sets the terms of trade — and how would a multinational know that the balance is shifting?**

Big works	Mearsheimer, J. J. (2001). *The Tragedy of Great Power Politics*. W. W. Norton. · Allison, G. (2017). *Destined for War: Can America and China Escape Thucydides's Trap?* Houghton Mifflin Harcourt.
Target figure	The **shares of world material capabilities** (CINC) of the major powers, 1816–2022, and the US–China crossover
Data you collect	Correlates of War, National Material Capabilities v7.0 — correlatesofwar.org · SIPRI Military Expenditure Database — sipri.org/databases/milex · WDI GDP (from Session 02)
Tableau skill	**Calculated fields** · ratios · **parameters** to choose a country · dual axis
Course themes	Inter-state anarchy and the geostrategic balance of the major powers that dictate international trade relations

3.1 A system with no government

The previous chapter described the international economic order as a set of rules, written by particular states and held in place by something. This chapter asks what that something is when it is not law. Inside a country, a contract is enforced by courts, and the courts are backed by a state that holds a monopoly of legitimate force. Between countries there is no such state. There is no world government that can make a rule binding, enforce a contract between two governments, or guarantee that a state which disarms will not be invaded. That absence is the starting point of the realist tradition in international relations, and it is the reason this course cannot treat geopolitics as a branch of regulation.

The session's question follows directly. In a system with no government above states, who sets the terms of trade — and how would a multinational know that the balance is shifting? The realist answer to the first half is blunt: the terms are set, in the end, by those who hold the capabilities others need, and they are revised when the distribution of capabilities changes. The second half is a measurement problem, and it is the subject of the practice round, which rebuilds the long-run distribution of material capabilities among the major powers and asks when, and on what measure, China overtook the United States.

The chapter proceeds in five steps. It defines anarchy and what follows from it; sets out the two main realist theories of how states respond, defensive and offensive; turns to power transition theory and its popular form, Graham Allison's "Thucydides's Trap"; examines how power is measured, and what the standard measure rewards and misses; and ends with what all this means for a firm that operates in both the rising and the ruling power.

3.2 Anarchy, and what it does not mean

Anarchy — in international relations, the absence of any authority above states capable of making binding rules, enforcing agreements between them, or guaranteeing their survival. It describes the *structure* of the system, not the level of violence within it.

The word is a trap for newcomers, because in ordinary language it means chaos. In realist theory it means nothing of the kind. Most of the time, most states are at peace with most others; they trade, sign treaties, join organisations and settle disputes by negotiation. Anarchy is compatible with all of that. What it denies is that any of it is *guaranteed* by a higher authority. A treaty is kept because the parties find it in their interest to keep it, not because a court can compel them; a trading relationship continues because both sides value it, not because a sovereign above them will punish defection.

Three consequences follow, in the realist reading, and each matters for firms.

The first is **self-help**. Because no one else guarantees a state's survival, each state must ultimately provide for its own security. It may ally with others, but alliances are arrangements of convenience, and a state that relies entirely on an ally has placed its survival in someone else's hands.

The second is **uncertainty about intentions**. A state can observe another's capabilities — its army, its industry, its population — but it cannot observe its intentions with confidence, and intentions can change faster than capabilities. A peaceful neighbour with a large army is a neighbour whose government could change. Because the cost of misjudging a hostile state is so high, states tend to plan for capabilities rather than for stated intentions.

The third is concern for **relative gains**. Because today's partner may be tomorrow's adversary, states care not only about how much they gain from cooperation, but about how much more the other side gains. A trade agreement that makes both countries richer but makes the rival richer faster shifts the balance of capabilities, and in an anarchic system that shift can be a threat.

Relative gains — the concern of a state with its gains from cooperation *relative to* those of its partner, rather than only with its own absolute gains, arising from the possibility that the partner's gains will be converted into capabilities used against it (Grieco 1988).

Joseph Grieco's 1988 article made this the central point of difference between realists and liberal institutionalists such as Keohane (Chapter 2): institutions can solve the problem of cheating, Grieco argued, but they cannot solve the problem that cooperation may help a rival more than oneself. Powell (1991) sharpened the argument by showing that the weight a state places on relative gains is not fixed but depends on circumstances — in particular, on whether the use of force is a live possibility between the two states. Where force is unthinkable, as between Canada and the United States, relative gains matter little and cooperation is easy; where it is conceivable, relative gains matter a great deal.

That is why the concept belongs in a course on international business. Relative-gains thinking is the logic by which trade becomes a security question. A sale of advanced semiconductors to a rival, or of the equipment to make them, may be profitable for the seller and beneficial for the buyer, and still be prohibited, because it narrows a capability gap the seller's government wishes to preserve. Export controls on advanced chips are not a failure of economic reasoning; they are the application of a different calculus, in which the question is not "are we better off?" but "are we better off *relative to them*?". A firm that sells into the gap between two powers' capabilities should expect its business to be judged by the second question.

3.3 Two realisms

Realists agree that anarchy shapes behaviour. They disagree about how much power states want, and the disagreement has direct implications for what to expect from the relationship between the United States and China.

3.3.1 Defensive realism: enough is enough

Kenneth Waltz's *Theory of International Politics* (1979) is the founding text of structural, or neo-, realism. Its claim is that the behaviour of states is shaped less by their domestic characteristics — their regimes, their leaders, their ideologies — than by the structure of the system they inhabit: anarchy, plus the **distribution of capabilities** among the units (Waltz 1979). States are alike in their basic function, which is to survive, and they differ in how well equipped they are to do it.

Balance of power — the distribution of capabilities among the major states of a system, and the tendency, in realist theory, of states to respond to a concentration of power by building their own capabilities or forming coalitions against the stronger state ("balancing").

In Waltz's version, states seek an *appropriate* amount of power: enough to be secure. Seeking more is usually self-defeating, because a state that grows too strong provokes others to balance against it, and ends up less secure than before. The system therefore tends towards equilibrium. Waltz also argued that a **bipolar** system — two great powers, as in the Cold War — is more stable than a multipolar one, because with only two principal actors there are fewer relationships to manage, fewer opportunities for miscalculation, and no uncertainty about who the main threat is.

Waltz's theory sits on an older idea that is essential for understanding how conflict can arise without anyone wanting it.

Security dilemma — the situation in which measures a state takes to increase its own security, such as arming or forming alliances, reduce the security of others, who respond in kind, leaving all less secure than before even though none intended aggression (Herz 1950; Jervis 1978).

John Herz named the dilemma in 1950. Robert Jervis's 1978 article gave it its modern form by asking when it is most severe. His answer turned on two variables: whether offensive and defensive weapons and postures can be distinguished, and whether offence or defence has the advantage. Where defensive preparations look defensive and defence is strong, states can secure themselves without threatening others, and the dilemma is mild. Where they cannot be told apart and offence is favoured, every precaution looks like preparation for attack, and the dilemma is acute.

The security dilemma is the most useful single idea in this chapter for a risk analyst, because it describes escalation without intent. A government imposes an export control to protect a technological lead; the other side reads it as an attempt at containment and responds with restrictions on a critical mineral; the first reads that as coercion and tightens the control. Each step is defensive in the mind of the state that takes it. A firm caught in the sequence does not need either government to want a confrontation in order to lose a market or a supplier.

3.3.2 Offensive realism: as much as possible

John Mearsheimer's *The Tragedy of Great Power Politics* (2001), the session's first big work, accepts Waltz's starting point but reaches a darker conclusion. It rests on five bedrock assumptions (Mearsheimer 2001):

1. The international system is **anarchic**.
2. Great powers inherently possess some **offensive military capability**.
3. States can never be certain about other states' **intentions**.
4. **Survival** is the primary goal of great powers.
5. Great powers are **rational actors**.

None of these, Mearsheimer notes, is individually alarming. Together they imply that great powers fear one another, cannot rely on anyone else for their security, and conclude that the best guarantee of survival is to be as powerful as possible relative to potential rivals. States are therefore not satisfied with an appropriate amount of power. They **maximise relative power**, and the ultimate goal is hegemony.

Global hegemony, however, is practically impossible, because of what Mearsheimer calls "the stopping power of water": large bodies of water make it extremely difficult to project land power across oceans. The realistic goal is therefore **regional hegemony** — domination of one's own region — together with preventing any other great power from dominating its region. The United States, on this account, is the only regional hegemon in modern history, having achieved dominance in the Western Hemisphere in the nineteenth century, and its foreign policy in the twentieth century can be read as a series of efforts to prevent the emergence of a hegemon in Europe or in North-East Asia.

The prediction for China follows mechanically. If China continues to grow economically, it will seek to dominate Asia as the United States dominates the Western Hemisphere, and the United States, which has spent a century preventing exactly that, will try to stop it. Mearsheimer's answer to the question of whether China can rise peacefully is no: the result will be intense security competition, with a considerable possibility of war.

	Defensive realism (Waltz)	Offensive realism (Mearsheimer)
How much power do states want?	enough for security	as much as possible, relative to others
What restrains them?	balancing by others; expansion is self-defeating	only the limits of their capabilities
Expected US–China outcome	managed competition	intense security competition

A caution belongs here, and the lecture states it twice. Realism predicts **competition**, not war. In Waltz's version it often predicts restraint, since overexpansion provokes balancing. Even Mearsheimer's version predicts security competition, of which war is one possible outcome among several. A risk analyst who reads realism as a forecast of war has misread it; one who reads it as a forecast that the economic relationship will be subordinated to the strategic one has read it correctly.

3.4 Power transition and the Thucydides's Trap

Balance-of-power theory treats an equal distribution of capabilities as the condition of stability. A different tradition reverses the claim.

Power transition theory — the theory, due to A. F. K. Organski, that peace is most secure when a dominant power holds a clear preponderance and that war is most likely when a dissatisfied challenger approaches parity with it (Organski 1958; Organski and Kugler 1980).

Organski's *World Politics* (1958) described the international system as a hierarchy rather than a balance. At the top sits a dominant power that built the existing international order and benefits from it; below are great powers, some satisfied with the order and some not. Industrialisation drives the rise of new powers, and their growth is uneven. Danger arises when a challenger that is *dissatisfied* with the order — that did not write its rules and does not benefit from them as much as it believes it should — draws close to the dominant power in capabilities. At that point the challenger has both the motive and, increasingly, the means to revise the order, and the dominant power has reason to act before its lead disappears. In empirical work, parity is often operationalised as the challenger reaching around 80 per cent of the dominant power's capabilities, a threshold the practice round places on the chart as a reference line.

Organski and Kugler (1980) took the theory to data in *The War Ledger*, and a substantial literature followed, including the essays collected in Kugler and Lemke (1996). The tradition connects to Robert Gilpin's account of hegemonic war, in which the order established by a dominant power becomes misaligned with the underlying distribution of power as others grow, until a war redistributes both (Gilpin 1981). It connects too to Paul Kennedy's history of the great powers, which argued that dominant powers tend to overextend their commitments relative to their economic base (Kennedy 1987).

Power transition theory has an obvious affinity with Chapter 2. There, the question was whether the rules of the economic order would survive the shrinking share of their authors. Here, the theory supplies a mechanism: a rising power that did not write the rules is likely to be dissatisfied with them, and as its capabilities approach those of the rule-writer, its ability to revise them grows.

3.4.1 Allison's sixteen cases

Graham Allison's *Destined for War* (2017), the session's second big work, brought the argument to a mass audience under a classical name. Thucydides, the historian of the war between Athens and Sparta in the fifth century BCE, wrote, in the translation Allison uses, that "it was the rise of Athens and the fear that this instilled in Sparta that made war inevitable" (Allison 2017).

From the archive · Thucydides, fifth century BCE

The Peloponnesian War began in 431 BCE and ended, after a long truce and its collapse, with the defeat of Athens in 404. Its historian, Thucydides, was an Athenian general who had been exiled after failing to prevent the loss of a city to Sparta's forces, and who spent his exile collecting the accounts of both sides.

The sentence Allison made famous comes early in the history, where Thucydides distinguishes the immediate grievances that both sides publicly cited — disputes over Corcyra and Potidaea, over alliances and trade — from what he calls the truest cause, the one least spoken of: the growth of Athenian power and the alarm it caused in Sparta. That distinction between the stated occasion and the structural cause is the analytical move every realist since has borrowed.

It is worth reading what Thucydides does *not* say. He does not say that rising powers always fight ruling ones; he says this war had this cause. He was writing about a particular conflict between two particular cities, one a maritime and commercial empire and the other a land power with a very different political system. The generalisation to a law of history is Allison's, and the critics' first objection is to the generalisation, not to Thucydides.

Allison and the Thucydides's Trap project at Harvard identify **sixteen** cases in the past five hundred years in which a rising power challenged a ruling one. **Twelve** ended in war. The four that did not — among them the United States overtaking Britain in the early twentieth century, and the Cold War rivalry between the United States and the Soviet Union — are presented as evidence that war is not inevitable, but that avoiding it requires deliberate and painful adjustment by both sides.

From the archive · Britain and the United States, 1895–1903

At the end of the nineteenth century the United States overtook Britain as the world's largest economy, and its navy was growing. The two powers had serious quarrels. In 1895 a boundary dispute between Venezuela and British Guiana brought a sharply worded American demand that Britain accept arbitration, and talk of war on both sides of the Atlantic. Britain accepted arbitration, and a tribunal issued its award in 1899. In 1901 the Hay–Pauncefote Treaty released the United States from an earlier agreement to share control of any canal across Central America, clearing the way for an American-controlled canal at Panama. In 1903 a tribunal settled the boundary between Alaska and Canada largely in the American favour, with the British member siding with the Americans against the Canadian members' objections.

In each dispute, the ruling power gave way. Britain, facing a rising Germany closer to home and overstretched across its empire, chose to accommodate the rising power in the Western Hemisphere rather than contest it. The case is the strongest evidence that a transition can be peaceful. It is also a reminder of the price: the accommodation was made, in part, at the expense of a third party — Canada — whose interests the ruling power judged less important than the relationship with the rising one.

For a firm, the analogy is uncomfortable and useful. A peaceful transition between great powers is not costless for everyone else. The terms on which it is reached may be paid by actors who were not at the table.

3.4.2 The critics

The Thucydides's Trap has been criticised on four grounds, and the lecture puts them to the room before showing them.

Case selection. The cases were assembled with knowledge of the outcome. What counts as a "rising" power, a "ruling" power, and a "challenge" is elastic enough that peaceful transitions — which leave less of a trace in history than wars — may be under-counted, inflating the proportion that ended in war. A ratio of twelve to sixteen is only as informative as the rule that produced the sixteen.

Nuclear weapons. No pre-1945 case involved two powers each capable of destroying the other. Nuclear deterrence changes the expected cost of great-power war so drastically that it is not clear any pre-nuclear case is informative about the present. The Cold War, the one transition-like rivalry between nuclear powers in the dataset, did not end in war between the principals.

Interdependence. The United States and China are bound by trade, investment and finance far more tightly than Athens and Sparta were. Liberal theory holds that interdependence raises the cost of conflict and so makes it less likely. The realist reply is that Britain and Germany were also major trading partners in 1914, and fought anyway. Session 04 takes up the debate with data.

Measuring "rise". The trap depends on knowing when a challenger is approaching parity, and that depends on how power is measured. Beckley (2018) argues that the conventional gross indicators — GDP, population, composite indices of capability

— systematically overstate the power of large, populous countries, because they ignore the costs a state bears in governing, feeding, policing and protecting its population. A country with a large GDP and a large population may have much less surplus available for projecting power than the gross figures imply. Beckley proposes measures that account for such costs, and on them the gap between the United States and China looks larger and the transition further off.

The last criticism is the one this session's practice round can examine directly, because it turns on the choice of measure.

3.5 Measuring power

Every theory in this chapter turns on a quantity — the distribution of capabilities, parity, relative power — that none of them can observe directly. Power is a latent concept. What can be observed are its supposed ingredients: people, money, industry, arms. The standard way of combining them in quantitative international relations was devised in the 1960s at the University of Michigan by J. David Singer and his collaborators on the Correlates of War project.

Composite Index of National Capability (CINC) — a yearly index of a state's material capabilities, equal to the unweighted average of its shares of the state-system total on six indicators: total population, urban population, iron and steel production, primary energy consumption, military personnel and military expenditure (Singer, Bremer, and Stuckey 1972).

The construction has three steps. For every state in the international system in a given year, take six indicators, grouped into three dimensions: demographic (total population, urban population), industrial (iron and steel production, primary energy consumption) and military (military personnel, military expenditure). For each indicator, compute the state's share of the system total that year. Average the six shares. The result is a number between 0 and 1; summed over all states, it comes to about 1 — not exactly, because a state with a missing component is averaged over the components it has. In the current release, the National Material Capabilities dataset version 7.0, the coverage runs from 1816 to 2022, one row per state-year.

The design has real virtues. It is transparent: anyone can recompute it. It is long: two centuries of comparable data, which is what allows Organski's and Mearsheimer's arguments to be confronted with evidence at all. And by using shares of the system total, it sidesteps the problem of comparing military spending in nineteenth-century pounds with twenty-first-century dollars — each year is compared only with itself. Mearsheimer's own measures of latent power in the *Tragedy* rely on similar ingredients: for the nineteenth and early twentieth centuries, he proxies a state's wealth with indicators of industrial output such as steel production and energy consumption.

3.5.1 What it rewards

The design also embodies a theory of power, and it is a theory of the 1960s. Two of the six components are population and two are heavy industry. CINC therefore rewards **size and smokestacks**. A large, populous, industrialising country — one that produces a great deal of steel and burns a great deal of energy — will score highly even if its technology is behind, its productivity low and its military poorly equipped.

The consequence is visible in the replication. CINC shows China passing the United States much earlier than military expenditure or GDP at market exchange rates do. One underlying reality, three measures, three dates. The practice round asks you to find each date for yourself; the point to take into it is that the date of the "transition" is a property of the measure, not only of the world.

3.5.2 What it misses

CINC is equally instructive for what it cannot see. It has no component for **technology** — the quality of weapons, the sophistication of industry, the capacity to innovate. It has no component for **alliances**, although a state with many capable allies commands far more power than its own figures suggest. It has nothing on **financial power**: the role of a state's currency in international payments, the depth of its capital markets, its control of the institutions through which money moves. And it has nothing on **control of networks** — the payment systems, cloud platforms and communications infrastructure that Farrell and Newman (2019) identify as instruments of coercion (Chapter 2). These are precisely the forms of power that reach firms most directly, and Sessions 08, 10 and 11 return to them.

The choice of measure is not a technicality, and Beckley's critique shows why. If power is gross resources, China caught up long ago. If it is net resources after the costs of a large population, or military capability weighted by quality, or control of the networks through which the world economy runs, the picture changes. Each measure is defensible. None is neutral.

3.6 Reading two centuries in one chart

The long-run chart of capability shares is a compressed history of the modern state system, and it is worth knowing what to expect before building it, so that the anomalies are recognisable as anomalies.

The first thing to expect is that the chart is not a chart of a fixed set of countries. The Correlates of War data describe the members of the state system in each year, and membership changes. China and Japan enter the data in 1860. Germany, coded `GMY`, disappears in 1945 and returns in 1991; in between, the Federal Republic and the Democratic Republic appear as separate states, `GFR` and `GDR`. Japan is absent from 1946 to 1951, under occupation. The Soviet Union, by contrast, is coded as `RUS` throughout, so that Russia, the USSR and Russia again form one continuous series. Each of these is a decision by the data's authors about what counts as the same state over time, and each produces a visible break or a visible continuity on the chart that has nothing to do with power and everything to do with coding. The codes themselves are a second trap: they look like ISO codes and are not, so that a join with the World Bank data of Chapter 2 on the three-letter code will silently fail for Germany, the United Kingdom and France.

The second thing to expect is that the components are incomplete. Missing values of the six components are coded as `-9`, not left blank. The index itself has no missing values, because it is averaged over whatever components exist, but any calculation that uses a component directly must exclude the `-9`s first or it will be wrong in a way that is easy to miss on a chart of shares.

The third is the substantive story. Read with these cautions, the chart shows a sequence that the theories of this chapter were built to explain: a nineteenth-century system in which European powers and then the United States hold the largest shares; the two World Wars as moments when the distribution is violently rearranged; a Cold War in which two powers dominate — Waltz's bipolarity, visible as data; and, from the late twentieth century, the rise of China's share. The practice round asks you to date the last of these yourself, and to see how the date moves when the measure does.

3.6.1 Bipolar, multipolar, or something else?

The chart also bears on a question that has occupied realists since the Soviet Union dissolved in 1991: what is the *polarity* of the system now? The 1990s were widely described as a unipolar moment, with the United States unmatched. Waltz expected that unipolarity would not last, because other states would balance against a power with no counterweight. Whether the present system is returning to bipolarity — the United States and China — or moving to a multipolar one with several significant powers is contested, and the answer depends, once again, on the measure.

The question matters to firms because the two configurations imply different environments. A bipolar system tends to produce two blocs with relatively clear membership, so that the firm's problem is choosing, or being made to choose, a side, and organising its operations to comply with two sets of rules. A multipolar system produces more hedging states — governments that trade with both sides, refuse to join either bloc, and bargain for advantage between them — and therefore more room for firms to locate operations in third countries that remain open to both. The geography of "connector" economies, which Session 04 examines with trade data, is a direct consequence of which of the two worlds firms are in.

3.7 How a firm would know the balance is shifting

The second half of the session's question — how would a multinational know that the balance is shifting? — has two answers, a structural one and a behavioural one.

The structural answer is the one the practice round builds: track the distribution of capabilities on several measures, and watch the ratios. A challenger's share approaching the dominant power's on one measure, while still far behind on others, is a transition in progress and a transition contested; it is the situation in which the security dilemma is most acute, because

each side can read the evidence as it wishes. Structural indicators of this kind, in the typology of Session 01, explain *why* tension is building. They do not say *when* it will matter.

The behavioural answer is to watch what states do with the rules. A shifting balance shows up first not in war but in the instruments of economic statecraft: export controls, investment screening, sanctions and counter-sanctions, restrictions on data, rules about where a firm may source a component. These are the measures by which states convert relative-gains concerns into policy, and they are observable — in official gazettes, in company disclosures, in the event datasets of later sessions — long before any military confrontation. For a firm, the most reliable signal that the balance is shifting is a change in which rules it is being asked to follow.

3.8 What it means for the multinational

The realist frame leads to five implications for a firm, each of which recurs later in the course.

The balance of power sets the rules of trade. Who can impose sanctions, export controls and market-access conditions depends on who holds capabilities others need: the largest market, the dominant currency, the critical technology. A firm's exposure to a state's economic statecraft is a function of how much of its business depends on capabilities that state controls.

A transition changes which rules you follow. A firm operating in both the United States and China increasingly faces two compliance regimes — and, since China adopted its own instruments of counter-sanction in 2020 and 2021, the possibility of penalties in one jurisdiction for complying with the other's rules. The question "is this transaction legal?" has acquired a second clause: "legal where, and under whose law?".

Relative gains turn commerce into security. Products that would make a rival stronger — advanced semiconductors, the tools that make them, artificial intelligence, batteries, aerospace — move from the trade ministry's jurisdiction to the security establishment's. For firms in these sectors, market access is no longer negotiated mainly on economic grounds, and commercial arguments about mutual benefit carry less weight than they did.

The date of the transition depends on the measure. A board slide that says "China has overtaken the United States" should say *on what*. On some measures it happened decades ago; on others it has not happened. The decision the slide is meant to inform — whether to expand, hedge, or relocate — depends on which kind of power is relevant to it.

Security dilemmas escalate without intent. A defensive measure by one government — an export control, a military exercise, an investment restriction — is read by the other as offensive and triggers a response that may fall on firms that had nothing to do with the original measure. A firm cannot assume that because neither government wants a confrontation, none will occur.

Mapped onto the five channels of this course, the balance of power runs through every one, but most directly through **market access** (which markets will remain open to firms from which bloc) and **assets and people** (where conflict, if it comes, will fall). It reaches **inputs** through export controls on critical goods, **finance and payments** through the dominant currency, and **data and operations** through the control of networks.

3.9 What the figure does not license

The target figure — the shares of world material capabilities of the major powers since 1816, with the US–China crossover — is the picture behind both big works. It does not license four conclusions.

CINC measures **material capabilities**, not influence, intentions or the likelihood of war. A crossover on the chart is not a prediction of conflict. Power transition theory says war is more likely near parity *if* the challenger is dissatisfied; dissatisfaction is not on the chart.

It **overweights size**. Population and heavy industry account for four of six components. The index cannot see semiconductors, alliances, the dollar or control of networks — the forms of power that matter most for the economic statecraft firms face.

It **ends in 2022**. The most recent shifts in the distribution of capabilities are not in it, and for a firm making decisions now the most recent years are the most relevant.

It describes **states**, not markets. A firm's exposure depends on where its revenue, suppliers, assets and data sit, not on the system's distribution of power. Two firms in the same country can face entirely different consequences from the same transition.

3.10 A short review of the literature

Realism's modern structural form begins with Waltz (1979), who explained state behaviour by anarchy and the distribution of capabilities and judged bipolarity the more stable configuration. The security dilemma that underlies much of it was named by Herz (1950) and given its analytical form by Jervis (1978), who related its severity to the offence–defence balance. Grieco (1988) made relative gains the realist objection to liberal institutionalism, and Powell (1991) showed that the weight states put on relative gains depends on whether force is a live option between them. Mearsheimer (2001) pushed the structural argument to its offensive conclusion — power maximisation and regional hegemony — and applied it to China.

A parallel tradition locates danger in hierarchy rather than balance. Organski (1958) introduced power transition theory; Organski and Kugler (1980) and the contributions in Kugler and Lemke (1996) tested it on data. Gilpin (1981) gave a systemic account of hegemonic war, and Kennedy (1987) a historical account of how great powers rise and decline relative to their economic base. Allison (2017) reframed the transition debate for the US–China relationship through sixteen historical cases.

Measurement runs through all of it. Singer, Bremer, and Stuckey (1972) introduced the Composite Index of National Capability and the Correlates of War data on which much of the quantitative literature rests. Beckley (2018) argued that gross indicators overstate the power of populous states and proposed net measures, which change the dating of the US–China transition. Farrell and Newman (2019) identified a form of power — control over network hubs — that no capability index was designed to capture.

The through-line is that every theory of the balance of power is also a claim about how to measure it, and that the measurement is where theories can be checked against the world — and where they most often disagree.

3.11 Into the practice round

The practice round rebuilds the distribution of material capabilities among the major powers from 1816 to 2022 using the Correlates of War data, then sets the US–China relationship against the parity and 80 per cent thresholds, and then measures the same transition with military expenditure and GDP. The steps are in the brief that follows. What this chapter adds is the reason each step matters.

When you sum CINC across all states and find that the total is not exactly one, you are seeing the consequence of averaging over the components a state has — a reminder that every index carries its construction with it. When you build the ratio of challenger to leader and set a reference line at 80 per cent, you are operationalising Organski's theory, and the line is only as meaningful as the theory. When you find the crossover year, and then find two different years on two other measures, you are reproducing Beckley's critique on your own screen.

Then write the catch, for your team's firm. A crossover on a capability chart does not tell a multinational that war is coming, that its Chinese revenue is at risk, or that its American suppliers are safe. It tells it that, on one measure that rewards size and heavy industry, the distribution of material capabilities has shifted. The question for the firm is which kind of power its own exposures depend on — markets, technology, currency, networks — and whether the measure on the chart can see it.

3.12 In the practice

> **The target.** The shares of world material capabilities (CINC) of the major powers, 1816–2022 — the picture behind Mearsheimer's and Allison's argument — and the year China passes the United States. Then: the same transition measured two other ways, dated two other years.

Work in your team. Everyone builds their own workbook; the team hands in one.

3.12.1 1 · Collect — 15 min

NMC v7.0 and the SIPRI workbook should already be in `data/raw/`; your Session 02 WDI extract too. If not: `../data/README.md`.

Provenance log: three lines — NMC (version 7.0, and the codebook's date), SIPRI (the file name gives the release), WDI (reused; same version as Session 02).

3.12.2 2 · Prepare — 20 min

3.12.2.1 NMC

1. **Connect** → **Text file** → `NMC-70-abridged.csv` (from `NMC-v7-abridged.zip`, inside `NMCv7.zip`). Drag it to the canvas.
2. Check the columns against the **codebook**. In v7.0 they are unchanged from v6.0: `stateabb` (three letters), `ccode` (number), `year`, the six components `milex`, `milper`, `irst`, `pec`, `tpop`, `upop`, then `cinc` and `version`. Missing component values are coded **`-9`**: before you use any component, exclude them (a filter `>= 0`), or every sum is wrong.
3. `year` should be a **number (whole)**, not a date: keep it that way for now.
4. Sanity check before you build anything: new sheet, `year` → Columns (continuous), `SUM(cinc)` → Rows, no filter. Every year should sum to **about 1** — in v7.0, between 1.000 and about 1.075. Which years are furthest from 1, and why? (The codebook's CINC section explains it: the average is taken over the components a state *has*.)

3.12.2.2 SIPRI

1. **Connect** → **Microsoft Excel** → the SIPRI workbook. Use the **`Constant (2024) US$`** sheet (US$ millions, constant 2024 prices).
2. Tick **Use Data Interpreter**: the sheet has five rows of notes above the header (`Country`, an unnamed column, `Notes`, `1949` … `2025`). Region headings (`Africa`, `North Africa`, …) remain as rows with no values: filter them out.
3. **Pivot** the year columns (as in Session 02). Rename to `Year` and `Milex`. Make `Year` a whole number with the same name as NMC's year field — it will be the link.
4. `Milex` arrives as text: SIPRI writes **`...`** for unavailable data and `xxx` for a country that did not exist (the notes at the top of the sheet; the `Footnotes` sheet explains the † § ¶ marks in `Notes`). Change to **Number (decimal)** and check what became null.
5. SIPRI identifies countries by **name**, not code. Write down the exact strings for China, the United States and Russia. This is a provenance-log entry.

3.12.3 3 · Replicate — 25 min

3.12.3.1 Sheet 1 · "Shares of world capabilities, 1816–2022"

1. `year` → Columns (continuous); `SUM(cinc)` → Rows.
2. `stateabb` → Colour. Filter to the major powers: `USA`, `UKG`, `FRN`, `GMY`, `RUS`, `JPN`, `CHN`, `ITA` — **check each code in the COW state list** (`COW-country-codes.csv`; see `../data/README.md`). Russia/USSR is one continuous series (`RUS`). Germany is not: `GMY` stops in 1945 and returns in 1991, with `GFR` and `GDR` in between. `JPN` also has a gap (1946–1951). Look for them, and decide what your line does there.
3. Highlight the US and China (strong colours), grey the rest.
4. Annotate 1945, 1991, and the year China passes the US.
5. Title with the finding; caption: *Correlates of War, NMC v7.0 (Singer, Bremer & Stuckey 1972), accessed ….*

3.12.3.2 Sheet 2 · "How close is the challenger?"

1. **Create → Parameter** `Challenger`: data type *String*, *List*, **Add values from** `stateabb`. Default `CHN`.
2. Calculated fields:

```
Challenger CINC = IF [stateabb] = [Challenger] THEN [cinc] END
US CINC         = IF [stateabb] = "USA" THEN [cinc] END
Power ratio     = SUM([Challenger CINC]) / SUM([US CINC])
```

3. `year` → Columns; `Power ratio` → Rows. Right-click the parameter → **Show Parameter**.
4. **Analytics → Reference Line**: constant **1** ("parity") and constant **0.8** ("power transition zone", Organski & Kugler).
5. Switch the parameter to `GMY`, `JPN` and `RUS`. When did each enter the transition zone against the US — or against the UK, if you also make the leader a parameter?

3.12.4 4 · Compare and break — 20 min

1. **Compare.** Put Sheet 1 beside the readings' claims. In which year does China's CINC first exceed the US's in your data? Does that date surprise you, given what you read?
2. **Break — change the measure of power.** Build two more ratios, China ÷ US:

   - **military expenditure** from SIPRI (constant dollars), and
   - **GDP** from your WDI extract (current US$ and PPP).

   Put them on one sheet with the CINC ratio by **blending** on the year: on a sheet that uses NMC, drag a SIPRI field onto the view; check that the link icon is active on the year field (*Data → Edit Blend Relationships* if Tableau did not find it). **Blending links fields only by name and type**: if the year is `year` in NMC, `Year` in SIPRI and `Year (n)` in your Session 02 WDI extract, rename them to the same name (right-click → *Rename*) and make all three whole numbers before you blend, or use *Edit Blend Relationships → Custom* to pair them by hand. Use a **dual axis** (right-click the second axis → *Dual Axis → Synchronize Axis*) so both ratios share one scale, with the parity line at 1.
3. **Read it.** Three measures, three transition dates (or no transition yet). Which date belongs in a board presentation, and for which decision?
4. **Extend.** Use the NMC components: which of the six drives China's CINC? Build one component's share (e.g. iron and steel) with a percent-of-total table calculation computed across states, restarting every year.

3.12.5 5 · The catch — 10 min

One slide:

Original	Ours	The catch
The distribution of power as the readings describe it	Sheet 1, and the three-measure ratio sheet	One sentence: what this figure does **not** license your team's multinational to conclude

A member drawn at random presents in two minutes.

3.12.6 Hand in

On ZoneCours before leaving, one per team: `.twbx`, provenance log, slide. Formative.

The one rule: your log records at least one place where the data disagreed with the readings — or with each other.

Team project milestone (Session 04): a one-paragraph footprint of your firm — its five most important countries, with sources.

3.12.7 What loses credit

- Joining NMC to WDI or SIPRI on names or on COW codes treated as ISO
- `SUM` of a ratio across years or states
- "China overtook the US in …" without saying on what measure
- A catch about the theory rather than about what the firm may conclude

Chapter 4

Trade, Conflict and Geoeconomic Fragmentation

Session 4 · Does trade keep the peace — and is the world now trading along geopolitical lines? Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **If states trade more with friends than with rivals, how does a multinational find out which side each of its markets is on — and how fast the lines are hardening?**

Big works	Gopinath, G., Gourinchas, P.-O., Presbitero, A. F., & Topalova, P. (2025). Changing global linkages: A new Cold War? *Journal of International Economics*, 153, 104042. doi:10.1016/j.jinteco.2024.104042 (IMF Working Paper WP/24/76) · Mansfield, E. D. (2021). International trade and conflict. In *The Oxford handbook of international political economy* (pp. 660–671). doi:10.1093/oxfordhb/9780198793519.013.27
Target figure	**Trade within versus between geopolitical blocs**, indexed to the period before Russia's 2022 invasion of Ukraine
Data you collect	UN General Assembly ideal points — Bailey, Strezhnev & Voeten, Harvard Dataverse doi:10.7910/DVN/LEJUQZ · IMF, International Trade in Goods by partner country (IMTS, formerly DOTS) — data.imf.org
Tableau skill	**Joins and relationships** on dyads · **groups** and **sets** for blocs · indexing to a base year
Course themes	The system of international economic order; inter-state anarchy and the balance of the major powers that dictate international trade relations

4.1 An old promise, and an old embarrassment

The idea that commerce keeps the peace is older than the discipline that now tests it. Montesquieu put it in a sentence in 1748 — peace is the natural effect of trade — and the sentence did a great deal of work for the next century and a half. Richard Cobden built a political career on it: the repeal of the Corn Laws in 1846 and the Anglo-French commercial treaty of 1860 that carries his name were sold, in part, as instruments of peace between nations that had spent much of the previous century fighting each other. The argument was not that merchants are nicer than generals. It was that exchange creates gains, that war destroys them, and that a state which stands to lose a great deal from fighting will think harder before it fights.

The argument's most famous restatement came at the worst possible moment.

From the archive · Angell, 1910

In 1910 Norman Angell, a British journalist, published *The Great Illusion* (Angell 1910). Its claim is usually remembered as "war has become impossible", which is not what it said. Angell's claim was narrower and more interesting: that in an economy of credit, contracts and cross-border investment, conquest no longer pays. A victor cannot seize the wealth of an industrial country, because that wealth consists of claims, trading relationships and productive organisation that collapse

when they are seized. The economic case for war, he argued, rested on an illusion — the belief that national wealth is a stock of things that can be carried off.

The book sold in enormous numbers and was translated widely. Four years later Britain and Germany — two of each other's most important trading partners, bound by finance, shipping and insurance as well as by goods — were at war. The war went on for more than four years and destroyed exactly the wealth Angell said it would destroy.

The episode is usually told as a refutation. It is better read as a lesson about which claim was refuted. Angell was largely right that the war would be ruinous for the victors as well as the vanquished. What failed was the inference from *war does not pay* to *states will therefore not make war*. Leaders in 1914 were not computing the net present value of conflict, or were computing it with expectations about the war's length and cost that turned out to be badly wrong. Angell received the Nobel Peace Prize in 1933, as the next European war was being prepared. The commercial peace has lived with the embarrassment of 1914 ever since, and every serious version of the argument is a version that tries to explain it.

This chapter takes the argument seriously, sets out the realist reply, asks what a century of evidence has settled, and then turns to the question that has replaced it. The old question was whether trade pacifies. The question that matters to a multinational in the 2020s is different: if states have begun to trade more with friends than with rivals, how does a firm find out which side each of its markets is on, and how fast the lines are hardening? That question has an empirical answer, it can be rebuilt from public data, and it is the target of this session's replication.

4.2 Four reasons trade might pacify

The modern version of the argument is less a single claim than a family of mechanisms, and they are worth keeping apart because they fail in different ways.

Commercial peace — the proposition that economic interdependence between two states lowers the probability that they will come into militarised conflict with each other. It is a claim about *dyads* — pairs of states — and it is probabilistic: it predicts fewer disputes, not none.

The first mechanism is **opportunity cost**. Conflict disrupts trade, and the more a dyad trades, the more it stands to lose. This is Angell's mechanism with the conquest argument stripped out, and it was formalised by Polachek (1980), who modelled conflict as an activity whose price rises with the gains from trade that it puts at risk. Its weakness is the weakness of 1914: it assumes leaders weigh those costs, and weigh them accurately.

The second is **signalling**. Interdependent states have a menu of costly actions short of force — threatening, restricting or suspending economic ties — that allow them to reveal how much they care about an issue. A state that is willing to lose trade to make a point is telling its rival something credible about its resolve, and credible information reduces the risk of miscalculation that rationalist accounts of war identify as its main cause. On this account trade matters not because it makes war expensive but because it makes communication possible.

The third is **domestic coalitions**. Trade creates exporters, importers, port operators, banks and their employees, all of whom have something to lose from a rupture with the partner. They lobby. A government contemplating confrontation faces an organised constituency on the other side, and the larger the trade, the larger the constituency. This mechanism is the one most familiar to a multinational, because the multinational is frequently the constituency.

The fourth is **institutions**. Trade agreements create committees, forums and dispute-settlement procedures through which grievances can be processed before they escalate, and they create repeated interaction among officials who come to know each other. Mansfield and Pevehouse (2000) found that pairs of states belonging to the same preferential trading arrangement are less prone to militarised disputes, and that the effect strengthens as trade between them grows — a result that has made institutions the most robust strand of the liberal argument.

Statistical work on dyads from the 1990s onwards gave these mechanisms empirical weight. Oneal and Russett (1997), using dyads from 1950 to 1985, titled their findings with some confidence — the classical liberals were right — and the programme culminated in Russett and Oneal (2001), which set interdependence alongside democracy and shared membership of international organisations as the three legs of a "Kantian" peace. Dyads that trade more relative to their GDP experience fewer militarised disputes. That correlation is real. What it means is a different question.

4.3 The realist reply: interdependence as exposure

The realist critique does not deny that trade creates gains. It denies that gains are what states care about most. In an anarchic system — the world of Session 03, with no authority above states to enforce agreements — a state's first concern is survival, and every economic relationship is evaluated for what it does to security.

Seen that way, interdependence is a polite word for **vulnerability**. If a state depends on a rival for food, fuel or critical components, the rival holds a lever over it. Kenneth Waltz drew the conclusion that great powers will try to limit such dependence, and that where they cannot, they may be tempted to secure supplies by other means (Waltz 1979). The relationship that liberals see as a bond, realists see as a hostage.

The argument has a precise early statement in economics, and it was written by someone watching it happen.

Albert Hirschman, a refugee economist writing during the Second World War, analysed how Germany in the 1930s had organised its foreign trade with the smaller states of south-eastern Europe (Hirschman 1945). The terms were often generous. Germany bought their agricultural exports in quantities no one else would, through bilateral clearing arrangements that tied payment to purchases of German goods. The result was that a large share of each small country's trade came to depend on a single partner, while each of them was a trivial share of Germany's. Hirschman separated two effects. The *supply effect* is the gain in resources that trade brings a state. The *influence effect* is the power a state acquires over partners for whom the relationship matters much more than it does to the state itself — because breaking it would cost them dearly and cost it little. The influence effect depends not on the volume of trade but on its *asymmetry* and its *concentration*, and to measure concentration Hirschman proposed an index that, after a later rediscovery by Herfindahl, became the standard measure of market concentration in competition policy. The intellectual ancestor of every modern dependency dashboard was designed to measure geopolitical leverage.

Two further realist arguments sharpen the point. The first concerns **relative gains**. In Session 03 we saw why a state in an anarchic system may reject a deal that makes it better off if the deal makes its rival better off by more. Trade usually benefits both partners, but rarely equally, and a state that expects to face its partner as an adversary must worry that today's gains will fund tomorrow's threat.

The second is Joanne Gowa's argument about **security externalities** (Gowa 1994).

Security externality — the effect of trade on the military potential of the partner. Because the gains from trade raise a partner's economic capacity, and economic capacity can be converted into military power, trade with an ally strengthens one's own side while trade with an adversary strengthens the other. States should therefore trade more with allies than with adversaries.

Gowa and Mansfield (1993) tested the prediction and found that alliances do increase trade between their members, and that the effect is stronger in bipolar than in multipolar systems — exactly where the security stakes of strengthening the wrong partner are clearest. The prediction matters for this chapter because it is, word for word, the hypothesis that the fragmentation literature now tests: that trade will reorganise itself along the lines of geopolitical alignment.

The most important refinement of the realist position came from Dale Copeland, and it rescues something from both sides. Copeland's claim is that what matters is not the level of trade today but **expectations of future trade** (Copeland 1996, 2015). A dependent state that expects trade to continue has every reason to keep the peace, because the liberal gains are real and are expected to keep flowing. The same state, at the same level of dependence, becomes dangerous when it comes to expect that it will be cut off — because then dependence is a vulnerability about to be exploited, and acting before the window closes can look rational. Interdependence is pacifying or provocative depending on what leaders believe about the future.

From the archive · Washington and Tokyo, 1941

Copeland's most cited illustration is the road to Pearl Harbor. Through the 1930s Japan depended on the United States for a large part of its oil and of other strategic materials, while it was waging war in China. As Japan advanced into Indochina, Washington tightened export controls step by step, and in the summer of 1941 the United States froze Japanese assets, which in practice cut Japan off from American oil; Britain and the Netherlands, which controlled the other accessible supplies, followed.

Japan's leaders now faced a dependence that had turned, in their reading, from an asset into a countdown: stocks of oil would run down, the navy would lose the capacity to act, and the terms on offer from Washington — withdrawal from

China — were ones they would not accept. In December 1941 they attacked the American fleet at Pearl Harbor and moved south against the oil-producing Dutch East Indies.

The episode does not show that trade causes war; Japan's expansion was the cause of the embargo before the embargo became a cause of the attack. What it shows is the mechanism Copeland identifies. The same trading relationship that had given Japan a reason for restraint gave it, once expectations of continued supply collapsed, a reason for haste. For a firm the lesson transfers directly: the moment of maximum danger in a dependent relationship is not when trade is high, but when one side comes to believe it is about to end.

4.4 What the dyads say, and why they disagree

By the end of the 1990s the two camps had produced findings that pointed in opposite directions. Oneal and Russett (1997) found interdependence pacifying. Barbieri (1996), using dyads from 1870 to 1938 and measures that distinguished the salience of a relationship from its symmetry, found that extensive interdependence was associated with *more* militarised disputes, not fewer, and later extended the case against the liberal view at book length (Barbieri 2002). Much of the disagreement turned out to be about measurement — trade relative to whose GDP, counted from whose customs records, over which years, against which definition of a dispute — which is a warning this course will repeat in every practice round.

Edward Mansfield's review in the *Oxford Handbook of International Political Economy* is the best short guide to where the debate now stands (Mansfield 2021). Four of its points organise everything that follows.

The first is that **the relationship runs both ways**. Conflict reduces trade: states that expect to fight trade less, and states that are fighting trade hardly at all. A negative correlation between trade and conflict is therefore what we would observe whether trade causes peace, peace causes trade, or both are driven by something else — a shared alliance, a shared regime type, geographical distance. This is reverse causality in its classic form, and it is fatal to any naive reading of a dyadic correlation. It returns in this session's replication in a new form: when trade between two blocs falls, we must ask whether the trade fell because the relationship deteriorated, or whether the deterioration is what the falling trade is measuring.

The second is that **institutions matter**. The finding of Mansfield and Pevehouse (2000) on preferential trading arrangements has proved more durable than the bare correlation between trade and peace, perhaps because institutions supply the information and the repeated interaction that the signalling and coalition mechanisms require.

The third is that **context matters**. The pacifying effect of trade differs with regime type, with the level of development, and with the distribution of power between the partners. A finding pooled across all dyads in all decades is an average over very different relationships, and an average can describe none of them well.

The fourth is the one that matters most for a business school: **the question has changed**. As great-power competition returned, research turned from asking whether trade pacifies to asking how states use trade as an instrument of conflict. That is the question of Sessions 08, 09 and 10 — cyber, economic diplomacy and sanctions — and it is the reason this session ends not with dyadic dispute data but with the geography of trade itself.

4.5 Bilateral ties, multilateral exits

One contribution reshaped the debate by asking which trade matters. Martin, Mayer, and Thoenig (2008) — economists rather than political scientists — separated two quantities that earlier work had often lumped together.

Bilateral versus multilateral openness — *bilateral* openness is how much two states trade with each other; *multilateral* openness is how much each trades with the rest of the world. The first raises what the pair loses from fighting each other. The second lowers it, because a state with many alternative partners can replace the trade a conflict destroys.

The logic is that of any outside option. If country A trades heavily with B and with almost no one else, a war with B is ruinous. If A trades heavily with B but also with fifty other countries, losing B is costly but survivable: the fifty others become substitutes. Their model predicts, and their empirical work supports, both effects at once: bilateral trade lowers the probability of a bilateral conflict, and multilateral openness raises it.

	Effect on the risk of conflict between A and B
Bilateral trade between A and B	**lowers** it: more to lose from fighting each other
Multilateral openness of A and B with everyone else	**raises** it: substitutes make losing the partner cheaper

The implication is uncomfortable for the liberal view and was meant to be. **Globalisation lowers the cost of bilateral conflict.** The very process that multiplied trading relationships after 1990 made each individual relationship less essential, and therefore made breaking any one of them cheaper. A world of many alternative suppliers is a world in which a state can afford to fall out with one of them.

There is a corollary for the multinational that is easy to miss. Martin, Mayer and Thoenig describe the cost of switching *for states*, aggregated across an economy. A government that decides to reduce its dependence on a rival can point to a world of substitutes. The firm whose particular supplier sits in that rival, whose tooling is calibrated to that supplier's components, whose certifications name that plant, does not get the aggregate substitution elasticity. It gets its own contracts, which do not become cheaper to break because the national economy has alternatives. Geopolitical decisions are taken at the level where switching is cheapest and paid for at the level where it is dearest.

4.6 From "does trade pacify?" to "how is trade used?"

The fourth of Mansfield's points has a theoretical counterpart that has become central to the study of geoeconomics. Farrell and Newman (2019) observe that globalisation did not produce a flat network of equal partners. It produced hubs: the dollar clearing system, the messaging network banks use for cross-border payments, a handful of firms that control critical technologies, a small number of undersea cables and cloud providers. States with jurisdiction over a hub acquire two kinds of power over everyone who must pass through it. The *panopticon* effect is the ability to see — to gather information from the flows that cross the hub. The *chokepoint* effect is the ability to deny — to cut a target off from the network.

This is Hirschman's influence effect, rebuilt for a networked economy. It also turns the liberal argument on its head. In the liberal story, dense economic ties reduce conflict by raising its cost. In the account of weaponised interdependence, dense ties are the instruments through which conflict is conducted. Interdependence does not stop states competing; it changes the terrain on which they compete, and the multinational, which builds and uses those networks, finds itself standing on that terrain.

The policy response has been to reorganise the networks themselves — to reduce exposure to a rival's chokepoints and to build one's own. That reorganisation has a name.

4.7 Fragmentation: three words, three policies

The vocabulary of the early 2020s distinguishes three policies that are often run together in public debate, and a firm that cannot tell them apart will misread its own exposure.

Friend-shoring — a term popularised in US policy debate in 2022, most prominently by the US Treasury Secretary — proposes to move supply chains towards trusted partners: to keep trading, but with friends. **De-risking** — the European Commission's term, adopted in 2023 to describe its approach to China — proposes to reduce dependence on a rival in critical goods and technologies while preserving trade overall. **De-coupling** is the end point that both of the others say they are trying to avoid: separate economic blocs with little exchange between them.

Geoeconomic fragmentation — a policy-driven reversal of global economic integration, in which states redirect trade, investment, technology and payments according to geopolitical alignment rather than cost (Aiyar et al. 2023). It is distinguished from ordinary changes in trade patterns by its cause: it is driven by security and alignment, not by prices or technology.

The definition, from an IMF Staff Discussion Note that set out the research agenda, contains a claim that is hard to test: that the cause is geopolitical. A fall in trade between two countries may be the result of a tariff, a sanction, a shift in comparative advantage, a change in exchange rates, or a firm's decision to build a factory closer to its customers. Fragmentation is a

hypothesis about *why* trade patterns change. What the data can show directly is only *whether* they change along geopolitical lines — whether the lines that matter for trade start to coincide with the lines that matter for foreign policy.

That narrower question is the one Gopinath and her co-authors set out to answer.

4.8 How Gopinath and co-authors draw the lines

Gopinath et al. (2025), first circulated as an IMF working paper (Gopinath et al. 2024), ask whether the global economy has begun to split into blocs since Russia's full-scale invasion of Ukraine in February 2022. Their design has three steps, and the replication rebuilds a simplified version of each.

First, define the blocs. Countries are grouped into a US-leaning bloc, a China-leaning bloc and a non-aligned group on the basis of their foreign-policy alignment as revealed by voting in the UN General Assembly. The choice is deliberate. Alliances are too sparse to classify most countries; trade itself cannot be used, because trade is the outcome; and General Assembly votes are cast by nearly every state on hundreds of resolutions a year, recorded publicly and comparable across countries. Section 2 of the paper states exactly which measure, which reference period and which cut-offs the authors use. Read it carefully, because the version built in the practice round is a simplification, and the size of the simplification is part of what you must report.

Second, compare trade within and between blocs before and after the invasion. The comparison is made in a gravity framework — the workhorse model of international trade, in which bilateral trade is explained by the economic size of the partners, the distance between them and a set of other frictions — so that the change attributed to bloc membership is the change left over once the usual determinants of bilateral trade have been accounted for. The paper also examines foreign direct investment, where the same question is asked of announced investment projects.

Third, look for connectors. Non-aligned countries whose trade with *both* blocs grew after 2022 are candidates for rerouted trade: goods that once moved directly between the blocs may now move through a third country, sometimes after some processing, sometimes after very little. Countries such as Mexico and Vietnam are the obvious candidates, and they appear in the lecture as illustrations.

The main finding is that after the invasion trade *between* the two blocs fell relative to trade *within* them, by a margin the paper estimates precisely and which you should take from the paper itself rather than from any summary, including this one; foreign direct investment shows a similar pattern. The authors are careful in two directions. The fragmentation they measure is real and is concentrated in strategic sectors. It is also small compared with the Cold War, when trade between the Western and Soviet blocs was a small fraction of what it is now. Both halves of that conclusion matter to a firm. The first says the lines are hardening. The second says that the world is a long way from two separate economies, and that a strategy built on the assumption of complete de-coupling is a strategy built for a world that does not yet exist.

The design deserves the scrutiny a referee would give it. Bloc membership is fixed at a reference date, so countries that drift — and some do — are classified by where they stood, not where they are going. The non-aligned group is large and heterogeneous; it contains countries that lean slightly towards one side and countries that genuinely balance between them. The before-and-after comparison is anchored on a single event, and anything else that happened in 2022 is a candidate explanation. None of this undermines the paper. It defines what the paper licenses, which is the subject of the final section.

4.9 Turning alignment into a number

The replication depends on a single variable that most students have never met, and it is worth understanding how it is made, because every choice in its construction travels into the figure.

A General Assembly roll call records, for each resolution, whether each state voted yes, no or abstained. The simplest way to measure the alignment of two states is to count how often they vote the same way. Agreement scores of this kind have been used since the 1960s, and their limits were documented by Signorino and Ritter (1999), who showed that the common measures confound genuine similarity with the base rate of agreement on resolutions everyone supports. A deeper problem is comparability over time. The agenda changes from one year to the next. If the Assembly votes on more resolutions about the Middle East in one year and more about decolonisation in another, a change in a state's agreement with the United States may reflect the agenda rather than the state.

Bailey, Strezhnev, and Voeten (2017) solve the problem with a spatial model. Each state is assumed to have a position — an *ideal point* — on an underlying dimension, and each resolution divides the dimension into the positions that favour it and those that oppose it. The model estimates both states' positions and resolutions' cut-points from the pattern of votes. Comparability over time comes from resolutions that are repeated in essentially identical form across years: because the same text is voted on again, it anchors the scale, and movement by a state relative to those anchors is movement by the state. Voeten (2000) had earlier shown that after the Cold War the main dimension of Assembly voting separates states by their stance towards the US-led liberal order — which is why the dimension can be read as alignment with, or against, the United States.

Ideal point — a state's estimated position on a single underlying dimension of UN General Assembly voting, derived from all its roll-call votes and designed to be comparable over time. Higher values indicate positions closer to those of the United States. It measures revealed foreign-policy alignment in the Assembly; it does not measure trade, alliance membership, military co-operation or friendship.

The data come from the Dataverse record maintained by Erik Voeten and co-authors (Voeten 2026). Version 39, released in July 2026, estimates ideal points by calendar year, where older releases used Assembly sessions; the file still carries a session column, and the calendar year is the session number plus 1945. That change of meaning is exactly the sort of detail that belongs in a provenance log, because a figure built on the older file and a figure built on the newer one can disagree for reasons that have nothing to do with geopolitics. In 2021, the reference year used in the practice round, the file covers 193 countries, with the United States at 2.41 and China at −0.41 on the scale.

An ideal point is a continuous variable. A bloc is a category. Getting from one to the other requires a rule, and the rule is a modelling choice. The practice round uses a simple one: a country is US-leaning if it is closer to the United States than to China by more than a chosen margin, China-leaning if the reverse holds, and non-aligned otherwise. The margin is a parameter, which means it can be moved — and when it moves, countries change bloc and the figure changes with them. A result that survives a range of reasonable margins is evidence of fragmentation. A result that appears at one margin and vanishes at another is evidence of the margin.

The trade side of the figure has its own decisions. The source is the IMF's International Trade in Goods by partner country, the dataset formerly known as the Direction of Trade Statistics (International Monetary Fund 2026b). It reports, for each reporting country and each partner, the value of goods exports and imports in US dollars. Two things about it shape the replication. The first is the **mirror problem**: A's reported exports to B are not equal to B's reported imports from A, because of timing, of valuation (exports are recorded free on board, imports including cost, insurance and freight), of re-exports and of confidentiality. A figure must pick one side and say which; the practice round uses exporter-reported exports throughout. The second is that the values are in **current US dollars**. When commodity prices rise, the dollar value of trade rises without a single extra barrel or tonne crossing a border, and 2022 was a year of very high energy and food prices. Trade "within" a bloc that includes large energy exporters can rise for reasons that have nothing to do with alignment.

The final step is the index. Each series — trade within the US-leaning bloc, within the China-leaning bloc, between the two, and involving the non-aligned — is divided by its own value in a base year and multiplied by a hundred. Indexing removes the enormous differences in level between the series and lets the eye compare their *trajectories*, which is what fragmentation is about. It also hands enormous weight to the base year. Index to 2021 and the post-COVID rebound of 2021 is built into the denominator; index to 2019 and it is not. The paper's choice and the replication's choice should be stated, and if they differ, the difference should be logged.

4.10 What fragmentation does to a multinational, channel by channel

The course follows five channels through which geopolitical risk reaches a firm. Fragmentation touches all of them, but not equally, and a firm that can say which channel matters most for its own business has already done most of the analysis.

Market access is the most direct. A firm whose markets sit in different blocs faces the prospect that the rules governing its sales — tariffs, export controls, procurement preferences, product standards — will diverge between them. The first practical step is unglamorous: classify the firm's five most important countries by alignment, as the team milestone of this session asks, and look not only at where they sit but at whether they are drifting. A market that is non-aligned today and drifting is a different exposure from one that is non-aligned and stable.

Inputs are where friend-shoring and de-risking bite. Fragmentation in the data is concentrated in strategic sectors — semiconductors, critical minerals, advanced machinery, energy — and so is the policy pressure. A firm making consumer goods and a firm making semiconductor equipment live, in effect, in two different worlds: the first may see little change beyond rising transport and compliance costs, the second may find its suppliers, customers and technology licences reorganised by law. The connectors matter here twice over. A supplier in a connector country may be the cheapest route around a bloc boundary. It may also be the next target of rules-of-origin enforcement and transshipment investigations, because governments that wish to make fragmentation stick have every reason to close the routes around it.

Finance follows the same lines. The between-bloc decline Gopinath and co-authors find in foreign direct investment is the investment counterpart of the trade finding, and it reaches firms through investment screening, restrictions on outbound investment in sensitive technologies, and the cost of capital for projects that straddle the boundary. Payments, too, run through hubs of the kind Farrell and Newman (2019) describe, and access to them is a matter of jurisdiction.

Assets and people are exposed in the connector countries and in the blocs' borderlands. A plant built to serve both blocs from a non-aligned location is a bet that the location will stay non-aligned and that both blocs will continue to accept goods from it. Staff who move between blocs face visa regimes, export-control rules on what technical knowledge they may carry, and in extreme cases detention risk.

Data and operations fragment as well: rules on data localisation, on cloud providers and on cross-border transfers increasingly follow the same alignment, and a firm whose systems are integrated across blocs may find that integration itself becomes a compliance problem.

Across all five channels the same distinction from Martin, Mayer and Thoenig applies. The aggregate economy has substitutes; the firm has contracts. When a government decides that de-risking is affordable because the country's imports can be sourced elsewhere, it is making a claim about averages. The firm's own exposure is whatever its supply chain happens to be, and its own substitutes may be few, slow and expensive. The distinction between de-coupling and de-risking matters here too. If the data show de-risking — selective reductions in strategic sectors with overall trade preserved — then a blanket strategy of exit from the other bloc overreacts. If they show de-coupling, a strategy of waiting overreacts in the other direction. On the evidence so far the picture is closer to the former, and the right response is sectoral: know which of your products sit in the strategic categories where fragmentation is concentrated.

A tempting but wrong conclusion is that non-aligned markets are safe because they are on neither side. They are where between-bloc trade is being rerouted, and they are therefore where enforcement, scrutiny and secondary sanctions will concentrate. Non-alignment is a position in a contest, not an exemption from it.

4.11 A short review of the literature

The commercial peace has a long pre-history in political philosophy and a short history as a testable proposition. Angell (1910) gave it its most influential popular form and its most notorious timing. The modern empirical literature begins with Polachek (1980), who modelled conflict as an activity priced by the gains from trade it endangers. Oneal and Russett (1997) and Russett and Oneal (2001) established the dyadic correlation between interdependence and peace, alongside democracy and international organisations; Barbieri (1996) and Barbieri (2002) challenged it with longer series and different measures, and much of the subsequent dispute concerned measurement as much as theory. Mansfield and Pevehouse (2000) showed that preferential trading arrangements dampen conflict, and Mansfield (2021) is the most recent and compact review of the whole field.

The realist tradition starts from Hirschman (1945), whose distinction between the supply and influence effects of trade, and whose concentration index, remain the foundation of the study of economic dependence as power. Waltz (1979) placed dependence within structural realism as a vulnerability states will seek to limit. Gowa and Mansfield (1993) and Gowa (1994) identified the security externalities of trade and showed that allies trade more with each other. Copeland (1996), developed at length in Copeland (2015), reconciled the two camps by making expectations of future trade the variable that decides whether interdependence restrains or provokes. Martin, Mayer, and Thoenig (2008) separated bilateral from multilateral openness and showed that globalisation can lower the cost of bilateral conflict.

The contemporary turn is from trade as a restraint to trade as an instrument. Farrell and Newman (2019) recast interdependence as a network with hubs that confer panopticon and chokepoint power. Aiyar et al. (2023) defined geoeconomic

fragmentation and set out its channels; Gopinath et al. (2024), published as Gopinath et al. (2025), provided the first systematic evidence that trade and investment have begun to reorganise along lines of foreign-policy alignment. The measurement of that alignment rests on a parallel literature: Signorino and Ritter (1999) on the pitfalls of agreement scores, Voeten (2000) on the post-Cold War structure of General Assembly voting, and Bailey, Strezhnev, and Voeten (2017) on dynamic ideal points comparable over time. Le Billon and Shykora (2020) extends the analysis from manufactured goods to commodities, showing how supply chains themselves carry conflict.

The through-line is a shift in what is being explained. The older literature asked whether economic ties prevent war. The newer one asks how the expectation of conflict reorganises economic ties — and therefore treats trade data not as a cause of peace but as a record of how states expect each other to behave.

4.12 What the figure will not tell the firm

The practice round rebuilds, from ideal points and IMF trade data, a version of the figure at the centre of Gopinath et al. (2025): trade within and between blocs, indexed to the period before the invasion. If the build goes well, the lines will separate after 2022 in the direction the paper reports. The temptation will be to read that separation as a finding about geopolitics. It is a finding about a classification, a currency and a date, and each of those carries part of the result.

It is a **correlation with a date**. Sanctions on Russia, the post-pandemic rebound and the surge in commodity prices all move trade values in 2022, and all of them move trade *between* some blocs more than *within* others. A simple index cannot separate them; the paper's gravity specification goes some way towards doing so, and the replication does not. **Bloc membership is a modelling choice**: move the margin and countries move between blocs, and if the gap between the lines moves with them, the gap is partly an artefact of where the lines were drawn. **Values are not volumes**: a bloc rich in energy exporters will show trade rising in 2022 because oil and gas were expensive, not because its members drew closer. And the figure is **aggregate**: a multinational's products may sit in a sector where fragmentation is far stronger than the average, or in one where it is absent.

What the figure *can* do is narrower and still useful. It can show whether the broad pattern Gopinath and co-authors report is visible in raw data, with transparent choices that anyone can reproduce and change. It can show which of a firm's markets sit in which bloc under a stated rule, and how fragile that classification is. It can identify the connectors through which between-bloc trade appears to be flowing, which is where both the opportunities and the enforcement risks are concentrated.

The question to carry into the practice round is therefore the one the course asks of every replication. Suppose the lines separate exactly as the paper says. What does that figure *not* license a multinational to conclude? It does not license the conclusion that geopolitics caused the separation, that the firm's own sector follows the aggregate, that its non-aligned markets are safe, or that the bloc boundaries drawn at one margin in one year will be the boundaries that matter in five. The answer you give — in one sentence, on the catch slide — is the part of the round that most resembles the judgement a risk analyst is paid to make.

4.13 In the practice

> **The target.** Gopinath, Gourinchas, Presbitero & Topalova (2025): trade **between** geopolitical blocs against trade **within** them, before and after Russia's invasion of Ukraine. You define the blocs yourself from UN General Assembly ideal points, then watch what happens to the finding when you move the line.

Work in your team. One extract per team; everyone builds their own workbook.

4.13.1 1 · Collect — 10 min

Ideal points and codebook from Dataverse; your IMF IMTS extract of bilateral goods exports, annual, 2015–latest. If not done: `../data/README.md`.

Provenance log: one line per file. For the ideal points, log **version 39** and the change to calendar-year estimation. For IMF, log the date of the extract and the exact indicator you chose.

4.13.2 2 · Prepare — 25 min

4.13.2.1 A · Classify countries into blocs (ideal points)

1. **Connect** → **Text file** → `IdealpointestimatesFP_2026FP.csv`. The **ISO 3-letter code**, the **year** and the **ideal point** are `iso3c`, `year` and `IdealPointFP` (see the codebook). Rename `iso3c` → `iso3` and `IdealPointFP` → `IdealPoint` (right-click → *Rename*): the steps below use those names.
2. New sheet. Filter `year` = **2021** — the last full year before the invasion. (Gopinath et al. may use a different reference: check, and log the difference.)
3. Read the ideal points of the US (`USA`) and China (`CHN`) off the data. Create two parameters, `US ref` and `CHN ref` (*Float*), and type those two values in.
4. Create a third parameter `Margin` (*Float*, start at **0.5**) and the calculated field:

```
Bloc =
IF ABS([IdealPoint] - [CHN ref]) - ABS([IdealPoint] - [US ref]) > [Margin] THEN "US-leaning"
ELSEIF ABS([IdealPoint] - [US ref]) - ABS([IdealPoint] - [CHN ref]) > [Margin] THEN "China-leaning"
ELSE "Non-aligned"
END
```

5. `iso3` → Rows, `Bloc` → Rows, `IdealPoint` → Text. Sort by ideal point. Does the classification look right for countries you know? Adjust `Margin` until you can defend it — and write down the value and why.
6. **Export the classification** as a derived file: *Worksheet* → *Export* → *Crosstab to Excel* (or *View Data* → *Export All* to CSV). Save it as `data/derived/blocs_2021.csv` — **not** in `raw/`. Add a provenance line: derived from the ideal points, `Margin` = …, base year 2021.

4.13.2.2 B · Relate trade to blocs — the dyad model

1. New data source: **Connect** → **Text file** (or Excel) → your IMF extract. Identify the exporter, partner, year and value columns — in the IMF's SDMX layout `COUNTRY`, `COUNTERPART_COUNTRY`, `TIME_PERIOD`, `OBS_VALUE` — and rename them `exporter`, `partner`, `year`, `value`. Keep only `INDICATOR` = `XG_FOB_USD`.
2. Remove aggregate partners ("World", "Advanced Economies", regions — codes starting with `G` or `TX`, e.g. `G001` World): filter them out, and log which.
3. In the data source pane, drag `blocs_2021.csv` onto the canvas: Tableau creates a **relationship**. Click the noodle → relate `exporter` = `iso3`.
4. Drag `blocs_2021.csv` onto the canvas **a second time**. Relate `partner` = `iso3`. Rename this copy **Partner blocs** and its `Bloc` field **Partner bloc**; rename the first copy's field **Exporter bloc**.
5. IMTS uses ISO3 codes, so the relationship works directly. If your download shows names instead, you need a concordance table — log it.

4.13.3 3 · Replicate — 25 min

4.13.3.1 Sheet 1 · "Trade within and between blocs"

1. The pair type:

```
Pair type =
IF [Exporter bloc] = "Non-aligned" OR [Partner bloc] = "Non-aligned" THEN "Involves non-aligned"
ELSEIF [Exporter bloc] = [Partner bloc] THEN "Within bloc"
ELSE "Between US- and China-leaning"
END
```

2. The index, 2021 = 100:

```
Trade index = SUM([value]) / WINDOW_SUM(SUM(IF [year] = 2021 THEN [value] END)) * 100
```

3. `year` → Columns as a **discrete** dimension (blue); `Trade index` → Rows; `Pair type` → Colour. Right-click `Trade index` → **Edit Table Calculation** → **Compute Using** → **year**, so each pair type is indexed to its own 2021.
4. Reference line at 100. Annotate 2022.
5. Title with the finding. Caption: both sources with versions, and *"Blocs: ideal-point distance to US vs China, 2021, margin = ..."*.

4.13.3.2 Sheet 2 · "Who sits where?" — the classification on a map

`iso3` (geographic role *Country/Region*, ISO 3166-1 alpha-3) on the map, coloured by `Bloc`. Mark your team firm's five main countries with a **group** (*Create* → *Group*, "Our markets").

4.13.4 4 · Compare and break — 20 min

1. **Compare.** Put Sheet 1 beside the paper's figure. Their data are quarterly and their estimate controls for other determinants of trade; ours is annual and raw. Does the direction match? Is the gap between "within" and "between" of a similar order? Write one difference and its likely cause.
2. **Break the bloc definition.** Move `Margin` (0.25, 1.0), re-export, refresh. Which countries change bloc? Does the gap between the lines survive? This is the most important answer of the round.
3. **Break the side of the trade.** If your extract has the mirror series (imports reported by the partner), rebuild the index with it. Same picture?
4. **Extend — connectors.** Create a **set** of non-aligned countries (*Create* → *Set*, *Top* tab: top 10 by growth in exports to *both* blocs, 2021 → latest — or pick by hand from a table). Which are they? Is any of them among your firm's markets?

4.13.5 5 · The catch — 10 min

One slide:

Original	Ours	The catch
Gopinath et al.'s within/between figure	Sheet 1, and the same chart with another `Margin`	One sentence: what this figure does **not** license your team's multinational to conclude

A member drawn at random presents in two minutes.

4.13.6 Hand in

On ZoneCours before leaving, one per team: `.twbx` (including the derived `blocs_2021.csv`), provenance log, slide. Formative.

The one rule: your log records at least one place where the data disagreed with the paper — and your slide says how your blocs differ from theirs.

4.13.7 What loses credit

- Joining on country names (Türkiye, Côte d'Ivoire, Korea…)
- Leaving "World" or regional aggregates among the partners
- An index computed across pair types instead of within each one
- A bloc threshold that is not reported on the slide

Chapter 5

International Institutions: Who Finances, Insures and Referees the Firm

Session 5 · The economic institutions that finance, insure and arbitrate business — old and new Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **When a multinational relies on an international institution to finance, insure or protect its business, is it relying on a rule — or on the states that buy and sell influence inside it?**

Big work	Kuziemko, I., & Werker, E. (2006). How much is a seat on the Security Council worth? Foreign aid and bribery at the United Nations. *Journal of Political Economy*, 114(5), 905–930. doi:10.1086/507155
Target figure	US aid to elected Security Council members **in event time** — years −3 to +3 around the start of a two-year term
Data you collect	UN Security Council elected members (United Nations); US foreign aid (Greenbook, ForeignAssistance.gov or its mirror) — or the authors' replication file from AidData
Tableau skill	**LOD expressions** (`FIXED`) · years relative to an event · event-time charts · reference bands
Course themes	The impact and role of international institutions in analysing the geopolitical risks on which multinationals base their decisions

5.1 A hotel in New Hampshire, and what it still decides

Ask a manager which international institutions matter to her firm and the first answer is usually the United Nations. Ask instead who lent the money for the port her goods leave from, who insured the plant her company built in a country whose currency may not be convertible next year, who will hear the case if that country's government takes the plant, and who decided which firms were allowed to bid for the contract to build the port in the first place — and the list becomes long, specific and almost entirely economic. The World Bank, its private-sector arm, its insurer, its arbitration centre; a regional development bank in Manila, Abidjan or London; and, increasingly, a bank in Beijing or Shanghai that did not exist when today's graduate students were in primary school.

This chapter is about those institutions. It treats them in two ways at once, because a multinational has to. They are **business infrastructure**: sources of long-term finance, guarantees, political risk insurance, contracts and dispute settlement in exactly the markets where private finance is scarcest and political risk highest. And they are **arenas of power**: bodies governed by states, whose votes are unequally distributed, whose rules are bargained, and whose decisions — as the session's big work shows — can be bought. A firm that relies on such an institution to manage its geopolitical risk is relying on a rule; it is also relying on the coalition of states that keeps the rule in force. The question that runs through the chapter is which of the two it is really relying on, institution by institution and issue by issue.

The story starts, as the order it describes did, in July 1944.

From the archive · Bretton Woods, 1944

In July 1944 delegations from forty-four countries met at a resort hotel in Bretton Woods, New Hampshire, to design the post-war monetary order. The war in Europe was not over. The two plans on the table were British and American: John Maynard Keynes proposed an International Clearing Union with its own unit of account and pressure on creditor as well as debtor countries to adjust; Harry Dexter White, for the US Treasury, proposed a more modest stabilisation fund to which members would contribute and from which they could draw. The American plan largely prevailed, for the reason a realist would give: the United States was the creditor, and the creditor wrote the rules.

The Articles of Agreement of the International Monetary Fund were adopted on 22 July 1944, alongside those of the International Bank for Reconstruction and Development, and entered into force in December 1945. The design choice that matters most for this chapter was made then and has never been reversed: each member's **quota** — its subscription to the Fund — determines both how much it can borrow and how many votes it has. Money and voice were tied together from the first day. Eighty years later the argument about whether the new institutions of Asia and the BRICS complement or contest the Bretton Woods order is, at bottom, an argument about that tie.

5.2 Why would states under anarchy build institutions at all?

Session 03 left us with a system that has no government above states, and Session 04 with trade reorganising itself along geopolitical lines. Yet the same system is thick with institutions. Two traditions explain why, and they make opposite predictions for a firm.

The liberal institutionalist answer, given its canonical form by Keohane (1984), is functional. States that would gain from cooperation often fail to achieve it because they cannot observe each other's behaviour, cannot make credible promises, and fear being exploited. Institutions lower the **transaction costs** of agreement, supply **information** about compliance, and lengthen the **shadow of the future** by turning one-off bargains into repeated games. Keohane's point was that this logic survives the decline of the hegemon that created the institutions: once built, they are cheaper to maintain than to replace, and states keep obeying them because reneging would be costly.

The realist answer, set out most sharply by Mearsheimer (1994), is that institutions reflect the distribution of power. The strong build them to serve their interests, and when an institution's rules collide with a powerful state's interests, the rules give way. On this view an institution has no independent effect on behaviour; it is a stage on which power is exercised, and its apparent successes are cases in which states would have behaved the same way without it.

International institution — a persistent set of rules, usually with an organisation to administer them, that states have agreed to and that prescribes behaviour, constrains activity and shapes expectations. The economic institutions in this chapter add a balance sheet: they lend, guarantee, insure or invest, and their votes are attached to capital.

Neither account is a strawman, and a firm does not need to choose between them in general. It needs to know **which institution, on which issue, behaves like which**. A MIGA guarantee, an IFC loan covenant and an ICSID award are rules that have, on the whole, been honoured; the WTO's Appellate Body is a rule whose enforcement collapsed when its most powerful member stopped appointing judges. That is an empirical question, and the rest of the chapter treats it as one.

A third idea is needed to read the last decade. Morse and Keohane (2014) observed that dissatisfied states do not only lobby inside an institution; they also go around it. They may move an issue to a different existing body — *regime shifting* — or build a new one — *competitive regime creation*. They called the result **contested multilateralism**.

Contested multilateralism — the use of multilateral institutions to challenge the rules, practices or missions of other multilateral institutions, either by shifting an issue to an existing alternative forum or by creating a new institution to compete with the old one (Morse and Keohane 2014).

The Asian Infrastructure Investment Bank was their leading example of the second strategy, and it is where this chapter ends up. To see why it was built, one has to start with the votes inside the institutions it was built beside.

5.3 The votes behind the money

The International Monetary Fund lends to governments, never to firms. It matters to a multinational because its programmes set the conditions in which the firm operates: whether the host currency floats, whether capital may leave, whether subsidies and public-sector wages are cut, whether the government can pay its suppliers. A country in an IMF programme has its macroeconomic policy partly written in Washington.

The Fund now has 191 members. Its governance follows the 1944 design: quotas determine votes. As of 30 April 2026 the United States held 16.49% of total voting power, Japan 6.14% and China 6.08%; the other founding BRICS each held less than 3% (International Monetary Fund 2026a). Article III of the Articles requires an 85% majority of total voting power for any change in quotas, so a member holding more than 15% can block a quota reform alone — and only one member does.

Voting power and capital — in a shareholder-governed institution, a member's votes are a function of its capital subscription (plus, usually, a small equal allocation of basic votes). Voting power is therefore a lagging indicator of economic weight: it moves only when members agree to reallocate capital, and those whose share would fall can often block the reallocation.

The lag is the hinge of the session. China's economy is far larger relative to the American economy than a vote of 6.08% against 16.49% suggests, and reform has been slow. The Sixteenth General Review of Quotas, approved in December 2023, raises total quotas by half but does so **equiproportionally**: every member's share stays where it was. Even that increase had not taken effect by October 2026 — the IMF had extended the deadline for members' consents to 15 November 2026 — and the realignment of shares was left to a Seventeenth Review for which, at the time of writing, guiding principles had been endorsed but no formula agreed. The same pattern holds at the World Bank: in the IBRD the United States held 15.98% of votes, Japan 6.88% and China 5.75% as of 30 June 2026 (World Bank Group 2026).

Two other institutions shape the conditions of business without lending a cent. The World Trade Organization, founded in 1995 with 166 members today, holds the tariff bindings and non-discrimination rules on which a firm's market access rests. Its Appellate Body has had no members since 30 November 2020, because appointments require consensus and the United States has withheld it; a subset of members now appeal through an interim arbitration arrangement instead. The rule still exists; whether it binds depends on which members have opted in. The OECD's relevance is quieter. Its Arrangement on Officially Supported Export Credits, in force since 1978, limits the terms that participating governments' export-credit agencies may offer, so that exporters compete on price and quality rather than on subsidised finance — and China is not a participant. Its Guidelines for Multinational Enterprises give anyone a route to complain about a firm's conduct through National Contact Points. And its Inclusive Framework negotiated the global minimum tax that now shapes where multinationals book profits.

5.4 The World Bank Group as the firm's toolkit

The World Bank is usually described as a development institution, which it is. For a multinational it is also five distinct services, run by five legally distinct institutions with overlapping but different memberships.

The **IBRD** (1944) lends to middle-income and creditworthy governments; the **International Development Association** (1960) gives grants and concessional credits to the poorest. Neither lends to firms. Both matter to firms because the money they lend is spent buying goods, works and services, and the World Bank's Procurement Framework, in force since July 2016, requires that much of that buying be done through open international competition, with notices published on United Nations Development Business and the Bank's own site. Procurement is the least glamorous and most overlooked business channel of the multilateral system; for an engineering, construction or equipment firm it is often the largest.

Procurement has a disciplinary side. Firms found to have engaged in fraud, corruption, collusion or coercion in Bank-financed contracts are **debarred**, and their names published. Since the Agreement for Mutual Enforcement of Debarment Decisions of 9 April 2010, a debarment of more than one year by the World Bank Group, the Asian Development Bank, the African Development Bank Group, the European Bank for Reconstruction and Development or the Inter-American Development Bank Group is enforced by the other four. A sanction at one bank becomes a sanction at five — which makes integrity compliance in one country a condition of market access in dozens.

The **International Finance Corporation** (1956) invests in the private sector directly: loans, many in local currency, equity stakes, guarantees, and the mobilisation of other investors. Its Managed Co-Lending Portfolio Program lets institutional investors — insurers, pension funds, sovereign funds — lend alongside IFC in a portfolio that mirrors its own. For a firm, IFC is a co-investor whose presence lowers the cost of other finance and signals to the host government that the project has a powerful friend.

The **Multilateral Investment Guarantee Agency** (1988) is the institution in this chapter most directly in the business of geopolitical risk. It insures cross-border investors and lenders against five political perils, and in fiscal year 2026 it issued a record US$10.6 billion of guarantees for sixty projects, bringing its gross exposure to US$41.4 billion (Multilateral Investment Guarantee Agency 2026). Since July 2024 the Group's guarantee products have been brought together in a single platform housed at MIGA.

Political risk insurance — cover for a cross-border investor or lender against losses caused by government action or political violence: currency inconvertibility and transfer restriction, expropriation, war and civil disturbance, breach of contract by the host government, and non-honouring of a sovereign financial obligation. Multilateral cover adds something private cover cannot: the insurer's shareholders include the host state.

That last clause is the deterrence argument for multilateral insurance. A host government that expropriates a MIGA-insured investor is not only in dispute with a foreign firm; it is in dispute with an agency of which it is itself a member, which will pay the claim and then seek recovery from it, and whose other members include the governments that decide its access to World Bank lending. The insurance works, in part, because the coalition behind it is large. It also means that the insurer is governed by states: in MIGA the United States held 14.81% of votes as of 30 June 2026 (World Bank Group 2026), and the cover a firm can buy reflects what that coalition is willing to underwrite.

The **International Centre for Settlement of Investment Disputes** (Convention in force 1966) provides the forum. A firm whose host state has consented — usually through a bilateral investment treaty — can bring a claim against the state before an arbitral tribunal rather than the state's own courts. ICSID had registered 1,118 cases by mid-2026, and of the cases registered in fiscal year 2026 a quarter concerned oil and gas and almost a fifth mining; more than three in five were brought under bilateral investment treaties (International Centre for Settlement of Investment Disputes 2026).

Investor–state arbitration — a procedure in which a foreign investor brings a claim directly against a host state before an international tribunal, under consent given in a treaty, a contract or a national investment law. The award is binding; whether it is paid depends on the state.

The forum is a rule a state can leave. Bolivia denounced the ICSID Convention in 2007, Ecuador in 2009 and Venezuela in 2012; Ecuador rejoined in 2021, and Honduras, which gave notice in 2024, rejoined in 2026. The investor protection a firm buys when it structures an investment through a treaty jurisdiction is real, but it is protection under a regime whose membership moves with the politics of host states.

5.5 Does influence inside an institution have a price?

Everything so far assumes that the institutions apply their rules to all comers. The realist suspicion is that they do not — that the votes in the tables above are used, and that influence is traded. The session's big work offers the sharpest test of that suspicion, and it does so in the institution whose decisions bind every firm in the world.

The UN Security Council has fifteen members: five permanent, with a veto, and ten elected by the General Assembly for two-year terms by regional group, five each year, without immediate re-election. Before 1966 there were six elected seats. Substantive decisions need nine votes and no permanent-member veto, and measures adopted under Chapter VII of the Charter — sanctions, arms embargoes, authorisations of force — bind every member state, and therefore every firm operating in one. The arithmetic gives the elected members leverage: the permanent five cannot pass anything without at least four of them.

Kuziemko and Werker (2006) asked whether that leverage is paid for. Their identification idea is the reason the paper is the session's big work. Rotation onto the Council follows regional queues and informal norms, not a country's need for aid in a particular year. So instead of comparing members with non-members — who differ in a hundred ways — they compare a country **with itself**, in the years before, during and after its term. If aid rises during the term and falls back afterwards, the seat is the likeliest explanation: poverty, alliances and politics do not rotate on a two-year cycle.

Event time — time measured relative to an event rather than by the calendar: year 0 is the first year of a Council term, −1 the year of the election, +2 the first year after the term. Re-indexing every country's series to its own event aligns cases that happened decades apart and lets one ask what typically happens around the event.

They found that US aid to a developing country rises substantially when it rotates onto the Council — by about 59% in the paper's headline estimate — and that aid through the UN rises too, more modestly. The increase is concentrated in the term years and disappears afterwards. And it is larger in years when the Council matters more, measured by how much the press covered it. That last result turns a pattern into a mechanism: if aid followed membership regardless of what the Council was doing, rotation might proxy for something else; that it tracks the **salience** of Council business is what makes vote buying the most plausible reading.

The distinction between finding and interpretation matters for the practice round: the paper shows that aid moves with membership and with salience; *bribery* is the authors' word for the most plausible mechanism, not something the data observe.

For this chapter the decisive step is what came next. The same research programme looked inside the economic institutions. Temporary Council members receive more World Bank projects (Dreher, Sturm, and Vreeland 2009a) and IMF programmes with fewer conditions attached (Dreher, Sturm, and Vreeland 2009b); Vreeland and Dreher (2014) assemble the evidence across channels. Kilby (2009) showed that World Bank disbursements respond less to a borrower's compliance with conditions when the borrower is aligned with the United States. Stone (2011) built a general account of how the dominant shareholder exercises **informal** influence over the IMF when its interests are at stake, while leaving the formal rules to operate otherwise. The voting tables of the previous section are not trivia. Who holds the votes shapes who gets the money — and the money decides where the contracts, the co-investment and the insurance are.

5.6 Regions with banks of their own

The World Bank model was copied, region by region, with a different balance of shareholders each time. The **Inter-American Development Bank** (1959) was the first; its private-sector arm, now IDB Invest, has operated since 1986, and in 2024 its governors approved a capital increase of US$3.5 billion tied to an "originate-to-share" model in which the bank originates loans and sells much of them on to private investors. The **African Development Bank** followed in 1964, the **Asian Development Bank** in 1966, the **European Bank for Reconstruction and Development** in 1991, to finance the transition of the former Soviet bloc. The **European Investment Bank** (1958) is the EU's own bank, owned by its member states, and the **Islamic Development Bank** (operations from 1975) serves the members of the Organisation of Islamic Cooperation, with separate arms for private investment, trade finance and export-credit and political risk insurance. Sub-regional banks fill in the map — CAF in Latin America, owned mainly by its borrowers; the Eurasian Development Bank in the post-Soviet space, founded by Russia and Kazakhstan.

Each has a private-sector window, a trade-finance programme that guarantees the letters of credit of local banks so that international banks will confirm them, and a procurement market. And each tells a geopolitical story through its shareholder register. Japan and the United States each held 15.6% of the Asian Development Bank's shares at the end of 2025; the United States holds about 30% of the votes in the Inter-American Development Bank; Nigeria is the largest single voter in the African Development Bank; Saudi Arabia holds the largest share of the Islamic Development Bank.

Three features of the regional banks matter to a firm in ways the World Bank's rules can obscure. The first is that **procurement eligibility is foreign policy**. The Asian and Inter-American banks restrict contracts to firms and goods from member countries; the EBRD's are open to firms from any country; the African Development Bank opens contracts financed by its concessional fund to all while restricting those financed by the bank itself to members. Whether a Canadian, Chinese or Turkish firm may bid on a road in Bangladesh or Bolivia depends on whether its home state paid in.

The second is that a development bank can become an **instrument of sanctions**. On 4 April 2022 the EBRD's Board of Governors suspended Russia's and Belarus's access to the bank's resources and decided to close its offices in Moscow and Minsk (European Bank for Reconstruction and Development 2022). Both remain shareholders. A firm that had counted on EBRD co-finance in either country lost it overnight, through a decision taken by votes in which the host state took part and lost.

The third is that **shareholding moves**, slowly. The EBRD has expanded into sub-Saharan Africa, with Benin, Côte d'Ivoire and Nigeria becoming countries of operation in 2025; the Asian Development Bank amended its charter for the first time,

with effect from March 2026, to lend more against the same capital; the Eurasian Development Bank cut Russia's share below half at the end of 2022. Each move redraws the map of where a firm finds support, and on whose terms.

Multilateral development bank — a bank owned by several states that borrows on capital markets against its members' paid-in and callable capital and lends, invests and guarantees for development purposes. Its rating, and therefore its cost of funds, depends on the creditworthiness of its shareholders and on its standing in the dollar and euro bond markets.

5.7 Building beside the old house: the new institutions

The new institutions of the last decade are a response to the lag between votes and weight. They come in three kinds: development banks, reserve pools, and the national policy banks of China.

From the archive · London and Beijing, March 2015

On 12 March 2015 the British government announced that it intended to join China's proposed Asian Infrastructure Investment Bank as a founding member — the first major Western economy to do so. Within weeks other European governments had followed, among them France, Germany and Italy. The United States and Japan stayed out, and are still out. On 29 June 2015 the bank's Articles of Agreement were signed in Beijing by fifty countries; they entered into force on 25 December 2015, and the bank opened on 16 January 2016 with fifty-seven founding members.

For this chapter the episode is a demonstration of Morse and Keohane's mechanism. China had sought a larger voice in the IMF and the World Bank for a decade, without a realignment of shares. It then built an institution in which it held the largest share by design — and invited the Europeans in. Once they were in, the new bank could not easily be dismissed as a Chinese instrument, and the old ones lost their monopoly on legitimacy in infrastructure finance. Competitive regime creation does not need to replace the incumbent to change the bargaining inside it.

The **Asian Infrastructure Investment Bank** has an authorised capital of US$100 billion. Its membership had reached 111 approved members by October 2026, 106 of them full members (Asian Infrastructure Investment Bank 2026). China holds 30.5% of subscriptions and 26.37% of votes; India, the second-largest member, holds 7.53% and Russia 5.92%. The Articles require a "super majority" — two-thirds of governors representing at least three-quarters of total voting power — for the most important decisions, among them capital increases and the election of the president. China's quarter of the votes is therefore a veto, as the United States' sixth of the IMF's is. The board is non-resident, a deliberate contrast with the full-time resident boards of the Bretton Woods institutions. The bank signed co-financing frameworks with the World Bank and the Asian Development Bank in its first months and co-financed many of its early projects with them; it lends to private as well as sovereign borrowers, and offers non-sovereign loans in a range of local currencies.

The **New Development Bank** was created by the five original BRICS under an agreement signed at Fortaleza on 15 July 2014 and is headquartered in Shanghai. Its initial subscribed capital of US$50 billion was divided **equally** among the founders, and voting power equals subscribed shares; the founders' combined votes may not fall below 55%, and non-borrowing members' combined votes may not exceed 20% (BRICS Governments 2014). Bangladesh, the United Arab Emirates, Egypt, Algeria and, in 2026, Uzbekistan have since joined; several other countries have been admitted but have not yet completed accession. Its 2022–2026 strategy set a target of 30% of financing in local currencies. The equal-shares rule is the most interesting design choice in the new architecture: it prevents China from dominating the bank by capital, at the price of limiting how much capital the bank can raise.

The reserve pools are less visible to firms but matter to the country risk they face. The **BRICS Contingent Reserve Arrangement**, signed on the same day in Fortaleza, commits US$100 billion — China 41, Brazil, India and Russia 18 each, South Africa 5 — to members facing balance-of-payments pressure; but only 30% of a member's maximum access is available without an IMF programme in place. The **Chiang Mai Initiative Multilateralisation** among the ASEAN members, China, Japan and Korea, in force since 2010, is a US$240 billion network of currency-swap commitments, 40% of which has been available without an IMF link since 2021; its surveillance unit, AMRO, became an international organisation in 2016. Both are regional insurance that defers, for most of its value, to the Fund they were meant to supplement.

IMF-delinked portion — the share of a regional financing arrangement that a member can draw without having an IMF programme in place. It measures how far the arrangement is an independent lender of last resort rather than a co-financier of the Fund: 30% for the BRICS Contingent Reserve Arrangement, 40% for the Chiang Mai Initiative Multilateralisation.

The third kind of new institution is not multilateral at all. China's two policy banks, the China Development Bank and the Export-Import Bank of China, both created in 1994, and the Silk Road Fund, established in December 2014, are the financial core of the Belt and Road Initiative announced in 2013. AidData's tracking of Chinese official finance counts US$1.34 trillion of loans and grants to 165 low- and middle-income countries between 2000 and 2021 (AidData 2023). Its terms differ from those of the multilateral banks in ways that matter: Horn, Reinhart, and Trebesch (2021) show that a large part of China's overseas lending has gone unreported to the international debt statistics, much of it on commercial rather than concessional terms and often collateralised; Dreher et al. (2022) trace its aims and effects project by project. For a non-Chinese firm the opportunities are case by case. Contracts on Chinese-financed projects have often gone to Chinese contractors, and the standards that govern them are set bilaterally rather than by the published policies of a multilateral board.

BRICS itself, finally, is not an institution: it has no charter and no secretariat, and its two treaty bodies are the NDB and the CRA. Its expansion matters to a firm as a map of alignment. Egypt, Ethiopia, Iran and the United Arab Emirates became members on 1 January 2024 and Indonesia in January 2025; Saudi Arabia is listed by the BRICS chairs but has never formally confirmed its membership, and its status is best described as ambiguous. A category of partner countries, created at the Kazan summit in October 2024, numbered ten by mid-2025. The New Delhi summit of September 2026 added no members or partners and asked for continued work on cross-border payments in local currencies — work, not a launched system.

5.8 Complement or counter-hegemony?

The question every analyst asks of the new institutions is whether they complement the Bretton Woods order or contest it. Put that way, it has no answer; split into four tests, it does.

On **votes**, the new banks give emerging economies the voice that quota reform at the Fund has not — and reproduce its structure. China's veto in the AIIB mirrors the American veto in the IMF; the NDB avoids a dominant shareholder only by fixing equal shares among the founders. A realist reading these tables would not be surprised.

On **standards**, the evidence leans towards complement. The AIIB adopted its own environmental and social framework and co-financed many of its first projects with the established banks; Wang (2017) and Humphrey (2022) both stress how much of the older banks' operating model the new ones borrowed. The contest is sharper outside the multilateral banks, in bilateral Chinese lending that sits outside the OECD export-credit rules and the cross-debarment agreement, on terms that are often confidential.

On **currency**, local-currency lending by the AIIB and the NDB reduces the currency mismatch that has ruined many infrastructure borrowers, which any development economist would call a complement. The BRICS work on local-currency settlement is something else: an effort to reduce members' dependence on the dollar and the payment chokepoints through which sanctions are enforced (Session 10).

On **sanctions**, the test came in March 2022. Within a week of Russia's full-scale invasion of Ukraine, the AIIB announced that all activities relating to Russia and Belarus were on hold and under review, and the NDB put new transactions in Russia on hold — Russia being one of its five founders. Both banks depend on dollar funding markets. When the choice was between a founding member and access to the market that funds them, they chose the market. That is the strongest single piece of evidence that the new multilateral banks are, for now, inside the system they were thought to contest.

The verdict, then, is **both**, and the reason is structural. The new institutions do not need to replace the old ones to change the bargaining inside them; they need only to exist as an **outside option**. Ikenberry and Lim (2017) frame the AIIB as institutional statecraft that pursues influence partly through the existing order rather than against it. Humphrey and Michaelowa (2013) had already shown, for Latin America, that borrowers with access to a bank controlled by borrowers prefer it to the banks controlled by creditors. A host state with several lenders can shop among them, and the conditions any one of them attaches bind less.

5.9 What the firm should do with all this

Across the course, geopolitical risk reaches a firm through five channels: market access, inputs, finance, assets and people, and data and operations. The economic institutions of this chapter touch all five, but they open four specific **business channels** — procurement, finance, insurance and dispute settlement — and each comes with its own geopolitical risk.

Procurement is a market-access channel. The projects financed by the World Bank, the regional banks, the AIIB and the NDB buy through published procedures; a firm that tracks project pipelines and procurement notices sees demand years before it reaches the market. The risk is eligibility: whether the firm may bid at all depends on its home state's shareholding, and a cross-debarment for misconduct in one country closes the channel everywhere.

Finance and co-investment belong to the finance channel. IFC, IDB Invest, the EBRD and the private-sector windows of the Asian and African banks lend and invest alongside firms, increasingly in local currency and increasingly by mobilising institutional investors. A development bank in the capital structure lowers the cost of other finance and raises the political cost to a host government of interfering. The risk is conditionality and its politics: the institution's own standards bind the project, and its decisions respond, as the previous sections showed, to the preferences of its largest shareholders.

Insurance and guarantees protect assets and people. MIGA, the World Bank Group's guarantee platform, the Islamic Development Bank's insurer and the trade-finance programmes of the regional banks turn political risk into a premium. The risk is that the cover is itself political: what a multilateral insurer will underwrite, and in which countries, reflects its shareholders' view of those countries.

Dispute settlement — ICSID and the treaty network behind it — protects assets against the host state. The new institutions have built nothing comparable. The risk here is exit: states denounce the Convention, terminate treaties, or decline to pay awards, and the protection is only as strong as the regime's membership.

Three further risks cut across the channels. **Sanctions exposure** runs through the institutions as well as around them: suspension at the EBRD, freezes at the AIIB and the NDB. **Alignment** is signalled by membership — BRICS status, AIIB or EBRD shareholding, a Belt and Road memorandum, a vote at the Council — and an analyst can read it. **Competing standards** mean that a project bankable by one lender may not be bankable by another.

An institution is a risk **mitigator** when the coalition that sustains it is stable, and a risk **source** when it is not. A firm's dashboard should track the coalition, not only the rule: who holds the votes, who sits on the Council, who has been suspended, who has joined which club.

A guarantee, a co-financing agreement or a treaty is a promise by a coalition of states. Read the coalition before the contract.

5.10 How the concept becomes data

Institutions leave good data: they publish their rules, members and votes. Voting power is the simplest series: the IMF publishes each member's quota and votes, the World Bank Group publishes country voting tables for each of its institutions, and the AIIB publishes each member's subscription and voting power (International Monetary Fund 2026a; World Bank Group 2026; Asian Infrastructure Investment Bank 2026). A comparison of China's and the United States' shares across the IMF, the IBRD and the AIIB, dated and sourced, is a one-sheet summary of the session's argument. Activity is the second series: MIGA's annual issuance and exposure, ICSID's caseload by sector and legal basis, the regional banks' commitments, AidData's project-level record of Chinese official finance. Membership and its changes — accessions, suspensions, denunciations — are the third, and the easiest to turn into events.

The fourth kind of data is the one the practice round uses: an institution's own rules, turned into a natural experiment. The rotation of elected Council members is set by the UN Charter and by regional custom. The UN publishes the list of elected members; the United States publishes its aid by country and year. Neither file contains the quantity Kuziemko and Werker study. It has to be built by joining the two on a country code, re-indexing every country-year to the start of a term, and expressing aid relative to the country's own pre-term level — which is where the session's Tableau skill comes in.

The level-of-detail expression in the practice computes each country's pre-term baseline once per term, whatever else is in the view. That is the event-study idea in one calculation: every country is its own control, and the comparison is made **within** each term before it is averaged **across** terms.

5.11 A short review of the literature

The theoretical frame is the exchange between Keohane (1984), for whom institutions persist because they reduce the costs of cooperation, and Mearsheimer (1994), for whom they reflect the distribution of power and constrain it little. Ikenberry (2011) reads the post-war order as an American project that bound its creator as well as others, which is why its institutions outlived the moment of American dominance. Morse and Keohane (2014) supply the concept that organises the last decade — contested multilateralism, through regime shifting and competitive regime creation.

On the politics inside the Bretton Woods institutions, Woods (2006) examines how the IMF and the World Bank's relations with their borrowers are shaped by their largest shareholders, and Stone (2011) models the informal influence of the dominant shareholder alongside the formal rules. The empirical programme on the price of influence starts with Kuziemko and Werker (2006) on US and UN aid to elected Security Council members, and extends to World Bank projects (Dreher, Sturm, and Vreeland 2009a), IMF conditionality (Dreher, Sturm, and Vreeland 2009b) and World Bank disbursements (Kilby 2009); Vreeland and Dreher (2014) synthesise it. Humphrey and Michaelowa (2013) show that shareholder composition shapes borrowers' choices among development banks.

On investment protection, Büthe and Milner (2008) find that international trade agreements raise FDI into developing countries by making policy commitments more credible, and Simmons (2014) shows that developing countries signed bilateral investment treaties, and tied their hands most tightly, when their bargaining position was weak — after which investors sued governments with surprising regularity. On the new institutions, Wang (2017) gives an early assessment of the AIIB and the NDB as complements to and potential competitors of the established banks; Ikenberry and Lim (2017) read the AIIB as Chinese institutional statecraft; Humphrey (2022) places both in a broader account of multilateral development banks under geopolitical change. On Chinese official finance, Dreher et al. (2021) build the project-level dataset and estimate its growth effects, Dreher et al. (2022) extend the analysis of its aims and impacts, and Horn, Reinhart, and Trebesch (2021) document the scale of lending absent from official debt statistics.

5.12 Into the practice round

The practice round rebuilds Kuziemko and Werker's evidence in event time: US aid to developing countries in the three years before, the two years of, and the two years after a Security Council term, each country measured against its own pre-term level. If the paper's pattern survives your data — flat before, higher during the term, back down afterwards — you will have seen, in a chart you built yourself, influence being priced inside an international institution.

Read the chart with the discipline of the earlier sections. It is evidence about **one** institution, **one** donor, and the decades the data cover. It does not show that the World Bank, MIGA, the AIIB or the NDB sell their decisions; the follow-up literature makes a case for some of them, with its own designs and its own limits. It does not tell a firm how a particular host government on the Council will vote, or whether a sanctions resolution will survive the next change of government in a permanent member. And because the aid series you will download is a later release than the one the authors used, in different constant dollars, your figure will not match theirs exactly; the log you keep is where that disagreement is recorded.

The question to ask at the end — what does this figure not license a multinational to conclude? — is the institutional version of the session's opening question. The chart shows that the coalition behind a rule can be bought. A firm that relies on the rule should therefore know who is in the coalition, what they are being offered, and how long the offer lasts.

5.13 In the practice

> **The target.** Kuziemko & Werker (2006): US aid to developing countries **in the years around a Security Council term** — flat before, higher during the two term years, back down after.

In your teams. One workbook per team; everyone builds at least one sheet.

5.13.1 1 · Collect — 15 min

You should arrive with route A or route B files in `data/raw/` (`../data/README.md`). Finish what is missing, and fill the provenance log — one line per file.

file	source	url	version / release	accessed	licence	notes
us_foreign_aid_country.csv	US State Dept (ForeignAssistance.gov, Country Summary)	foreignassistance.gov/data/ *or the mirror*	*"frozen 2024-12-19"*	*today*	free	*fiscal years; obligations; constant_amount; economic + military combined*
unsc_elected_members.xlsx	United Nations	main.un.org/securitycouncil/…	*your download*	*today*	free	*your transcription; name changes handled*

Route A teams also need a **term table**. Open the replication file, filter on Security Council membership = 1, show country × year as a text table, export it (*Worksheet → Export → Crosstab to Excel*), and mark the first year of each two-year run as `term_start`. Ten minutes; it gives you the same term table as route B.

5.13.2 2 · Prepare — 20 min

1. **Connect** to the aid file and to `unsc_elected_members.xlsx`. In the Country Summary, filter `Transaction Type Name` = *Obligations* (a data-source filter), exclude `Fiscal Year` = `1976tq`, then make `Fiscal Year` a **number** and rename it `Year`; rename `constant_amount` to `Aid`.
2. Drag the aid table to the canvas, then **double-click it** to open the physical layer, and drag the term table next to it: **inner join** on `Country Code` = `Iso3` (or on country name only if you have checked every match — see the trap list in `TABLEAU-TOOLKIT.md`). Route A teams: use the replication file's own country, year and aid fields. > Why a join, not a relationship? The next step is a row-level calculation that uses `Year` from > one table and `Term start` from the other. Relationships do not allow that.
3. Check the row count before and after the join and write both in your log. The join should multiply rows: each country-year now appears once per term of that country.
4. Keep only aid **recipients** (drop regional aggregates such as "World", "Africa Region" — every `Country Name` ending in "Region"), and check that aid is in **constant** dollars (`constant_amount`, not `current_amount`). If it is not, say so in the log — the ratio to baseline below limits the damage, but does not remove it.

5.13.3 3 · Replicate — 25 min

5.13.3.1 Calculated fields

```
[Event time]               = [Year] - [Term start]

[Baseline aid]             = { FIXED [Iso3], [Term start] :
                               AVG( IF [Event time] >= -3 AND [Event time] <= -1
                                    THEN [Aid] END ) }

[Aid relative to baseline] = [Aid] / [Baseline aid]      // row level: one value per country-year-term
```

Route A teams: if aid is in logs in the file, use `[Log aid] - [Baseline log aid]` instead of a ratio.

5.13.3.2 Sheet 1 · "Aid in event time"

1. Filter `[Event time]` to **−3 … +3** (make it a dimension, discrete).
2. `[Event time]` → Columns; `[Aid relative to baseline]` → Rows, then change its aggregation to **AVG** (right-click the pill → *Measure* → *Average*). Because the ratio is computed row by row — one row per country-year-term — the average at each event year is the average **across terms**.
3. Mark type **Line**, then add **Circle** marks on a dual axis so each year is visible.
4. **Analytics → Reference Band** from 0 to 1 on `[Event time]`, shaded, labelled "On the Council".
5. Constant **reference line at 1.0**, labelled "Pre-term baseline".
6. Title it with the finding; caption with both sources, versions and access dates.

5.13.3.3 Sheet 2 · "How many terms?"

`[Event time]` → Columns, `COUNTD([Iso3])` → Rows. The number of terms behind each point. Is the sample balanced across event years? If not, why?

5.13.4 4 · Compare and break — 20 min

1. **Differ.** Put your Sheet 1 beside the paper's event-time evidence. Same shape? Same size? Write one difference and its most likely cause (release of the aid data, years covered, the sample of countries, constant vs current dollars, levels vs logs).
2. **Break one decision.** Replace the **mean** across terms by the **median** (a quick change of aggregation). Does the term-year bump survive? What does that tell you about whether a few large cases drive it?
3. **Break another.** Restrict to terms starting **after 1990**. Does the pattern survive the end of the Cold War?
4. **Extend — salience.** The paper finds the effect is larger when the Council matters more. Build a crude proxy: tag the years 1990–91 (Gulf War), 2002–03 (Iraq) as *high salience* with a calculated field, and colour Sheet 1 by it. What do you see — and why is two episodes far too few to conclude?

5.13.5 5 · The catch — 10 min

Original	Ours	The catch
The paper's event-time evidence	Your Sheet 1	What it does **not** license a multinational to conclude — for instance about how durable a UN sanctions resolution is, or what a host government on the Council will do

A member drawn at random presents in two minutes.

5.13.6 Hand in

Upload to ZoneCours before leaving: the `.twbx` and the provenance log. Formative: commented, not marked; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

Team project. Is any of your firm's five key countries — or its home country — on the Council now, or about to be? Add it to your footprint notes.

Chapter 6

Private Authority, Standards and Global Regulation

Session 6 · Who writes the rules a multinational actually follows? Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Most of the rules that govern a multinational's daily operations were not written by a government. Who wrote them, with whose power — and does trade spread them up or down?**

Big work	Prakash, A., & Potoski, M. (2006). Racing to the bottom? Trade, environmental governance, and ISO 14001. *American Journal of Political Science*, 50(2), 350–364. doi:10.1111/j.1540-5907.2006.00188.x
Companion readings	Abbott & Snidal (2009), *The governance triangle* · Mattli & Büthe (2003), *Setting international standards* · Brunsson, Rasche & Seidl (2012), *The dynamics of standardization* · Green (2013), *Rethinking Private Authority*
Target figure	**The diffusion of ISO 14001** across countries — a map and small multiples over time — and each country's adoption against the adoption level of its **export markets**
Data you collect	ISO Survey of certifications (ISO 14001, ISO 9001); IMF bilateral exports (IMTS); World Bank GDP (WDI)
Tableau skill	**Maps** · geocoding by ISO code · fixing unmatched countries · the **Pages** shelf · small multiples
Course themes	The impact and role of international institutions — here private and hybrid ones — in the geopolitical risks on which multinationals base their decisions

6.1 Count the rules in a telephone

Pick up a mobile phone and ask how many rules it obeys. Its radio speaks a language specified by 3GPP, a partnership of standards organisations in which the engineers of competing telecommunications firms draft the technical specifications for each generation of mobile networks. Its wireless connection follows a family of specifications developed under the IEEE, a professional association of engineers. Its charging port follows a specification written by an industry forum. The factory that assembled it is probably certified to ISO 9001 for quality management and ISO 14001 for environmental management; its data centre partners to ISO/IEC 27001 for information security. The company that sold it reports its emissions under the Greenhouse Gas Protocol, a standard developed by a research institute and a business council, and its suppliers of tin, tantalum, tungsten and gold carry out due diligence on conflict minerals under frameworks developed by an intergovernmental organisation and audited by private firms.

Now ask how many of those rules were written by a government. Very few, and in most cases none. Some have since been made compulsory or referenced in procurement and trade law, but they were drafted and revised by engineers, auditors, industry associations and non-governmental organisations. The multinational that sells the phone spends far more of its compliance effort on these rules than on any treaty.

Session 05 looked at intergovernmental institutions — the organisations states build and in which states vote. This chapter is about the other half of global governance, the rules that come from somewhere other than states, and about why that half is increasingly geopolitical.

Private authority — the capacity of non-state actors (firms, industry associations, standards bodies, non-governmental organisations) to make rules that others treat as binding, even though no state has enacted them and no court may enforce them directly.

For a multinational, private authority is two things at once. It is a **risk**: a standard written elsewhere, by actors the firm does not control, can become a condition of access to a market. And it is a **lever**: a firm that shapes a standard shapes the market the standard governs, and may write its own technology, its own processes and its own costs into the rules its competitors must follow. The question this chapter asks is the one in the session's title: who writes the rules a multinational actually follows, with whose power — and does trade spread those rules up or down?

6.2 Where private authority comes from

For most of the twentieth century, private authority in international affairs was close to a contradiction in terms: international relations studied states, and rule-making was what states did. The concept entered the field in the 1990s, as globalisation multiplied the transactions that crossed borders faster than states could agree on rules to govern them.

Susan Strange made the argument most sharply. In *The Retreat of the State* she contended that markets, firms and a variety of non-state actors had come to exercise authority over outcomes — prices, investment, technology, credit — that had once been the preserve of governments, and that the study of international political economy had to follow power wherever it actually lay (Strange 1996). Cutler, Haufler, and Porter (1999) gave the concept its name in an edited volume on private authority in international affairs, focusing on the inter-firm arrangements — industry norms, business associations, private regimes — through which firms govern their own transactions. Hall and Biersteker (2002) extended it to a broader set of actors, from credit-rating agencies to transnational NGOs and even illicit networks, and asked a question that has never been fully settled: what makes private authority *authority*, rather than simply market power or persuasion?

The usual answer is **legitimacy combined with consequences**. A rule has authority when those it governs accept that it ought to be followed, and when failing to follow it has costs. Private rules acquire those costs through markets rather than courts. A supplier that is not certified loses the contract; a forest company whose timber is not certified loses access to the retailers whose customers demand certification; a firm whose emissions report does not follow the recognised protocol loses credibility with investors.

"Voluntary" does not mean "optional". A standard required by a firm's buyers, insurers or investors is, for that firm, as compulsory as a law — and sometimes more so, because a law can be challenged in court and a lost customer cannot.

Voluntary describes how a standard is adopted — no state compels it. It does not describe how much choice the firm has once its customers have adopted it.

6.3 The governance triangle

If rules can come from states, firms and NGOs, it helps to have a map. Kenneth Abbott and Duncan Snidal provide one (Abbott and Snidal 2009). They study what they call regulatory standards-setting schemes — arrangements that write standards for business conduct, monitor compliance, and sometimes certify it — and place each scheme in a triangle whose three corners are the three types of actor that can write rules.

Regulatory standards-setting (RSS) scheme — in Abbott and Snidal's terms, an arrangement through which states, firms, NGOs or combinations of them set standards for the conduct of firms, and often monitor or certify compliance. Its position in the governance triangle is given by the share each of the three actor types has in its governance.

Schemes near a **corner** are run by one type of actor. An industry code of conduct, drafted by firms for firms, sits near the firm corner. A certification scheme created by an activist organisation sits near the NGO corner. A traditional intergovernmental rule sits near the state corner. Schemes in the **interior** are hybrids. The Forest Stewardship Council, founded in 1993, brings together environmental organisations, social organisations and firms in a multi-stakeholder structure, and sits on the edge

between NGOs and firms. The UN Global Compact, launched in 2000 by the UN Secretary-General as a set of principles to which firms commit, brings states, firms and civil society together and is often placed near the centre. ISO itself is harder to place: its members are national standards bodies, some of them government agencies and many of them private associations whose committees are staffed largely by industry experts — which is why the placement of ISO standards is a useful argument to have in class rather than a fact to memorise.

The triangle's most important feature is not on the diagram. Abbott and Snidal stress the **shadow of the state**: even schemes run entirely by private actors are shaped by the possibility that states will regulate if private actors do not, and by states' decisions to endorse, require, reference or ignore private schemes. Firms write codes of conduct partly to pre-empt legislation; NGOs create certification schemes partly because they have despaired of getting legislation; and states frequently pick up private standards and give them legal force, at which point the scheme moves across the triangle without a word of its text changing.

From the archive · Brussels, 2022

For years, mobile phones in Europe were sold with a variety of incompatible chargers. The European Commission encouraged manufacturers to converge voluntarily, and a memorandum of understanding among major producers in 2009 led much of the market towards a common connector — but not all of it, and not permanently, as connectors evolved and one large manufacturer kept its own.

Meanwhile an industry body, the USB Implementers Forum, had written the specification for a new reversible connector, USB Type-C. It was a private standard in the purest sense: drafted by firms, adopted voluntarily, and spreading because device makers found it convenient.

In 2022 the European Union adopted a directive amending its Radio Equipment Directive to require a common charging port (European Parliament and Council of the European Union 2022). From the end of 2024, new mobile phones and a range of other portable devices sold in the EU had to charge through USB Type-C, with laptops following in 2026. A specification written by an industry forum became, by a single legislative act, a legal requirement for access to a market of several hundred million consumers. Because few manufacturers wish to design one version of a phone for Europe and another for everywhere else, the EU rule had consequences far beyond the EU.

In the governance triangle the specification did not change at all. What changed was its position: it moved from the firm corner towards the state corner, carried by the shadow of the state that had been hanging over it for more than a decade. And it illustrates in one case both of the chapter's main arguments — that private standards can become conditions of market access, and that the jurisdiction which turns them into law extends its rules well beyond its borders.

For a firm, the triangle is useful because a scheme's position predicts its behaviour. A scheme near the firm corner is easier for firms to change, and correspondingly less trusted by NGOs and regulators. A scheme with strong NGO participation is more legitimate in the eyes of activist consumers and harder for any single firm to steer. A scheme that a state has begun to reference in law is a scheme that may become mandatory, and a firm that ignores it is betting against the shadow.

6.4 Who writes an international standard?

The triangle tells us what kinds of actor write rules. It does not tell us whose rules win when actors from different countries meet in the same room. That is the question Walter Mattli and Tim Büthe ask of the two principal international standards bodies for products, ISO and the International Electrotechnical Commission (Mattli and Büthe 2003).

Some institutional background makes the question concrete. ISO was founded in 1947, and its members are national standards bodies — one per country. Standards are drafted in technical committees and their subcommittees and working groups, where experts nominated by the national bodies — overwhelmingly engineers employed by firms — negotiate the text. Each national body takes positions on drafts and votes on them. The process looks consensual and technical, and in one sense it is. But the experts who sit at the table, the national positions they carry, and the timing with which they bring proposals all determine whose existing practice becomes the international standard. A firm whose product already conforms to the standard that emerges has a head start; a firm whose product does not must redesign.

Mattli and Büthe compare three explanations of who wins. On the first, **technological rationality**, the technically best solution wins whoever proposes it — the engineers' view of themselves. On the second, the **primacy of power**, the standard favours the states with the largest markets — the realist's view of everything. Mattli and Büthe argue for a third.

Institutional complementarity — in Mattli and Büthe's account, the fit between a country's domestic standard-setting institutions and those of the international body. A hierarchical, coordinated national system, which speaks with one voice and passes information efficiently between firms and the national delegation, matches the structure of ISO and IEC and wins more influence than a fragmented, decentralised one, regardless of market size.

Their evidence pointed to the advantage of the European national standards bodies, which are hierarchical and coordinated, over the fragmented, decentralised system of the United States, in which many private standards developers compete and no single body aggregates national preferences. The result was striking because it cut against the realist expectation: the country with the largest market was not the most effective at shaping international product standards. Influence over "technical" rules depends on how a country and its firms are organised to take part, not only on how big they are. Büthe and Mattli (2011) later generalised the argument to accounting standards as well as product standards, showing that the same logic of institutional fit shapes who wins in the private bodies that write rules for global financial reporting.

The lesson for a firm is practical and somewhat uncomfortable. Participation is costly — it takes the time of senior engineers over years — and its benefits are diffuse and delayed. It is easy to conclude that someone else will look after the standard. But if a competitor's national standards body is better organised than one's own, and that competitor's engineers are in the room, the standard will reflect their technology, and the firm that stayed away will pay to comply with it. In standards, absence is a decision.

6.5 Standards as geopolitics

For most of the post-war period, the politics of standards was the politics of firms and of national standards bodies, and it rarely reached foreign ministries. That has changed. The competition over telecommunications standards, over battery and charging standards, over the rules for artificial intelligence and for the flow of data, has become a competition between blocs, each trying to make its domestic rules the world's.

The reason is that standards confer advantages that outlast the meeting at which they were agreed. A standard that incorporates patented technology generates licence revenue for as long as it is used; one that matches a country's industrial practice lowers its firms' costs and raises its competitors'; one that embeds choices about data access or encryption shapes what governments can see. And standards are sticky, so early influence compounds. When governments came to see technical standards as a component of technological leadership, and technological leadership as a component of national power, standard-setting acquired the stakes of geopolitics — and the bodies in which standards are written, which had been designed for engineers, found themselves hosting a contest between states.

The European Union's version of this contest has been described most fully by Anu Bradford as the **Brussels effect** (Bradford 2020). The EU rarely sets out to impose its rules on the world. It regulates its own market. But when a large market sets a strict rule, firms that sell in it face a choice: produce two versions of their product, one for the strict market and one for everyone else, or produce one version that meets the strictest rule and sell it everywhere. Where production cannot easily be divided — because of economies of scale, because the product is a single global platform, or because maintaining two compliance systems costs more than over-complying in lax markets — firms adopt the strict rule globally, and then often lobby their home governments to adopt it too, so that their less global competitors bear the same costs. Bradford identifies the conditions under which the effect operates: a large market, the regulatory capacity to write and enforce rules, a preference for strict rules, regulation of targets that cannot easily leave the market (consumers rather than mobile capital), and non-divisibility of production. Data protection, chemical safety, competition law and product standards are her main cases; the common charger is a small, clean example.

The Brussels effect is a mechanism of *unilateral* regulatory globalisation, but it shares a logic with private standards. In both cases, rules travel along the channels of trade, carried by firms that find it cheaper to adopt one rule everywhere than to maintain many. That shared logic is precisely what the big work of this session tests.

6.6 Of, by, as: what standards do to organisations

Standards are usually associated with stability and sameness: a standard exists so that things will be the same everywhere. Nils Brunsson, Andreas Rasche and David Seidl argue that this view misses most of what is interesting about them (Brunsson, Rasche, and Seidl 2012). They organise research on standards around three perspectives.

The first is **standardisation of organisations**: standards that tell firms how to be organised. ISO 9001 and ISO 14001 are the archetypes. They do not specify what a product must be like or how much pollution a plant may emit. They specify that a firm must have a management *system* — documented procedures, objectives, responsibilities, internal audits, management reviews and a commitment to continual improvement. The second is **standardisation by organisations**: the bodies that write standards, how they work, and how they compete with each other for adopters. The third is **standardisation as organisation**: standards as a way of organising markets and fields without formal hierarchy, coordinating the behaviour of thousands of firms that have no contract with each other and no common owner.

Their central point is that standards are **dynamic**. They are revised, contested, interpreted differently in different places, and — most important for anyone using certification as evidence — **decoupled** from practice.

Decoupling — the separation between the formal structures an organisation adopts to signal conformity to external expectations and the practices it actually follows. A firm certified to a management standard may change its documents and its audits while leaving its operations largely as they were.

The idea is older than the study of standards. Meyer and Rowan (1977) argued that organisations adopt formal structures because those structures are regarded as legitimate in their environment, not necessarily because they make the organisation more efficient, and that organisations protect themselves by decoupling those structures from the technical core of their work. A certificate can be a ceremony.

Whether ISO 14001 certificates are mostly ceremony is an empirical question, and the evidence is mixed in an informative way. Potoski and Prakash (2005) found that US facilities certified to ISO 14001 reduced their pollutant emissions somewhat more than comparable uncertified facilities — the standard's "weak sword", in their phrase, could still cut, because third-party audits and the reputational stakes of certification gave it some force. Christmann and Taylor (2006), studying firms in China, found that whether certification was implemented substantively or symbolically depended on the firm's circumstances, and in particular on whether its customers monitored what it actually did. Certification backed by demanding buyers was more likely to change practice; certification sought only for its signal was more likely to be decoupled. The implication for a multinational is direct. If it relies on its suppliers' certificates as evidence of environmental performance, the certificate's value depends on whether anyone — the multinational included — looks behind it.

6.7 Agents and entrepreneurs

Private authority does not all arise in the same way, and its origin matters for how durable it is. Jessica Green draws the key distinction (Green 2013).

Delegated versus entrepreneurial private authority — *delegated* authority exists when states hand rule-making or implementation to private actors, who act as their agents; *entrepreneurial* authority exists when private actors create rules on their own initiative and others choose to adopt them.

Green's field is carbon governance, and it supplies clean illustrations of both. Under the Clean Development Mechanism of the Kyoto Protocol, private bodies were accredited to validate and verify emission-reduction projects: they exercised authority that states had delegated to them, within rules states had written. The Greenhouse Gas Protocol, by contrast, was created by a research institute and a business council without any state mandate, and became the dominant framework for corporate carbon accounting because firms, investors and eventually regulators adopted it.

The distinction tells a firm what to watch. Delegated authority is as durable as the delegation: the state that handed it over can take it back, rewrite it or abolish the programme, and when the Kyoto mechanisms gave way to new arrangements under the Paris Agreement, the authority of the bodies that served them changed with them. Entrepreneurial authority is as strong as the number of its users. It grows through adoption and network effects, and it can be displaced by a competing standard that attracts more adopters — or captured by a state that writes it into law, at which point it becomes something closer to delegated authority after the fact.

Benjamin Cashore's analysis of forest certification shows how entrepreneurial schemes gain the authority to make rules (Cashore 2002). He calls them non-state market-driven governance systems: they have no state backing, and their authority derives from the market — from retailers and consumers who demand certified products, and from the firms in the supply chain who must therefore decide whether to comply. The Forest Stewardship Council, created by environmental groups with stricter rules, was soon met by industry-backed schemes with laxer ones, and the competition between them became a

contest over which would be accepted as legitimate by the buyers whose purchases conferred authority. Private authority, on this view, is not given; it is won, one supply chain at a time, and it can be lost the same way.

6.8 Race to the bottom, or trading up?

With the concepts in place, the session's central question can be stated precisely. Trade intensifies competition among producers in different countries. Does that competition push standards down or up?

The worry is the **race to the bottom**. If firms can produce anywhere and sell everywhere, governments compete for investment and exports, and one way to compete is to lower the costs that regulation imposes. Environmental standards are an obvious candidate: lax rules make production cheaper, and countries that compete hard for exports may tolerate pollution that richer or less trade-dependent countries would not. The argument predicts that greater openness to trade will be associated with weaker environmental governance.

The alternative was named by David Vogel (Vogel 1995).

California effect — the tendency of trade to spread *stricter* rules from strict to less strict jurisdictions. When a large, rich market imposes demanding standards on the products sold in it, producers elsewhere who wish to sell there adopt those standards, and may then press their own governments to adopt them too. Trade becomes a mechanism of "trading up".

The name comes from American environmental history. California, with severe air-quality problems and a large car market, set vehicle emission standards stricter than federal ones; manufacturers that wanted to sell in California built cars to meet them, and other states followed. Vogel contrasted it with the **Delaware effect** — the race to the bottom in corporate chartering law, in which American states competed to attract company registrations by offering laxer rules, and Delaware won. The two effects are not contradictory. They describe different regulatory targets. Where the rule governs a *product* that must cross a border to be sold, the importing market can impose it, and trade tends to spread strict rules upward. Where the rule governs a *process* or a *legal form* that can be relocated, jurisdictions compete for the mobile target, and trade can push rules down.

Environmental management standards sit in an interesting middle position. ISO 14001 governs a process — how a plant manages its environmental impacts — and not a product. No importing government can require it at the border. But buyers can, and that is where the question becomes empirical.

From the archive · Rio and Geneva, 1992–1996

ISO 14001 was born of a particular moment. In the run-up to the UN Conference on Environment and Development in Rio de Janeiro in 1992, business leaders organised to present a constructive business response to global environmental concerns, and international standardisation was one of the avenues they explored. ISO and the IEC set up an advisory group on the environment to consider whether international standards for environmental management were needed.

They were not starting from nothing. The British Standards Institution had published BS 7750, a national standard for environmental management systems, in 1992, building on the management-system model that the ISO 9000 quality standards, first published in 1987, had made familiar to firms worldwide. In 1993 the European Union adopted its own Eco-Management and Audit Scheme. ISO created a new technical committee for environmental management, TC 207, in 1993, and in 1996 it published ISO 14001.

The standard reflected its parentage. Like ISO 9001, it specified a management system rather than a performance level: a certified firm commits to comply with applicable law and to improve continually, but the standard sets no emission limit of its own. That design made it adoptable anywhere, by firms in any sector and any regulatory environment — and it made it, from the first, vulnerable to the charge that certification and performance could diverge. It was revised in 2004 and again in 2015, and it became the most widely adopted environmental management standard in the world.

The episode illustrates Mattli and Büthe's argument in miniature. The international standard grew out of a European national standard and a European regulatory scheme, drafted in a committee where coordinated national bodies with a working model to propose were well placed to shape it. Firms in countries with that model already in place adopted the international standard more easily — a pattern documented by Delmas (2002), who traced how differences in institutional environments made adoption faster in Europe than in the United States.

Aseem Prakash and Matthew Potoski put the race-to-the-bottom hypothesis to a direct test (Prakash and Potoski 2006a). ISO 14001 is a useful instrument because it goes beyond what many domestic laws require, is adopted voluntarily, and is counted, country by country, every year. Using a panel of 108 countries over seven years, they asked whether a country's adoption of ISO 14001 depends on adoption in its **export markets** — the countries to which it sells.

Their finding is that trade linkages **encourage** ISO 14001 adoption **when a country's major export markets have themselves adopted it**. Exporters to markets where certification is widespread certify more; exporters to markets where it is rare do not. Trade, on this evidence, does not drive environmental governance down. It transmits the standards of customers to their suppliers — a California effect operating through private rather than public regulation, with buyers rather than governments doing the imposing. The argument was developed at book length in Prakash and Potoski (2006b), which treats voluntary programmes like ISO 14001 as "green clubs" whose value to members depends on the credibility of the club's standards and the visibility of membership to outsiders.

The mechanism is consistent with earlier work on the diffusion of quality certification. Guler, Guillén, and Macpherson (2002) showed that ISO 9000 certificates spread across countries through trade and investment ties with countries where certification was already common, alongside the influence of states and of imitation among firms. Christmann and Taylor (2001) found, among firms in China, that environmental self-regulation was more likely where firms were linked to multinational owners or exported to developed-country customers. Across these studies the picture is consistent: private standards travel along supply chains, from buyer to supplier, and the supply chains themselves determine where they go.

For the firm, the conclusion is one of the most practical in the course. A private standard can become a de facto condition of access to a market without a single law changing. Monitoring "regulatory" risk therefore means monitoring not only what governments legislate but what one's customers' home markets adopt.

6.9 Turning a certificate into data

The practice round rebuilds the two figures that carry Prakash and Potoski's argument: the diffusion of ISO 14001 across countries over time, and each country's adoption against the adoption level of its export markets. Every step of the construction involves a choice that affects what the figure shows.

The data on certification come from **the ISO Survey of Management System Standard Certifications**, which counts valid certificates by country and year for the main management-system standards. Two features of the survey should temper any reading of it. It counts *certificates* issued by certification bodies, not firms or sites, and its coverage of certification bodies has changed across editions, so that a sudden jump in one country in one year may reflect a change in reporting rather than a wave of adoption. And from 2025 it is compiled from the IAF CertSearch database, a change of method that belongs in a provenance log. The 2015 revision of ISO 14001 adds a further wrinkle: as certificates to the old version expired and were reissued, transition years can show dips that have nothing to do with firms abandoning the standard.

A count of certificates is not an adoption rate. A large economy has more certificates because it has more firms, and a map of raw counts is mostly a map of economic size. The practice round therefore **normalises** by GDP, expressing adoption as certificates per billion US dollars of output. That is one reasonable choice among several — per firm, per unit of manufacturing output, per unit of exports would each tell a somewhat different story — and the scaling Prakash and Potoski themselves use should be checked against the paper. If it differs, the difference is logged, not hidden.

The trade side requires a measure of **export-market adoption** for each country: how much ISO 14001 adoption there is, on average, among the countries it sells to. The natural measure is a weighted average of partners' adoption rates, with each partner weighted by its share of the country's exports. Bilateral exports come from the IMF's International Trade in Goods by partner country (International Monetary Fund 2026b), the dataset already used in Session 04. The weighting is the substance of the argument, not a technical detail. The California effect predicts that what matters is adoption among the markets a country actually sells to, in proportion to how much it sells to them. An unweighted average of all partners would give a country's smallest customers as much influence as its largest and would test a different, much weaker hypothesis.

A scatter of own adoption against export-market adoption can show whether the two move together, not why; the paper's panel model with controls and the replication's descriptive cloud differ precisely there, and the final section returns to the gap.

The most instructive check is a **placebo**. ISO 9001, the quality-management standard, is demanded by buyers everywhere and has no environmental content at all. If adoption of ISO 9001 tracks export-market adoption of ISO 9001 just as closely

as ISO 14001 tracks its own, then the export-market pull may have little to do with environmental governance and a great deal to do with the general practice of supply chains demanding certificates of their suppliers. That would not overturn the finding that trade spreads private standards upward. It would change what the finding means — from "trade spreads environmental governance" to "trade spreads certification", which is a different and less encouraging claim.

6.10 What private authority means for the multinational, channel by channel

The five channels through which the course traces geopolitical risk all run through private authority, and in each the firm is both a rule-taker and a potential rule-maker.

Market access is the most direct channel. A certificate required by buyers is a condition of entry as real as a tariff, and it is set by the buyers and, behind them, by their home markets. When a large market writes a private standard into law, as the EU did with the common charger, the condition of access hardens and its reach extends through the Brussels effect to markets far beyond the one that legislated. A firm that monitors only government regulation will see these changes late. The practical discipline is to track what one's customers' home markets adopt, both privately and in law, because the evidence of this chapter says that is where one's own future requirements will come from.

Inputs are where the multinational becomes the imposer rather than the imposed upon. A firm that requires its suppliers to certify is exercising private authority over them, and the evidence on decoupling says that the value of that requirement depends on whether the firm monitors behind the certificate. A supplier's ISO 14001 certificate is evidence that a management system exists. It is not, by itself, evidence that the supplier pollutes less, and a firm that treats it as such is exposed when the gap between certificate and practice becomes public.

Finance increasingly runs on private standards too — carbon accounting, sustainability reporting frameworks, credit ratings — each written by a private or hybrid body, each conditioning access to capital, and several being absorbed into law, which moves them across the triangle and changes who can change them.

Assets and people are exposed through the standards that govern labour and human rights in supply chains. A plant certified under a scheme that later loses legitimacy is an asset whose value in the eyes of customers has fallen overnight.

Data and operations are where standards have become most explicitly geopolitical. Telecommunications standards, information-security standards, rules for artificial intelligence and for cross-border data flows are now contested between blocs. A firm whose products embed one bloc's standards may find them excluded from the other's procurement, and a firm whose engineers sit on the committees writing the next generation of a standard is not only shaping its own costs but taking part, whether it likes it or not, in a contest between states.

Across the five channels, the strategic choice is the same, and it has three options. The firm can **adopt** a standard written elsewhere, accepting the cost of compliance as the price of access. It can **shape** the standard, investing in participation in the bodies that write it — the lesson of Mattli and Büthe is that this investment pays only if the firm and its national body are organised to use it. Or it can **resist**, backing a competing standard or lobbying against legal adoption — a bet that the rival standard will win enough users, or that the shadow of the state will not fall. The choice depends on where the scheme sits in the governance triangle, on whether its authority is delegated or entrepreneurial, and on whether the firm's own customers are among those who confer it.

6.11 A short review of the literature

The study of private authority in international political economy begins with Strange (1996), who argued that authority over economic outcomes had shifted from states to markets and firms, and with Cutler, Haufler, and Porter (1999) and Hall and Biersteker (2002), who gave the concept its name and extended it from inter-firm regimes to NGOs, rating agencies and other non-state actors. Abbott and Snidal (2009) organised the resulting landscape into the governance triangle and introduced the shadow of the state; Green (2013) distinguished delegated from entrepreneurial private authority and traced both in carbon governance; Cashore (2002) showed how non-state, market-driven schemes such as forest certification gain the authority to make rules through the purchasing decisions of firms and consumers.

The politics of who writes international standards is the subject of Mattli and Büthe (2003), whose argument for institutional complementarity over technological rationality and raw power was generalised in Büthe and Mattli (2011) to the private

regulation of products and financial reporting. Bradford (2020) documented the unilateral mechanism by which strict EU rules become global through firms' preference for a single production standard; the EU's common-charger directive (European Parliament and Council of the European Union 2022) is a recent and transparent instance of a private specification hardened into law.

On standards as organisation, Brunsson, Rasche, and Seidl (2012) set out the three perspectives — of, by and as organisation — and the dynamism of standards in practice, drawing on the institutionalist insight of Meyer and Rowan (1977) that formal structure may be decoupled from activity. Whether ISO 14001 changes behaviour is addressed by Potoski and Prakash (2005), who find modest emission reductions among certified US facilities, and by Christmann and Taylor (2006), who show that substantive rather than symbolic implementation depends on monitoring by customers.

On the race to the bottom, Vogel (1995) named the California effect and set it against the Delaware effect. Prakash and Potoski (2006a) provided the cross-national test at the centre of this session, finding that trade spreads ISO 14001 from export markets to exporters, and Prakash and Potoski (2006b) developed the theory of voluntary programmes as clubs. The diffusion mechanism is corroborated for quality standards by Guler, Guillén, and Macpherson (2002), for Chinese firms by Christmann and Taylor (2001), and for the differing paths of Europe and the United States by Delmas (2002). The through-line is that private rules travel along supply chains, and that the power to write them lies with those organised to take part.

6.12 What the map will not tell the firm

The practice round produces a map of ISO 14001 adoption that changes as the years advance, small multiples of its diffusion by region, and a scatter of each country's adoption against the adoption of its export markets. If the build goes well, the scatter will slope upward, as Prakash and Potoski's argument predicts. The temptation will be to read that slope as proof that trade spreads environmental governance. It is evidence of something narrower.

The scatter is a **correlation**, not a test. Countries whose export markets certify heavily are richer, more industrialised and more embedded in multinational supply chains than others, and any of those differences could produce the slope on its own; the paper's panel model controls for some of them, the cloud of points controls for none. Certificates are **not performance**: the evidence on decoupling says that a certified plant may or may not pollute less than an uncertified one, and the figure says nothing about which. The data carry the scars of their **collection**: certification-body coverage changes between survey editions, the 2015 revision produces transition dips, and the switch to IAF CertSearch is a change of method. And the measure depends on choices — normalising by GDP rather than by firms or exports, weighting partners by exports rather than equally — that the comparison step asks you to break deliberately.

What the figures *can* show is whether the broad pattern the paper reports is visible in public data with transparent choices; how unevenly a private standard has diffused across the world; and, through the ISO 9001 placebo, whether the pull of export markets is specific to environmental governance or general to certification. Each of these matters to a firm deciding whether to certify, whether to require certification of its suppliers, and how much to trust the certificates it receives.

The question for the catch slide is the course's usual one, applied to private rules. Suppose the scatter slopes upward exactly as the paper says. What does it *not* license a multinational to conclude? It does not license the conclusion that a certified supplier is environmentally better than an uncertified one, that certification caused a country's export success, that a standard voluntary today will stay voluntary, or that the rules the firm's markets follow were written by anyone the firm can vote for. The answer you give — one sentence — is a judgement about where authority lies, and it is the judgement a multinational must make every time it decides whether to adopt, shape or resist a rule it did not write.

6.13 In the practice

> **The target.** Prakash & Potoski (2006): ISO 14001 spreads across countries, and a country adopts more when its **export markets** have adopted more. Two views: the diffusion (a map and small multiples over time), and the trade channel (own adoption against export-market adoption).

In your teams. One workbook per team; everyone builds at least one sheet.

6.13.1 1 · Collect — 10 min

Your three files should already be in `data/raw/` (`../data/README.md`). Finish the provenance log:

file	source	url	version / release	accessed	licence	notes
iso_survey_1400[1].xlsx *or* iso_survey_transcribed.xlsx	ISO / IAF CertSearch	…	*survey edition*	*date*	*free / account*	*full tables or transcription; coverage changes?*
imf_imts_exports.csv	IMF	data.imf.org — IMTS	*download date*	*date*	free	*exports FOB, US$; partner codes used*
wdi_gdp.csv	World Bank	api.worldbank.org	*last updated*	*date*	CC BY 4.0	

6.13.2 2 · Prepare — 20 min

1. **ISO table.** If years are columns, **pivot** them (select the year columns → right-click → *Pivot*), rename to `Year` and `Certificates`. Add or check `iso3`. Keep 14001 and 9001 as two columns, or as a `Standard` dimension — your choice, logged.
2. **WDI GDP.** Pivot the year columns too. Keep `Country Code` (it is ISO3) and the value. From the `Metadata_Country_…` file in the same zip, bring in **Region**, and drop the aggregates (World, regions, income groups): they are the rows with an empty `Region`.
3. **Join** ISO to GDP on `iso3` **and** `Year` (both join clauses). Check the row count.
4. Calculated field:
   ```
   [ISO 14001 per $bn GDP] = [Certificates 14001] / ([GDP current US$] / 1e9)
   ```
5. Log: which countries did not match, and why.

6.13.3 3 · Replicate — 30 min

6.13.3.1 Sheet 1 · "ISO 14001 spread" — a map

1. `iso3` → *Geographic Role* → *Country/Region*; double-click it. Check the **unknown** indicator, bottom right; fix with *Map* → *Edit Locations*.
2. Mark type **Map**; `[ISO 14001 per $bn GDP]` → Colour, **sequential** palette. Consider a log-like scale: the distribution is very skewed (try *Edit Colors* → *Stepped* with 5 steps).
3. `Year` → **Pages**. Press play. Title it with what you see.

6.13.3.2 Sheet 2 · "Diffusion by region" — small multiples

`Year` → Columns (continuous), `Region` → Rows, `MEDIAN([ISO 14001 per $bn GDP])` → Rows. One small line per region. Which region adopted first? Which caught up?

6.13.3.3 Sheet 3 · "The pull of export markets" — the trade channel

This is the core of Prakash and Potoski: does a country's adoption track its **export markets'** adoption?

1. New data source. Start from `imf_imts_exports.csv` (reporter × partner × year × exports). In the IMF layout reporter and partner are `COUNTRY` and `COUNTERPART_COUNTRY`, in ISO3 codes (rename them `Reporter iso3` and `Partner iso3`; `TIME_PERIOD` → `Year`, `OBS_VALUE` → `Exports`). Drop aggregate partners (codes starting with `G` or `TX`). If your download shows names instead, build a concordance. Kosovo is `KOS` at the IMF but `XKX` at the World Bank.
2. **Join** the ISO + GDP table to it **twice** (Tableau lets you add the same table again; rename the copies):
   - as **Partner**: `Partner iso3 = iso3` and `Year = Year`
   - as **Reporter**: `Reporter iso3 = iso3` and `Year = Year`
3. Calculated fields:

```
[Weighted partner adoption]  = [Exports] * [Partner ISO 14001 per $bn GDP]

[Export-market adoption]     = SUM([Weighted partner adoption]) / SUM([Exports])

[Own adoption]               = { FIXED [Reporter iso3], [Year] :
                                   AVG([Reporter ISO 14001 per $bn GDP]) }
```

   The `FIXED` matters: the reporter's own value is repeated on every partner row, so it must not be summed.
4. Scatter: `[Export-market adoption]` → Columns, `AVG([Own adoption])` → Rows, `Reporter iso3` → Detail, `Year` → Colour (or filter to one year). **Log scales** on both axes (right-click axis → *Edit Axis* → *Logarithmic*). Add a trend line (*Analytics* → *Trend Line* → *Linear*) — read it as a description, not a test: Session 09 returns to what its R^2 means.

6.13.4 4 · Compare and break — 20 min

1. **Differ.** Your scatter versus the paper's argument. The paper estimates a panel with controls; you have drawn a cloud. What can your cloud show, and what can it not?
2. **The placebo.** Rebuild Sheet 3 with **ISO 9001** — the quality standard, demanded by buyers everywhere, with no environmental content. If the export-market pull is just as strong, what does that say about *why* firms certify to 14001?
3. **Break the weighting.** Replace the export-weighted average with an **unweighted** average of partners (`AVG` instead of the weighted ratio). Does the relationship survive? Why should weighting matter for the California-effect argument?
4. **Break the normalisation.** Raw counts instead of per $bn GDP on Sheet 1. Which countries jump to the top — and why is that misleading?

6.13.5 5 · The catch — 10 min

Original	Ours	The catch
Prakash and Potoski's finding	Your Sheet 3 (and Sheet 1)	What it does **not** license a multinational to conclude — for instance that a certified supplier is environmentally better, or that adoption *caused* export success

A member drawn at random presents in two minutes.

6.13.6 Hand in

`.twbx` + provenance log on ZoneCours. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

Team project. Which private standards do your firm's buyers or home market require? Which of its key countries are below the median adoption of its export markets? Note it for the dashboard.

Chapter 7

Measuring Risk: How a Political Judgement Becomes a Number

Session 7 · How a political judgement becomes a number — and what is lost on the way Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Every geopolitical risk score a multinational buys or builds is a chain of choices: which texts, which words, which coder, which weights. Can you see the chain — and defend each link?**

Big work	Hassan, T. A., Hollander, S., van Lent, L., & Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4), 2135–2202. doi:10.1093/qje/qjz021
Companion works	Baker, Bloom & Davis (2016), *Measuring Economic Policy Uncertainty*, QJE, doi:10.1093/qje/qjw024 · OECD/JRC (2008), *Handbook on Constructing Composite Indicators*, doi:10.1787/9789264043466-en · Krippendorff (2018), *Content Analysis*, 4th ed.
Target figures	(a) **Average firm-level political risk over time**, against the Economic Policy Uncertainty index · (b) **your own coded index**, with its intercoder agreement · (c) **a composite exposure index** for your team's firm, with weights the reader can move
Data you collect	firmlevelrisk.com firm-quarter file; EPU (global); 40 news headlines from GDELT that **your team codes**; GPR country indices and Worldwide Governance Indicators for the composite
Tableau skill	**Tableau Prep** flows (or the data source pane) · unions · cleaning text-coded data · **normalisation** · **parameters as weights**
Course themes	Methodology: content analysis, coding, big data, statistical treatment — the Synthesis outcome

7.1 The thermometer in the wrong room

Every geopolitical risk score that reaches a board has a pedigree, and the pedigree is usually hidden. A consultancy reports that a country's risk has "risen to 71"; a bank's dashboard turns a cell from amber to red; an index published by a central bank spikes in the week an invasion begins. Each of these numbers is the last link of a chain that started with a concept nobody can observe directly — the probability that politics will damage a firm's business — and passed through a sequence of choices about texts, words, coders, units, weights and scales. The choices were made by someone, for reasons, and most of them could have been made differently.

This chapter is about seeing that chain, so that a score one is sold can be audited and a score one builds can be defended. Two words carry most of the weight, and they are worth fixing at the start.

Validity — the degree to which an indicator measures the concept it claims to measure. Is the share of newspaper articles mentioning war threats a measure of geopolitical *risk*, or of editors' attention to geopolitics, or of the number of journalists posted to a war zone?

Reliability — the degree to which the same procedure, applied by someone else to the same material, yields the same number. Would a second coder, following the same rules, put the same article in the same category? Would the same dictionary, run next year on the same archive, return the same count?

The two are independent in one direction only. A measure can be perfectly reliable and entirely invalid: a thermometer that everyone reads identically, hanging in the wrong room. It cannot be valid without being reasonably reliable, because a measure that returns different numbers for the same object is measuring, in part, the person who applied it. The history of measurement in the social sciences is largely a history of trading one against the other — cheap, mechanical procedures that are perfectly reliable and of uncertain validity, against slow human judgement that captures meaning and disagrees with itself.

7.2 The chain, link by link

Write the chain out once, so that its parts have names. A measure of geopolitical risk passes from a **concept** ("geopolitical risk") to an **indicator** (the share of articles that discuss geopolitical tensions), drawn from a **source** (a fixed set of newspapers), processed by a **coding rule** (a list of words grouped into categories), summarised by an **aggregation** (a monthly share, rescaled so that a base period averages 100) and delivered in a **presentation** (a line with spikes labelled "9/11" and "Iraq invasion"). Every one of the risk measures in this course can be written in that form, and every link is a place where the number can part company with the concept.

At the concept link, the danger is that nobody has said what is being measured. "Political risk" in the business literature has meant expropriation, contract repudiation, currency inconvertibility, civil unrest, regulatory change and, more recently, sanctions and export controls — things with different causes, different probabilities and different remedies. An indicator that pools them may be a reasonable summary of the newspapers and a poor guide to any one of the decisions it is meant to inform.

At the indicator and source links, the danger is selection. Newspapers report what is new and dramatic, in the countries their readers care about, in the language they print in. The ten newspapers behind the Caldara–Iacoviello index are American, British and Canadian; a war in a country with no correspondents may barely register. That is a property of the index, not a flaw, but a user must carry it into every inference.

At the coding link, the danger is the gap between words and meaning: a rule that counts "war" cannot tell a war from a price war or a trade war.

At the aggregation link, the danger is that choices about scaling and weighting, innocuous one at a time, decide the result. A share of articles behaves differently from a count when the number of articles changes; an average across countries weighted by GDP tells a different story from an unweighted one. And at the presentation link — the one most often forgotten — the labels on the spikes tell the reader what to see, and the absence of labels on the troughs tells them what not to.

From the archive · Inferences from enemy propaganda, 1941–1945

The systematic analysis of text as evidence about political intentions did not begin with computers. During the Second World War, analysts in the United States — among them a group at the Federal Communications Commission's foreign broadcast monitoring service — read German radio propaganda not for what it said but for what its saying implied. If the propaganda ministry began preparing the public for a particular kind of news, what did the regime expect to happen? If a theme disappeared from broadcasts, what had changed in the regime's own calculation?

Alexander George later reconstructed this work and asked how well the analysts' inferences had held up once the German archives could be checked (George 1959). His account made two points that every risk analyst should know. The first is that the useful inferences were rarely about the *manifest* content — the frequency of a word — and usually about what the content revealed of the speaker's situation and purposes, which required a model of the speaker. The second is that the analysts were most often wrong when they forgot that the text had been produced *for* an audience, by an actor with reasons

to mislead it. A newspaper is not a propaganda ministry, and an earnings call is not a broadcast to a wartime public. But both are texts written by people with purposes, and an index built from them inherits those purposes.

7.3 Content analysis: from reading to inference

The discipline that formalised these practices is content analysis. Its early textbook definition called it a technique "for the objective, systematic, and quantitative description of the manifest content of communication" (Berelson 1952). The modern definition, Krippendorff's, is broader and more demanding.

Content analysis — a research technique for making replicable and valid inferences from texts (or other meaningful matter) to the contexts of their use (Krippendorff 2018). Replicable: a second analyst following the same procedure should reach the same result. Valid: the inference must hold about the context, not merely about the texts.

The shift from "description" to "inference" matters. Nobody builds a geopolitical risk index because they want to know how often the word "missile" appears in newspapers. They want to infer something about the world the newspapers describe — and, in the case of a multinational, about the likelihood that the world will change in ways that touch the firm. The texts are evidence; the context is the object.

Krippendorff breaks the procedure into steps, and they map cleanly onto both the published indices and the coding exercise in this session. **Unitising** decides what a single unit is: an article in the GPR and EPU indices, a transcript in Hassan and colleagues' measure, a headline in this session's own coding. **Sampling** decides which units enter: every article in ten newspapers, every earnings call of listed firms that hold one, forty headlines drawn at random from a news query. **Recording** or coding assigns each unit to a category, by a dictionary or by human coders following a protocol. **Reducing** summarises the codes — counts, shares, monthly averages. **Inferring** draws the conclusion about the context: geopolitical risk is high this month; this firm faces unusual political risk this quarter; the countries in which a firm operates are in the news for military build-ups rather than for elections.

7.3.1 Manifest and latent content

The distinction that does most of the work in practice is between what a text *says* and what it is *about*. Manifest content is on the surface: the words, their frequency, their co-occurrence. Latent content requires interpretation: whether a headline reports a threat or an act, whether a statement is ironic, whether a reference to war is literal. Dictionaries read manifest content; human coders can read latent content. The trade-off between them is the trade-off between reliability and validity seen from another angle. A dictionary is perfectly reliable — run it twice and it returns the same count — and blind. A human coder sees meaning, and two human coders see slightly different meanings.

The best measures use both. They encode the analyst's judgement in a dictionary that a machine can apply to millions of documents, and then audit the machine with humans who read a sample and check whether the dictionary's classifications match what the texts are about. That design — a dictionary audited by people — is common to all three of the indices this session examines, and the quality of the audit is the best single guide to the quality of the index.

7.4 Three dictionaries, three theories of risk

The three measures on the table share a family resemblance and differ in ways that matter.

	GPR	EPU	PRisk
Text	newspaper articles	newspaper articles	earnings-call transcripts
Unit counted	articles containing geopolitical terms	articles containing economy *and* policy *and* uncertainty terms	political bigrams near synonyms of risk or uncertainty
Dictionary	authors' eight categories	authors' term sets, audited by hand	learned from political vs non-political texts
Scaled by	all articles	all articles	transcript length

	GPR	EPU	PRisk
Level	world; 44 countries	country; global aggregate	firm × quarter

7.4.1 Geopolitical risk as newspaper attention

Caldara and Iacoviello define geopolitical risk as the threat, realisation and escalation of adverse events associated with wars, terrorism and tensions between states that affect the normal and peaceful course of international relations (Caldara and Iacoviello 2022). Their index counts, each month, the share of articles in a fixed set of newspapers that contain words from eight categories: war threats, peace threats, military build-ups, nuclear threats, terror threats, beginning of war, escalation of war and terror acts. The first five form a sub-index of *threats*, the last three a sub-index of *acts*. The series is rescaled so that its average over a base period equals 100, and national indices are built by requiring that the article also mention the country.

The theory of risk embedded in this procedure is explicit: geopolitical risk is what informed newspaper readers in a few large democracies are being told to worry about. The split between threats and acts is not a technical detail. It distinguishes the risk that something will happen from the fact that it has, and the two behave differently — threats can rise for months before an act, and an act can resolve uncertainty as well as create it. The coding protocol of this session borrows the eight categories, with a rule that acts beat threats.

7.4.2 Policy uncertainty as a three-way conjunction

Baker, Bloom and Davis built the Economic Policy Uncertainty index on a different logic (Baker, Bloom, and Davis 2016). An article counts only if it contains a term from each of three sets: one about the economy, one about policy and one about uncertainty. The conjunction is the clever part. A filter on "uncertainty" alone would capture weather and sport; a filter on "policy" alone would capture every legislative report. Requiring all three narrows the count to articles that discuss uncertainty about economic policy, at the cost of missing articles that discuss it without using any of the listed words.

What distinguishes the EPU work methodologically is the audit. The authors had human readers code a large sample of articles and compared the human judgements with the dictionary's, tuning the policy term set to reduce errors in both directions — articles the dictionary flagged that humans judged irrelevant, and articles humans judged relevant that the dictionary missed. The audit does not make the index perfect. It makes its error rates *known*, which is the more useful property. The global index used in this session is a GDP-weighted average of national indices; the data file offers two weightings, current-price and purchasing-power-parity GDP, and the choice between them is a small instance of the aggregation decisions discussed below.

7.4.3 Political risk as what managers discuss

Hassan, Hollander, van Lent and Tahoun asked a question the newspaper indices cannot answer: how much political risk does an *individual firm* face, and does it matter for what the firm does (Hassan et al. 2019)? Their text is not newspapers but the transcripts of quarterly earnings conference calls, in which senior managers present results and answer questions from analysts. The choice is deliberate. An earnings call is the place where a firm's own leadership, under questioning from people paid to find weaknesses, talks about what worries it.

Their dictionary is not written by hand. It is *learned*. The authors took a body of political texts and a body of non-political texts and identified two-word combinations — bigrams — that are frequent in the first and rare in the second. A bigram such as a reference to a particular kind of legislation is political; a bigram such as a reference to a production process is not. They then counted, in each transcript, the political bigrams that appear close to a synonym of "risk" or "uncertainty", weighted them by how distinctively political they are, and divided by the length of the transcript. The result, PRisk, is a firm-quarter measure of the share of the conversation devoted to political risk.

Dictionary method — a measurement procedure that assigns texts to categories by counting words or phrases from a pre-specified list. The list may be written by the analyst (GPR, EPU) or learned from labelled texts (PRisk). Either way it is a theory of what the concept looks like in language, and its validity depends on whether that theory holds in the texts to which it is applied.

The proximity requirement deserves a moment. Counting political bigrams alone would measure how much a firm talks about politics — which may reflect opportunity as much as risk, or simply a chief executive's interests. Requiring that the political language appear near the language of risk and uncertainty turns a measure of political *attention* into a measure of political *risk*. It is the same logic as EPU's three-way conjunction, applied at the level of a sentence rather than an article. The authors also construct the companion measures in the data file — non-political risk, sentiment and risk by political topic — so that a user can ask whether a firm's political risk moves independently of its overall risk and its overall mood.

The dependence of a dictionary on its domain is not a hypothetical worry. Loughran and McDonald showed that the general-purpose word lists then in common use classified much of the ordinary vocabulary of corporate filings as negative — words such as "liability", "tax" or "cost" that in a financial report are neutral descriptions — and built a finance-specific list in response (Loughran and McDonald 2011). Tetlock's earlier work on newspaper pessimism and stock prices had shown that a dictionary measure of tone could carry real information about markets (Tetlock 2007). The two papers together make the point that runs through this chapter: a dictionary can work, but only for the texts it was built for, and its transfer to a new domain is an empirical question, not an assumption.

7.5 What the firm-level measure found

Hassan and colleagues validated PRisk in three ways, and the structure of the validation is as instructive as the result. First, the aggregate of PRisk across firms moves over time with the newspaper-based EPU index, which is what one would expect if both capture the same underlying variation in political uncertainty. Second, the measure picks out the topics one would expect at the times one would expect — firms talk about political risk in a given domain when policy in that domain is in motion. Third, and most persuasively, firms with high PRisk *behave* as theory predicts firms facing political risk should behave: they cut investment and hiring, and they become more active politically, lobbying more and donating more to politicians. A measure that predicted nothing about behaviour would be a curiosity. A measure that predicts the behaviour the concept implies has passed a test of validity that no amount of reliability could substitute for.

The finding that surprised the authors, and that matters most for a multinational, concerns where the variation lies. Most of the variation in political risk is not common to all firms at a given time, nor common to all firms in a sector; it is *specific to the firm*. Two companies in the same industry, in the same quarter, can face very different political risk, because they sell to different governments, depend on different regulations, source from different countries or hold different licences.

The implication is uncomfortable for anyone who buys a country-risk score. An aggregate index — however carefully built, however well audited — measures what is common. The political risk that actually reaches a particular firm is mostly what is *not* common. A board that tracks a national index and believes it is tracking its own exposure is watching the weather report for a region while the risk is in the plumbing of its own building.

This is where the five channels used throughout the course earn their place. Political risk reaches a multinational through **market access** (who may sell what to whom), **inputs** (what can be bought, and from where), **finance** (who will lend, insure, settle), **assets and people** (what can be seized, damaged or detained) and **data and operations** (what can be stored, transferred or shut down). An index built from newspapers measures, at best, the climate in which these channels operate. A firm-level measure such as PRisk measures how much a firm's own managers are worried, which integrates the channels from the inside. Neither tells the firm which channel is exposed. That is the job of the team composite index, built at the end of this chapter, which is assembled channel by channel for a named firm.

7.6 Two coders and a protocol

The obvious measure of reliability is **percent agreement**: the share of items on which two coders chose the same category. It is intuitive and misleading. Two coders who never read the headlines and simply assigned codes at random, in proportion to how often each category occurs, would agree often — and very often if one category dominates. In a sample where four headlines in five are not about geopolitics at all, two coders who both write "not geopolitical" for every item will agree on most of them, while having measured nothing.

Cohen's coefficient corrects for this by comparing observed agreement with the agreement that chance alone would produce, given each coder's own distribution of codes (Cohen 1960).

Cohen's kappa — a measure of agreement between two coders on a nominal scale, corrected for chance: $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is the observed share of agreements and p_e the agreement expected if the two coders assigned categories independently at their own marginal rates. A kappa of 1 is perfect agreement; 0 is agreement no better than chance; negative values are worse than chance.

The expected agreement is computed from the margins of the agreement table. If coder A puts $n_{A,k}$ of N items in category k and coder B puts $n_{B,k}$ there, the chance that both pick k for the same item, if they choose independently, is the product of their shares; summing over categories gives

$$p_e = \sum_k \frac{n_{A,k}}{N} \cdot \frac{n_{B,k}}{N}.$$

Take forty headlines, collapsed into three categories — not geopolitical, threat, act — with coder A's codes in the rows and coder B's in the columns:

	B: 0	**B: threat**	**B: act**	**A total**
A: 0	**20**	3	1	24
A: threat	2	**8**	0	10
A: act	0	1	**5**	6
B total	22	12	6	40

The coders agree on the diagonal, $20 + 8 + 5 = 33$ items, so $p_o = 33/40 = 0.825$. Chance agreement is $(24 \cdot 22 + 10 \cdot 12 + 6 \cdot 6)/40^2 = 684/1600 = 0.4275$. Kappa is therefore $(0.825 - 0.4275)/(1 - 0.4275) \approx 0.69$. An agreement rate that sounds excellent — better than four in five — shrinks by almost a fifth once the agreement the coders would have reached without reading is removed. The shrinkage is driven by the dominant category: because both coders put more than half the headlines in "not geopolitical", much of their agreement there was to be expected.

What counts as good enough is a convention, and the conventions differ. Landis and Koch proposed a set of labels that have become widely used, under which values between 0.61 and 0.80 are "substantial" (Landis and Koch 1977). Krippendorff, for his alpha — a generalisation that handles more than two coders, missing data and ordinal or interval scales — asks for at least 0.800 before data are relied on, and accepts values from 0.667 only for tentative conclusions (Krippendorff 2018; Hayes and Krippendorff 2007). By the first convention the worked example is respectable; by the second it would support only a tentative claim.

7.6.1 What disagreement is for

The worst thing to do with disagreement is to average it away — to take the mean of two codes, or to pick one coder's version at random. Disagreements are where the concept is fuzzy, and they are information. Read each one. Some will be errors: a coder misread a headline, or forgot a decision rule. Others will reveal that the protocol did not anticipate a case — a headline about sanctions, a cyber operation, a military exercise that might be a build-up or might be routine. Those are the valuable ones. The right response is to **revise the protocol**, write down the new rule, and then adjudicate.

The protocol used in this session contains a deliberate gap of exactly this kind. Sanctions, export controls and trade wars are not among Caldara and Iacoviello's eight categories, so the protocol tells coders to assign them to "not geopolitical" and flag them. For a multinational in the 2020s, that is a strange place for them to be. A headline about export controls on semiconductors may matter more to a firm's inputs channel than a headline about a distant military exercise. The gap is not a mistake in the GPR index, which measures what it says it measures; it is a reminder that an index built for one purpose — tracking the risk of war and terrorism as perceived by newspaper readers — may not be the index a firm needs.

7.7 Big data: detection, not measurement

The coding exercise draws its headlines from the GDELT Project, which monitors news media worldwide in many languages and turns them into searchable records. It is the most visible example of what is usually called big data in political risk: a source that promises *coverage* — every country, every day — and *speed*, without a research assistant reading anything.

The promise is real, and the traps are four. The first is **selection**. GDELT records what is reported, by the sources it monitors, in the languages it processes. Its coverage of sources has changed over time, so a rising count of articles about a country may mean more sources rather than more events. The second is **duplication**: a single wire-service story is republished by hundreds of outlets, and a naive count treats each republication as evidence. The third is **machine coding**: automated classification of events misreads irony, quotation and historical reference, and errs in ways that are not random — it is worse in some languages than in others, and worse for some kinds of event. The fourth is the most seductive. "N = all" is never all. A very large sample drawn by a biased process is still biased, and the large number only makes the estimate more confidently wrong.

The fourth trap has been formalised. Meng showed that the error of an estimate from a large non-random sample depends on the correlation between being in the sample and the quantity being measured, and that even a tiny such correlation can make a sample of millions worth less than a small random sample (Meng 2018). The parable everyone cites is the story of a search-based model of influenza that, after an initial success, substantially overestimated flu prevalence for long stretches, because the behaviour generating the data changed while the model did not (Lazer et al. 2014). Neither example is about geopolitics; both describe exactly what can happen to an event count built from online news.

More data is not better measurement. A dataset of millions of machine-coded news records is a sample drawn by a process nobody controls, measured with error nobody has fully characterised. Forty headlines drawn at random from a well-defined query and coded by two people with a written protocol can be more *valid* for a specific question — not because forty is more than millions, but because you know what the forty are.

The working rule for a risk analyst follows. Use big data for **detection**: something is happening, here, now, and it is worth a closer look. Use small, carefully coded data for **measurement**: how much is happening, compared with what, and whether it has changed. The two jobs are complementary. A monitoring system that flags a spike in news about a firm's supplier country is doing detection well. A board paper that reports that "risk in the supplier country has doubled" on the strength of the same spike is doing measurement badly.

7.8 Composite indicators and the arithmetic of judgement

The third measurement task of the session is the one that a multinational is most likely to perform itself: combining several indicators into one number. Country-risk ratings, governance indices, supplier-risk scores and the team's own exposure index are all composite indicators. They are attractive for the same reason they are dangerous: they compress several dimensions into one figure that a board can read at a glance, and they hide, in the compression, a series of decisions that determine the ranking.

The standard guide is the handbook produced by the OECD and the European Commission's Joint Research Centre (OECD and European Commission Joint Research Centre 2008). It sets out ten steps, from the theoretical framework to the presentation of results. Five of them decide most of the outcome.

The **theoretical framework** says which dimensions of risk the index captures and why. Without one, the index becomes a collection of whatever data were available — which is to say, an index whose content was chosen by the data suppliers rather than by the analyst. For the team composite, the framework is the five channels: each indicator must be defended as a measure of exposure through a named channel, or it does not belong.

Missing data is where indices quietly lie. The country GPR indices cover 44 countries; the country EPU indices about twenty. A composite that drops every country missing an indicator will cover only large, well-documented economies — and the countries it silently drops are, very often, the risky ones, about which fewer people write. Imputing a value is a claim about a country nobody measured; flagging the gap is honest and less tidy. Whatever is chosen should be visible in the output.

Normalisation puts indicators measured in different units on a common scale. The simplest method, min–max, maps each indicator to the interval from 0 to 1:

$$x^* = \frac{x - \min x}{\max x - \min x} \qquad \text{or, where higher is safer,} \qquad x^* = \frac{\max x - x}{\max x - \min x}.$$

The second form matters more than it looks. The Worldwide Governance Indicators' measure of political stability is scored so that higher means *more* stable; in a risk index it must be inverted, or a stable country will score as risky. The indicators also changed scale in their 2025 edition, moving to a 0–100 score and recalculating their history, so that values from different editions cannot be mixed; the methodology behind the indicators is set out in Kaufmann, Kraay, and Mastruzzi (2011). Other normalisations — z-scores, ranks, distance to a reference country — produce different rankings from the same data, because they treat outliers and gaps differently. Min–max is sensitive to the extremes: a single country with an extraordinary value compresses everyone else into a narrow band.

Weighting and aggregation combine the normalised indicators. The usual form is a weighted average,

$$CI = \frac{\sum_j w_j \, x_j^*}{\sum_j w_j},$$

and the weights can be equal, derived statistically (from principal components, say), or set by experts. Each choice is defensible and none is neutral. Equal weights are a judgement that every dimension matters equally, which is rarely what anyone believes. Statistical weights reward the indicators that move together, which may simply be the redundant ones. Expert weights encode the expert.

The aggregation rule carries a further assumption that is easy to miss.

Compensability — the property of an aggregation rule under which a high score on one dimension can offset a low score on another. Additive aggregation is fully compensatory: a country with excellent governance and an active war can average to "medium". Geometric aggregation is partially compensatory, penalising unbalanced profiles; non-compensatory rules rank countries without allowing any trade-off at all.

For a risk index, full compensability is often the wrong assumption. A firm does not experience the average of a country's risks; it experiences the worst of them that touches its operations. Excellent contract enforcement does not compensate for a plant inside a war zone. Additive aggregation is still the default, because it is simple and transparent, but an analyst who uses it should be able to say why the dimensions can trade off — or at least show the profile behind each score so that the reader can see when they cannot.

From the archive · The Human Development Index, 1990 and 2010

The most influential composite indicator of the past half-century was designed to argue with a single number. The first *Human Development Report*, published by the United Nations Development Programme in 1990 under the direction of Mahbub ul Haq, proposed the Human Development Index as an alternative to ranking countries by income per head (United Nations Development Programme 1990). It combined three dimensions — a long and healthy life, knowledge and a decent standard of living — each normalised against minimum and maximum values, and averaged them with equal weights. Its authors were explicit that the index was a simplification built to change a conversation, and it did.

Twenty years later the 2010 report changed the aggregation from an arithmetic to a geometric mean of the three dimension indices and revised the indicators of education and income (United Nations Development Programme 2010). The reason given was compensability: under the arithmetic mean, a country could make up for poor health with high income, one for one; under the geometric mean, poor performance in one dimension weighs more heavily. The change re-ranked countries. Ravallion then showed that the new formula implied trade-offs between longevity and income that few would accept if they were stated openly — in effect, valuations of an extra year of life that varied enormously with a country's income, and were very low in the poorest countries (Ravallion 2012).

The episode is a compact lesson in everything this section argues. The weights and the aggregation rule were value judgements; changing them changed the ranking; and the implicit prices they carried became visible only when someone did the algebra. A firm's exposure index embeds trade-offs of the same kind — between, say, sanctions exposure and political instability — and the board deserves to see them stated.

7.9 Sensitivity: the ranking that one weight overturns

The last of the decisive steps is the one most often skipped. **Sensitivity analysis** asks how much the results move when the choices change: a different normalisation, different weights, a different treatment of missing data, a different aggregation rule. Saisana, Saltelli and Tarantola set out how to do this systematically, by treating each methodological choice as an uncertain input and examining the distribution of ranks it produces (Saisana, Saltelli, and Tarantola 2005). The simplest version needs no special software. Record the ranking under equal weights; then under each indicator alone; then count how many countries move by more than a few places. A country that stays near the top under every reasonable choice is robustly high-risk. A country whose rank swings from fifth to twenty-fifth depending on one weight is not "fifth"; it is a country about which the index has no firm opinion.

This is why the practice round builds the team composite with weights a reader can move. The point is not interactivity for its own sake. It is that a composite which hides its weights is an opinion dressed as a measurement, and a composite whose weights are visible and adjustable is an argument the reader can test.

A minimal sensitivity report for any composite: the ranking under the chosen weights; the ranking under equal weights; the ranking under each indicator alone; the number of units whose rank changes by more than a stated threshold; and a sentence on the unit most sensitive to the weights and why.

7.10 A short review of the literature

Content analysis as a social-science method was codified in the mid-twentieth century. Berelson (1952) gave the classic definition centred on manifest content, and George (1959) reconstructed the wartime analysis of enemy propaganda that showed both the power and the dangers of inference from text. Krippendorff (2018) is the standard modern treatment, recasting the method around replicable and valid inference and setting out the reliability apparatus; Hayes and Krippendorff (2007) make the case for Krippendorff's alpha as a general standard. The chance-corrected agreement coefficient is Cohen (1960), and the labels most often attached to its values are Landis and Koch (1977).

Automated text analysis in political science and economics has grown into a large literature. Grimmer and Stewart (2013) is the essential warning and guide for political texts — their principle that automated methods amplify rather than replace human reading, and must be validated for every application, is the spine of this chapter. Gentzkow, Kelly, and Taddy (2019) survey text as data in economics. In finance, Tetlock (2007) showed that dictionary measures of media tone carry information about markets, and Loughran and McDonald (2011) showed how badly general-purpose dictionaries misread specialised documents.

The three indices at the centre of the session are Caldara and Iacoviello (2022) for geopolitical risk, Baker, Bloom, and Davis (2016) for economic policy uncertainty and Hassan et al. (2019) for firm-level political risk; the last is notable both for its learned dictionary and for its demonstration that most political risk is firm-specific. On big data, Lazer et al. (2014) is the cautionary tale and Meng (2018) the formal account of why sample size does not rescue a biased selection process.

Composite indicators have their own methodological literature. OECD and European Commission Joint Research Centre (2008) is the practitioner's handbook and Saisana, Saltelli, and Tarantola (2005) the statement of sensitivity and uncertainty analysis as a quality test. Kaufmann, Kraay, and Mastruzzi (2011) document the Worldwide Governance Indicators, a composite of composites. The Human Development Index (United Nations Development Programme 1990, 2010) and the critique of its implicit trade-offs in Ravallion (2012) are the clearest case study of how aggregation rules carry value judgements.

7.11 What the figure will not tell the firm

The practice round produces three figures — the firm-level measure against the newspaper-based one, the team's own coded index and the draft composite for the team's firm — and each comes with a sentence it does not license. The first comparison is two indicators of a similar concept built from different texts, dictionaries and levels of aggregation; where they part company, the question is which link of which chain explains the gap.

The line of average firm-level political risk set against policy uncertainty does not tell a particular multinational how much political risk *it* faces. It is an average over listed firms that hold earnings calls in English, which is mostly to say American listed firms, over a period that ends in early 2022. The most important finding of the paper it replicates is that most of the variation in political risk is specific to the firm — and that is precisely the variation an average removes. A board can use the line to see when political uncertainty was high for corporate America as a whole; it cannot use it to see its own exposure, which would require scoring its own calls, or at least those of its closest peers.

The team's coded index, with its kappa, does not license a claim about how geopolitical the news about a country is. It is forty headlines from a three-month window of an online news monitor whose sources are weighted towards English-language media, coded against a protocol that by design places sanctions and export controls outside geopolitics. What it licenses is a statement about how reliably two trained readers can apply that protocol — and a list of the places where the protocol needs to change before anyone relies on it.

The composite exposure index does not license a ranking presented as fact. It licenses a ranking *under stated weights*, together with the weight that would overturn it. The question to carry into the practice round, and into every risk score a multinational is sold, is the five-part audit this chapter has built: what concept, from which sources, coded by whom or by what, normalised and weighted how, and how sensitive to each of those choices. A score that can answer all five is a measurement. A score that cannot is somebody's opinion, with decimals.

7.12 In the practice

> **The targets.** (a) Hassan et al. (2019): **average firm-level political risk over time**, against the EPU index. (b) **Your own coded index** — 40 headlines, two coders, a kappa. (c) A **composite exposure index** for your team's firm, with weights the board can move — your team's draft.

Today has three short builds instead of one long one. Keep to the clock.

Part	Minutes	The five steps, compressed
(a) Firm-level risk vs EPU	30	collect · prepare · replicate · compare
(b) Your coded index	20	agreement · adjudicate · count
(c) A composite with moving weights	30	select · normalise · weight · test
The catch	10	one slide, three panels

Data: `../data/README.md`. Provenance log as always — one line per file, including your own `coding_sheet.xlsx` ("produced by the team, protocol of Session 07").

7.12.1 (a) Firm-level political risk against EPU — 30 min

7.12.1.1 Prepare — choose a route

The firm file is **firm** × **quarter**; EPU is **monthly**. Both must reach a common **quarter** grain.

The quarter identifier in the firm file is a string such as `2010q1` in the vintages we have seen — **check yours**. Convert it to a date:

```
[Quarter date] = MAKEDATE( INT(LEFT([date], 4)),
                           (INT(RIGHT([date], 1)) - 1) * 3 + 1,
                           1 )
```

And for EPU (`Year` and `Month` in the current file; `Year` may arrive as text because of the citation rows at the bottom, hence the `INT`):

```
[Quarter date] = DATETRUNC('quarter', MAKEDATE(INT([Year]), INT([Month]), 1))
```

Route 1 · Tableau Prep (if your licence includes it)

1. *Input* `firmquarter_2022q1.csv` → *Clean* step: add `[Quarter date]` → *Aggregate* step: group by `[Quarter date]`; `AVG(PRisk)`, `COUNTD([gvkey])`.
2. *Input* the EPU workbook → *Clean*: add `[Quarter date]`, drop rows with no value → *Aggregate*: group by `[Quarter date]`; `AVG([GEPU_current])` (or `GEPU_ppp` — log which).
3. *Join* the two on `[Quarter date]` (inner) → *Output* as `.hyper`. Open it in Tableau Desktop.
4. Every step is a visible, saved decision: export the flow (`.tfl`) with your hand-in.

Route 2 · the data source pane (Tableau Desktop or Tableau Public edition)

1. Connect to both files. Drag the firm file to the canvas, then the EPU sheet next to it.
2. In the relationship dialog, choose **Create Relationship Calculation** on each side and enter the two `[Quarter date]` formulas above.
3. Relationships aggregate each table at its own level: `AVG(PRisk)` is averaged over firms, `AVG(GEPU_current)` over the months of the quarter — exactly what you want.

7.12.1.2 Replicate — Sheet "Firm-level political risk and EPU"

1. `[Quarter date]` → Columns as continuous **Quarter**.
2. `AVG(PRisk)` → Rows; the EPU measure → Rows; **dual axis**, *not* synchronised (different units).
3. Annotate three peaks; check what happened in those quarters.
4. Title with the finding; caption with both sources, vintages and access dates.

7.12.1.3 Compare

- Do the two series move together? Where do they part company — and which one would you trust for a firm, and which for a country?
- Add `COUNTD` of firms as a third view. Does the number of firms change over the period? Why does that matter for an average?
- Filter `hqcountrycode` (2-letter codes) to exclude `US`. How many firms are left?

7.12.2 (b) Your coded index — 20 min

Connect to `coded/coding_sheet.xlsx`.

1. **The agreement matrix.** `coder_A` → Rows, `coder_B` → Columns, `CNT(id)` → Text and Colour: a **highlight table**. *Analysis → Totals → Show Row and Column Grand Totals.*
2. **Percent agreement** in Tableau:
   ```
   [Agree] = IF [coder_A] = [coder_B] THEN 1 ELSE 0 END      →   AVG([Agree]) = p_o
   ```
3. **Kappa, by hand.** From the grand totals, compute $p_e = \sum_k (n_{A,k}/N)(n_{B,k}/N)$ and $\kappa = (p_o - p_e)/(1 - p_e)$. Write both on your slide. If categories are sparse, also compute kappa on three collapsed categories (0 / threat / act) — and say which you report.
4. **Adjudicate.** The adjudicator reads every disagreement and fills `final`, with a one-line reason. List the **protocol changes** you would make.

5. **The index.** From `final`: the share of headlines in each category, and the share coded geopolitical (≠ 0) by month. Mark `econ` comments separately: how many headlines about sanctions or export controls did the GPR categories force into 0?

7.12.3 (c) A composite exposure index with moving weights — 30 min

7.12.3.1 Select

Start from a **spine**: a sheet listing your firm's key countries plus comparators — **8 to 12 countries**, with `iso3`. Every input is **left-joined** to it, so missing values are visible rather than silently dropped.

Indicator	Source	Direction	Preparation
Country GPR	`GPRC_xxx` columns	higher = riskier	pivot the `GPRC_` columns (not `GPRHC_`; values are % of articles); extract the code with `RIGHT([Pivot Field Names], 3)`; average the last 12 months
Country EPU	`All_Country_Data.xlsx`	higher = riskier	pivot country columns; add `iso3` (the columns are country names such as `UK`, `US`, `Korea`, and China appears three times); average the last 12 months
Political stability	WGI 2025, sheet `pv`, `Governance score (0–100)`	**higher = safer**	latest year; join on `Economy (code)`
Distance in UN voting from the firm's home country	Session 04 ideal points	higher = riskier	`ABS([Ideal point] – [Home ideal point])`, latest year

Each input must be one row per country before it is joined to the spine — otherwise the min and max below are taken over months, not countries. Reduce each input first: an *Aggregate* step in Tableau Prep, or — in Desktop — a small sheet (country × the averaged value) exported with *Worksheet → Export → Crosstab* into a derived file in `data/derived/`, logged as such. Raw files are never edited.

7.12.3.2 Normalise

Min–max across the spine countries, with a `FIXED` expression over the whole table:

```
[GPR n]   = ([GPR]   – {MIN([GPR])})   / ({MAX([GPR])}   – {MIN([GPR])})
[WGI n]   = ({MAX([WGI])} – [WGI])     / ({MAX([WGI])}   – {MIN([WGI])})      // inverted
```

…and the same for EPU and voting distance.

7.12.3.3 Weight

Create four **parameters**, `w GPR`, `w EPU`, `w WGI`, `w Vote` — integers 0 to 5, default 1 — and show their controls. Missing values must not count as zero risk, so reweight over the indicators that exist:

```
[Composite] =
  ( IF NOT ISNULL([GPR n])  THEN [w GPR]  * [GPR n]  ELSE 0 END
  + IF NOT ISNULL([EPU n])  THEN [w EPU]  * [EPU n]  ELSE 0 END
  + IF NOT ISNULL([WGI n])  THEN [w WGI]  * [WGI n]  ELSE 0 END
  + IF NOT ISNULL([Vote n]) THEN [w Vote] * [Vote n] ELSE 0 END )
  /
  ( IF NOT ISNULL([GPR n])  THEN [w GPR]  ELSE 0 END
  + IF NOT ISNULL([EPU n])  THEN [w EPU]  ELSE 0 END
  + IF NOT ISNULL([WGI n])  THEN [w WGI]  ELSE 0 END
  + IF NOT ISNULL([Vote n]) THEN [w Vote] ELSE 0 END )
```

That is a **decision**, not a fix: write it in your log ("composite renormalised over available indicators").

Sheet: countries on Rows, sorted by `SUM([Composite])`, bars, with `RANK(SUM([Composite]))` as a label.

7.12.3.4 Test

Record each country's rank under **equal weights**, then under each indicator **alone** (weight 1, others 0). In a small table: how many countries change rank by more than two places? Which country is most sensitive to the weights — and why?

7.12.4 The catch — 10 min

One slide, three panels:

(a)	(b)	(c)
PRisk vs EPU: what an average of US-listed firms does **not** tell your firm	Your kappa and the protocol change you would make	Your composite's ranking — and the weight that would overturn it

A member drawn at random presents in two minutes.

7.12.5 Hand in

`.twbx` (+ `.tfl` if you used Prep), `coding_sheet.xlsx` with `final` filled, and the provenance log, on ZoneCours. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper — for (a), or for the GPR categories meeting your headlines in (b).

Team project milestone. Part (c) is your **draft composite index**. Bring it, revised, to the dashboard review in Session 10.

Chapter 8

Cyber Geopolitics: Restraint, Rivals and Weaponised Networks

Session 8 · Restraint, rivals and weaponised networks Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Is cyberspace a new battlefield for great powers — or a new way for old rivals to hurt each other, with multinationals as collateral?**

Big work	Valeriano, B., & Maness, R. C. (2015). *Cyber War versus Cyber Realities: Cyber Conflict in the International System*. Oxford University Press. doi:10.1093/acprof:oso/9780190204792.001.0001
Companion	Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79. doi:10.1162/isec_a_00351
Target figure	Incidents by year, method and severity — and the **initiator × target matrix** of rival dyads
Data you collect	Dyadic Cyber Incident and Campaign Data (DCID) v2.0, doi:10.7910/DVN/CQOMYV; EuRepoC Global Dataset v1.3, doi:10.5281/zenodo.14965395
Tableau skill	Highlight tables · dyad matrices (initiator × target) · filter and highlight actions
Course themes	The evolution of cyber geopolitics and digital threats in the decision-making architecture of multinational firms

8.1 The war that was supposed to happen

For most of the past three decades, the public conversation about cyberspace and international politics has been conducted in the future tense. A "cyber Pearl Harbor" would switch off a country's power grid; a handful of skilled operators would humble a great power; the next war would be won or lost before a shot was fired. The scenarios were vivid, and they were not absurd: modern economies do run on networked systems, and those systems can be broken. What the scenarios lacked was a record. They were claims about what *could* happen, offered in the absence of systematic evidence about what *had*.

The opposite caricature is just as easy to state. On this view cyber operations are simply crime and espionage conducted with better tools — a nuisance for security departments, of no strategic consequence, and no business of a course on geopolitics. It, too, is a claim about evidence, and it, too, is usually made without any.

This chapter takes the question to the data. It does so through the work of Brandon Valeriano and Ryan Maness, who built a dataset of cyber incidents between rival states and asked what the record actually shows (Valeriano and Maness 2015), and through the argument of Henry Farrell and Abraham Newman that the networks globalisation built are not flat, and that states with jurisdiction over their hubs can turn them into instruments of coercion (Farrell and Newman 2019). The two works answer different questions. The first asks how states use offensive cyber operations against each other. The second

asks how states use the architecture of the global economy — including its digital architecture — to watch and to deny. A multinational needs both answers, because it sits inside the architecture and in the path of the operations.

The question for the firm is narrower and more practical than the question for the strategist. It is not whether cyberspace is a new domain of war. It is *through which channels inter-state rivalry in cyberspace reaches a firm's operations, and how its decision architecture should absorb it*. The answer, this chapter argues, is that the firm is rarely the intended target of strategic cyber operations, frequently their collateral, and increasingly a piece of the infrastructure over which states compete.

8.2 Three activities that look alike

Three kinds of activity are routinely run together under the heading "cyber attack", and the first task is to separate them.

Espionage steals information: state secrets, negotiating positions, intellectual property, the personal data of officials. Its typical actors are intelligence services and their contractors, and its typical intrusion — into a ministry, a defence supplier or a law firm advising on a merger — is designed never to be noticed. **Disruption and degradation** interrupt or damage systems: a denial-of-service attack that floods a ministry's website, malware that wipes the disks of a company's computers, an operation that manipulates an industrial control system. Its actors are military and intelligence units and the proxies they sponsor. **Crime** extracts money: ransomware that encrypts a hospital's records, fraud against a bank's payment systems. Its actors are criminal groups.

Cyber operation — the use of computer code or network access by an organised actor to obtain information from, disrupt, degrade or manipulate a target's information systems. Espionage, disruption and crime are all cyber operations; they differ in purpose and actor, not in technique. The same intrusion may serve any of the three.

The boundaries blur, and the blurring is itself a geopolitical fact. States tolerate criminal groups that operate from their territory against foreign targets, hire them, shelter them, and occasionally use their tools. An intrusion that begins as espionage leaves access that can later be used for disruption. A ransomware campaign run by a criminal group may serve a state's purposes by sowing disorder in a rival's economy, without the state having ordered it. When a ransomware attack by a criminal group believed to operate from Russia forced the shutdown of the Colonial Pipeline in the United States in May 2021, it was crime in its motive and geopolitics in its consequences: the question of what the Russian state knew, tolerated or could have prevented became a matter between governments.

For a multinational the distinction matters because the three activities call for different responses. Espionage is a threat to the firm's intellectual property and its negotiating position, and the defence is largely technical and organisational. Disruption is a threat to its operations, and the defence includes resilience — the ability to keep working when systems fail. Crime is a threat to its cash and its reputation, and it is the one where law enforcement, insurers and the firm's own willingness to pay become part of the calculation. A risk register that lists "cyber" as a single line has not yet done the first piece of analysis.

From the archive · Estonia, 2007

In April 2007 the Estonian government moved a Soviet-era war memorial, the Bronze Soldier, from the centre of Tallinn to a military cemetery. The decision provoked protests by members of the country's Russian-speaking minority and an angry response from Moscow. Over the following weeks a wave of distributed denial-of-service attacks hit Estonian government websites, banks, newspapers and political parties. Estonia, among the most digitised societies in Europe, found its online services intermittently unreachable.

Estonian officials pointed to Russia. The Russian government denied involvement, and the public record never established a chain of command; much of the traffic came from botnets and from individuals mobilised on Russian-language forums. The episode thus contained, in compressed form, nearly every feature that later analysts would identify in state cyber conflict: an existing political rivalry, an operation of low technical sophistication and limited physical harm, high political salience, and attribution that remained contested.

Its institutional legacy was substantial. NATO's Cooperative Cyber Defence Centre of Excellence was established in Tallinn the following year, and under its auspices a group of international lawyers produced the *Tallinn Manual*, an attempt to state how existing international law applies to cyber operations (Schmitt 2013). The episode was widely described at the time as the first cyber war. With hindsight it looks more like the first clear case of what Valeriano and Maness would call restraint: a rival state's displeasure expressed through disruption that stopped well short of destruction.

8.3 The restraint thesis

Valeriano and Maness built their dataset around a simple design choice that shapes everything they found. They did not try to catalogue every cyber operation in the world. They took pairs of states with a history of enduring hostility — **rivals**, in the sense the international-conflict literature has used for decades — and recorded the cyber incidents between them. The unit is the incident, nested in campaigns; the population is rival dyads; the evidence is public. The dataset has since been extended as the Dyadic Cyber Incident and Campaign Data, whose version 2.0 covers incidents beginning between 2000 and 2020 and is the replication data for this session (Maness et al. 2022).

Cyber incident (in DCID) — an individual cyber operation launched by one state against another within a rival dyad, with a start date, a method, a target type, an objective and a severity score. Incidents can be grouped into **campaigns**, sustained series of operations with a common purpose. The unit is the incident between rivals, not the operation in general.

The choice of rivals as the population is a theoretical commitment, not a convenience. The study of international conflict has long found that a small number of pairs of states account for a large share of militarised disputes, and that these enduring rivalries have their own dynamics: hostility persists, crises recur, and each side interprets the other's moves in the light of a long history. If cyber operations are a new instrument of old conflicts, rivalries are where they should appear; if they are something genuinely new — an instrument that lets any state hurt any other cheaply — they should appear outside rivalries too. Restricting the data to rivals makes the first hypothesis testable and the second invisible — a limitation that the comparison dataset in the practice round makes concrete.

From that record they drew four propositions.

The first is **restraint**. States use cyber operations far less often, and far less destructively, than the "cyber war" narrative predicts. Most incidents in the data are of low severity: defacements of websites, denial-of-service attacks, espionage. Operations that destroy physical property or take lives are rare. The image of cyberspace as a domain of constant, escalating attack does not survive contact with a systematic count.

Cyber restraint — the empirical pattern, and the theoretical argument, that states with offensive cyber capabilities use them sparingly and at low severity against their rivals, holding back the most destructive options. Restraint is a claim about *strategic* severity, not about frequency of nuisance or about commercial cost.

The second is that cyber conflict follows **regional rivalries**. The dyads that account for most of the incidents are those that were already hostile before anyone had a network: the United States and China, Russia and its neighbours and the West, Israel and Iran, India and Pakistan, the two Koreas. Cyber operations do not create new conflicts; they are a tool in old ones. That finding matters for anyone who expected the low cost of entry to cyber operations to flatten the international hierarchy. The states that are most active in cyberspace are, by and large, the states that were already contending.

The third proposition explains the first: **why** states hold back. The fourth is that cyber operations have **limited coercive effect**. They rarely change the target's behaviour. They are better at signalling displeasure and at gathering intelligence than at compelling a rival to do something it would otherwise not do — a conclusion that the same authors developed at greater length in later work on cyber strategy (Valeriano, Jensen, and Maness 2018).

The restraint thesis did not arrive in an empty field. Thomas Rid had argued, provocatively, that cyber war in the sense of an act of force that is violent, instrumental and political had never occurred, and that what was called cyber war was better understood as sabotage, espionage and subversion — old activities in a new medium (Rid 2012, 2013). Erik Gartzke made a related argument from the logic of force: an attack that cannot be followed up by conventional power changes nothing durable, so cyber operations are likely to be adjuncts to force rather than substitutes for it (Gartzke 2013). Lucas Kello took the opposite position, arguing that the cyber revolution genuinely strained the concepts of international relations because it blurred war and peace into a condition of permanent, sub-threshold "unpeace" (Kello 2013, 2017). Valeriano and Maness's contribution was to bring data to an argument that had been conducted largely through concepts and anecdotes, and the data came down closer to Rid and Gartzke than to the prophets of catastrophe.

8.4 Why states hold back

The reasons for restraint are worth taking one at a time, because each has an analogue in the firm's own calculation.

Escalation. A destructive cyber operation against a rival invites a response, and the response need not stay in cyberspace. A state that switches off a rival's grid has made a move whose consequences it cannot control. In a relationship where both sides have conventional and sometimes nuclear forces, the incentive to stay below the threshold that would provoke them is strong.

Revelation. A cyber capability depends on a vulnerability the target has not fixed. Using it reveals it, and once revealed it can be patched — in the target's systems and everyone else's. The most sophisticated capabilities are therefore wasting assets, worth more held in reserve than spent.

Blowback. Malicious code, once released, can be captured, analysed and reused. A weapon that can be reverse-engineered and turned back on its maker, or on the maker's allies and firms, is a weapon to be used with care.

Access. The value of being inside a rival's networks is often greater than the value of damaging them. A state that can read a rival's communications has every reason not to do anything that would alert the rival to its presence. Espionage, in this sense, restrains disruption.

These incentives are not the whole story. Ben Buchanan has argued that intrusions made for defensive purposes — to understand a rival's capabilities, or to pre-position against a future threat — look identical, from the target's side, to preparation for attack, and that this creates a security dilemma in which each side's prudence alarms the other (Buchanan 2017). Restraint in the use of destructive operations is compatible with constant, mutual intrusion, and the intrusions themselves are a source of instability.

The war in Ukraine has tested the restraint thesis harder than anything in the dataset, which ends in 2020. Russian cyber operations against Ukraine have been extensive since 2014, and they intensified around the full-scale invasion of February 2022. The best systematic study of the earlier period found that cyber operations had little discernible effect on events on the battlefield (Kostyuk and Zhukov 2019), which is consistent with the fourth proposition: cyber operations are weak instruments of compulsion even in war. Whether the restraint thesis holds *as of today* is an open question, and it is the right question to ask of the replication. A pattern measured to 2020 is evidence about 2000–2020; extending it requires data the dataset does not contain.

The most famous exception proves the rule in an instructive way. The operation later known as Stuxnet, discovered in 2010, sabotaged centrifuges at Iran's uranium-enrichment plant at Natanz by manipulating the industrial control systems that ran them. It was widely attributed to the United States and Israel, neither of which has formally acknowledged it. It is the clearest case in the public record of code causing physical destruction to a rival's strategic asset, and it fits the restraint thesis better than the catastrophe narrative: it was narrowly targeted, designed to look like malfunction rather than attack, and aimed at delaying a programme rather than at provoking a war. It also illustrates blowback. The code escaped the plant's isolated network, spread to computers around the world, and was analysed in detail by security researchers, so that its techniques became available to everyone. For firms, the lesson was that industrial control systems — the machinery of plants, pipelines and utilities — were now a target class, and that the code designed to attack one of them could end up on their own networks.

Low strategic severity is not low commercial cost. An operation that barely registers on a scale of harm between states — a wiper that destroys no infrastructure and kills no one — can halt a company's production for weeks. The restraint thesis is about what states do to each other. The firm's question is what happens to the firms that happen to be in the way.

8.5 Weaponised interdependence

Valeriano and Maness study offensive operations. Farrell and Newman study something quieter and, for a multinational, probably more consequential: the way in which the networks of the global economy can be used by the states that sit at their centre (Farrell and Newman 2019).

Their starting observation is that the networks globalisation built — for finance, for communications, for the internet's physical and logical infrastructure — are not flat meshes in which every node is equal. They are highly asymmetric, with a few hubs through which most traffic passes. Network economics explains why: hubs are efficient, and efficiency attracts more traffic, which makes the hub more central still. The internationalisation of the 1990s was conducted on the assumption that this centralisation was a technical fact with no political content. Farrell and Newman's argument is that the assumption was wrong. A state that has jurisdiction over a hub can use it.

Panopticon effect — the use of jurisdiction over a network hub to *see* what flows through it: to gather information about the activities of other states, firms and individuals whose traffic passes through the hub.

Chokepoint effect — the use of jurisdiction over a network hub to *deny* others access to it: to cut a bank out of a payment system, a firm out of a software supply, a country out of a market for critical components.

The two effects need each other. The panopticon tells a state where to apply the chokepoint, and the chokepoint gives the panopticon its teeth. Farrell and Newman's cases make both concrete. The disclosures by Edward Snowden in 2013 showed the extent to which American intelligence agencies had used the concentration of global internet traffic and data storage in American firms and on American territory to collect information at scale — the panopticon. The disconnection of sanctioned Iranian banks from the SWIFT financial-messaging network in 2012, after the European Union adopted the necessary legal measures, showed how a hub located in one jurisdiction could be used to deny a whole country's banking system its normal connection to the world — the chokepoint.

The implication for a multinational is uncomfortable in two ways. The first is that the firm's exposure to geopolitical risk depends on where the hubs it uses are located and whose law applies to them. A firm whose payments clear through one jurisdiction, whose data are stored with providers subject to another, and whose software updates arrive from a third has three sets of exposure to the coercive choices of three states, most of which have nothing to do with the firm.

The second is more uncomfortable still: **the firm itself can be the hub**. A cloud provider, a payment network, a telecommunications operator, a maker of widely used software — each is infrastructure that its home state can watch through or weaponise, and that rival states therefore have reason to treat as a target, to exclude, or to replace with a domestic alternative. A firm in that position is not a bystander to geopolitical competition. It is part of the terrain. This is the bridge from this session to sanctions in Session 10, where the chokepoint becomes a legal instrument, and to space in Session 11, where the hubs are in orbit.

8.6 Attribution: evidence about evidence

Every dataset of state cyber operations rests on an act of attribution. Somebody has to say who did it. And attribution in cyberspace has four properties that make it unlike the identification of the attacker in almost any other kind of conflict event.

Attribution — the identification of the actor responsible for a cyber operation, at three levels: the machines used, the individuals or groups who operated them, and the state, if any, that directed or sponsored them. Public attribution is the decision to announce the third.

It is **slow**. Forensic analysis takes weeks or months; public attribution by governments often comes a year or more after the event. It is **probabilistic**. Technical indicators — the code, the infrastructure, the working hours of the operators, the language settings left in a file — can be faked, borrowed or reused, and a careful operator plants false ones. It is **political**. Governments decide when to attribute publicly, to whom, and with how much of their evidence, and they decide in the light of their relations with the accused state and their wish to protect their own sources. And it is **selective**. What is attributed publicly is what someone wanted known.

Rid and Buchanan put the point sharply: attribution is what states make of it (Rid and Buchanan 2015). It is not a technical fact waiting to be discovered but a judgement, assembled from technical, operational and strategic evidence, and communicated for a purpose. Lindsay offered a partial corrective: for the most consequential operations the set of capable actors is small, and the attribution problem is therefore less severe than it is for routine intrusions — which is one reason deterrence of major attacks remains possible (Lindsay 2015). Both views lead to the same conclusion about data.

An event dataset of cyber operations is a record of public attributions, not a record of operations. That sentence is the most important caveat of the practice round. It has predictable consequences for what the data show. States that attribute publicly — mostly Western democracies, with active security services and a free press — appear more often as victims than states that do not. Operations by allies against each other are rarely attributed in public and are therefore nearly invisible. Operations that succeeded in remaining undetected are, by definition, absent. A dyad with no incidents in the data may be a dyad with no operations, or a dyad in which neither side chose to say anything.

8.7 When the firm is collateral

The restraint thesis concerns the intentions of states. The cost to firms is driven by something else: the technical reach of what states release.

From the archive · NotPetya, June 2017

On 27 June 2017, the eve of a Ukrainian public holiday, malware spread through the update mechanism of a Ukrainian accounting program widely used by businesses operating in the country. It looked like ransomware, displaying a demand for payment, but its encryption could not be reversed: it was a wiper dressed as a ransom note. Within hours it had spread beyond Ukraine through the internal networks of any company that ran the software at a Ukrainian site. In February 2018 the governments of the United States and the United Kingdom, among others, publicly attributed the operation to the Russian military.

The apparent target was Ukraine. The victims included some of the largest firms in the world. The shipping company Maersk lost the use of systems across its global operations. The pharmaceutical company Merck and the food company Mondelez reported very large losses, as did logistics and manufacturing firms in many countries. None of them had a quarrel with anyone; they had a subsidiary, an office or a supplier in Ukraine, running a program everybody used (Greenberg 2019).

The aftermath carried a second lesson. When the companies claimed under their property insurance, insurers invoked the **war exclusions** that such policies had carried for generations, arguing that an operation by a state's military was an act of war. The litigation lasted years and did not go uniformly the insurers' way; in Merck's case, New Jersey courts held that the traditional exclusion, written with armed conflict in mind, did not cover the attack. The market's response was to write explicit exclusions for state-backed cyber operations into its contracts, which the Lloyd's market required from 2023. The effect is that a firm's cyber cover may now end exactly where geopolitics begins.

Three lessons follow for the firm. The first is that **proximity in cyberspace is technical, not political**. Firms were hit by NotPetya because they ran a piece of software in a country at war, not because anyone chose them. A firm's exposure to a state's cyber operations depends on its network architecture — what it connects to, what it updates from, what its subsidiaries can reach — as much as on its political profile.

The second is that **insurance is not a backstop**. A risk transfer that excludes the scenario the board is most worried about does not transfer that risk. The question to ask of a cyber policy is not whether it covers "cyber attacks" but what it says about operations attributed to states — and who decides the attribution.

The third is that **the board owns it**. Securities regulators increasingly require listed firms to disclose material cyber incidents and to describe how their boards oversee cyber risk. In the United States, rules adopted in 2023 require listed companies to report material incidents promptly and to describe their risk management and governance of cybersecurity in their annual reports (U.S. Securities and Exchange Commission 2023). The geopolitical part of cyber risk — the part that arises from the rivalry of states — is thereby made a matter of corporate governance, whatever the firm's own view of it.

8.8 Digital sovereignty and the splintering network

Weaponised interdependence has a defensive mirror. States that fear the panopticon and the chokepoint of others respond by asserting control over the networks within their own jurisdiction: where data must be stored, who may access them, which foreign equipment may be installed, which vendors may operate. The vocabulary varies — digital sovereignty, data localisation, trusted vendors, technological self-reliance — and the effect is the same. The single global network on which the internationalisation of the 1990s assumed firms could operate is splintering into jurisdictions with different rules.

Three instruments matter most for a multinational. **Data-localisation and transfer rules** require that certain data be stored within a country or restrict their movement abroad; the invalidation by the Court of Justice of the European Union in 2020 of the framework that governed transfers of personal data from the European Union to the United States showed how quickly a legal basis for a firm's data architecture can disappear. **Security reviews and vendor bans** exclude foreign suppliers from critical networks; the restrictions imposed on Huawei's telecommunications equipment by the United States and a number of its allies from 2019 onward forced operators to redesign and in some cases to strip out networks they had already built. **Access laws** give governments the power to compel firms under their jurisdiction to hand over data, including data

stored abroad; the United States' CLOUD Act of 2018 is the best-known example, and it sits awkwardly beside European data-protection law.

For the firm, these instruments generate three kinds of cost. **Duplication**: separate data centres or cloud regions per jurisdiction, separate supply chains for network equipment, separate compliance regimes. **Conflicting obligations**: one state's law requires a disclosure that another's forbids, and the firm must choose which to break. **Vendor risk**: a supplier banned in one market may be mandatory, or at least dominant, in another, so that a single global architecture becomes impossible. None of these costs arises from a cyber operation. All of them arise from the geopolitics of cyberspace.

8.9 Where cyber risk sits in the firm

The five channels through which geopolitical risk reaches a multinational help locate cyber risk precisely. It runs most obviously through **data and operations**: the operation that halts production, the localisation rule that forces a redesign. It runs through **assets and people**, when intellectual property is stolen or staff in a jurisdiction are compelled to provide access. It runs through **finance**, when insurers exclude state-backed operations and when payment hubs become chokepoints. It runs through **inputs**, when a vendor is banned or a software supply chain is compromised — NotPetya travelled through a software update. And it runs through **market access**, when a firm is excluded from a market as a security risk, or when its home state's use of the firm as a hub makes it unwelcome abroad.

Inside the firm, the risk is distributed across several actors, and the distribution is part of the problem. The **board** sets risk appetite and oversees disclosure; its distinctive question is *which state actors would want our data or our downtime, and why?* The **chief information security officer** turns threat intelligence into controls and owns incident response, but typically reports through a technology function with limited geopolitical analysis. The **chief risk officer** is the natural integrator of cyber risk with country, sanctions and supply-chain risk — the view the team project takes. The **insurer** prices the risk and, increasingly, draws the line at state-backed operations. And **regulators** impose disclosure, localisation and vendor rules from outside. A firm in which each of these actors holds a piece of the picture and nobody holds the whole has a decision architecture designed for crime and espionage, not for geopolitics.

What changes on Monday, for a firm with a plant in a country that sits in one of the densest cells of the rivalry matrix? Not, in the first instance, its technology. The first change is to the map: an inventory of which systems at that site connect to which systems elsewhere, which software is updated from which vendors, and which data cross which borders — because it is along those connections that an operation aimed at a government travels into a firm. The second is to segmentation: a subsidiary in a contested jurisdiction whose network can reach the group's core systems is a bridge for anyone who compromises it, and the cheapest geopolitical control a firm can buy is often a wall between the two. The third is to contracts: what the insurance policy says about operations attributed to states, what the cloud provider's terms say about government access in each jurisdiction, and what suppliers have agreed about notifying the firm when they are compromised. The fourth is to governance: who in the firm is responsible for asking, before an investment rather than after an incident, which states would want the firm's data or its downtime. None of these is a technical fix. All of them follow from treating cyber risk as a geopolitical exposure rather than an IT problem.

8.10 From concept to data

The replication target is the descriptive picture at the heart of Valeriano and Maness's book: incidents by year, by method and by severity, and the matrix of who targets whom. Rebuilding it from DCID version 2.0 is mostly a matter of reading the codebook carefully, and the codebook repays the reading because it is where the concepts of this chapter become columns.

The unit is one incident between an initiating state and a target state within a rival dyad. The dyad's two states are recorded as names, ordered by their Correlates of War codes, so that the first state of the pair is not the initiator; the initiator is recorded separately, as a code, and the target is whichever state of the dyad did not initiate. The method is coded on a scale that runs from vandalism through denial of service and network intrusion to network infiltration, with sub-codes for kinds of malicious software. Severity is coded on a scale the codebook defines from 0 to 10, of which the released file uses only the values 0 to 6 — a discrepancy worth logging and, if the codebook's scale applies, consistent with restraint. The type of target distinguishes private and non-state targets from governmental ones, civilian and military, and the objective distinguishes

disruption, short- and long-term espionage and degradation. The September 2022 release contains 433 incidents with start dates from September 2000 to January 2021.

Two properties of the data follow directly from the arguments above. The first is that the population is rival dyads. A cell of the initiator × target matrix that is empty does not mean "no cyber conflict"; it means "no rivalry in the data", because pairs of states that are not coded as rivals never enter. The second is that the incidents are public attributions. The matrix is therefore a picture of who has been publicly *accused* of operating against whom — itself a geopolitical artefact, shaped by which governments attribute and when.

The comparison dataset, the European Repository of Cyber Incidents, makes both properties visible (Zettl-Schabath et al. 2025). It records incidents with state and non-state initiators, including criminal, hacktivist and unattributed operations, and in its version 1.3 it contains 3,414 incidents with start dates from 2000 to 2024 — several times the size of DCID. Restricting it to incidents attributed to state or state-affiliated actors leaves 1,014; restricting it to states alone leaves 329. The gap between those numbers, and between all of them and DCID's 433, is a measure of how much the answer to "how much cyber conflict is there?" depends on the definition of the population and of attribution.

8.11 A short review of the literature

The debate over whether cyberspace is a domain of war was framed in the early 2010s. Rid (2012) and Rid (2013) argued that cyber war had not taken place and would not, reclassifying cyber operations as sabotage, espionage and subversion; Gartzke (2013) argued from the logic of force that cyber operations would complement rather than replace conventional power; Kello (2013) and Kello (2017) argued, against both, that the cyber revolution unsettled the conceptual foundations of international relations. The legal question of how existing international law applies to cyber operations is set out in Schmitt (2013).

Systematic evidence arrived with Valeriano and Maness (2015), whose dataset of incidents between rival states supported a thesis of restraint and regional rivalry. The dataset has been extended through successive releases, of which Maness et al. (2022) is the current public version; Maness and Valeriano (2016) examine whether cyber incidents shift the broader conflict and cooperation between rivals, and Valeriano, Jensen, and Maness (2018) develop the argument that cyber operations are weak instruments of coercion. Kostyuk and Zhukov (2019) tested cyber operations in the war in eastern Ukraine against battlefield events and found little effect. Buchanan (2017) sets out the cybersecurity dilemma created by intrusions that are defensive in purpose and threatening in appearance.

On attribution, Rid and Buchanan (2015) treat it as a political judgement assembled from technical and strategic evidence, and Lindsay (2015) argues that its difficulty is overstated for consequential operations. On the economic architecture of coercion, Farrell and Newman (2019) is the foundational statement of weaponised interdependence, with its panopticon and chokepoint effects. Wider datasets that include non-state actors, such as Zettl-Schabath et al. (2025), now allow the rival-dyad picture to be set against the full record of attributed incidents, and the firm's experience of state operations is documented most vividly in Greenberg (2019).

8.12 What the matrix will not tell the firm

The practice round rebuilds the incident picture and the initiator × target matrix, and then sets them against a wider dataset. The dashboard that results is informative. It shows that cyber conflict between rivals concentrates in a few dyads, that most incidents are of low severity, and that the methods have shifted over two decades. It is tempting to read it as a map of where a multinational's cyber risk lies. It is not, and the reasons are the arguments of this chapter.

The matrix does not license a firm to conclude that it is safe in a country whose cells are empty. An empty cell means no publicly attributed incident between *rivals*; it says nothing about operations that were never attributed, about operations between states that are not coded as rivals, or about criminal groups, which account for much of the cyber harm firms actually suffer. Nor does a dense cell tell a firm that it is a target. The incidents in that cell were aimed mostly at governments, and the firm's exposure to them depends on whether it is technically in the path — whether it runs software, holds data or provides infrastructure that an operation against the government might traverse. NotPetya would have appeared in the data as one incident in one dyad; its cost fell on firms in dozens of countries.

Nor does the evidence for restraint license complacency. It is a pattern in state behaviour to 2020, measured on a scale of harm between states. It does not bound the commercial cost of a low-severity operation, and it does not tell a firm what has happened since a large-scale war began in Europe. The question to carry into the practice round is the one the dashboard cannot answer by itself: given where interstate cyber conflict concentrates, which of the firm's networks, suppliers and hubs lie in its path — and what has the firm agreed with its insurer about the operations that its government will one day attribute to a state?

8.13 In the practice

> **The target.** Valeriano & Maness's descriptive picture of interstate cyber conflict: incidents **by year and method**, the **distribution of severity**, and the **initiator × target matrix** of rival dyads — rebuilt from DCID v2.0, then confronted with the wider EuRepoC record.

Work in your project teams. Everyone builds their own workbook; the team hands in one slide.

8.13.1 1 · Collect — 15 min

If you did the pre-session, both files are in `data/raw/`. If not:

- **DCID v2.0** — doi:10.7910/DVN/CQOMYV → file `Master_DCID_2.0_Release_public_Sep2022.tab` → *Access File → Comma Separated Values (Original File Format)*. The codebook is not on Dataverse: *Codebook 2.0* is on ryanmaness.com/cyber-conflict-dataset.
- **EuRepoC v1.3** — doi:10.5281/zenodo.14965395 → `eurepoc_global_dataset_1_3.csv` and `eurepoc_codebook_1_2.pdf`.

Provenance log — one line per file:

file	source	url	version / release	accessed	licence	notes
Master_DCID_2.0_Release_public_Sep2022.tab	Maness, Valeriano et al.	doi:10.7910/DVN/CQOMYV	V2.0, Sep 2022	*today*	CC0, cite	*original CSV behind the .tab; 433 rows*
eurepoc_global_dataset_1_3.csv	EuRepoC	doi:10.5281/zenodo.14965395	v1.3 (16 Mar 2025)	*today*	CC BY-NC 4.0	*3,414 rows*

8.13.2 2 · Prepare — 15 min

1. **Connect → Text file** → the DCID CSV. Look at the grid.
2. Find the columns (checked in the September 2022 file — see `../data/README.md`): `initiator`, `StateA`, `StateB`, `Dyadpair`, `interactionstartdate`, `method`, `severity`, `targettype`. **There is no target column**, and `StateA`/`StateB` are the dyad, not initiator and target. Write the exact names in your log.
3. `initiator` is a **COW code** (16 values: 2 US, 365 Russia, 710 China, 630 Iran, 731 North Korea, 770 Pakistan, 666 Israel, 750 India, 732 South Korea…). `Dyadpair` joins the COW codes of `StateA` and `StateB` (`2365` = 2 + 365; State B's code is always the last three digits). Build names for both sides:

```
[Initiator] = IF STR([initiator]) = RIGHT(STR([Dyadpair]), 3)
              THEN TRIM([StateB]) ELSE TRIM([StateA]) END
[Target]    = IF STR([initiator]) = RIGHT(STR([Dyadpair]), 3)
              THEN TRIM([StateA]) ELSE TRIM([StateB]) END
```

Two incidents (313 and 335) name an initiator outside the dyad — check them by hand.

4. `method` and `targettype` are **codes**. Alias them from the codebook: `method` 1 vandalism, 2 denial of service, 3 network intrusion, 4 network infiltration (4.1–4.4 are infiltration sub-types — group them, or round down with `FLOOR([method])`); `targettype` 1 private / non-state, 2 government non-military, 3 government military.
5. `interactionstartdate` is month/day/year text (`2/1/2008`). If Tableau has not read it as a date, create `Start Year` = `YEAR(DATEPARSE("M/d/yyyy", [interactionstartdate]))`.
6. Record every recode in the log.

8.13.3 3 · Replicate — 25 min

8.13.3.1 Sheet 1 · "Incidents per year, by method"

1. `Start Year` → **Columns** (discrete is fine: one bar per year).
2. `COUNTD([Cyberincidentnum])` (or the table's record count) → **Rows**.
3. Method → **Colour**. Stacked bars.
4. Title states the finding. Caption: *Source: DCID v2.0 (Maness, Valeriano et al., 2022), accessed …*

8.13.3.2 Sheet 2 · "How severe?"

1. Severity → **Columns** (discrete), count → **Rows**. A bar chart of the distribution.
2. Add a **reference line** at the median severity (*Analytics → Median with Quartiles* or a constant you computed).
3. Annotate the highest-severity incidents: what were they?

8.13.3.3 Sheet 3 · "Who targets whom" — the dyad matrix

1. `[Initiator]` → **Rows**, `[Target]` → **Columns** (the calculated fields from step 2.3).
2. Count → **Colour**, then *Show Me → highlight table*. Add count to **Label**.
3. Sort initiators by total count (right-click the pill → *Sort → Field → Count → Descending*).
4. Which five cells hold most incidents? Do they match the book's regional rivals?

8.13.3.4 Dashboard · link them

1. *New Dashboard* → drag Sheet 3 on the left, Sheets 1 and 2 on the right.
2. *Dashboard → Actions → Add Action → Filter…* — source sheet: Sheet 3; target sheets: 1 and 2; run on **Select**; clearing the selection **shows all values**.
3. Click the US–China cell, then the Russia–Ukraine cell. Write down how their histories differ.

8.13.4 4 · Compare and break — 20 min

1. **Differ.** Put your Sheets 1–2 beside the book's descriptive figures. DCID v2.0 runs to 2020 (one incident starts in January 2021) and has been revised since 2015. Write one difference, and its likely cause.
2. **Break the population.** Load EuRepoC as a second data source. Rebuild Sheet 1 on it — first with **all** incidents, then filtered to incidents attributed to **state** actors (field `initiator_category`; read the codebook to decide which values — `State` only, or also `State affiliated actor`). EuRepoC's `start_date` is day.month.year text: `DATEPARSE("dd.MM.yyyy", [start_date])`. How many times larger is EuRepoC? What does the difference tell you about what DCID leaves out?
3. **Restraint, tested.** In DCID, what share of incidents sits in the lowest severity levels? (*Quick Table Calculation → Percent of Total* on Sheet 2.) Does the evidence support restraint *as of 2020*? What would you need to know about 2022 onward?
4. **Extend — the firm.** Using `targettype` = 1 (*private / non-state*), isolate incidents that hit non-government targets. Which dyads hit firms most? Name a sector you would worry about.

8.13.5 5 · The catch — 15 min

One slide (a dashboard exported as an image):

Original	Ours	The catch
The book's incident picture	Your dashboard	One sentence: what this matrix does **not** license a multinational to conclude

A hint: it is a matrix of *publicly attributed* incidents between *rivals*. A member drawn at random presents in two minutes.

8.13.6 Hand in

On ZoneCours before leaving: your `.twbx` and provenance log. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the book — a dyad that is missing, a method with a different label, a severity that shifted between versions.

8.13.7 What loses credit

- Comparing DCID with *all* of EuRepoC without saying so
- Empty cells read as "no cyber conflict" rather than "no rivalry in the data"
- Codes (1, 2, 3) left on the chart instead of aliased labels
- A catch about the book rather than about what the firm may conclude

Chapter 9

Economic Diplomacy Redefined: Two-Level Games, Embassies and the Firm

Session 9 · Two-level games, digital diplomacy, and embassies as export promotion Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **When states compete for markets, investment and technology, is the multinational the object of diplomacy, its instrument — or a diplomat in its own right?**

Big work	Rose, A. K. (2007). The foreign service and foreign trade: Embassies as export promotion. *The World Economy*, 30(1), 22–38. doi:10.1111/j.1467-9701.2007.00870.x
Companions	Putnam, R. D. (1988). Diplomacy and domestic politics: The logic of two-level games. *International Organization*, 42(3), 427–460. doi:10.1017/S0020818300027697 · Bjola, C., & Manor, I. (2018). Revisiting Putnam's two-level game theory in the digital age: Domestic digital diplomacy and the Iran nuclear deal. *Cambridge Review of International Affairs*, 31(1), 3–32. doi:10.1080/09557571.2018.1476836
Target figure	Bilateral **exports against the number of diplomatic posts** the exporter keeps in the partner, log scale, with a trend line
Data you collect	Lowy Institute Global Diplomacy Index (`DataDownload.xlsx`); IMF bilateral goods trade (IMTS); World Bank WDI GDP
Tableau skill	Log scales · trend lines, and what their R^2 and p-value do and do not say · building the zeros a post-level file omits
Course themes	The redefinition of economic diplomacy and the implications for multinational firms

9.1 From serving firms to screening them

For most of the period covered by Session 02 — the decades in which the liberal economic order was built out and then globalised — economic diplomacy had a settled meaning and a settled direction. Diplomats negotiated trade agreements, opened markets, promoted exports and courted investors. Trade commissioners in embassies found distributors for small exporters; ministers led delegations of chief executives on state visits; export credit agencies insured contracts that private insurers would not touch. The state served the firm, and the firm's success abroad was counted as the state's success.

That meaning has not disappeared, but it has been joined by others, and the direction of travel has partly reversed. The same governments that promote their firms abroad now screen foreign firms at home, restrict what their own firms may sell and to whom, and draw companies into public quarrels conducted on social media. A multinational today is still the beneficiary of economic diplomacy. It is also, increasingly, its object and its instrument — and sometimes a diplomat in its own right.

Economic diplomacy — the use of the instruments of foreign policy for economic ends, and of economic instruments for foreign-policy ends. The first covers trade negotiation, export and investment promotion; the second covers sanctions, export controls, aid and investment screening used to pursue security or political goals.

Commercial diplomacy — the part of economic diplomacy that serves particular firms and sectors: trade commissioners, market intelligence, introductions, advocacy in disputes, trade missions, export credit and investment promotion. Its unit of account is the firm, not the economy.

Three shifts have redefined the field. The first is **securitisation**: trade, investment and technology are now treated as matters of national security, so that a firm faces screening, export controls and "trusted vendor" rules not only from rival states but from its own side. The second is the move **from promotion to coercion**: the networks that were built to promote exports — financial messaging, payment systems, technology supply chains — are the same networks that states now use to deny access, as Session 08's chokepoints showed and Session 10's sanctions will show in detail. Home- state support can become home-state constraint overnight. The third is **digitalisation**: diplomacy is increasingly conducted in public, on social media, addressed to domestic and foreign audiences at once, and firms are pulled into it — through boycotts, hashtags and official denunciations — without having chosen to be.

This chapter takes the redefinition in three steps. It begins with the theory that best captures how international bargaining is shaped by domestic politics, Robert Putnam's two-level game, and with the argument of Corneliu Bjola and Ilan Manor that digital diplomacy has changed how that game is played. It then turns to the empirical question at the heart of the replication: whether the most traditional instrument of commercial diplomacy, the embassy, actually promotes exports, as Andrew Rose found. Finally it turns to the relationship between states and firms directly — bargaining, screening and the choice a firm faces about when to lean on its home state.

9.2 Two boards at once

Putnam's article begins from a frustration with theories of international relations that treated the state as a unitary actor, and with theories of domestic politics that ignored the international environment (Putnam 1988). His alternative is to model an international negotiation as two games played simultaneously.

At **Level I**, negotiators representing their governments bargain with each other to reach a tentative agreement. At **Level II**, each negotiator's domestic constituencies — legislatures, parties, interest groups, bureaucracies, public opinion — decide whether to ratify it. Ratification need not be formal; any domestic process that can kill an agreement counts. An agreement survives only if it is acceptable at both levels, on both sides.

Win-set — for a given Level II constituency, the set of possible Level I agreements that it would ratify. An agreement is possible only where the two sides' win-sets overlap. The size of each win-set depends on the distribution of power and preferences at home, on the domestic institutions that govern ratification, and on the strategies of the negotiators themselves.

Three lessons follow, and each has a direct application to firms.

The first is that **larger win-sets make agreement more likely**. If many possible deals would be ratified at home, the negotiators have more room to find one that the other side can also accept. But a large win-set also leaves a negotiator open to being pushed: the other side knows that concessions can be extracted without endangering ratification.

The second, and the most counter-intuitive, is that **a small win-set can be a bargaining asset**. A negotiator who can say, credibly, "my parliament will never accept that" has a reason to refuse that the other side cannot argue away. Thomas Schelling had made a version of this point about the power of binding oneself; Putnam located the binding in domestic politics. The credibility matters. A constraint that is invented for bargaining purposes is a bluff, and bluffs can be called.

The third is that **negotiators play both boards at once**. A skilled negotiator does not simply report the constraints of Level II; he or she works to shape them — building support at home for a deal that is coming, and reaching across to the other side's Level II to make its ratification easier. Putnam called the latter "reverberation": a move at Level I that changes the balance of opinion in the other country's domestic politics.

Putnam's framework also explains two features of international agreements that otherwise look irrational. The first is **involuntary defection**: an agreement that a negotiator sincerely accepted fails at Level II because the domestic coalition behind it collapses. The second is the role of **side-payments and issue linkage**: a negotiator who cannot win ratification on the merits of a deal can enlarge the win-set by compensating the losers at home, or by linking the deal to a concession on

another issue that brings a new constituency into the coalition. Both have a direct bearing on firms. The firms and sectors that stand to lose from an agreement are the ones a government must compensate or overrule; the firms that stand to gain are the ones it mobilises. And the domestic institutions that govern ratification — a legislature's voting rules, the role of sub-national governments, the timing of elections — determine how large the win-set can be, which is why a firm's analysis of a negotiation that affects it should begin with the other side's ratification process rather than with the negotiators' public statements.

9.2.1 Where the multinational sits

Draw the two-level game on a board — two negotiators facing each other at Level I, each standing on a circle of domestic constituencies at Level II — and ask where a multinational belongs. The answer is that it belongs in *several* circles at once. At home, an export-oriented firm or industry is one of the constituencies whose support enlarges the win-set for a trade agreement; firms lobby, testify and fund campaigns, and governments count their support before signing. Abroad, the same firm's subsidiaries, plants and employees are part of the *other* country's Level II, where they may be assets (a large employer can argue for an agreement locally) or hostages (a visible foreign firm is a convenient target for retaliation if negotiations go badly).

The double position changes what a firm should expect from negotiations that concern it. A firm whose home government is negotiating with a host government on its behalf is not a spectator. It is part of the domestic politics that determine what its own negotiator can accept, and part of the foreign domestic politics that determine what the other side can accept. The question it must ask is which of its positions — lobbyist at home, employer abroad, symbol of its home country — the negotiators will use, and whether it can control that use.

The renegotiation of the North American Free Trade Agreement between 2017 and 2018 shows the game at work, with firms on every board. The United States imposed tariffs on Canadian and Mexican steel and aluminium in June 2018, on national-security grounds, while the negotiation was under way — a Level I move aimed squarely at the other sides' Level II, where steel and aluminium producers and their workers would press their governments to settle. Canada's system of supply management in dairy was a constraint at its own Level II, defended by a well-organised domestic constituency, and the eventual agreement opened only a limited part of the dairy market. Automobile manufacturers, whose supply chains crossed all three borders many times, were Level II actors in all three countries at once, and the new rules of origin for vehicles were written with their production networks in view. For each firm involved, the question was not only what the agreement would say but which domestic coalition in which country would decide whether there was an agreement at all.

9.3 The two-level game goes digital

Putnam wrote in 1988, about a world in which Level II was reached mainly through legislatures, interest groups and the press. Bjola and Manor revisit his model three decades later through the negotiations over Iran's nuclear programme, which culminated in the Joint Comprehensive Plan of Action of July 2015 (Bjola and Manor 2018). Their argument is that **digital diplomacy** — ministries, embassies and leaders communicating directly with publics on social media — gives negotiators a new instrument for acting on Level II, both their own and the other side's.

The mechanics are straightforward. Before social media, a negotiator who wanted to build domestic support for a deal worked through intermediaries: parliamentary allies, friendly editors, interest groups. A negotiator who wanted to reach the other side's public had few channels at all. Digital platforms remove the intermediaries. Foreign ministers and their staff can address their own public directly, framing the negotiation as it happens, and can address the other side's public in its own language, bypassing its government. In the Iran negotiations, Bjola and Manor show, governments used these channels to shape how the negotiation was perceived at home and abroad, attempting to widen the win-sets on which ratification depended while the bargaining was still going on.

Two implications matter for multinationals. The first is **speed**. Domestic opinion now moves within the negotiation rather than after it, so that the outcome a firm plans around can change faster, and for reasons visible to anyone who follows the right accounts. The second is **exposure**. A firm's brand can be pulled into a digital campaign by either government — praised as a model of cooperation, denounced as a symbol of the other side, or made the subject of a boycott that a ministry amplifies. Neither role is chosen by the firm.

The Iran agreement also illustrates the fragility of win-sets over time, which is the deeper lesson for planning. The agreement was concluded and ratified in the sense Putnam meant; it then lost its domestic support in one of the countries that had made it. The United States withdrew in 2018 and reimposed sanctions, including sanctions aimed at foreign firms that continued to do business with Iran. European companies that had re-entered the Iranian market after 2015 found that the domestic politics of a third country — the United States — determined whether they could stay. The two-level game, for a multinational, is played on more boards than two.

9.4 Do embassies sell?

The most traditional instrument of commercial diplomacy is the diplomatic post itself: the embassy, high commission or consulate, with its commercial section, its trade commissioners and its network of contacts. Governments spend substantial sums maintaining these networks and justify part of the cost by their contribution to trade. Rose asked whether the justification holds (Rose 2007).

His design combined, in a single recent cross-section, data on the embassies and consulates that 22 large exporting countries maintain in some 200 destinations with their bilateral exports, using the framework that economists have used to explain bilateral trade since Jan Tinbergen first applied it in the 1960s (Tinbergen 1962).

Gravity model — the empirical relationship in which bilateral trade between two countries rises with their economic size and falls with the distance between them, by analogy with Newton's law of gravitation. Modern versions add measures of trade costs — common language, shared borders, colonial ties, trade agreements — and controls for each country's overall openness to trade with all partners.

The model's empirical success is remarkable — few relationships in economics fit cross-country data as well — and its theoretical foundations were put on a firm footing by Anderson and van Wincoop, who showed that what matters is not only the trade cost between two partners but each partner's trade costs with everyone else, which they called multilateral resistance (Anderson and Wincoop 2003). Head and Mayer's survey is the practical guide to estimating it (Head and Mayer 2014). For Rose's purposes the gravity model is a way of asking a precise question: *holding constant the size of the partners and the distance between them, and the other standard determinants of trade, do countries export more to places where they keep more diplomatic posts?*

His answer was yes. Exports are higher where a country keeps more posts, and the association survives the standard gravity controls. Bilateral exports rise by approximately six to ten per cent for each additional consulate abroad, and the effect varies by exporter and is non-linear: consulates have smaller effects than the creation of an embassy (Rose 2007). The size should be read off his tables, by exporter and type of post, rather than remembered from a summary.

9.5 Three objections

The replication tests the result against three objections. Each is a general problem in inferring the effect of a policy from a cross-section, and each will recur in the evaluation of sanctions in Session 10.

9.5.1 Reverse causality

Countries open posts where they already trade. A foreign ministry deciding where to open a new consulate looks, among other things, at where its exporters are active and where its nationals travel and invest. If posts follow trade, then a cross-section in which posts and trade go together tells us that ministries are sensible, not that embassies generate exports. The two explanations predict the same snapshot.

They predict different *dynamics*. If posts promote exports, then pairs of countries in which a post is opened should see faster subsequent growth in exports than comparable pairs in which it is not. If posts follow exports, then export growth should come first and the post should follow. Telling the two apart requires data over time, and even then requires that the timing of openings not be driven by anticipated growth — a ministry that opens a consulate because it expects a market to boom will produce a post that "precedes" growth without causing it. Rose took the objection seriously and argued that his

evidence was consistent with posts having an effect, but a cross-section cannot settle the question. That is the catch of the replication, and its extension step is a first, rough attempt at the dynamic comparison.

9.5.2 Omitted variables

Large, rich and nearby partners receive more posts *and* more trade, for reasons that have nothing to do with each other. The gravity controls are designed to absorb exactly this, but a raw scatter of exports against posts does not include them. The replication's simplest correction — dividing exports by the partner's GDP — removes the most obvious confounder and shows how much of the raw relationship was simply size. Other confounders are harder: a shared language, a large diaspora, a history of colonial ties or a security alliance may raise both the number of posts and the volume of trade.

9.5.3 Measurement

A post is not a budget. A small consulate with two staff and a large embassy with a commercial section of fifty count the same in a tally of posts, and an honorary consulate run by a local businessman may count too. The type of post matters, which is why the replication separates embassies from consulates. The direction of the measurement problem is not obvious. Consulates are often placed in commercial cities precisely to serve trade; yet Rose found the creation of an embassy to matter more than an additional consulate, which may say as much about what an embassy signals as about what it does.

9.5.4 The zeros

There is a fourth problem, which the data file makes unavoidable. The Lowy Institute's file lists posts that exist. A country with no post in a partner does not appear — the zero is missing, not zero — and must be constructed. Once constructed, the zeros create a second difficulty: the logarithm of zero is undefined, so on a chart with logarithmic axes the partners with no posts fall off the plot. Dropping them silently changes the sample in a way that is correlated with the outcome, because the partners with no posts are disproportionately the small and distant ones with little trade. The same problem in the trade variable — pairs of countries that do not trade at all — led Santos Silva and Tenreyro to argue that the standard practice of estimating gravity equations in logarithms is biased, and to propose an estimator that keeps the zeros in (Santos Silva and Tenreyro 2006). The lesson for a chart is simpler: say how many observations a log scale has removed, and show what the relationship looks like with and without them.

A trend line with a high R^2 and a tiny p-value does not show that embassies cause exports. It shows a strong association in a cross-section. The R^2 measures how closely the points cluster around the line; the p-value measures how surprising the slope would be if there were no association at all. Neither says anything about which way the causation runs, or whether a third variable drives both.

9.6 The rest of the toolbox, and what the evidence says

Embassies are one instrument among several. The literature on the others is useful both for what it finds and for how it finds it, because each study faces the same objections as Rose.

State visits and trade missions. Nitsch examined the travel of the heads of state of France, Germany and the United States from 1948 to 2003 and found that state and official visits are associated with exports higher by about 8 to 10 per cent; when he turned to a difference-in-differences design to address reverse causality, the effect on export growth was strong but short-lived (Nitsch 2007). Head and Ries studied Canada's trade missions, which during the 1990s took the prime minister abroad with large delegations of provincial premiers and business leaders. Their conclusion was sobering. Canada does trade more than normal with the countries to which it sent missions, but the missions do not seem to have caused the increase: in their preferred specification, with country-pair fixed effects, the effects are small, negative and mainly insignificant (Head and Ries 2010).

From the archive · Team Canada, 1994

In November 1994 the Canadian prime minister, Jean Chrétien, led a delegation to China that included most of the provincial premiers and hundreds of business representatives. The mission was branded "Team Canada", and it became the model for a series of similar missions over the following years, to destinations in Asia, Latin America, Europe and elsewhere. Each mission announced a large total of contracts and agreements signed during the visit; according to the Canadian government, the missions generated tens of billions of dollars in new business deals (Head and Ries 2010).

The announced totals were the wrong evidence. Critics pointed out that many of the agreements were memoranda of understanding, that some were deals that had been negotiated before the trip and timed to be signed during it, and the missions were sent to markets that were already growing. Head and Ries's later evaluation used the missions as a natural test of whether high-profile commercial diplomacy changes trade flows, and compared Canada's trade with mission destinations before and after the missions against the trade it would otherwise have been expected to have. They found that the missions had small, negative and mainly insignificant effects.

The episode is a clean illustration of the reverse-causality objection to Rose. Governments direct their most visible commercial diplomacy at the markets they expect to matter — which are, very often, the markets that would have grown anyway. A ministry that evaluates its missions by the deals announced during them, or by the subsequent growth of trade with the destinations, will always conclude that the missions succeeded.

Export promotion agencies. Lederman, Olarreaga and Payton surveyed national export promotion agencies and found that, on average, their budgets are associated with higher exports, with diminishing returns as budgets grow and with considerable differences in effectiveness depending on how the agencies are organised (Lederman, Olarreaga, and Payton 2010). **The body of evidence as a whole.** Moons and van Bergeijk assembled 32 empirical studies published between 1986 and 2011 into a meta-analysis and found a positive and significant effect of economic diplomacy on trade and investment — with the exception of state visits — and reported effects that are sensitive to the design of the individual studies (Moons and Bergeijk 2017). The fair summary is that commercial diplomacy probably helps at the margin, that the size of its effect is hard to pin down, and that the most visible instruments are the most vulnerable to the selection problems that inflate naive estimates.

9.7 Triangular diplomacy

The two-level game puts firms inside the domestic politics of states. A second tradition puts them across the table. John Stopford and Susan Strange argued that the changing structure of the world economy had turned diplomacy into a triangle (Stopford, Strange, and Henley 1991; Strange 1992). Alongside the familiar **state–state** diplomacy of governments, there is **state–firm** diplomacy, in which governments bargain with multinationals over the terms of investment — where a plant will be built, what share of inputs will be sourced locally, what will be exported — and **firm–firm** diplomacy, in which companies form alliances, joint ventures and consortia that shape markets as surely as treaties do. Governments, on this account, compete for world market shares through the firms they attract, and firms compete partly by bargaining with governments.

The bargaining has a dynamic, and Raymond Vernon described it before Stopford and Strange formalised the triangle (Vernon 1971). Before an investment is made, the firm holds the cards: it can choose among countries, and each host must offer terms good enough to win. Once the investment is made — once the mine is dug, the plant built, the pipeline laid — the firm's capital is sunk and cannot be moved. The host's bargaining power rises and the firm's falls. Terms that were agreed when the firm was mobile can be renegotiated when it is not.

Obsolescing bargain — the tendency for a multinational's bargaining power over a host government to decline after its investment is sunk, so that the terms agreed at entry become vulnerable to renegotiation, higher taxation or, in the limit, expropriation. The bargain "obsolesces" because the conditions that made it acceptable to the host no longer hold.

From the archive · The nationalisation of Chilean copper, 1971

In the 1960s the large copper mines that dominated Chile's exports were owned by two American companies, Anaconda and Kennecott. The terms on which they operated had been renegotiated more than once during the decade, in each case with the Chilean state taking a larger share — first through taxation, then through partial state ownership under the "Chileanisation" policy of President Eduardo Frei. In July 1971, under President Salvador Allende, the Chilean Congress unanimously approved a constitutional amendment nationalising the large copper mines. The compensation formula allowed the government to deduct "excess profits" earned in previous years, and the deductions largely eliminated what the companies would otherwise have been owed.

The episode is the textbook case of the obsolescing bargain, and it was a geopolitical event as much as a commercial one. The companies' home government treated the nationalisation as a matter of foreign policy, and the dispute became entangled with the wider hostility of the United States to Allende's government in the years before the coup of 1973. Kobrin's study of expropriations across developing countries from 1960 to 1979 placed Chile's in a wider wave, in which natural resources — where capital is most sunk — were prominent (Kobrin 1984). For a firm, the lesson has two halves: sunk capital weakens bargaining power over time, and a dispute over that capital can quickly stop being a dispute between a firm and a host and become one between two states.

9.8 Screening: diplomacy aimed at firms

The newest element of economic diplomacy reverses the direction of the oldest. Commercial diplomacy helps a state's firms abroad; **investment screening** examines foreign firms at home, and blocks or conditions their acquisitions on grounds of national security.

The United States has had a screening body, the Committee on Foreign Investment in the United States (CFIUS), since the 1970s, but its reach and visibility have grown sharply. In 2006 the acquisition by Dubai Ports World of a British company that operated terminals at several American ports, approved under the existing procedures, provoked a political storm in Congress; the company agreed to divest the American operations, and the following year Congress tightened the law. A further reform in 2018 widened the committee's jurisdiction to non-controlling investments in businesses involving critical technologies, critical infrastructure and sensitive personal data. The European Union adopted a framework regulation for the screening of foreign direct investment in 2019 (Regulation 2019/452), which created a mechanism for member states and the Commission to share information and comment on each other's cases. Canada added a national-security review to the *Investment Canada Act* in 2009, and has since used it in a widening range of sectors.

The widening is the important part. Screening began as a defence matter — who may own a supplier of military equipment. It now reaches data, critical minerals, infrastructure, semiconductors and artificial intelligence, which is to say much of the modern economy. For a multinational this is geopolitical risk through the **market access** and **assets** channels at once: an acquisition can be blocked, a completed one unwound, and a firm's ownership structure — who its shareholders are, where its parent is incorporated — becomes a determinant of what it may buy. It is also, in Putnam's terms, a Level II phenomenon: screening decisions are made by bodies that answer to domestic politics, and they are subject to the same reverberation as any other part of a negotiation between states.

9.9 When to lean on the home state

Commercial diplomacy is not only promotion. When a firm's investment is threatened by a host government — an expropriation, a cancelled licence, a discriminatory regulation — its home state's embassy is one of the places it turns. Gertz, studying the commercial work of American embassies, showed that diplomats routinely intervene on behalf of their firms in disputes with host governments, and argued that commercial diplomacy is therefore in part an instrument of political-risk management: a way of raising the cost to a host of mistreating a foreign investor (Gertz 2018).

That protection has a price, and the price has risen with the redefinition this chapter describes. In a dispute that remains commercial, home-state backing helps: the host weighs the diplomatic cost of mistreatment against the gain. In a confrontation between states, being seen as an instrument of one of them can make the firm a target of the other. A firm that accepts its ambassador's offer to raise its mining concession with a host president has gained an advocate and has also, in the host's eyes, become part of its home state's foreign policy. If relations between the two states deteriorate, that association will not be forgotten. Blackwill and Harris describe a world in which states use economic instruments systematically for geopolitical ends (Blackwill and Harris 2016); in such a world, the line between being protected by one's home state and being used by it is thin.

Canada's experience after December 2018 shows how quickly a firm can move from beneficiary to hostage. When Canadian authorities arrested a senior executive of Huawei in Vancouver at the request of the United States, China detained two Canadian citizens and, in March 2019, suspended the export permits of major Canadian canola exporters, citing quality concerns. The exporters had done nothing in the dispute; they were simply the most visible Canadian commercial interest in China that could be squeezed without large cost to China itself. Their home state could protest, and did, but it could

not restore their market access without conceding on an issue that was not theirs. In Putnam's terms, they had become instruments for acting on Canada's Level II — a constituency whose losses were meant to raise the domestic cost of the government's position.

The decision can be framed with the course's five channels. Leaning on the home state may protect **market access** and **assets** in a dispute, but it may endanger **market access** in a rival's market, **inputs** sourced from a rival, **finance** from banks that fear sanctions, and the safety of **people** posted in a country where the firm has become a symbol. And it may draw the firm into the **data** demands that come with being seen as part of a state's infrastructure. The question is not whether a firm should ever accept help. It is whether the firm has thought about what the help will mean if relations between the two states change — and whether it is prepared to be, in the eyes of its host, its government's instrument.

9.10 From concept to data

The replication asks whether a country's exports to a partner rise with the number of diplomatic posts it keeps there, using the most comprehensive public record of diplomatic networks. The Lowy Institute's Global Diplomacy Index records the networks of 66 countries and territories — G20, OECD and Asian states, with data collected up to November 2023 — at the level of the individual post. Each row of its 2023 sheet is one post, about 7,700 in all, with the sending country, the host country, the city and the type of post, which distinguishes embassies and high commissions from consulates, permanent missions and other offices. Earlier sheets allow the 2023 networks to be compared with those of 2017, which covers 60 senders.

Three features of the data carry the arguments of this chapter into the replication. The first is that the file is at the wrong level: the gravity question is about pairs of countries, and the file is about posts. Turning posts into a count per pair is a reduction in Krippendorff's sense, and like every reduction it discards information — the city, the type of post — that the analysis may later need. The second is the missing zeros. The pairs in which a country keeps no post do not appear, and must be built from the list of partners in the trade data; without them, the analysis can only compare partners with one post against partners with several, which is not Rose's question. The third is the exports themselves, taken from the IMF's bilateral trade statistics as reported by the exporter, and scaled in one variant by the partner's GDP — the simplest available correction for the most obvious omitted variable.

The type of post deserves a moment of its own. Embassies and high commissions are the standard bilateral posts; consulates and consulates-general are usually in commercial or populous cities other than the capital, and are the posts most directly concerned with trade and with citizens abroad. Permanent missions are a different thing altogether: a mission to the United Nations in New York or Geneva, or to an international organisation elsewhere, is located in the host country but is accredited to the organisation, not to the host. Counting it as a bilateral post would credit Switzerland or the United States with diplomatic attention that is really directed at multilateral bodies. Deciding which types to count is a measurement decision of exactly the kind Session 07 asked to be made explicitly.

Two further features are worth knowing before the first chart is drawn. The Lowy file identifies countries by name only, with inconsistent spellings, and the trade and GDP data use codes, so every join depends on a concordance built by hand. And a join done carelessly, on country fields that do not uniquely identify a pair, can multiply rows and inflate exports without any error message. Both are data problems; both, unnoticed, would change the slope of the trend line more than any of the substantive objections.

9.11 A short review of the literature

The theory of how domestic politics shapes international bargaining is Putnam (1988), whose two-level game and win-set remain the standard vocabulary; Bjola and Manor (2018) extend it to digital diplomacy through the Iran nuclear negotiations. On the relationship between states and firms, Vernon (1971) introduced the dynamic later called the obsolescing bargain, Kobrin (1984) documented the wave of expropriations that gave it empirical content, and Stopford, Strange, and Henley (1991) and Strange (1992) recast diplomacy as a triangle of state–state, state–firm and firm–firm relations. Blackwill and Harris (2016) describe the return of geoeconomics — the systematic use of economic instruments for geopolitical ends — that has redefined the field.

The empirical evaluation of commercial diplomacy rests on the gravity model, introduced by Tinbergen (1962) and given its modern theoretical foundation by Anderson and Wincoop (2003); Head and Mayer (2014) is the practical guide and Santos Silva and Tenreyro (2006) the warning about logarithms and zeros. Rose (2007) is the central study of diplomatic posts and exports. Nitsch (2007) examines state visits, Head and Ries (2010) Canada's trade missions, Lederman, Olarreaga, and Payton (2010) export promotion agencies, and Moons and Bergeijk (2017) synthesises the econometric evidence in a meta-analysis. Gertz (2018) shows commercial diplomacy at work in investment disputes, connecting the promotion of firms abroad to the management of political risk.

9.12 What the trend line will not tell the firm

The practice round produces a scatter of exports against diplomatic posts, on logarithmic axes, with a trend line and a model summary. It will almost certainly slope upwards, and the R^2 will be respectable. The temptation is to read it as Rose's finding confirmed: posts promote exports.

The trend line does not license that conclusion, for a multinational or for a foreign ministry. It does not show that opening a post in a market will raise exports there, because posts are opened where exports are expected, and the snapshot cannot tell the two directions apart. It does not show that the relationship is about diplomacy at all until the size of the partner has been removed, and it may show something quite different once embassies and consulates are separated. And if the zeros have been dropped — silently, by a logarithmic axis — it does not show the relationship for the partners where the question is most interesting: the markets in which a country has no presence yet.

For a firm, the more useful reading is the reverse of the naive one. The density of a home state's diplomatic network in a market is a measure of how much that state already cares about the market — a measure of the political attention a firm can expect when it needs help, and of the political attention it will attract when relations sour. The question to carry into the practice round is the one this chapter has framed: in the markets that matter to the team's firm, is the firm the object of its home state's diplomacy, its instrument, or its hostage — and what would change that answer?

9.13 In the practice

> **The target.** Rose (2007)'s central relationship: for each exporter–partner pair, **exports** against the **number of diplomatic posts** the exporter keeps in the partner — on log scales, with a trend line — rebuilt with the Lowy Global Diplomacy Index (2023 data) and IMF trade.

Work in your project teams. Everyone builds their own workbook; the team hands in one slide.

9.13.1 1 · Collect — 15 min

In `data/raw/` (see `../data/README.md`):

- `DataDownload.xlsx` — Lowy Global Diplomacy Index;
- IMF IMTS — **exports of goods, FOB, US$, 2023**, for Canada and at least four other Lowy countries × all partners;
- WDI `NY.GDP.MKTP.CD` — GDP, current US$.

Provenance log — one line per file, including **which side** of the trade flow you used (exporter- reported exports) and the IMF indicator name exactly as shown.

9.13.2 2 · Prepare — 25 min

This is the hard part today. Take your time.

1. **Lowy.** Connect to `DataDownload.xlsx`, sheet **2023**. The columns for sending country, host country and post type are `COUNTRY`, `POST COUNTRY` and `TYPE OF POST`. Filter out posts that are not bilateral: `TYPE OF POST` = `Permanent Mission` (missions to the UN, EU, OECD…). Look at what `Other` holds (representative and trade offices, e.g. in Taipei) and log your choice.

2. **IMF.** Add the IMF file as a **second table** on the canvas. Keep only the five exporters and the year 2023; drop aggregate partners ("World", regional totals — codes starting with `G` or `TX`). Rename `OBS_VALUE` → `Exports`.

3. **Concordance.** Lowy has country names only; the IMF file has ISO3 codes (`COUNTRY`, `COUNTERPART_COUNTRY`). Make a small sheet `name → ISO3` for the Lowy `COUNTRY` and `POST COUNTRY` names (watch the typos listed in the data README) and add ISO3 to Lowy.

4. **Relate, don't join.** Drag the IMF table first, then the Lowy table, and create a **relationship** on `exporter ISO3 = sending ISO3` **and** `partner ISO3 = host ISO3`. A relationship keeps the export totals at their own grain; a plain join would repeat each export value once per post and inflate it.

5. **Build the zeros.** Calculated field `Posts`:

   ```
   ZN(COUNT([POST CITY]))
   ```

   (`POST CITY` is never empty in the 2023 sheet). Partners with exports but no post now get 0.

6. **Make logging possible.** Calculated field `Posts + 1` = `[Posts] + 1`. Write this decision in the log: it is the same compromise most gravity papers make, and it has consequences.

7. **Exports per partner GDP.** Add WDI GDP (relationship on partner ISO3, year 2023) and create `Exports / partner GDP (‰)` = `SUM([Exports]) / SUM([GDP]) * 1000`.

9.13.3 3 · Replicate — 20 min

9.13.3.1 Sheet 1 · "More posts, more exports"

1. `Posts + 1` → **Columns** (as a measure; aggregation `SUM` at the dyad level).
2. `Exports` → **Rows**.
3. Exporter ISO3 **and** partner ISO3 → **Detail** — one mark per pair.
4. Exporter → **Colour**.
5. Right-click each axis → *Edit Axis* → **Logarithmic**.
6. *Analytics* → **Trend Line** → *Linear*. Right-click the line → **Describe Trend Model**. Note the slope, R^2 and p-value (one line per exporter if colour splits them — try both ways).
7. Title states the finding; caption cites Lowy (2024 edition, 2023 data) and IMF IMTS, with access dates.

9.13.4 4 · Compare and break — 20 min

1. **Differ.** Rose used an earlier period and a different set of exporters. Is your slope steeper or flatter than the relationship he reports? Name one reason.
2. **Break: size.** Replace `Exports` by `Exports / partner GDP (%)`. What happens to the slope and the R^2? What did the raw chart owe to the sheer size of partners?
3. **Break: type of post.** Filter `TYPE OF POST` to **`Embassy / High Commission`** only, then to **`Consulate / Consulate-General`** only. Which carries the relationship? Does that match Rose?
4. **Break: the zeros.** Exclude partners with `Posts = 0`. How much does the trend line move? Why should a careful analyst never drop them silently?
5. **Extend — reverse causality** (fast teams). Add the Lowy **2017** sheet and IMF exports for

2017. For each pair, compute the change in posts and the growth of exports 2017→2023. Do pairs that **gained** posts show faster export growth — or did posts open where exports had already grown? What would you need to tell the two apart?

9.13.5 5 · The catch — 10 min

Original	Ours	The catch
Rose's posts–exports relationship	Your Sheet 1	One sentence: what this trend line does **not** license a multinational — or a foreign ministry — to conclude

A member drawn at random presents in two minutes.

9.13.6 Hand in

On ZoneCours before leaving: **`.twbx`** and provenance log, including your concordance fixes. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

9.13.7 What loses credit

- A **join** that inflated exports, unnoticed (check one pair's total against the IMF file)
- Zeros dropped without saying so
- "$R^2 = 0.6$, so embassies explain 60% of exports" — they explain nothing; they *co-vary*
- Unmatched countries silently missing from the chart

Chapter 10

Economic Sanctions: Do They Work, and Whom Do They Hit?

Session 10 · Do they work — and what do they do to the firms caught between sender and target? Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Sanctions have become the instrument of first resort. Are they an efficient mechanism for regulating geopolitical risk — or a source of it?**

Big works	Hufbauer, G. C., Schott, J. J., Elliott, K. A., & Oegg, B. (2007). *Economic Sanctions Reconsidered* (3rd ed.). Peterson Institute for International Economics. · Felbermayr, G., Kirilakha, A., Syropoulos, C., Yalcin, E., & Yotov, Y. V. (2020). The Global Sanctions Data Base. *European Economic Review*, 129, 103561. doi:10.1016/j.euroecorev.2020.103561
The critique	Pape, R. A. (1997). Why economic sanctions do not work. *International Security*, 22(2), 90–136. doi:10.1162/isec.22.2.90
Target figure	**Active sanctions cases per year, by type**, 1950–today — and the **share judged successful**, by objective
Data you collect	Global Sanctions Data Base, Release 5 (by request); TIES v4.0 (Morgan, Bapat & Kobayashi, 2014)
Tableau skill	Year scaffolds and range joins · Gantt charts for episodes · stacked percent bars · quick table calculations
Course themes	The efficiency of economic sanctions as mechanisms for regulating geopolitical risks

10.1 The instrument of first resort

When a government wants to do something about a foreign state's behaviour and is not prepared to go to war over it, the menu is short. It can protest, which costs nothing and changes nothing. It can withdraw an ambassador, which is visible and mostly symbolic. Or it can impose economic sanctions: restrict what its firms and banks may sell to, buy from, lend to or invest in the offending state, its government, its companies or its leading citizens. Over the last three decades the third option has moved from the end of the menu to the top. Sanctions are now what governments reach for first, in response to invasions and coups, nuclear programmes and election interference, human-rights abuses and cyberattacks. The Global Sanctions Data Base, whose descriptive picture the practice round rebuilds, documents both the rise in the number of active cases since the end of the Cold War and the change in what those cases are made of (Felbermayr et al. 2020).

The session's question is whether this instrument is an efficient mechanism for *regulating* geopolitical risk — a way of raising the cost of aggression without the cost of war — or a *source* of it. The two are not mutually exclusive, and the chapter argues that, for a multinational, the second matters more than the first. A firm does not need to know whether the sanctions on a given state will achieve their political goal. It needs to know whether they will cut it off from a customer, a

supplier, a bank or an asset; how far the effective perimeter of the measures extends beyond their legal text; and what the target can do to it in return.

Economic sanctions — restrictions on economic relations imposed by one or more *senders* on a *target* — a state, a government, a firm or an individual — in order to change its behaviour, weaken it, constrain what it can do, or signal disapproval. The instrument is the restriction; the objective is political.

That definition contains the whole debate in compressed form. Its last clause lists four objectives, and the answer to "do sanctions work?" depends on which of them you hold the instrument to. A measure that fails to change a regime's policy may still succeed in making its weapons programme slower and more expensive, or in telling allies and voters at home that something has been done. A large part of the disagreement in the literature, as we shall see, comes from asking the same question of cases that were never trying to do the same thing.

10.2 A very old weapon

Economic coercion is older than the states that now practise it. Thucydides records the Megarian decree, by which Athens barred the merchants of Megara from the harbours of its empire, among the grievances that preceded the Peloponnesian War; Napoleon's Continental System tried to defeat Britain by closing the continent to British trade, and was undone by smuggling and by the costs it imposed on France's own allies. Both episodes contain, in miniature, the problems this chapter returns to: sanctions hurt the sender's own merchants, they invite evasion through third parties, and their effect on the target's political decisions is uncertain at best.

The modern idea — sanctions as an organised, collective alternative to war — dates from the League of Nations. Article 16 of its Covenant obliged members to sever trade and financial relations with any member that resorted to war in breach of its commitments. The idea was tested once, and the test is the founding case of the literature.

From the archive · The League of Nations and Italy, 1935

Italy invaded Ethiopia in October 1935. Within days the League's Council found that Italy had resorted to war in breach of the Covenant, and the members agreed a package of measures that came into force in November: an arms embargo, a ban on loans and credits to the Italian government, a ban on imports of Italian goods, and an embargo on the export to Italy of a list of raw materials.

What was left off the list mattered more than what was on it. Oil was not embargoed; nor were coal and steel, and the Suez Canal, through which Italy supplied its army in East Africa, stayed open. The great powers that would have had to enforce a wider embargo feared driving Italy into Germany's arms, and the United States, not a member, was outside the system. Italian forces took Addis Ababa in May 1936, and the League lifted the sanctions in July.

The lesson drawn at the time was that collective sanctions had failed. The lesson worth drawing now is subtler. The measures were designed by senders who wanted to be seen to act while protecting their own interests, and the gap between the announced objective — to stop a war — and the instrument actually deployed was visible to everyone, including the target. Every later debate about whether sanctions "work" is, in part, a debate about whether the measures studied were ever designed to.

The United Nations Charter, drafted a decade later, put the idea on a firmer legal footing: Article 41 allows the Security Council to decide on "measures not involving the use of armed force", including the "complete or partial interruption of economic relations", and makes them binding on all members. For most of the Cold War the veto kept the provision almost idle. The Council imposed mandatory sanctions on Southern Rhodesia after its unilateral declaration of independence in 1965 and an arms embargo on South Africa in 1977, and little else. Sanctions in this period were overwhelmingly unilateral, and overwhelmingly American. The end of the Cold War unblocked the Council, and the 1990s were later called the "sanctions decade". The instrument has not looked back.

10.3 Four ways to describe a sanction

Before anything can be counted it must be classified, and the classification already embeds a theory of what sanctions are for. The lecture uses four dimensions, and the GSDB codes three of them directly.

The **sender** may be one state acting alone, a group of states acting together — the European Union is the most important plurilateral sender, its measures adopted by unanimity in the Council — or the United Nations Security Council, whose decisions bind every member. The distinction matters to a firm because it determines how many jurisdictions it must satisfy and how much room there is for a competitor from a non-sending state to take its place.

The **type** describes the instrument: restrictions on trade (exports, imports, or both, partial or complete), on finance (loans, investment, payments, asset freezes), on travel, on arms, on military assistance, and a residual category. The GSDB records each as a separate indicator, so a single case can be of several types at once — which is why, in the practice round, the stacked count of cases by type exceeds the number of cases.

The **scope** distinguishes measures aimed at a whole economy from measures aimed at named persons, entities or sectors.

Comprehensive versus targeted sanctions — *comprehensive* sanctions restrict all, or nearly all, economic relations with a target country; *targeted* or "smart" sanctions restrict relations only with named individuals and entities (asset freezes, travel bans), with specific sectors (energy, finance, defence), or in specific goods (arms, dual-use items, luxury goods). The distinction is about who is meant to bear the cost.

The **objective** is what the sender says it wants. The GSDB distinguishes changing a policy, destabilising a regime, resolving a territorial conflict, preventing a war, ending one, fighting terrorism, protecting human rights and restoring or defending democracy. Many cases have more than one, and the stated objective is not always the real one. A sender that announces "policy change" may be content with "punishment", and one that announces "democracy" may care more about signalling to its own electorate. Coders work from the public record, so the objective variable is a record of what senders said, not of what they wanted.

10.4 Do they work? Coding success

The modern empirical literature begins with one book in three editions. Gary Clyde Hufbauer and Jeffrey Schott, later joined by Kimberly Ann Elliott and Barbara Oegg, assembled every sanctions episode of the twentieth century they could document and asked of each a question that sounds simple: did it succeed (Hufbauer et al. 2007)?

Their answer to the problem of defining success has become the field's reference point, so it is worth stating precisely. Each episode is scored on two components. The first is the **policy result**: how far the outcome the sender sought was actually achieved, on a scale from failure to success. The second is the **contribution of sanctions**: how far the sanctions, as opposed to other factors — military force, a change of government for unrelated reasons, a negotiation that would have happened anyway — contributed to that outcome. Each component is scored from 1 to 4, and the two are multiplied into a success score. A case counts as a success when the product reaches 9 or more: that is, when the outcome was at least partly achieved *and* sanctions made at least a modest contribution to it.

Sanctions success (Hufbauer et al.) — an episode in which the sender's policy goal was at least partly achieved and sanctions made at least a modest contribution to that achievement, operationalised as a product of the two scores, each from 1 to 4, of 9 or more. Success is a joint judgement about an outcome and a cause.

On this definition, roughly a third of the episodes in the study count as at least partial successes. The rate varies systematically with the circumstances. Sanctions succeed more often when the goal is modest — the release of a prisoner rather than the overthrow of a regime — when the target is economically small and weak relative to the sender, when target and sender were previously on friendly terms and trade heavily, and when the measures are imposed quickly and decisively rather than ratcheted up slowly, giving the target time to adapt. They succeed less often when the goal is major, when the target is an adversary, and when the target can turn to other suppliers and buyers.

That last condition was formalised by Bryan Early as *sanctions busting*: third countries whose firms step into the trade the sanctions forbid, for profit or for strategic reasons, and so supply the target with the alternatives that make resistance affordable (Early 2015). It is the reason multilateral sanctions are not straightforwardly more effective than unilateral ones.

The more senders there are, the fewer alternatives remain for the target, but the harder it is to agree on anything strong, and the more likely the package resembles the League's of 1935.

The "about a third" figure has had a long public life. It appears in policy debates as a fact about the world: sanctions work a third of the time. It is nothing of the kind. It is a fact about one coding of one set of cases, produced by one team's judgements on two four-point scales, and changing any of those three elements changes the number.

10.5 Pape's audit, and why the debate never closed

The most consequential demonstration of this came from Robert Pape, who in 1997 went back through the cases Hufbauer and his colleagues had counted as successes in their second edition and asked whether each one survived scrutiny (Pape 1997). His conclusion was that most did not. In many of the claimed successes, he argued, the outcome had been achieved by the use or threat of military force rather than by economic pressure; in others the target had made no real concession, or the concession could not be attributed to sanctions at all. By his reckoning only five of the forty claimed successes held up, and he concluded that economic sanctions rarely achieve major political goals and should not be expected to.

Elliott replied that the glass was half full rather than empty (Elliott 1998). Sanctions, she argued, had never been claimed to be a substitute for force in pursuit of major goals, and judged against the modest goals for which they are typically used they did better than Pape allowed; moreover, their record had improved as they were used more often in concert, after the Cold War. Pape answered that sanctions still did not work (Pape 1998). The exchange has been taught ever since as a debate about effectiveness. It is better understood as a debate about coding: about what counts as a concession, how much a contribution must be, and what the counterfactual is.

The counterfactual problem is the deep one, and David Baldwin had set it out a decade earlier (D. A. Baldwin 1985). To say that sanctions failed is to say that some other instrument would have done better. If the alternative was war, a sanction that achieved nothing at little cost may still have been the better choice; if the alternative was diplomacy, a sanction that achieved a little at great cost may have been the worse one. Effectiveness, on Baldwin's account, is always comparative and always relative to cost, and a success rate computed without reference to the alternatives answers a question no policymaker faces.

A second line of argument asks *how* sanctions are supposed to work, and finds the naive answer wanting. Johan Galtung, studying the sanctions on Rhodesia, described what he called the naive theory: economic pain causes political disintegration, which forces the government to comply (Galtung 1967). The Rhodesian evidence suggested the opposite. Pressure from outside produced a rally around the government, a reorganisation of the economy towards self-sufficiency, and a set of domestic winners — smugglers, import substitutes, those who controlled scarce foreign exchange — with an interest in the sanctions continuing. Jonathan Kirshner later generalised the point: sanctions work, when they work, by changing the relative power of groups within the target, and their effect therefore depends on who inside the target bears the cost and who profits from it (Kirshner 1997).

The lesson for an analyst. "Do sanctions work?" has no answer independent of how a case, an objective and a success are coded, and of what the comparison is. Every number in this debate is a coding decision. The practice round makes several of them — which objective counts when a case has several, whether ongoing cases enter the denominator — and each one moves the share judged successful.

10.6 The cases we never see

There is a deeper problem with any success rate computed from imposed sanctions, and it was put most clearly by Daniel Drezner. His first book began from a puzzle: states impose sanctions most often on their adversaries, yet adversaries are the targets least likely to concede, because they expect future conflict and fear that any concession will be exploited; allies concede more readily, but are sanctioned less often (Drezner 1999). The second step of the argument concerns the threat (Drezner 2003). If a target that would concede does so as soon as sanctions are threatened, the episode ends before any measure is imposed. The cases that reach imposition are disproportionately those in which the target had decided to resist.

Selection effect of threats — the bias that arises when sanctions most likely to succeed do so at the threat stage and are therefore never imposed. Samples of imposed sanctions over-represent hard cases, so imposed sanctions look less effective

than the instrument is. The most significant effects of economic coercion may be the ones that never appear in a list of sanctions.

The implication for measurement is direct. A dataset that records only impositions cannot estimate the effectiveness of the instrument, because it has selected its cases on a variable correlated with the outcome. This is the reason the Threat and Imposition of Economic Sanctions dataset exists. T. Clifton Morgan, Navin Bapat and their colleagues recorded threats as well as impositions, first for 1971–2000 (Morgan, Bapat, and Krustev 2009) and then, in the version used in the practice round, for 1945–2005 (Morgan, Bapat, and Kobayashi 2014). In TIES a case begins with a threat or an imposition, whichever comes first, and the dataset records whether sanctions were in fact imposed. A substantial share of its cases never were.

The selection argument also cuts the other way, and an honest analyst should say so. Imposition is a costly signal of the sender's seriousness, and some of its effects may come from the signal rather than from the economic pain. None of this can be read off a count of cases. It can only be argued, with evidence about each episode.

10.7 From comprehensive to smart

The sanctions decade of the 1990s produced the instrument's most damaging case, and that case changed how sanctions are designed.

From the archive · Iraq, 1990–2003

Four days after Iraq invaded Kuwait in August 1990, the Security Council adopted Resolution 661, imposing comprehensive sanctions: a ban on all trade with Iraq except medical supplies and, in humanitarian circumstances, foodstuffs. The war of 1991 expelled Iraq from Kuwait; the sanctions stayed, tied now to the disarmament of Iraq's weapons programmes.

They stayed for more than a decade, and their humanitarian consequences became the central fact of the debate about them. Malnutrition, the collapse of water and health services, and a sharp rise in mortality were documented by UN agencies and independent researchers, although the size of the excess mortality and how much of it to attribute to sanctions rather than to war damage and the regime's own choices were — and remain — contested. In 1995 the Council created the Oil-for-Food Programme (Resolution 986), allowing Iraq to sell oil under UN supervision to pay for food and medicine. The programme relieved the shortages and was in turn corrupted: the regime used surcharges and kickbacks from the firms that bought its oil and sold it goods, and the subsequent inquiry named companies from many countries.

The regime survived the sanctions intact. Those who bore their cost were the population and, through the programme's scandals, the firms that had traded with the regime under UN licence.

The Iraqi experience made it difficult for senders to defend comprehensive sanctions on a whole population, and the response was the turn to *targeted* measures. In a series of intergovernmental processes around the turn of the century — at Interlaken on financial sanctions, at Bonn and Berlin on arms embargoes and travel bans, at Stockholm on implementation — governments, the UN Secretariat and experts worked out how to aim sanctions at those responsible: freeze the assets and ban the travel of named leaders and their financiers, embargo arms and the specific goods that sustain a war or a weapons programme, restrict specific sectors.

Targeted sanctions created problems of their own. Listing a person under UN or EU measures is an executive act with severe consequences for the person listed, and in the *Kadi* judgment of 2008 the European Court of Justice annulled the EU's implementation of a UN listing because it did not respect the listed person's rights of defence. Listing procedures, review mechanisms and an Ombudsperson for one of the UN regimes followed. For firms the lesson was that lists change, are litigated, and may be annulled — and that a listing in one jurisdiction may not exist in another.

Whether targeting works better is an empirical question with a mixed answer. The most comprehensive study of UN targeted sanctions distinguishes three purposes — to coerce a change in behaviour, to constrain the target's ability to act, and to signal a norm — and finds that the measures do better at constraining and signalling than at coercing (Biersteker, Eckert, and Tourinho 2016). On humanitarian cost the evidence is clearer for the comprehensive kind: Dursun Peksen finds that sanctions are associated with a deterioration of the target government's respect for physical-integrity rights, and that the effect is stronger for comprehensive measures (Peksen 2009). The targets of sanctions, in other words, tend to pass the cost down.

10.8 The financial turn

The second change in the design of sanctions since 2000 has been the rise of financial measures, and it is the change that matters most to a multinational.

Henry Farrell and Abraham Newman explained it in a framework the course met in Session 08 (Farrell and Newman 2019). Global economic networks — payment systems, financial messaging, correspondent banking, the internet's infrastructure — are not flat. They have hubs, and the hubs are disproportionately under the jurisdiction of a few states, above all the United States. A state with jurisdiction over a hub can exploit it in two ways: as a *panopticon*, using the information flowing through the hub to see what others are doing, and as a *chokepoint*, denying others access to the hub and therefore to the network. International dollar payments clear, in the end, through banks with access to the US financial system; banks around the world communicate payment instructions through SWIFT, a Belgian cooperative subject to European law. Cutting a bank off from either is a sanction of a different order from an embargo on its country's exports.

This is why targeted does not mean contained. A financial sanction is aimed at named banks, but every firm that pays or is paid through those banks is affected, and the banks of third countries — which need access to the dollar system more than they need any single customer — will cut off a sanctioned client long before the law requires them to. Sanctions on banks, insurers and shippers propagate down every supply chain that depends on them. The measures of 2022 against Russia showed the reach of the instrument: the freezing of a large part of the Russian central bank's foreign reserves, the disconnection of major Russian banks from SWIFT, and a price cap on Russian oil enforced not by blockade but through the Western insurers and shipping services the oil trade relied on.

10.9 The firm in the middle

Sanctions are aimed at states, governments and the people who run them. They are implemented by firms. Governments write the rules; banks, exporters, shippers and insurers decide, transaction by transaction, whether a given payment, shipment or contract falls within them, and bear severe penalties if they decide wrongly. The firm is therefore not a bystander to sanctions policy. It is its executor, and its executor of last resort.

10.9.1 Lists, ownership and the perimeter

Each sending jurisdiction maintains its own legal architecture. In the United States, sanctions programmes rest mostly on the International Emergency Economic Powers Act of 1977 and are administered by the Treasury's Office of Foreign Assets Control, whose Specially Designated Nationals list is the most consulted list in global commerce. In the European Union, restrictive measures are adopted by Council decision under the Common Foreign and Security Policy and given effect by Council regulations that bind every person and firm in the Union. In Canada, they rest on the *Special Economic Measures Act*, the *United Nations Act* and the *Justice for Victims of Corrupt Foreign Officials Act*. The lists overlap but do not coincide, and a firm operating across jurisdictions must satisfy all of them.

The list is only the beginning of the perimeter. Under the US "50 percent rule", an entity owned, directly or indirectly, 50 percent or more in aggregate by listed persons is itself treated as blocked, whether or not it appears on any list; EU guidance extends the restrictions to entities *owned or controlled* by listed persons, with control judged on a range of indicators. A firm must therefore screen not only its counterparties but their owners, and their owners' owners, through chains of holding companies that may be designed to obscure them.

Compliance perimeter — the set of counterparties, transactions, goods and jurisdictions a firm must treat as restricted in order to comply with every sanctions regime that applies to it. It is wider than any list: it includes entities owned or controlled by listed persons, goods restricted by end use, and the jurisdictions whose rules reach the firm through its nationality, its currency, its banks or its inputs.

10.9.2 Secondary sanctions and blocking statutes

Primary sanctions bind the sender's own nationals and transactions on its territory. Secondary sanctions go further: they threaten third-country firms with penalties — exclusion from the sender's market, its financial system, or its government contracts — if they deal with the target, even where no person or transaction of the sender is involved.

Secondary sanctions — measures that penalise third-country persons for dealing with a sanctioned target, by restricting their own access to the sender's market or financial system. They make a sender's foreign policy binding on firms that are not its nationals, by forcing them to choose between the sender and the target.

The tension this creates between allies is old. When the United States legislated against foreign firms investing in Cuba, Iran and Libya in 1996, the European Union adopted a *Blocking Statute* forbidding EU firms to comply with the US measures. When the United States withdrew from the Iran nuclear agreement in 2018 and reimposed secondary sanctions, the EU updated the Statute. European firms nevertheless left Iran in large numbers. The choice a secondary sanction forces is not a legal one but a commercial one: the US market and the dollar are worth more to almost any multinational than the target's market is, and no blocking statute changes that arithmetic. The case of BNP Paribas, which pleaded guilty in the United States in 2014 to processing transactions for sanctioned parties and paid a penalty without precedent at the time, settled the question for many compliance departments: dollar transactions with sanctioned parties anywhere in the world fall within US reach.

10.9.3 Overcompliance and de-risking

Penalties are severe, rules are complex, lists change without notice, and the cost of a mistake is asymmetric: a bank that refuses a lawful payment loses a fee, a bank that processes an unlawful one may lose its licence. The rational response is to go further than the law.

Overcompliance and de-risking — the withdrawal of firms, and above all banks, from whole countries, sectors or categories of client that the law does not close, because the expected cost of an error exceeds the revenue at stake. The effective scope of sanctions is set by compliance departments as much as by governments.

The evidence that this matters is now substantial. Matthieu Crozet and Julian Hinz, studying the sanctions on Russia after 2014 and the Russian counter-sanctions, find that the trade losses borne by the sending countries fell largely on products that were not embargoed at all — "friendly fire" that they attribute mainly to the financial measures and the uncertainty they created (Crozet and Hinz 2020). A firm selling unrestricted goods to an unrestricted customer may still be unable to get paid, insured or shipped. The question the lecture puts to the room captures it: if your compliance team tells you the new measures do not legally require you to exit, and your bank tells you it will no longer process your payments to that country, who has just set your sanctions policy?

10.10 Russia 2022: the exit decision

The full-scale invasion of Ukraine in February 2022 produced the largest and fastest expansion of sanctions against a major economy since the Second World War, and it confronted hundreds of Western multinationals with a decision most had never modelled: whether to leave a large market that the law, for many of them, did not require them to leave. Public pressure, employee pressure, the practical impossibility of making payments and moving goods, and the expectation of further measures pushed many to suspend, scale back or exit.

Exit turned out to be harder than announcing it. Russia's countermeasures made the sale of a foreign-owned business subject to approval by a government commission, required sellers to accept a steep discount on the valuation, and added a payment to the state budget; buyers were frequently local managers or firms close to the state. In 2023 a presidential decree allowed the temporary state management of foreign-owned assets, and the Russian subsidiaries of Danone and Carlsberg were placed under it within weeks. The firms that waited lost the option of selling on their own terms.

The four options — stay, suspend, sell, walk away — each have a cost and a residual risk. McDonald's sold its Russian restaurants to a local licensee, which reopened them under a new name. Renault sold its controlling stake in AvtoVAZ to a state entity for a symbolic rouble, keeping an option to buy it back within six years — a sale designed to preserve an option on a return. Others suspended operations and paid for idle assets while hoping the situation would change. Others stayed,

citing obligations to employees and customers or the essential nature of their products, and accepted the reputational cost and the risk of further sanctions or of seizure.

The right answer depends on the firm, not on the country. Ask the room to choose for a consumer-goods firm and then for an energy firm, and the answers diverge: the first has brands, local staff and a domestic supply chain that a buyer can run; the second has assets that cannot be moved, partners that may be sanctioned, and a product the target state regards as strategic. Country-level exposure tells you that a decision must be taken. It does not tell you which one.

10.11 Through which channels sanctions reach the firm

The course organises the ways geopolitical risk reaches a multinational's profit and loss account into five channels, and sanctions run through all of them.

Market access. Export bans and import bans close markets, either directly — the goods may not be sold — or indirectly, when customers are listed or owned by listed persons. Dual-use export controls, which increasingly operate alongside sanctions, close specific product lines to specific end users.

Inputs. A supplier that is sanctioned, or that sits in a sanctioned jurisdiction, can no longer be paid. Counter-sanctions and export restrictions imposed by the target close supply from the other side. The firm's exposure here is often deeper in the supply chain than it can see.

Finance and payments. This is where financial sanctions bite first: the correspondent bank that will not process a payment, the letter of credit that cannot be confirmed, the insurer that will not cover a cargo. It is also the channel through which overcompliance operates.

Assets and people. Assets in a target country can be frozen by the sender if they are linked to listed persons, or seized, placed under temporary management, or sold only at a forced discount by the target in retaliation. Employees in the target country, and the firm's obligations to them, are part of the exit calculation and part of its cost.

Data and operations. Software licences, cloud services, maintenance contracts and technical support may be restricted services in themselves; screening a customer base against changing lists is an operational burden; and a sanctioned partner on a shared platform can force the firm to cut off its access.

The lecture's list of what a risk dashboard should track maps onto these channels. The firm's own **exposure** — revenues, assets and suppliers in sanctioned or at-risk jurisdictions — is the starting point. The **pipeline** of threats and announced measures is the TIES lesson applied: the cases that matter most to a firm are the ones that have not yet been imposed. The **propagation** risk sits in the banks, insurers and logistics providers whose own exposure could cut the firm off. And the **counter-measures** the target can take against foreign firms on its soil are, as Russia showed, the risk that determines what an exit costs.

10.12 From episodes to a dataset

The Global Sanctions Data Base was built to answer, at scale, questions that case studies cannot: how the use of sanctions has changed over time, what types and objectives dominate, and — joined to trade data — what sanctions do to trade flows (Felbermayr et al. 2020). Its unit is the *case*: a sanctions episode defined by its senders, its target, its start and end, its types and its objectives, with an outcome coded for each objective. Its producers have released it repeatedly, extending the period and revising earlier cases; the third release added the measures following the pandemic and the escalating sanctions on Russia (Syropoulos et al. 2024), and the practice round works with the fifth.

Turning episodes into a time series of "active cases" requires a decision about what being active means. An episode with a start year and an end year is active in every year in between, so the practice round expands each case into one row per active year, with a year scaffold and a range join. Three decisions are buried in that construction. Cases that have not ended must be given an end — and the distributed file gives ongoing cases an end year that is the last year of the data, which is not what its codebook says. A case of several types appears once in each type, so a stacked chart of types overstates the number of cases. And a case is a case whether it consists of a single listing or of a comprehensive embargo, so the number of active cases measures the frequency of the instrument's use, not its intensity.

The outcome coding carries the success debate into the data. Each objective of each case receives an outcome — total success, partial success, a negotiated settlement, failure, or ongoing — and the share judged successful depends on which objective a multi-objective case is counted under and on whether the ongoing cases, which by definition have not succeeded yet, enter the denominator. These are not technical details. They are Hufbauer's and Pape's disagreement reproduced in two calculated fields.

TIES, set beside the GSDB, supplies what the GSDB cannot: the threats that were never imposed. The two datasets differ in their unit of observation, their period, their coding of senders and targets and their definition of an outcome, so they cannot be merged case by case. What they can do is be compared, year by year, to give an order of magnitude to the selection effect: how much of economic coercion happens before any measure is imposed.

10.13 A short review of the literature

The empirical study of sanctions begins with Hufbauer et al. (2007), whose three editions established the case list, the two-component coding of success and the "about a third" finding; Pape (1997) audited the claimed successes and found few that survived, Elliott (1998) replied, and Pape (1998) rejoined. D. A. Baldwin (1985) had already argued that effectiveness must be judged relative to the alternatives and their costs, which reframes the whole exchange. The mechanism literature runs from Galtung (1967), whose study of Rhodesia dismantled the "naive theory" that pain produces compliance, to Kirshner (1997), who located the effect of sanctions in the redistribution of power among groups within the target.

The strategic turn came with Drezner (1999), who explained why sanctions are imposed most on the adversaries least likely to concede, and Drezner (2003), who identified the selection effect of threats. The data response was TIES, which records threats as well as impositions (Morgan, Bapat, and Krustev 2009; Morgan, Bapat, and Kobayashi 2014). Early (2015) showed how third-country sanctions busters sustain targets, which helps explain why multilateral sanctions do not always outperform unilateral ones.

On design, Biersteker, Eckert, and Tourinho (2016) evaluate UN targeted sanctions against the purposes of coercing, constraining and signalling, and Peksen (2009) documents the human-rights cost of sanctions to the populations of target states. Farrell and Newman (2019) explain the rise of financial sanctions through the chokepoints of global networks. The GSDB (Felbermayr et al. 2020) and its later releases (Syropoulos et al. 2024) made it possible to study sanctions at the scale of trade data, and Crozet and Hinz (2020) used that kind of data to show that much of the trade cost of the Russia sanctions after 2014 fell on products that were not sanctioned at all — the quantitative trace of overcompliance.

10.14 What the figure will not tell the firm

The practice round rebuilds the GSDB's descriptive picture: the number of active sanctions cases per year by type, and the distribution of outcomes by objective. It will show the rise of the instrument since the end of the Cold War and the changing mix of its types. It will also show the decisions underneath: the end year given to ongoing cases, the double counting of multi-type cases, the first-objective rule, the denominator of the success share.

The question to carry into the round is the one every round ends on: what does this figure not license a multinational to conclude? It does not license a conclusion about the firm's *exposure*. A count of cases naming a country says how often senders have used the instrument against it, not how much of the firm's revenue, supply or payments the measures touch; a single financial sanction on a country's largest banks can matter more to a firm than a dozen travel bans. It does not license a conclusion about the *perimeter*, which is drawn by ownership rules, secondary sanctions and the firm's own banks rather than by the list of cases. It does not license a conclusion about the *pipeline*, because the cases that would matter most next year are threats today, and the GSDB does not record threats. And the share of cases judged successful says nothing about whether a sanction will be lifted soon enough to make waiting the cheaper option.

The figure is the right place to start a sanctions view on a team dashboard, because it shows where the instrument has been used and how its use has changed. It is the wrong place to stop. The firm's own sanctions risk lives in its footprint, its counterparties' owners, its banks, and the countermeasures of the states it operates in — and none of those is in the case list.

10.15 In the practice

> **The target.** The Global Sanctions Data Base's descriptive picture (Felbermayr et al., 2020): **active sanctions cases per year, by type**, from 1950 to today, and the **distribution of outcomes by objective** — rebuilt from Release 5, then confronted with the **threats** in TIES.

Work in your project teams. The last **20 minutes** are the team dashboard skeleton review.

> **Licence.** GSDB data may not be redistributed. Hand in on **ZoneCours only**. Do not publish this workbook to Tableau Public.

10.15.1 1 · Collect — 5 min

Everything should already be in `data/raw/`: the GSDB files as e-mailed, TIES v4.0, both codebooks, and `years.csv` (see `../data/README.md`). Fill the provenance log — for the GSDB, the **release number** and the **date of the e-mail**.

10.15.2 2 · Prepare — 20 min

1. **Connect → Text file / Excel →** the GSDB **case-level** file. Confirm in the codebook, and log, the names used below: `case_id`, `sanctioned_state`, `sanctioning_state`, `begin_year`, `end_year`, the six **type** columns, `objective`, `success`.
2. **Pivot the types.** The six types are separate 0/1 columns (`trade`, `financial`, `travel`, `arms`, `military`, `other`): select them in the grid → right-click → **Pivot**. Rename `Pivot Field Names` → `Type`, `Pivot Field Values` → `Type flag`. Filter later on `Type flag = 1`.
3. **Build the years.** Add `years.csv` to the canvas and **join** it to the GSDB table with two clauses:
   - `Year` >= `begin_year`
   - `Year` <= *join calculation* `IFNULL([end_year], 2025)` (click the right-hand field of a clause → *Create Join Calculation*). Each case now has one row per active year — and, after the pivot, one per type. (In the R5 file ongoing cases already carry `end_year` = 2025; the `IFNULL` is a guard for releases that leave it empty.)
4. **Check the grain.** Note the row count before and after. Write in the log what one row now is.
5. **Outcomes.** Outcomes are **not** indicator columns: `success` is one text field holding one code per objective, in the order of `objective` (e.g. `objective` = `human_rights,democracy`, `success` = `failed,ongoing`). Take the first objective and its outcome:

```
Objective  =  TRIM(SPLIT([objective], ",", 1))
Outcome    =  CASE TRIM(SPLIT([success], ",", 1))
              WHEN "success_total"   THEN "Success"
              WHEN "success_part"    THEN "Partial success"
              WHEN "nego_settlement" THEN "Negotiated settlement"
              WHEN "failed"          THEN "Failure"
              WHEN "ongoing"         THEN "Ongoing"
              ELSE "Not coded" END
```

   Log the decision: cases with several objectives count here under their **first** one only.

10.15.3 3 · Replicate — 25 min

10.15.3.1 Sheet 1 · "Active sanctions cases per year, by type"

1. `Year` → **Columns** (continuous).
2. `COUNTD([case_id])` → **Rows** — *count distinct*, never `SUM`, because the pivot and the scaffold multiply rows.
3. Filter `Type flag = 1`; `Type` → **Colour**; mark type **Area** or **Bar**.
4. Note in a caption: a case with several types is counted once **in each** type, so the stacked total exceeds the number of cases. Then build a version with no colour to show the true total.
5. Annotate 1990–91, 2014 and 2022. What explains the shape?

10.15.3.2 Sheet 2 · "How do cases end, by objective?"

1. Use the **case-level** grain: duplicate the data source **without** the year scaffold, or use `{FIXED [case_id] : MIN([Outcome])}`.
2. `Objective` → **Rows**, `COUNTD([case_id])` → **Columns**, `Outcome` → **Colour**.
3. *Quick Table Calculation* → *Percent of Total*; *Edit Table Calculation* → *Compute using* `Outcome` so each objective sums to 100%.
4. Exclude ongoing cases from the denominator in a second version. How much does the success share move? (That is a coding decision too.)

10.15.3.3 Sheet 3 · "One target's history" — a Gantt chart

1. Filter `sanctioned_state` to **one** target (Iran, Russia, Cuba, North Korea…) — with a calculated filter `CONTAINS([sanctioned_state], "Iran")`, because some cases list several targets.
2. `begin_year` → **Columns** (continuous); `case_id` (or `sanctioning_state` + `case_id`) → **Rows**.
3. Mark type **Gantt Bar**; calculated field `Duration` = `IFNULL([end_year], 2025) - [begin_year]` → **Size**.
4. `Type` → **Colour**. What does the target's history look like — layers added, never removed?

10.15.4 4 · Compare and break — 10 min

1. **Differ.** Put Sheet 1 beside the GSDB paper's figure. Release 5 runs years further and has added and recoded cases: write one difference.
2. **Threats vs impositions.** Open TIES. Using the manual, separate **threatened-only** cases (`imposition` = 0) from **imposed** cases (`imposition` = 1) and count each per `startyear` (1945–2005). Put the counts beside GSDB impositions for the same years. What does the gap say about Drezner's selection effect?
3. **Extend — the firm.** For your team's firm, list its top five countries. How many active GSDB cases name each as the sanctioned state in the latest year?

10.15.5 5 · The catch — 10 min

Original	Ours	The catch
The GSDB paper's figure	Your Sheet 1 (and Sheet 2)	One sentence: what these charts do **not** license a multinational to conclude about *its* sanctions exposure

A member drawn at random presents in two minutes.

10.15.6 6 · Team dashboard skeleton review — 20 min

Have your team-project workbook open. The instructor checks: four datasets from three indicator families; the composite index; at least one channel view; the watchlist with thresholds. You leave with three written comments.

10.15.7 Hand in

On **ZoneCours only**: `.twbx` and provenance log. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

10.15.8 What loses credit

- `SUM` of anything after the scaffold join
- Stacked type totals presented as the number of cases
- Success shares without saying whether ongoing cases are in the denominator
- The GSDB workbook published on Tableau Public

Chapter 11

Space Geopolitics: The Orbital Commons and the New Space Race

Session 11 · The orbital commons and the new space race Pre-session slides · Lecture slides · Practice slides · Pre-session page · Session folder · Data instructions

> **Satellites now carry the position, timing, connectivity and imagery a multinational runs on. When orbit is crowded, contested and dominated by one operator, who holds the risk — and what can the firm do about it?**

Big works	ESA Space Debris Office. (2026). *ESA's annual space environment report* (Issue 10.0, 1 May 2026; GEN-DB-LOG-00288-OPS-SD). European Space Agency. PDF · Samson, V., & Brett, K. (Eds.). (2026). *Global counterspace capabilities: An open source assessment*. Secure World Foundation. swfound.org
Companion readings	Pekkanen, S. M. (2019). Governing the new space race. *AJIL Unbound*, 113, 92–97. doi:10.1017/aju.2019.16 · The Outer Space Treaty (1967) · Farrell & Newman (2019), from Session 08
Target figure	ESA's **evolution of the number of objects in geocentric orbit by object class**, 1957–2025 (Fig. 2.1), with the 2007 Chinese ASAT test, the 2009 Iridium–Cosmos collision and the 2021 Russian ASAT test annotated — and the **concentration** of who puts satellites up, then and now
Data you collect	CelesTrak **SATCAT** (`satcat.csv`, the public satellite catalogue); optionally Jonathan McDowell's **GCAT** launch log
Tableau skill	**Running sum** (table calculation) on a pivoted event table · **Herfindahl–Hirschman index with LOD expressions** · **dashboard actions** across data sources
Course themes	Inter-state anarchy and the balance of the major powers; cyber geopolitics and digital threats in the decision-making architecture of multinational firms

11.1 Infrastructure overhead

Few multinationals own a satellite. Almost all of them depend on one, and most do not know which. A container line tracks its ships by satellite and talks to their crews by satellite when they are out of range of land. A mining company runs its remote sites over satellite links. An agribusiness forecasts harvests from imagery, an insurer settles flood claims from it, and a buyer of commodities checks the stockpiles of its suppliers with it. Above all, almost every firm depends on satellites for *time*. Telecommunications networks, electricity grids, data centres and trading venues synchronise their clocks to the

signals of global navigation satellite systems, because those signals carry time from atomic clocks in orbit to any receiver on the ground, cheaply and everywhere.

Three services carry most of this dependence. **Positioning, navigation and timing** comes from four global systems, each run by a state: GPS by the United States, Galileo by the European Union, BeiDou by China and GLONASS by Russia. **Connectivity** comes from operators in geostationary orbit, 36,000 kilometres up, and since 2019 from constellations of thousands of satellites in low Earth orbit, a few hundred kilometres up. **Earth observation** comes from state agencies and from a growing number of commercial imaging firms.

Sessions 08 and 10 showed how networks become instruments of power: a network's hubs are under someone's jurisdiction, and whoever has jurisdiction can watch the traffic or cut it off. This session's network is physically above the firm. Its hubs are a handful of states and, increasingly, a handful of companies, and its environment is degrading in a way no terrestrial network's does — through debris that stays where it was created for years or centuries. The scale of the change is recent. The European Space Agency's 2026 environment report notes that the exponential growth in the number of new payloads continued in 2025, driven by a higher launch frequency, rideshare launches and the deployment of large constellations by commercial operators (ESA Space Debris Office 2026).

The session's question is who holds the risk when orbit is crowded, contested and dominated by one operator — and what the firm can do about it. The answer the chapter builds is that the firm holds more of it than it thinks, that most of it arrives through the reversible and unspectacular threats rather than the destructive ones, and that dependence on a satellite service is, by the construction of space law, dependence on a state.

11.2 A domain written for two superpowers

The law of outer space was written at speed, during the Cold War, by and for two states that were then the only ones able to get there. Sputnik's launch in October 1957 settled one question before anyone had negotiated it: the Soviet satellite passed over every country on Earth, and no state protested that its airspace had been violated. The principle that satellites may overfly territory — which the United States, planning reconnaissance satellites of its own, had every interest in establishing — became the foundation of the domain.

The rest of the foundation was laid in the Outer Space Treaty, opened for signature in January 1967 and in force by October of that year (United Nations 1967). It remains the constitution of the domain, and the articles that matter to a firm can be read in two pages. Article II declares that outer space "is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means": no one can own an orbit. Article IV forbids placing nuclear weapons or other weapons of mass destruction in orbit, and reserves the Moon and other celestial bodies for peaceful purposes — but it says nothing to prohibit conventional weapons in orbit, or the testing of weapons that destroy satellites from the ground. Article VI makes states internationally responsible for national activities in space "whether such activities are carried on by governmental agencies or by non-governmental entities", and requires those activities to be authorised and continually supervised by "the appropriate State". Article VII makes the launching state liable for damage caused by its objects. Article IX requires states to act with "due regard" to the interests of others and to consult before activities that could cause harmful interference — the closest the Treaty comes to traffic rules, without anyone to enforce them.

Orbital commons — Earth orbit treated as a resource that no one may appropriate (Outer Space Treaty, Article II) and from whose use no one can easily be excluded, but whose use by one actor degrades it for others — through congestion of orbits and spectrum and through debris. The commons is legal in origin and physical in its consequences.

Article IV's silence on conventional weapons was not an accident; neither superpower wanted to give up the option. Its prohibition on nuclear weapons in orbit, by contrast, had been learned the hard way.

From the archive · Starfish Prime, 1962

On 9 July 1962 the United States detonated a thermonuclear warhead of about 1.4 megatons some 400 kilometres above Johnston Island in the Pacific. The test, Starfish Prime, was intended to study the effects of a nuclear explosion at high altitude. Its most lasting effect was one its planners had underestimated: electrons from the explosion were trapped in the Earth's magnetic field, creating an artificial radiation belt that persisted for years.

The satellites then in low orbit were few, and several were damaged. Telstar 1, the first commercial communications satellite — built by AT&T's Bell Laboratories and launched the day after the test — suffered radiation damage to its electronics and

failed within months; Britain's first satellite, Ariel 1, was also damaged. A single test by one state had degraded an orbital environment shared by everyone, and the first commercial satellite was among the casualties.

The Partial Test Ban Treaty of 1963 prohibited nuclear tests in outer space, and the Outer Space Treaty four years later prohibited nuclear weapons in orbit. Neither prohibited the next form of the same problem: a conventional weapon that destroys a satellite and fills its orbit with fragments.

11.3 Liability, registration and the right to transmit

The Outer Space Treaty is the constitution; three further layers turn it into rules a firm encounters.

The first is liability and registration. The Liability Convention of 1972 elaborated Article VII: a launching state is absolutely liable for damage its space objects cause on the surface of the Earth or to aircraft in flight, and liable for fault for damage elsewhere in space. The Registration Convention of 1975 requires launching states to register their objects with the United Nations. The two together mean that every object in orbit has, in law, a state that answers for it — long after it has stopped working.

From the archive · Cosmos 954, 1978

On 24 January 1978 a Soviet ocean-surveillance satellite, Cosmos 954, re-entered the atmosphere over the Northwest Territories of Canada. It was powered by a small nuclear reactor, and its re-entry scattered radioactive debris over a wide stretch of the Canadian north. Canada and the United States mounted a search and clean-up operation, Operation Morning Light, that lasted for months.

Canada presented a claim to the Soviet Union under the Liability Convention and general principles of international law, for the cost of locating, recovering and cleaning up the debris. The claim was settled in 1981, by protocol, for three million Canadian dollars — about half the amount claimed. It remains the only claim ever brought under the Convention.

The case shows both the strength and the limits of the regime. Liability attaches to a state, and a state paid. But it took three years, the settlement was negotiated rather than adjudicated, and the claimant was a state. A firm whose satellite is destroyed by a fragment from another state's anti-satellite test has no standing under the Convention: it must persuade its own state to claim, against a launching state that may deny responsibility, for damage whose cause it may be unable to prove.

The second layer is spectrum. A satellite is useless without radio frequencies to command it and to carry its service, and frequencies and orbital positions are coordinated by the International Telecommunication Union through its Radio Regulations. An operator's state files a satellite network with the ITU and must coordinate it with networks filed earlier, so priority goes, broadly, to the first to file. The incentive is to file early and large — sometimes for "paper satellites" that may never fly — and the result is that orbital slots and frequencies are a scarce asset allocated by a procedure most firms never see.

The third layer is national licensing. Article VI's "authorization and continuing supervision" becomes, in practice, a set of permits: to launch, to use spectrum, to operate a remote-sensing system, to export a satellite or its components, to land a service in a national market. This is where a firm meets space law directly, and where the balance of power enters most visibly — through the export-control lists discussed below and through decisions about which foreign constellations may sell service in which countries.

11.4 The physics of a commons

Law makes orbit a commons in one sense: no one may own it. Physics makes it a commons in the stricter sense economists use. A common-pool resource is one from which exclusion is difficult and whose use by one actor subtracts from what is left for others. Garrett Hardin's parable of the herdsmen on a common pasture, each adding animals because he captures the whole gain while bearing only a share of the cost, is the classic statement of why such resources tend to be overused (Hardin 1968). Elinor Ostrom showed that real communities often escape the tragedy through rules they design and enforce themselves — but under conditions of clear boundaries, monitoring and graduated sanctions that Earth orbit does not meet (Ostrom 1990). Brian Weeden and Tiffany Chow applied her framework to space and found most of its design principles missing (Weeden and Chow 2012).

What turns congestion into a tragedy in orbit is a mechanism identified by Donald Kessler and Burton Cour-Palais in 1978 (Kessler and Cour-Palais 1978). Objects in the same orbital shell occasionally collide; each collision produces fragments; the fragments raise the density of objects in the shell and therefore the rate of further collisions. Below a critical density, the atmosphere drags enough debris down to keep the population in check. Above it, collisions generate fragments faster than drag removes them, and the debris population grows on its own.

Kessler syndrome — the self-sustaining growth of the debris population in an orbital shell once the density of objects passes a threshold at which collisions create fragments faster than atmospheric drag removes them. Past the threshold, the population grows even if nobody launches again.

Altitude matters enormously, because the drag that cleans orbits weakens with height. Debris a few hundred kilometres up re-enters within years; debris above about 800 kilometres can remain for decades or centuries. A fragmentation in a high, busy shell is therefore a long-lived degradation of an asset shared by every operator in it. Two decades ago, simulations by NASA's debris specialists already suggested that the low-orbit debris population would keep growing through collisions even if all launches stopped (Liou and Johnson 2006).

Economists have since modelled the incentives. Nodir Adilov, Peter Alexander and Brendan Cunningham show that, because each operator ignores the collision risk its satellites and debris impose on others, open access leads to more launches and more debris than would be efficient (Adilov, Alexander, and Cunningham 2015). Akhil Rao, Matthew Burgess and Daniel Kaffine estimate that an internationally harmonised fee on each satellite in orbit — making operators pay for the risk they impose — would greatly increase the long-run value of the space industry (Rao, Burgess, and Kaffine 2020). No such fee exists. Mitigation guidelines adopted through the UN and by space agencies ask operators to limit debris and to remove satellites from busy orbits at the end of their lives; compliance is voluntary and uneven.

For the firm: an orbit is a shared infrastructure whose quality is set by its least careful user. A destructive test by one state, or a rocket stage that explodes years after launch, raises the collision risk — and the insurance price — for every operator in that shell, and therefore for every customer of those operators.

11.5 Three readings of the new space race

Who will govern this commons, and in whose interest? International relations offers three readings of what has come to be called the new space race, and they do not exclude one another.

The first is classical geopolitics carried upwards. Everett Dolman's *Astropolitik* takes Mackinder's logic — whoever controls the pivot controls the world — and applies it to orbit: whoever controls low Earth orbit controls access to the rest of space, and through it can dominate the Earth below (Dolman 2002). On this realist reading the United States should seize and hold that control while it can, and space is a territory to dominate, its orbits and transfer routes the chokepoints of a new strategic geography. It is Session 03's anarchy, with orbits in place of straits.

The second reading is institutional and commercial. Saadia Pekkanen describes the new space race as driven by three intertwined trends (Pekkanen 2019). *Democratisation*: many more states, and many non-state actors, now operate in space. *Commercialisation*: firms no longer only supply governments but strike out on their own, with their own capital and customers. *Militarisation*: states increasingly treat space as a domain of warfare. Binding them together is the fact that about 95 percent of space technologies, on her account, are dual-use. With treaty-making in decline, the domain is governed less by new international law than by national legislation, contracts, and a few old principles. On this reading space is a market, and the firms in it are part of the security problem.

The third reading puts the Earth back at the centre. Bleddyn Bowen argues that spacepower matters for its effects on Earth: orbit is a "cosmic coastline" from which states support — and threaten — what happens on land, at sea and in the air (Bowen 2020). Satellites are valuable because they make terrestrial forces see, communicate and navigate; that is also why they are targets when terrestrial power is contested.

Read together, the three explain why the same constellation can be a consumer broadband business, a military communications network and an instrument of foreign policy all at once. Pekkanen supplies the dual use; Dolman and Bowen supply the strategic stakes. For a multinational the practical conclusion is the same under all three: a commercial satellite service is never only commercial.

11.6 Counterspace: what states can do to satellites

If satellites are targets, the question is what can be done to them, by whom, and with what consequences for the firms that use them. The Secure World Foundation's annual open-source assessment tracks the capabilities of thirteen countries to deceive, disrupt, deny, degrade or destroy space systems, in five categories (Samson and Brett 2026).

Counterspace capabilities — the means of deceiving, disrupting, denying, degrading or destroying space systems: direct-ascent missiles, co-orbital systems, electronic warfare, directed energy and cyber operations. They differ in whether their effects are reversible and in who bears them first — the satellite's owner, every operator in the orbit, or the users on the ground.

Direct-ascent weapons are missiles launched from the Earth to destroy a satellite. Their effects are irreversible and, at most altitudes, create debris that threatens every operator in the shell. **Co-orbital** systems are satellites that approach another to inspect it, interfere with it or attack it. **Electronic warfare** jams or spoofs signals — the uplink to a satellite, its downlink to users, or navigation signals on the ground — and its effects end when the jamming stops. **Directed-energy** weapons use lasers or microwaves to dazzle or damage sensors. **Cyber** operations attack satellites, ground stations or user terminals, and can disable a service without touching a satellite.

The assessment's central finding is the one that matters most for business: many states have researched and tested destructive capabilities, but only non-destructive ones are actively used against satellites in current military operations. Four states have destroyed one of their own satellites with a missile launched from the ground: the United States in 1985 — and, in 2008, by intercepting a failing satellite at low altitude — China in 2007, India in 2019 and Russia in 2021. In April 2022 the United States declared that it would not conduct destructive direct-ascent tests, and in December 2022 the UN General Assembly called on all states to make the same commitment.

What firms have actually suffered is the reversible kind. On 24 February 2022, the day Russia's full-scale invasion of Ukraine began, a cyberattack on the ground segment of Viasat's KA-SAT network disabled thousands of user modems in Ukraine and across Europe, including those used for the remote monitoring of thousands of wind turbines in Germany. The European Union, the United Kingdom and the United States attributed the attack to Russia. The satellite was untouched; the service failed for its customers, many of them nowhere near the war. Since then, the jamming and spoofing of navigation signals around conflict zones has become routine. In 2024 Finnair suspended its flights to Tartu, in Estonia, for several weeks because interference made the airport's satellite-based approach procedures unusable.

The lesson for an analyst. Destructive tests make headlines and debris. Jamming, spoofing and cyberattacks on ground segments make *losses* — and the losses fall on the customers, not on the satellite. A firm's counterspace exposure is measured on the ground, in the regions where its ships, aircraft and sites depend on signals from orbit.

11.7 The debris record, and how it is counted

The evidence on the state of the orbital environment comes, for this session, from ESA's Space Debris Office, which has published an annual *Space Environment Report* since 2017. The 2026 edition, with data to the end of 2025, is the session's big work (ESA Space Debris Office 2026), and its method deserves reading before its figures.

ESA counts two kinds of objects. *Catalogued* objects are those tracked and published in catalogues: the US public catalogue, the Russian Vimpel catalogue and, for the early decades, the United Kingdom's historical RAE tables. *Asserted* objects are known to exist by design — a satellite released from a dispenser, a component separated as planned — even if not yet catalogued. The two are cross-correlated in ESA's DISCOS database and classified into object classes. Objects too small or too erratic to be tracked are outside the report's scope.

On that basis, the report counts 44,964 objects in Earth orbit at the end of 2025, of which 16,946 are payloads, and 25,060 objects in low Earth orbit (ESA Space Debris Office 2026). It observes that, ever since the start of the space age, there has been more debris in orbit than operational satellites. And its most important finding for anyone who thinks the debris problem is the weapons tests is this: *fragmentations other than collisions* — explosions, mostly of old rocket stages and batteries — are the dominant source of debris. Non-deliberate fragmentations have averaged 9.8 a year over the last two decades, and 2024 alone added more than 3,000 catalogued fragments, mostly from propulsion-related break-ups.

The report's Figure 2.1, the number of objects in geocentric orbit by object class since 1957, is the practice round's target. It shows the cumulative result of seven decades of launches, explosions, collisions and re-entries, and three events stand out on it. On 11 January 2007 China destroyed its Fengyun-1C weather satellite with a ground-launched missile, at about 850 kilometres; the test remains the largest single source of catalogued debris in the public catalogue, and most of its thousands of fragments are still in orbit. On 10 February 2009 the active Iridium 33 and the defunct Cosmos 2251 collided at about 780 kilometres — the first accidental collision between two intact satellites, Kessler's mechanism observed. And on 15 November 2021 Russia destroyed Cosmos 1408 at roughly 470 to 490 kilometres, producing some 1,800 catalogued fragments; at that altitude almost all re-entered within three years, and ESA attributes the 2022 spike in re-entering debris to the test.

The contrast between the first and third events is a lesson in altitude. Both were deliberate destructions of a satellite by its own state; one polluted a high orbit for decades, the other a low orbit for a few years. A count of objects treats them alike.

What the figure does not measure. It counts objects, not risk: a ten-centimetre fragment and a five-tonne rocket stage each count once, and the objects too small to catalogue — the majority of the fragments able to disable a satellite — do not appear at all. It is drawn from catalogues that themselves improve: when sensors get better, the count rises with nothing new having happened in orbit. And it shows the *environment*, not the exposure of any operator or customer, which depends on which shells their satellites use.

11.8 One operator

The new space race has a feature the Cold War one did not. In the first years of the space age, concentration was extreme because only two states could launch anything; it fell for decades as Europe, Japan, China, India and many others acquired the capability, and as commercial operators multiplied. It has risen again — not because states have dropped out, but because one company has launched more satellites than everyone else together. SpaceX's Starlink, first launched in May 2019, grew in six years into the majority of the world's operating satellites. The practice round measures how large a majority, and finds that the concentration of who launches payloads has returned to Cold War levels, with a firm rather than a superpower at the top.

Concentration needs a measure, and the course uses the one antitrust authorities use. Its origins are in geopolitics: Albert Hirschman devised an index of the concentration of a country's trade in 1945, to show how Nazi Germany had used the structure of trade to make its smaller partners dependent on it (Hirschman 1945). Orris Herfindahl reinvented it a few years later for industrial concentration, and it carries both their names.

Herfindahl–Hirschman index (HHI) — the sum of the squared market shares of all participants, with shares in percent, so that it runs from near 0 (many small participants) to 10,000 (one participant with everything). It weights large shares heavily: one participant with 75 percent contributes 5,625 on its own. US merger guidelines treat markets above 2,500 (U.S. Department of Justice and Federal Trade Commission 2010) or, in the 2023 revision, above 1,800 (U.S. Department of Justice and Federal Trade Commission 2023) as highly concentrated.

Concentration matters to a firm because of what it implies about dependence, and Starlink in Ukraine shows what that dependence means. Within days of the 2022 invasion, Starlink terminals became a backbone of Ukraine's military and civil communications. The company's decisions about where its service would work — including, by its owner's account, a refusal to enable it for an attack near Crimea in 2022 — became questions of war policy, and governments moved to contract for the service directly.

Farrell and Newman's framework, met in Session 08, applies without modification (Farrell and Newman 2019). A constellation is a hub. Its operator sees who connects and from where — a panopticon — and can switch off a user, a region or a country — a chokepoint. Weaponisation requires two conditions: that the network be concentrated, and that a state can direct the firm that runs the hub. Article VI of the Outer Space Treaty guarantees the second, because every operator has a licensing state that is internationally responsible for its activities and must supervise them. A firm that depends on one operator depends on that operator's state as well.

States are responding as Session 03 would predict: by building their own. The European Union's IRIS² secure-connectivity programme, established by Regulation (EU) 2023/588, China's Qianfan and Guowang constellations, and the many national launch programmes are all, in part, answers to dependence on one American firm. Export controls draw the same line from

the other side. Spacecraft and their components sit either on the US Munitions List, under Category XV of the International Traffic in Arms Regulations, or — for most commercial satellites since 2014 — under the Export Administration Regulations, in the "9x515" entries; a non-US manufacturer whose satellite contains a controlled US component needs US permission to sell it to some customers. The balance of power reaches into the bill of materials.

Navigation is the older case of the same pattern, and the one on which almost every firm already depends. GPS was built by the US military. Its opening to the world followed a tragedy: after the Soviet Union shot down Korean Air Lines Flight 007, which had strayed into Soviet airspace, in September 1983, the United States announced that GPS would be made available for civilian use once it was operational. For years the civilian signal was deliberately degraded — "Selective Availability" — until the degradation was switched off in May 2000. The service on which the world's networks and logistics came to rely was, and remains, a gift that one state chose to give and could in principle withdraw or degrade, regionally or globally. Europe's decision to build Galileo, and China's to build BeiDou, were in large part refusals to accept that dependence. For a firm, the existence of four global systems is a genuine diversification — but only if its receivers use more than one, and only against the failure or withdrawal of a system, not against a jammer in the next port.

11.9 Through which channels, and what the firm can do

Space risk reaches the profit and loss account through the same five channels as every other risk in the course, and naming the channel is the first step towards a decision.

Through **market access**, export controls and licensing decide whether a satellite, a terminal or a service can be sold or switched on in a given market, and spectrum and landing rights decide whether a constellation may serve a country at all. Through **inputs**, a firm that buys connectivity for its ships or mines from one constellation, or that times its networks from one state's navigation system, has a single point of failure with a sovereign behind it. Through **finance and payments**, trading venues and payment systems time-stamp transactions from navigation signals, and the insurance capacity for space assets — a thin market — shrinks after large losses, so that a fragmentation event can double a premium. Through **assets and people**, ships and aircraft operating in jammed or spoofed regions are rerouted, delayed, or misreport their positions, with consequences for safety and for sanctions compliance alike. Through **data and operations**, a cyberattack on a ground segment cuts the firm's links, as the KA-SAT modems showed, and the loss of an imagery feed leaves a crop forecast or a supply-chain monitor blind.

A dashboard view of space risk should therefore track four things, each with a threshold that triggers a named decision. *Dependence*: the share of a critical service — connectivity, imagery, timing — supplied by one operator or one state's system; above a set share, say half from one operator, the decision is to contract a second. *Interference*: official warnings of navigation-signal interference, which aviation authorities publish, in the regions where the firm's ships, aircraft or sites operate. *Environment*: destructive tests and fragmentation events in the orbital shells used by the firm's providers, and the yearly indicators in ESA's report. *Rules*: changes in export-control lists, licensing conditions and ITU filings that affect the firm's suppliers or markets.

The strategies available each have a cost and a limit. A firm can buy **multi-orbit, multi-operator** connectivity — geostationary and low-orbit links from two providers — at the price of integration, but the second operator may share the first one's licensing state. It can use **multi-constellation receivers** that combine GPS, Galileo and others, which is cheap and protects against the failure of one system, but a local jammer can still blind them all. It can invest in **terrestrial back-ups** — holdover clocks and time delivered by fibre, cellular and fibre links — which requires capital and is unavailable at sea or at remote sites. It can buy **insurance**, in a thin market whose capacity shrinks after large losses. And it can practise **compliance by design**, classifying every space-related component it buys or sells under the relevant export-control regimes, at the cost of staff and delay, knowing that the rules change with the balance of power.

The question the lecture puts to the room ties these together. A shipping subsidiary's only broadband at sea is one low-orbit constellation; the operator's government has been asked by an ally to switch the service off over a strait the firm's ships use. What does the firm do this quarter? The answer turns on the distinction between dependence on a *service* and dependence on a *state*. A second operator under the same licensing state diversifies the first and not the second.

11.10 From a catalogue to a stock

The practice round rebuilds ESA's figure from the public catalogue maintained by CelesTrak from US Space Force data, and the construction is a lesson in how a concept — the number of objects in orbit — becomes data.

The catalogue does not record a stock. It records objects: one row for each catalogued object over its whole life, with a launch date and, if it has re-entered, a decay date. There is no "year" in the file. An object is in orbit in a given year if it was launched by then and had not decayed by then, so the stock in each year is the running sum of launches minus the running sum of decays — a stock rebuilt from two flows. The construction is general: much of what a risk analyst needs to know is a stock that no one publishes, and has to be rebuilt from the events that change it.

The construction exposes the catalogue's decisions. Debris carries the launch date of its parent object, not the date of the event that created it: the fragments of China's 2007 test are dated to the launch of Fengyun-1C in 1999, so a debris line built on launch dates moves in the wrong year. "Not decayed" is not the same as "in Earth orbit", because some objects orbit the Sun, the Moon or a Lagrange point. The owner field records a country, organisation or company code, not the operator: every Starlink satellite is simply American, and the former Soviet Union and Russia share one code. Operational status describes today only, so the file cannot say how many satellites were active in any past year. And the public catalogue is not the environment: it omits classified objects and everything too small to track, which is one reason ESA's totals, drawn from more catalogues and asserted objects, are higher.

Concentration is built from the same file by computing each owner's share of the payloads launched in a year and summing the squares. Here the owner-code problem bites twice: concentration by *state* hides the fact that one company dominates the American share, and concentration by *operator* must be reconstructed from object names. Each choice yields a different index, and each answers a different question — about the balance of power among states, or about the dependence of customers on firms.

11.11 A short review of the literature

The legal foundation is the Outer Space Treaty (United Nations 1967), whose Articles II, VI and VII make orbit unappropriable, place every private operator under a responsible state, and attach liability to the launching state. The governance problem it leaves open is that of a commons. Hardin (1968) states why open-access resources are overused and Ostrom (1990) the conditions under which communities escape the tragedy; Weeden and Chow (2012) apply Ostrom's design principles to orbit and find most of them absent. The physical mechanism is Kessler and Cour-Palais (1978), with Liou and Johnson (2006) showing by simulation that the low-orbit debris population would grow even without new launches. The economics of the externality is set out by Adilov, Alexander, and Cunningham (2015), and Rao, Burgess, and Kaffine (2020) estimate the gains from orbital-use fees.

The strategic literature divides along the lines of the session's three readings. Dolman (2002) carries classical geopolitics into orbit; Bowen (2020) argues that spacepower matters for its terrestrial effects; Pekkanen (2019) describes the democratisation, commercialisation and militarisation of a domain in which most technology is dual-use and treaty-making has stalled. The empirical record of threats is the Secure World Foundation's annual assessment (Samson and Brett 2026), and that of the environment ESA's annual report (ESA Space Debris Office 2026).

For the firm, the bridge is Farrell and Newman (2019): a concentrated constellation under a state's jurisdiction meets both conditions for weaponised interdependence. The measure of concentration is Hirschman (1945)'s index, conceived to analyse how trade structures create political dependence, and its antitrust thresholds are those of the US merger guidelines (U.S. Department of Justice and Federal Trade Commission 2010, 2023).

11.12 What the figure will not tell the firm

The practice round rebuilds ESA's count of objects in orbit by class, annotated with the 2007 test, the 2009 collision and the 2021 test, and adds the concentration of who launches payloads, from Sputnik to today. It will show the steady accumulation of debris, the steps that the three events cut into the record, the explosion of payloads since the large constellations began,

and a concentration index that falls for decades and then climbs back. It will also show the decisions: the dating of debris, the filtering of objects that do not orbit the Earth, the reading of owner codes, the definition of an active satellite.

The question to carry into the round is what this figure does not license a multinational to conclude. A rising count of objects does not tell a firm that its services are at greater risk, because risk depends on the size of objects, the shells its providers use, and the manoeuvring capability of their satellites — none of which the count contains. The absence of a spike does not mean the absence of a threat: the counterspace effects firms have actually suffered — jamming, spoofing, cyberattacks on ground segments — leave no trace in a catalogue of objects. And concentration by owner code is not the firm's dependence. A firm may buy its connectivity from a small operator in an unconcentrated market and still depend on one state's navigation system for its time, or buy from the dominant operator through a reseller and not know it.

The figure is the right place to begin a space-infrastructure view on a team dashboard, because it shows the condition of the commons and the shape of the new space race. The firm's own exposure begins elsewhere: with the list of satellite services it buys, the operators and states behind each, and the regions where its ships, aircraft and sites depend on signals that someone else can switch off.

11.13 In the practice

> **The target.** ESA's *Space Environment Report 2026*, Figure 2.1: the **number of objects in geocentric orbit by object class**, from 1957 to the end of 2025, with the 2007 Chinese ASAT test, the 2009 Iridium–Cosmos collision and the 2021 Russian ASAT test annotated — rebuilt from the public catalogue. You add the **concentration** of who puts payloads into orbit, year by year, and who operates the active satellites today — and wire the views into **one dashboard with an action**.

The last practice round of the term. The last twenty minutes of Session 10 reviewed your team dashboard's skeleton. Most skeletons still lack a space-infrastructure channel and any interaction between views. Leave today with both: the dependence view you build is designed to be dropped into your team dashboard, for the satellite services your own firm relies on.

Work in your project teams. Everyone builds the workbook; the team hands in one.

11.13.1 1 · Collect — 15 min

You arrive with `satcat.csv` (one download per team, pre-session). Use these fifteen minutes to **check**, not to download again:

1. **The snapshot.** What is the latest `LAUNCH_DATE` in the file? Log it with your download date: the catalogue changes every day.
2. **The codes.** Open the format documentation and the status codes. Which `OBJECT_TYPE` values are present? Which `OPS_STATUS_CODE` values count as **active**? What does a blank status mean?
3. **The trap you found before class.** Filter `OBJECT_NAME` = `FENGYUN 1C DEB`. How many rows? Which `LAUNCH_DATE`? Write the gap to 11 January 2007 in your log.

Provenance log — one line per file, as always:

file	source	url	version / release	accessed	licence	notes
satcat.csv	CelesTrak / US Space Force public catalogue	celestrak.org/pub/satcat.csv	*latest; LAUNCH_DATE; Last-Modified*	*date*	free (usage policy)	*debris dated by parent launch*

file	source	url	version / release	accessed	licence	notes
ESA report	ESA Space Debris Office	sdo.esoc.esa.int/publications/	Issue 10.0, *May 2026*	*date*	public	*the original: Fig. 2.1, Table 3.1*
launch.tsv *(optional)*	J. C. McDowell, GCAT	planet4589.org/space/gcat	*"Updated" line*	*date*	free	

11.13.2 2 · Prepare — 15 min

You build **two data sources from the same file**: one reshaped into events (for the stock of objects in orbit), one left as it is (for concentration).

11.13.2.1 Data source A · "SATCAT events"

1. *Connect* → *Text file* → `satcat.csv`. Check the types: `LAUNCH_DATE` and `DECAY_DATE` must be **dates** (click the type icon above the column if they arrive as strings), `NORAD_CAT_ID` a number.
2. **Data source filter** (top right of the data source page → *Filters* → *Add*): `ORBIT_CENTER` = `EA`. Objects around the Sun or the Moon never decay and would stay "in orbit" for ever.
3. **Pivot the two dates.** Select `LAUNCH_DATE` and `DECAY_DATE` in the grid → right-click → **Pivot**. Rename `Pivot Field Names` → `Event` and `Pivot Field Values` → `Event date`. Each object now has two rows: its launch and its decay (empty if it is still in space).
4. Calculated fields:

```
Change  =  IF [Event] = "LAUNCH_DATE" THEN 1 ELSE -1 END
Year    =  YEAR([Event date])
```

5. **Check the grain.** Note the row count before and after the pivot, then exclude the null `Event date` rows (decays that have not happened). What is one row now? Write it in the log.

11.13.2.2 Data source B · "SATCAT objects"

6. *Data* → *New Data Source* → the same `satcat.csv`, **not** pivoted. Data source filter `OBJECT_TYPE` = `PAY` — a **data source** filter, not a sheet filter (step 3 explains why).
7. Calculated fields:

```
Launch year  =  YEAR([LAUNCH_DATE])
Power        =  CASE [OWNER] WHEN "US" THEN "United States"
                             WHEN "PRC" THEN "China"
                             WHEN "CIS" THEN "USSR / Russia"
                             ELSE "Rest of the world" END
Active       =  [OPS_STATUS_CODE] = "+" OR [OPS_STATUS_CODE] = "P" OR [OPS_STATUS_CODE] = "B"
                OR [OPS_STATUS_CODE] = "S" OR [OPS_STATUS_CODE] = "X"
Operator     =  IF STARTSWITH([OBJECT_NAME], "STARLINK") THEN "Starlink (SpaceX)"
                ELSEIF STARTSWITH([OBJECT_NAME], "ONEWEB") THEN "OneWeb"
                ELSEIF STARTSWITH([OBJECT_NAME], "KUIPER") THEN "Kuiper (Amazon)"
                ELSEIF STARTSWITH([OBJECT_NAME], "QIANFAN") THEN "Qianfan"
```

```
                ELSEIF STARTSWITH([OBJECT_NAME], "HULIANWANG") THEN "Guowang"
                ELSE "Other: " + [OWNER] END
```

`CIS` is CelesTrak's code for the USSR **and** Russia; `OWNER` is a registry-style code, not the operator — log both decisions.

11.13.3 3 · Replicate — 35 min

11.13.3.1 Sheet 1 · "Objects in Earth orbit, by type" — the target figure (data source A)

1. `Year` → **Columns**: right-click → *Convert to Dimension*, keep it **continuous** (green).
2. `SUM([Change])` → **Rows**. You see net additions per year — a jagged line.
3. Right-click the pill → *Quick Table Calculation* → **Running Total**. Now you see the **stock**.
4. `OBJECT_TYPE` → **Colour**. *Edit Table Calculation* → *Compute using* → *Specific Dimensions* → tick `Year` only, so the running total restarts for each type. Mark type **Area** (stacked).
5. Filter `Year` ≤ 2025 to match ESA's end point. **Never filter out early years** with a normal filter: it runs before the table calculation and the running total would restart from zero. (To hide early years, use a table-calculation filter such as `LOOKUP(MIN([Year]), 0)`.)
6. **Annotate** 2007, 2009 and 2021 (*right-click* → *Annotate* → *Point*). Look at the debris band. Where are the steps?
7. Title stating the finding; caption with source, snapshot date and "objects dated by launch date".

11.13.3.2 Sheet 2 · "Payloads launched per year, by power" (data source B)

8. `Launch year` → Columns (continuous dimension), `CNT([NORAD_CAT_ID])` → Rows, `Power` → Colour, mark type **Bar**. Where does the Cold War end, and where does the new race begin?

11.13.3.3 Sheet 3 · "How concentrated is the new space race?" — the HHI (data source B)

9. Create:

```
Owner share  =  {FIXED [Launch year], [OWNER] : COUNT([NORAD_CAT_ID])}
             / {FIXED [Launch year] : COUNT([NORAD_CAT_ID])}
HHI piece    =  10000 * [Owner share] / {FIXED [Launch year] : COUNT([NORAD_CAT_ID])}
```

   Why the second line: each owner's squared share must be counted **once**, but the owner appears on as many rows as it has payloads. Spreading `share × 10,000` over the year's rows means that summing an owner's rows gives exactly `share² × 10,000`.

10. `Launch year` → Columns, `SUM([HHI piece])` → Rows, mark type **Line**. That sum **is** the HHI. Check: 1957 must give **10,000** (two Soviet payloads). Reference lines at **2,500** and **1,800**, labelled.

11. Why the data source filter in step 6: a `FIXED` expression is computed **before** ordinary filters. Had you filtered `OBJECT_TYPE` on the sheet, the shares would still include debris and rocket bodies. (Adding the filter to **context** also works.)

11.13.3.4 Sheet 4 · "Who launched that year?" (data source B)

12. `OWNER` → Rows, `CNT([NORAD_CAT_ID])` → Columns, sorted descending, `Operator` → Colour; a *Top 10 by count* filter on `OWNER`. This is the **target** of the action.

11.13.3.5 Sheet 5 · "Who operates the active satellites today?" (data source B)

13. Filter `Active` = True and `ORBIT_CENTER` = `EA`. `Operator` → Rows, `CNT([NORAD_CAT_ID])` → Columns, sorted; *Quick Table Calculation → Percent of Total*. Caption: "status as of the snapshot date; blank status excluded".

11.13.3.6 Dashboard · "Space infrastructure exposure"

14. *New Dashboard*. Sheet 1 top left, Sheet 3 top right, Sheet 4 bottom left, Sheet 5 bottom right.
15. **Action 1.** *Dashboard → Actions → Add Action → Filter*. Source: Sheet 1. Target: Sheet 4. Run on: *Select*. Clearing the selection: *Show all values*. Filter: **Selected fields** → source field `Year` → target data source *SATCAT objects*, target field `Launch year`. Because the two sheets use different data sources, map the fields explicitly.
16. **Action 2.** The same, with Sheet 3 as source (same data source as Sheet 4: field `Launch year` → `Launch year`).
17. Click 1965 on Sheet 3, then 2025. Sheet 4 should show who launched that year. If it shows nothing, the field mapping is wrong: fix it, and log it.

11.13.4 4 · Compare and break — 15 min

1. **Differ.** Put Sheet 1 beside ESA's Figure 2.1 and Table 3.1 (end of 2025: 44,964 objects, of which 16,946 payloads). Give your totals by type and **one** reason for each gap — which catalogues ESA adds, what it calls "unidentified", what the public catalogue leaves out.
2. **Break: when does debris appear?** Your debris band does not step up in 2007, 2009 or 2021. Re-date the four big events' debris to the date the event happened. In data source A:

```
Appearance year  =  IF [Event] = "LAUNCH_DATE" AND CONTAINS([OBJECT_NAME], "FENGYUN 1C DEB") THEN 2007
                    ELSEIF [Event] = "LAUNCH_DATE" AND (CONTAINS([OBJECT_NAME], "COSMOS 2251 DEB")
                         OR CONTAINS([OBJECT_NAME], "IRIDIUM 33 DEB")) THEN 2009
                    ELSEIF [Event] = "LAUNCH_DATE" AND CONTAINS([OBJECT_NAME], "COSMOS 1408 DEB") THEN 2021
                    ELSE [Year] END
```

   Duplicate Sheet 1 and replace `Year` by `Appearance year`. How big is each step now? What about the thousands of other fragments whose event you did not re-date?
3. **Break: flag or firm?** On Sheet 5, replace `Operator` by `OWNER`, and compute the HHI of active satellites both ways. Which number describes your firm's **dependence** better?
4. **Extend.** For your **team's firm**, name the satellite services it relies on (GNSS timing, connectivity at sea or in remote sites, imagery). Find their operators in the file — `NAVSTAR` (GPS), `GSAT0…` (Galileo, coded `ESA`), `BEIDOU`, `[GLONASS-M]` — and add the one that matters to Sheet 5. Optional, with GCAT: orbital launch attempts per year (`LaunchCode` starting `O` or `D`) by launch-provider state.

11.13.5 5 · The catch — 10 min

One slide, three panels:

Original	Ours	The catch
ESA's Figure 2.1	Your dashboard	What it does **not** license your firm to conclude

The model catch, if you need one to beat: *a count of objects is not a count of risk — the fragments able to disable a satellite are mostly too small to catalogue, and the public catalogue dates debris by its parent's launch, so the ASAT tests vanish from a chart built naively on it; and concentration by flag hides dependence on one firm.* Beat it — say something specific to your firm.

A member drawn at random presents in two minutes.

11.13.6 Hand in

Upload to ZoneCours before leaving: the `.twbx`, the provenance log, and the slide. Formative: commented, not marked; counts towards participation. (The SATCAT is freely available, so the workbook may also go to Tableau Public — credit CelesTrak in the caption.)

The one rule: the log records at least one place where the data disagreed with the paper — a total that does not reach ESA's, debris dated to the wrong year, an owner code that is not the operator.

11.13.7 Into the team dashboard

Before Session 12, re-point Sheet 5 at **your firm's** satellite services, keep the action, and add the dependence view to your **channel views** with a threshold — for example *more than half of a critical service from one operator → contract a second one*. A board member should be able to click a year and see who controlled access to orbit then, and see in one screen who the firm depends on now.

11.13.8 What loses credit

- A running total that restarts because early years were filtered out
- Objects around the Sun or the Moon counted as "in orbit"
- An HHI that is not 10,000 in 1957 — shares computed after a sheet filter, or squared on every row
- `OWNER` presented as the operator, or blank status counted as inactive
- A dashboard action that silently filters to nothing
- A catch about ESA rather than about your firm

Chapter 12

From Indicators to Decisions

Session 12 · A geopolitical risk assessment for one multinational Presentation-day slides · Session folder

Can you make a board act on your assessment — and say what would change your mind?

12.1 The question the course began with

The course opened with a question: how does a multinational firm know how much geopolitical risk it faces — and what should it do about it? Eleven sessions later the honest answer to the first half is that it never knows, in the sense of holding a number that settles the matter, and that the course's work has been to show why. The answer to the second half is more encouraging. A firm can organise what it does not know so that someone acts on it, at the right moment, for reasons that can be stated in advance and checked afterwards. That is what a risk function is for, and it is what the team dashboards presented in the final session try to do for one real multinational each.

This closing chapter draws the threads together. It does not introduce a new big work or a new dataset. It takes the two closing readings — Stephen Kobrin's review of political risk, published in 1979, and Witold Henisz's measure of political constraint, published in 2000 — and uses them to frame what the eleven sessions established: that geopolitical risk is a contingency rather than an event; that it is measured through families of indicators that disagree; that every famous figure is a stack of decisions; that the system a firm operates in moves slowly and then suddenly; that the risk reaches the firm through identifiable channels; and that indicators do not decide anything until someone attaches a threshold, a decision and an owner to them. It ends with the question every team was asked in its defence, which is also the question with which any analyst should end: what would change your mind?

12.2 The event is not the risk

Frank Knight drew the distinction that every risk analyst still uses, even when they do not name it (Knight 1921). *Risk*, in his sense, is uncertainty that can be measured: the outcomes are known and their probabilities can be estimated from experience, so the uncertainty can be priced, insured or diversified. *Uncertainty* proper is the rest: situations unique enough, or new enough, that no frequency distribution exists. Most of what the course has called geopolitical risk is Knightian uncertainty in that sense. There is no actuarial table of invasions, no frequency distribution of secondary sanctions on a given country, no base rate for a constellation operator's government ordering it to switch off a region. The indices the course rebuilt are attempts to make the second kind of uncertainty behave a little more like the first — to give it a scale, a history and a comparison — and it is important to remember that they do not succeed entirely.

Kobrin's contribution, in a review written as the field of political risk assessment was taking shape, was to insist on a second distinction (Kobrin 1979). The literature he surveyed used "political risk" in two ways: some writers meant political *events* — instability, violence, changes of government — and others meant the *effects* of politics on business — expropriation, restrictions, changes in the rules. Kobrin argued that the first is not the risk. Political instability in a country is a property of that country; whether it affects a particular firm's operations depends on the firm's industry, its assets, its partners and its relationship with the government. A coup may leave a consumer-goods importer untouched and ruin a mining concession; a peaceful election may produce a government that rewrites the tax code for one sector. Risk lies in the *contingency* between the event and the firm.

Political risk (Kobrin) — not the political event itself, but the contingency that political events or conditions will affect a particular firm's operations — its revenues, costs, assets and freedom to act. The same event carries different risks for different firms; a measure of the event is not a measure of the risk.

The course widened Kobrin's frame in one direction. His concern was largely with host-country risk: what the government of the country in which a firm invests might do to it. Geopolitical risk, as the course defined it in Session 01, includes the risk that originates *outside* the host country — in Washington, Beijing, Brussels or Moscow — and reaches the firm through its suppliers, customers, banks and networks. A sanction imposed by the firm's home state on the country where it has a subsidiary, an export control on a component it buys, a jammer operating in a strait its ships use: none of these is a decision of the host government, and all of them reach the firm through the contingencies Kobrin described.

From the archive · Iran, 1979

Kobrin's review appeared in the spring of 1979, in the first months of the revolution that ended the Shah's rule in Iran. Few events have taught multinationals more about the gap between country analysis and firm risk. Iran had been a favoured destination for Western investment and contracting through the 1970s, rich with oil revenue and governed by a regime regarded as a firm ally of the United States. Its collapse cancelled contracts, stranded assets and exposed staff to danger within months.

The year ended with a second lesson. After the seizure of the US embassy in Tehran in November 1979, President Carter used the International Emergency Economic Powers Act — the statute on which most of the sanctions programmes of Session 10 still rest — to freeze Iranian government assets under US jurisdiction. Many American firms had claims against Iran. Their recourse came only through the Algiers Accords of January 1981, which ended the hostage crisis and created the Iran–United States Claims Tribunal in The Hague to arbitrate them. The firms were paid, eventually, out of the frozen assets: their risk had been settled by a negotiation between states in which they had no seat.

The episode is commonly cited as a reason so many firms built or expanded formal political risk units in the years that followed. Kobrin's own study of how American firms organised political assessment appeared three years later (Kobrin 1982). The lesson both drew was the one this course has tried to teach: the analysis of a country is the beginning of the assessment of a firm's risk, not the end.

Every dashboard presented in the final session is a claim about Kobrin's contingency. The footprint — where the firm's revenue, assets, suppliers and people are — is where that claim starts. A dashboard that begins with a map of the world coloured by a country index has not yet begun to assess the firm's risk; one that begins with the firm's own disclosures, and asks which of its countries, suppliers and channels the index bears on, has.

12.3 Six families that disagree

Session 01 sorted the indicators of geopolitical risk into six families, and the sorting has carried the course. *Event-based* indicators count things that happened, coded from reports: conflicts, cyber incidents, sanctions episodes. *Text-based* indicators measure how much the press or firms talk about a risk. *Expert-based* indicators are scores assigned by analysts or by surveys. *Structural* indicators describe slow-moving features of the system: the distribution of material capabilities, the pattern of trade dependence. *Market-based* indicators are prices that embed risk, such as sovereign credit-default swap spreads. *Firm-level* indicators capture risk as firms themselves perceive and disclose it.

Family of indicators — a class of measures of geopolitical risk that share a source of information and therefore a characteristic strength and blind spot: event-based (concrete, but only what was reported and coded), text-based (timely, but measures attention rather than exposure), expert-based (covers latent concepts, but opaque and sticky), structural (explains why, rarely when), market-based (forward-looking, but mixes risk with risk appetite), firm-level (speaks to the decision, but self-reported).

The most cited text-based measure is the one Session 01 rebuilt. Dario Caldara and Matteo Iacoviello count the share of articles in leading newspapers that mention geopolitical tensions, wars and terrorism, and show that their index rises with the events that history books record and that high values are followed by lower investment and employment (Caldara and Iacoviello 2022). Its sibling, the economic policy uncertainty index of Scott Baker, Nicholas Bloom and Steven Davis, applies the same logic to economic policy (Baker, Bloom, and Davis 2016). Session 07 went down to the level of the firm with the measure of Tarek Hassan and his colleagues, built from the share of conversation in firms' earnings calls devoted to political topics and risk, which shows that political risk is largely firm-specific rather than aggregate (Hassan et al. 2019).

Each of these is ingenious, carefully validated and limited by its source: newspapers in a few countries, in one language, attending to what their readers care about; earnings calls of listed firms, speaking to investors.

The families disagree, and the disagreement is information. When the text-based indices are calm and the market-based ones signal danger, someone with money at stake knows or fears something the press has not yet written about. When an expert rating stays unchanged through a crisis that the event data record in detail, the rating is sticky or the raters see a resilience the events do not. When a structural indicator shows a power transition under way and every short-run indicator is quiet, the structural one is telling you *why* something may happen and the others are telling you, correctly, that it has not happened yet. A risk assessment that relies on one family inherits that family's blind spot. This is why the team brief required at least four datasets from at least three families: not because more is better, but because triangulation is the only defence against a blind spot one cannot see from inside one family.

12.4 Every figure is a stack of decisions

The design of the course rested on a claim: that a chart in a famous work is a set of decisions, and that one sees the decisions only by remaking them. Eleven practice rounds put the claim to the test, and each one found the decisions where the claim said they would be.

A share of world GDP depends on whether output is valued at market exchange rates or at purchasing-power parity, and on which deflator and base year are used. A share of world capabilities depends on which components enter the index and how they are weighted. Trade "within" and "between" blocs depends on where the line between blocs is drawn, which is itself an estimate from voting data. A count of sanctions cases per year depends on how a case with no end is ended and how a case of several types is counted. A count of objects in orbit depends on whether debris is dated by the event that created it or by the launch of its parent. None of these decisions is wrong. Each is defensible, and each is a choice someone made, usually for reasons stated in a codebook that few readers open.

The one rule of every practice round — that each deliverable must document at least one place where the data disagreed with the paper — was a way of making the decisions visible. Series are revised; countries are recoded; a release adds cases the paper did not have; a variable's meaning changes between the codebook and the distributed file. A replication that matches the original perfectly is less informative than one that does not, if the difference is understood and explained. The analyst who notices and records the disagreement has learned something about the instrument that the analyst who reproduces the chart has not.

The same holds for the dashboards themselves. A dashboard is not a window onto the firm's risk; it is an argument about it, assembled from decisions of exactly the kind the practice rounds exposed. Which countries enter the footprint, and on what measure of exposure — revenue, assets, employees, suppliers? Which indicator stands for which risk, and from which release? How are missing values treated, and how are countries named so that the joins work? A board shown a dashboard sees none of this unless the analyst shows it. The appendices the memo required — a provenance log for every dataset, a list of data decisions — are where the dashboard's argument is made inspectable, and an analyst who cannot produce them is asking the board to trust the picture rather than the reasoning behind it.

The catch slides, read together. Each practice round ended with one slide: original, ours, and the catch — what the figure does *not* license a multinational to conclude. Read in sequence, the eleven catches form the course's list of what each indicator cannot tell a firm: that attention is not exposure, that shares are not levels, that capabilities are not intentions, that votes are not trade, that counts are not intensity, that a catalogue is not the environment. That list is the most useful thing to take from the course into the take-home examination, and into a risk function.

12.5 Structure as well as events: Henisz on constraint

Most of the indicators the course rebuilt measure what is happening: events, attention, prices, perceptions. Henisz's measure of political constraint, the second closing reading, measures something different: how easily what is happening could change (Henisz 2000).

The idea comes from the theory of veto players, which George Tsebelis later developed into a general account of policy stability (Tsebelis 2002). A government that wants to change a policy — a tax, a regulation, a contract with a foreign investor

— must secure the agreement of every actor with the power to block it: the executive, each chamber of the legislature, the courts, sub-national governments. The more such actors there are, and the more their preferences differ, the harder it is to change anything. Henisz built from this an index, POLCON, that combines the number of independent branches of government with veto power and the alignment of their partisan composition into a measure, between zero and one, of the feasibility of policy change. He found, across a large panel of countries, that greater political constraint was associated with higher economic growth (Henisz 2000), and in later work with more investment in infrastructure — the kind of long-lived, immobile asset most exposed to a government changing its mind (Henisz 2002).

Political constraint (POLCON) — a measure of the feasibility of policy change, built from the number of independent veto players in a political system and the alignment of their preferences. Low constraint means a government can change policy quickly, in either direction; high constraint means change is slow, which protects existing commitments and delays new ones.

The measure matters for the closing argument in three ways. First, it is the structural complement to every event- and text-based indicator of the course. An index of attention tells a firm that tension is rising; a measure of constraint tells it how fast that tension could become a change in the rules. Second, constraint cuts both ways, and a firm must ask which way it cuts *for it*. Low constraint means a government can expropriate quickly, but also that it can grant a licence, settle a dispute or lift a restriction quickly; high constraint protects an existing investment and slows the policy change a new entrant needs. Third, constraint connects the closing readings to Session 08's account of weaponised interdependence. The second condition for a network hub to become a chokepoint is that a state can direct the firms that run it (Farrell and Newman 2019). Whether a government can in practice order a bank, a cloud provider or a satellite operator to cut off a customer — quickly, without a court or a legislature stopping it — is a question about political constraint. The same measure that tells an investor how exposed it is to its host government tells a customer how exposed it is to its supplier's.

Henisz's work also carries a lesson Kobrin would recognise. Constraint is a property of a country; its effect on a firm depends on the firm's position. A firm whose contract was negotiated with the current coalition is protected by constraint; a firm that needs the next coalition to change the law is obstructed by it. The indicator is structural, but the risk it measures is still a contingency.

The bargain between a firm and its host government also changes over time, and in a direction that Raymond Vernon identified half a century ago (Vernon 1971). Before a multinational invests, the host government needs its capital, technology and market access, and the firm can extract favourable terms. Once the plant is built or the mine is sunk, the balance shifts: the firm's assets are hostage, its technology has been transferred or copied, and the government can revise the terms with less to lose. Vernon called this the obsolescing bargain, and the wave of expropriations of the 1960s and 1970s that Kobrin documented (Kobrin 1980) gave it its evidence. Constraint slows the obsolescence; it does not stop it. For a risk function the implication is that the assets-and-people channel is not a fixed exposure measured once at the time of investment. It grows as the firm's bargaining power declines, and the analysis that justified the investment must be redone as the bargain ages — which is one more reason a watchlist must be revisited rather than filed.

12.6 The system moves slowly, then suddenly

Sessions 02 to 06 built the system in which multinationals operate: an international economic order resting on a distribution of power among states that has no government above them, partly governed by public institutions and by private authority. The structural indicators of those sessions — shares of world output, shares of material capabilities, the alignment of states in UN votes, the diffusion of standards — share a characteristic. They move slowly. A power transition takes decades; the reorganisation of trade into blocs shows up in data over years; the authority of an institution erodes before anyone announces its collapse.

The danger for a risk function is that slow-moving indicators are easy to ignore, because nothing in them demands a decision this quarter. The history of the twentieth century and of the last decade suggests that slow structural change is released in sudden events: the system bends for years and then breaks in a week. The interwar order of the League, examined in Session 10's archive, looked intact until it was tested; the integration of Russia into European energy and financial markets looked like interdependence until February 2022 revealed it as exposure. The structural indicator does not tell the firm *when*. It tells the firm *where* the break, when it comes, will be most costly — where interdependence has become dependence, where a bloc boundary runs through its supply chain, where an institution it relies on has lost the support of the states that made it.

This is why the team dashboards combine families. The structural indicators locate the fault lines; the event-based and text-based ones signal movement along them; the market-based ones show whether anyone with money at stake has noticed. None of them alone is a warning system. Together, with thresholds, they can be.

12.7 Channels: where the risk meets the profit and loss account

Sessions 08 to 11 took four instruments through which geopolitical risk now reaches the firm most directly: cyber operations, economic diplomacy, sanctions and space. Across them the course used five channels to trace how a risk becomes a cost.

Market access is the firm's ability to sell: closed by sanctions, export controls, licensing decisions, boycotts and the loss of a diplomatic relationship, opened by commercial diplomacy and trade agreements. **Inputs** are what the firm buys: components and raw materials whose supply is concentrated in one country, services — connectivity, timing, cloud computing — supplied by one operator under one state's jurisdiction. **Finance and payments** are the plumbing: correspondent banks, payment networks, insurance, the currency of settlement — the channel through which financial sanctions and overcompliance operate. **Assets and people** are what the firm has in place: plants, mines, ships, inventories and employees exposed to expropriation, temporary management, conflict or interference. **Data and operations** are the systems the firm runs on: networks open to cyberattack, software and services that can be withdrawn, imagery and signals that can be denied.

Risk channel — the route by which a geopolitical development reaches a firm's revenues, costs, assets or ability to operate: market access, inputs, finance and payments, assets and people, data and operations. Naming the channel converts a statement about the world into a statement about the firm, and points to the decision that could reduce the exposure.

The channels matter because they turn Kobrin's contingency into something a firm can analyse. A statement that "tension between the United States and China is rising" is about the world. A statement that "forty percent of our revenue is in a market exposed to bloc realignment, our key component comes from a single supplier in a country subject to export controls, and our payments in that market clear through banks that would be cut off by a financial sanction" is about the firm — and each clause points to a decision.

12.7.1 When risks compound

The most demanding element of the team brief was the analysis of interactions: how two risks compound or offset for *this* firm. It is demanding because indicators are built one at a time, and dashboards display them side by side, which invites the reader to add them up. Risks do not add. They interact.

They compound when one risk removes the remedy for another. A firm that depends on a single supplier in a country that is drifting out of its home state's bloc faces an input risk and an alignment risk; if the same country is a plausible target of sanctions, the sanction would cut the input at the moment the alignment drift made a replacement supplier in a friendly country most expensive. A firm whose ships depend on one satellite constellation faces a dependence risk; if the operator's state is a sender of sanctions against a country whose waters the ships cross, the dependence becomes an exposure to a foreign-policy decision. They offset when one exposure hedges another: a firm with plants on both sides of a bloc boundary loses one market in a rupture but may keep the other, and a firm whose suppliers are spread across states with different alignments can shift sourcing as the boundary moves.

Interactions are where the analyst's knowledge of the firm, rather than of the world, does the work. No index can be built for them in advance, because they depend on the particular combination of the firm's footprint and the system's fault lines. That is why the presentations reserved two of their twelve minutes for them, and why the defence returned to them most often.

12.8 The composite index, and who chooses the weights

Every team built a composite exposure index for its firm's main countries, with the method of Session 07: select indicators, normalise them to a common scale, weight them, and aggregate. The handbook the session relied on sets out each step and the choices it involves (OECD and European Commission Joint Research Centre 2008).

Composite index — a single measure built by combining several indicators after normalisation and weighting. Its ranking of units depends on which indicators are chosen, how they are normalised and how they are weighted — and the weights are not technical parameters but statements of what matters, and how much.

The lesson of building one is that the composite is the least neutral number on the dashboard. Normalisation decides whether a country's score is relative to the best and worst in the sample or to a fixed benchmark, and therefore whether adding a country changes everyone else's score. Weighting decides how much a conflict risk counts against a sanctions risk or a dependence risk, and no data can settle that, because the answer depends on the firm's exposure through each channel and on how much loss of each kind it can bear. Equal weights are a choice too: the choice to treat every indicator as equally important, which is rarely what anyone believes.

This is why the team dashboards exposed the weights as parameters the board could move, and why the presentations required one live move of a weight. A ranking that does not change when the weights move within a plausible range is robust, and the board can rely on it. A ranking that reverses when the weight on one indicator moves from a third to a half is telling the board that the decision depends on a judgement about priorities, not on the data — and that judgement belongs to the board, not to the analyst. Showing the sensitivity is not a confession of weakness. It is the analyst handing the decision to the person who must own it.

12.9 Indicators do not decide

The third point of the closing address was the bluntest. An indicator without a threshold and an owner is decoration. A dashboard can display every indicator the course built, beautifully, and change nothing in the firm, because nothing on it tells anyone what to do or when.

Watchlist — a short list of indicators, each paired with a threshold that triggers a named decision and with the person who owns that decision. It converts monitoring into commitment: the decision is made in advance, conditionally, so that when the threshold is crossed the question is no longer whether to act but whether the threshold was right.

The format the teams were asked to use has four columns: the indicator one watches, the number that triggers, the decision that follows — what happens on Monday — and the person who signs. Each column does work. The indicator must be one the firm can actually observe, at the frequency it needs, from a source it has logged. The threshold must be a number, set before the event, so that it cannot be rationalised away when the moment comes; it may be revised, but revising it should be a decision with a record. The decision must be specific — contract a second operator, cap new commitments in a market, pre-position inventory, start the approval process for a sale — not "monitor the situation" or "review our exposure". And the owner must be a person with the authority to take the decision, because a threshold crossed without an owner is crossed in silence.

From the archive · Shell's scenarios, 1972–1973

The most famous case for deciding in advance comes from the oil industry. In the early 1970s the planning group of Royal Dutch Shell, led by Pierre Wack, concluded that the forecasting methods on which the company based its investment decisions assumed a continuation of the conditions of the previous two decades — cheap oil, controlled by the companies — that the politics of the producing countries no longer guaranteed. Instead of a single forecast, the group produced scenarios, among them one in which producing governments took control of supply and prices rose sharply.

Wack's later account emphasised that the scenarios at first changed little: managers read them as forecasts to accept or reject, not as alternative worlds whose signposts they should watch. The group's work became effective only when it was recast to change how managers thought about their business, and to make them identify in advance what they would do in each world. When the producers' embargo and the price rises of 1973 arrived, Shell's managers had, in Wack's telling, already thought through what such a world would require of them (Wack 1985).

The lesson the course takes from the episode is not that scenarios predict. It is that a risk analysis changes behaviour only when it is tied, in advance, to the signals that would reveal which world is arriving and to the decisions each world calls for. A scenario without signposts and a dashboard without thresholds fail in the same way.

The best dashboards in the final session were the ones that would make someone do something on a given Monday. They did not have the most indicators or the most elaborate composite. They had three indicators the firm could observe, three thresholds set in advance, and three decisions with names attached.

12.9.1 Why firms wait, and when they should not

There is a rational reason firms delay decisions in the face of geopolitical risk, and it deserves to be stated, because a watchlist must be designed against it. Nicholas Bloom showed that a rise in uncertainty leads firms to pause investment and hiring: when the future is less predictable, the option to wait and see becomes more valuable, and the economy slows as many firms exercise it at once (Bloom 2009). Waiting is often right. But the sanctions and space sessions both showed cases where waiting destroyed options rather than preserving them — the firms that delayed their exit from Russia until the target's countermeasures had closed the door on a sale at their own terms; the firm with one connectivity provider that waits until the service is switched off to contract a second. The value of waiting depends on whether the options remain open while one waits. A good threshold is set at the point where they begin to close.

12.10 Where the risk function sits

A dashboard is an instrument; a risk function is an organisation that uses one. The course has said little about organisation, but the closing argument implies three things about it.

The first is that geopolitical risk assessment cannot live only in a central team. Kobrin's contingency is discovered in the firm's operations — in procurement, which knows where the inputs come from; in treasury, which knows which banks clear the payments; in the country managers, who know what the host government is likely to do. A central function can build the indicators and the method; the exposure must be mapped with the people who run the business. The footprint slide of every presentation was, in effect, an argument that the team had done this mapping, from the firm's own disclosures, before reaching for any index.

The second is that the function must report to people who can decide. Ian Bremmer and Preston Keat argued that the most damaging political risks are in the "fat tail" — rare, high-impact events that standard models underweight — and that firms need political knowledge integrated into strategic decisions, not appended to them (Bremmer and Keat 2009). The memo to the board, with its three risks and what to do about them, was the course's version of that integration: a document addressed to the people who own the decisions, written so that they can act on it.

The third is that the function must keep a record. The provenance log, which every practice round required, is not an academic formality. It is what allows a firm, a year later, to know which version of which dataset produced the number on which it acted, and therefore to learn whether the number, the threshold or the decision was wrong. A risk function that cannot reconstruct its own past judgements cannot improve them.

12.11 What would change your mind?

The defence that followed each presentation asked questions of many kinds, but the one that mattered most was the last: what would change your recommendation? It is the question that separates an assessment from an opinion.

An assessment that nothing could change is not an assessment. A team that recommends reducing exposure to a market should be able to say which indicator, moving by how much, would make it recommend the opposite — and should be able to say it before the indicator moves. This is the same discipline as the threshold, applied to the analyst rather than to the firm. It protects against the most common failure of risk analysis, which is not ignorance but commitment: the analyst who has argued for a view reads every new piece of evidence as confirmation, and the firm that has invested on a view finds reasons to stay.

It also returns the course to its method. Every practice round asked what a famous figure did not license a multinational to conclude. The defence asked the same question of the team's own work. The rubric rewarded answers that conceded what the data could not show and said what would change the recommendation, because those are the answers that tell a board how much weight to put on the recommendation in the first place.

12.12 A short review of the literature

The study of political risk in international business took shape in the 1970s, in the shadow of the wave of expropriations that Kobrin (1980) documented. Kobrin (1979) reviewed the early literature and argued that risk lies in the contingency between political events and a firm's operations, not in the events themselves; Fitzpatrick (1983) surveyed the same field and found its definitions of risk equally unsettled; and Kobrin (1982) studied how firms actually organised the assessment. The distinction between measurable risk and unmeasurable uncertainty on which the field rests is Knight (1921)'s.

The institutional strand begins from veto players (Tsebelis 2002). Henisz (2000) turned the idea into the POLCON index of political constraint and related it to growth, and Henisz (2002) to investment in infrastructure, the asset class most exposed to policy reversal. The measurement strand is more recent and largely text-based: Baker, Bloom, and Davis (2016) on economic policy uncertainty, Caldara and Iacoviello (2022) on geopolitical risk, and Hassan et al. (2019) on political risk as firms themselves discuss it. Bloom (2009) established the mechanism by which uncertainty, once measured, depresses investment, and OECD and European Commission Joint Research Centre (2008) codified the construction of composite indicators that most practical risk scores rely on.

The geopolitical turn of the last decade reframed the field around networks and dependence: Farrell and Newman (2019) explain how states weaponise the hubs of global networks, which turns a firm's suppliers and service providers into channels of state power. On the practice of deciding under such uncertainty, Wack (1985) remains the classic account of scenarios tied to decisions, and Bremmer and Keat (2009) the practitioner's case for integrating political knowledge into strategy.

12.13 After the last session

The course ends where a risk function begins. Each team leaves with a dashboard for one multinational, a memo to its board, and a method it has defended in public. Each student leaves with eleven replications behind them, each documenting at least one place where the data disagreed with the paper, and eleven catch slides that together list what each indicator cannot tell a firm.

The take-home examination tests the whole of this under the conditions in which the work is actually done: an unseen scenario, a source from which to collect the data, a firm, a time limit, and one's own notes. It will ask for the same moves the course has practised. Find the data at source and record where it came from. Notice where it disagrees with what one expected. Build the view that answers the question. Say which family of indicators it belongs to, and what that family cannot see. Name the channel through which the risk reaches the firm. And end, as every practice round ended, with what the figure does not license the firm to conclude, and what the firm should do anyway.

That last clause is the course in miniature. Geopolitical risk will never be known with the confidence of a price or the regularity of an actuarial table. The analyst's job is not to remove that uncertainty but to make the decisions inside it visible — the decisions in the data, the decisions in the method, and the decisions the firm will take when a threshold is crossed — to the person who must act. A dashboard that does that is finished. One that does not is a picture of the world.

12.14 On the day

12.14.0.1 A geopolitical risk assessment for one multinational

What happens	Each team presents and defends its geopolitical risk dashboard for its multinational
Assessment	Presentation and defence: **5 of the 20 points** of the team work. Memo and published dashboard (15 points) due **one week later**

Closing readings	Kobrin, S. J. (1979). Political risk: A review and reconsideration. *Journal of International Business Studies*, 10(1), 67–80. doi:10.1057/palgrave.jibs.8490631 · Henisz, W. J. (2000). The institutional environment for economic growth. *Economics & Politics*, 12(1), 1–31. doi:10.1111/1468-0343.00066
Deck	`INTE60019A-S12-Presentations.qmd`
Brief	`assessments/team-project/`

There is no pre-session reading beyond your own work, no practice round and no Risk Brief.

Lecture notes: this session is also Chapter 12 of the course book.

12.14.1 Run of show

Each team has **20 minutes**: **12 minutes of presentation**, the dashboard driven live, then **8 minutes of defence** — questions from the instructor and from the room.

Time	
0:00–0:05	Order of presentation drawn at random; the two-team "panel" for each slot announced
0:05–2:50	Teams 1–8, 20 minutes each, back to back. A five-minute break after team 4
2:50–3:00	Closing address: *from indicators to decisions*

With more than eight teams the session runs in two rooms in parallel, or is extended by agreement; the order and room are posted on ZoneCours the week before.

The panel. For each presentation, two other teams act as the **board**. Each prepares one question in advance from the published dashboard link (sent 48 hours before) and asks it in the defence. The quality of a team's board question counts towards its members' participation.

Who answers. The instructor addresses at least one question to a member **drawn at random**. Every member must be able to defend every part of the dashboard and the method behind it.

12.14.2 What the twelve minutes must contain

Minutes	What the board must see
0–2	The firm and its **footprint**: where its revenue, assets, suppliers and people are
2–5	The **composite exposure index** for its main countries — indicators, normalisation, weights — and one live move of a weight parameter to show what the ranking is sensitive to
5–9	**Two channel views**, driven live: e.g. sanctions, critical inputs, bloc alignment, cyber

Minutes	What the board must see
9–11	The **interaction**: how two risks compound or offset for *this* firm
11–12	**Decisions and the watchlist**: three indicators, three thresholds, three named decisions

Overrun is stopped at thirteen minutes.

12.14.3 Rubric (20 points — consistent with the team-work brief)

Criterion	Points	Assessed	Full marks look like
Conceptual grounding	3	memo + presentation	Risks framed with the course's concepts — order, anarchy, institutions, diplomacy, sanctions, interdependence — not newspaper vocabulary
Data and provenance	3	memo	Four or more datasets collected at source, from at least three families of indicators, every one logged; decisions named
Method of the composite index	4	memo	Indicators justified; normalisation and weighting explained; sensitivity to weights shown
Analysis of interactions	3	memo	How risks compound or offset — e.g. sanctions *and* input concentration *and* bloc alignment
Recommendations	2	memo	Decisions with triggers and owners, not "monitor the situation"
Presentation and defence	**5**	**today**	Twelve minutes; the dashboard driven live; questions answered by any member, including the one drawn at random

Presentation and defence (5 points), in detail:

Points	
2	Clarity and time: the five blocks above, in twelve minutes, the dashboard driven rather than screenshotted
2	Defence: answers that engage the question, concede what the data cannot show, and say what would change the recommendation

Points	
1	Random-member answer: the member drawn answers competently without help

Individual accountability (the peer-rating multiplier) is described in the team-work brief.

12.14.4 The closing address — from indicators to decisions

Ten minutes, after the last team. Its content is in the deck; in summary:

1. **Kobrin (1979) was right about the hard part.** Political risk is not the event — it is the *contingency* that the event affects *this* firm's operations. Every dashboard presented today is a claim about that contingency.
2. **Henisz (2000) on constraint.** The POLCON index measures how many independent veto players constrain a government. Low constraint means policy can change fast — in either direction. It is the structural complement to every event- and text-based indicator of the course.
3. **Indicators do not decide.** A watchlist is useful only if each indicator has a threshold and an owner. The best dashboards today were the ones that would make someone *do* something on a given Monday.
4. **What the twelve weeks established.** Geopolitical risk is measured through families of indicators that disagree; each figure in a famous work is a stack of decisions; and the analyst's job is to make those decisions visible to the person who must act.

12.14.5 After today

One week after Session 12	Memo (2,500 words + appendices) and the final published Tableau Public dashboard, on ZoneCours; confidential peer ratings
Final examination period	**Take-home examination** — 40%, individual, 3 hours of work in a 48-hour window. Brief and specimen: `assessments/take-home/`

Prepare for the take-home by re-reading the **catch** slides from your eleven practice rounds: together they are the course's list of what each indicator cannot tell a firm.

← Previous session · Syllabus

Appendix A

The replications

Each of Sessions 01–11 pairs one **big article or book** with the **data it is built on**. Before class you read the work and locate its source. In the 90-minute practice round your team:

1. **Collects** the data from the original producer and logs its provenance;
2. **Prepares** it in Tableau — pivot, join, recode — recording each decision;
3. **Replicates** the target figure as closely as the data allow;
4. **Compares** yours with the original, **breaks** one decision, and **extends** it in one direction;
5. Presents **one slide — original · ours · the catch** — in two minutes, by a member drawn at random.

Every source, its access conditions and its traps: `DATA-SOURCES.md`.

A.1 Article–data map

Session	Big work	Data you collect	Target figure	Compare · break · extend
01 · What is geopolitical risk?	Caldara & Iacoviello (2022), *Measuring Geopolitical Risk*, AER · doi:10.1257/aer.20191823	GPR index	The monthly GPR index since 1985, spikes annotated; the historical index since 1900	Threats vs acts · monthly vs yearly averages · one country index against the global
02 · The international economic order	Baldwin (2016), *The Great Convergence*, Belknap/Harvard UP	World Bank WDI: GDP (current US$ and PPP), trade % GDP	The **G7 share of world GDP**, 1970–today — the "great convergence" after 1990	Current dollars vs PPP · Baldwin's "Industrialising Six" · world trade/GDP and the "slowbalisation" after 2008
03 · Anarchy and the balance of power	Mearsheimer (2001), *The Tragedy of Great Power Politics*; Allison (2017), *Destined for War*	COW National Material Capabilities v7.0 (CINC); SIPRI military expenditure	**Shares of world capabilities** of the major powers, 1816–2022, and the US–China crossover	CINC vs military spending vs GDP — three measures, three different power transitions · a parameter to choose the challenger

Session	Big work	Data you collect	Target figure	Compare · break · extend
04 · Trade, conflict and fragmentation	Gopinath, Gourinchas, Presbitero & Topalova (2025), *Changing Global Linkages: A New Cold War?*, JIE · doi:10.1016/j.jinteco.2024.104042; Mansfield (2021)	UNGA ideal points (Voeten); IMF bilateral goods trade (IMTS)	**Trade within vs between geopolitical blocs**, indexed to before the 2022 invasion of Ukraine	Bloc definitions: thresholds on ideal-point distance · mirror trade (exports vs imports) · one "connector" country
05 · International institutions	Kuziemko & Werker (2006), *How Much Is a Seat on the Security Council Worth?*, JPE · doi:10.1086/507155	UNSC elected members; US foreign aid (ForeignAssistance.gov Country Summary — economic and military combined) or the AidData replication file	**US aid to elected members in event time**, years −3 to +3 around a Security Council term	Current vs constant dollars · mean vs median · years with and without salient Council business
06 · Private authority and standards	Prakash & Potoski (2006), *Racing to the Bottom? Trade, Environmental Governance, and ISO 14001*, AJPS · doi:10.1111/j.1540-5907.2006.00188.x; Abbott & Snidal (2009); Mattli & Büthe (2003)	ISO Survey (ISO 14001, ISO 9001); IMF bilateral exports; WDI GDP	**The diffusion of ISO 14001** across countries (map and small multiples), and adoption against the adoption level of each country's export markets	Counts vs per-$ of GDP · ISO 9001 as the placebo · trade-weighted vs unweighted export markets
07 · Measuring risk: text, coding, big data, indices	Hassan, Hollander, van Lent & Tahoun (2019), *Firm-Level Political Risk*, QJE · doi:10.1093/qje/qjz021; Baker, Bloom & Davis (2016), QJE; OECD/JRC (2008) *Handbook on Constructing Composite Indicators*	firmlevelrisk.com firm-quarter file; EPU; a corpus of news items from GDELT that **you code yourselves**	**Average firm-level political risk over time** against EPU — then **your own coded index**	Intercoder agreement on your corpus · normalisation choices · weights as parameters: how much does the ranking move?
08 · Cyber geopolitics	Valeriano & Maness (2015), *Cyber War versus Cyber Realities*, OUP · doi:10.1093/acprof:oso/9780190204792.001.0001; Farrell & Newman (2019)	DCID v2.0; EuRepoC v1.3	**Incidents by year, method and severity**, and the **initiator × target matrix** of rival dyads	DCID (state rivals) vs EuRepoC (all incidents) · severity distribution — evidence of "cyber restraint"? · incidents hitting the private sector

Session	Big work	Data you collect	Target figure	Compare · break · extend
09 · Economic diplomacy	Rose (2007), *The Foreign Service and Foreign Trade: Embassies as Export Promotion*, World Economy · doi:10.1111/j.1467-9701.2007.00870.x; Bjola & Manor (2018); Putnam (1988)	Lowy Global Diplomacy Index; IMF bilateral exports; WDI GDP	**Exports against the number of diplomatic posts** in the partner, log scale, with trend line	Raw vs GDP-adjusted · embassies vs consulates · reverse causality: posts follow trade?
10 · Economic sanctions	Hufbauer, Schott, Elliott & Oegg (2007), *Economic Sanctions Reconsidered*, PIIE; Felbermayr et al. (2020), *The Global Sanctions Data Base*, EER · doi:10.1016/j.euroecorev.2020.103561; Pape (1997)	GSDB Release 5 (by request); TIES v4.0	**Active sanctions cases per year by type**, 1950–today, and the **share judged successful** by objective	Threats (TIES) vs impositions · success by objective · a Gantt chart of one target's sanctions history
11 · Space geopolitics	ESA Space Debris Office (2026), *ESA's Annual Space Environment Report*, Issue 10.0 · PDF; Samson & Brett (Eds., 2026), *Global Counterspace Capabilities*, SWF; Pekkanen (2019)	CelesTrak SATCAT; GCAT launch log (optional)	**Objects in geocentric orbit by object class**, 1957–2025 (ESA Fig. 2.1), with the 2007, 2009 and 2021 events annotated; the **HHI of who launches payloads**	Debris dated by parent launch vs by event · concentration by flag (`OWNER`) vs by firm (operator) · public catalogue vs ESA's catalogued + asserted objects
12 · Team presentations	Kobrin (1979); Henisz (2000)	All of the above; POLCON; WGI	**Your team's dashboard** for one multinational	Defended live, in front of the room

A.2 Why replicate figures, not regressions

This is not an econometrics course. The figure at the heart of a big work is where its argument meets the reader — the image that ends up in board presentations and ministerial speeches. Rebuilding it forces every decision the authors made into the open: which countries, which years, which deflator, which coding rule, which normalisation. A multinational's risk team makes the same decisions every time it builds a dashboard. Learning to see them is the point.

A.3 Citing

Cite the work **and** the dataset, with version and access date. They are distinct contributions, often by different people, and the dataset has usually been revised since the work was published.

Appendix B

Data sources

Every dataset used in the course: who produces it, where to collect it, what it costs you to get it, and the trap it holds. **Collect at source, every time**: the course repository contains no data.

Access status was last checked on **16 October 2026**. Sources move: if a link has changed, the producer's landing page is canonical, and the change goes in your provenance log.

> **Lead times.** Two sources need action **days before** the session: the **Global Sanctions Data Base** (request form, data e-mailed within about 24 hours — request it after Session 08) and the **ISO Survey** (requires a free IAF CertSearch account — log in at iafcertsearch.org/services/iso-survey; create it after Session 04).

B.1 The registry

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
1	**Geopolitical Risk (GPR) index**	Caldara & Iacoviello	matteoiacoviello.com/gpr.htm	`data_gpr_export.xls` (one sheet, `Sheet1`; monthly `GPR`, `GPRT`, `GPRA`, historical `GPRH`; 44 country series `GPRC_XXX` — percent of articles, from 1985 — and historical `GPRHC_XXX`; variable notes in the `var_name`/`var_label` columns)	1900 – last month	free	01, 07, 12

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
2	**World Development Indicators**	World Bank	DataBank WDI	`NY.GDP.MKTP.CD` (GDP, current US$), `NE.TRD.GNFS.ZS` (trade, % GDP), `NY.GDP.MKTP.PP.CD` (GDP, PPP) — per-indicator CSV: `api.worldbank.org/v2/en/indicator/{CODE}?downloadformat=csv` (4 lines above the header; `Indicator Code`, not `Series Code`; missing = blank)	1960–2025	free, CC BY 4.0	02, 03, 09
3	**National Material Capabilities v7.0**	Correlates of War	correlatesofwar.org/data-sets/national-material-capabilities	`wp-content/uploads/NMCv7.zip` (codebook `NMC_Documentation_v7.pdf` + nested `NMC-v7-abridged.zip` → `NMC-70-abridged.csv`)	1816–2022	free; cite Singer, Bremer & Stuckey (1972)	03
4	**Military Expenditure Database**	SIPRI	sipri.org/databases/milex	`SIPRI-Milex-data-1949-2025_v1.2.xlsx` (revised 27 April 2026)	1949–2025	free for non-commercial use	03
5	**UN General Assembly ideal points**	Bailey, Strezhnev & Voeten	Harvard Dataverse doi:10.7910/DVN/LEJQZ	`Idealpointestimates_2025_2026.csv` (country-year: `ccode`, `year`, `IdealPointFP`; `session` = `year` − 1945), `AgreementScores.csv`, codebook	1946–2025	free	04, 12

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
6	**International Trade in Goods by partner (IMTS, formerly DOTS)**	IMF	data.imf.org — IMF.STA:IMTS	exporter × partner × year: indicator `XG_FOB_USD`; columns `COUNTRY`, `COUNTERPART_COUNTRY`, `TIME_PERIOD`, `OBS_VALUE` (US dollars); ISO3 codes except `KOS`, `WBG`; aggregates coded `G…/TX…` (Download button)	from 1948 for many countries	free	04, 06, 09
7	**Kuziemko & Werker replication data**	AidData	AidData replication page	"Download Data" link (a Google Drive file; required a Google sign-in when checked in October 2026 — contents unverified)	1946–2001	free	05
8	**UN Security Council elected members**	United Nations	main.un.org/securitycouncil — elected members	tables, copied into a sheet: country × term	1946–today	free	05

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
9	**US foreign aid (Greenbook)**	US State Dept / ForeignAssistance.gov	foreignassistance.gov/data · frozen mirror (December 2024; use its Dropbox zip): foreignassistance.andrewheiss.com	Country Summary `us_foreign_aid_country.csv` — ISO3 in `Country Code`; filter `Transaction Type Name` = Obligations; use `constant_amount`. Economic and military **combined** — the split is only in the Greenbook xlsx on the mirror (`greenbook.html`, FY 1946–2020)	official 1946–2026 · mirror 1946–2024	free; the official site has been unstable since 2025 — use the mirror if it fails, and log it	05
10	**ISO Survey of certifications**	ISO, via IAF CertSearch	iso.org/the-iso-survey.html · IAF CertSearch — ISO Survey	ISO 14001 and ISO 9001 certificates by country and year; from 2025 compiled from IAF CertSearch data (file format unverified)	1993–2024	free; IAF CertSearch account required	06
11	**Firm-level political risk**	Hassan, Hollander, van Lent & Tahoun	firmlevelrisk.com/download	`firm_level_risk_2022Q1.csv` (firm × quarter `PRisk`, `NPRisk`, sentiment, topic risks; **tab-separated despite `.csv`**; `hqcountrycode` is ISO **2**-letter)	2002Q1–2022Q1	free; no longer updated	07

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
12	**Economic Policy Uncertainty**	Baker, Bloom & Davis	policyuncertainty.com · doi:10.1093/qje/qjv041	`media/Global_Policy_Uncertainty_Data.xlsx` (`Year`, `Month`, `GEPU_current`, `GEPU_ppp` — no `GEPU` column), `media/All_Country_Policy_Uncertainty_Data.xlsx` (sheet `EPU`, one column per country, named in English)	1997–present, monthly (some countries from 1985; the US from 1900 in a separate file)	free, CC BY 4.0	07
13	**GDELT**	The GDELT Project	gdeltproject.org	news articles, via the DOC 2.0 API (CSV columns `URL`, `MobileURL`, `Date`, `Title`; one request every 5 seconds) or the web "Summary" tool	the DOC API searches a rolling last 3 months — use `timespan=12weeks`	free	07
14	**Dyadic Cyber Incident and Campaign Data (DCID) v2.0**	Maness, Valeriano, Hedgecock, Macias & Jensen	Harvard Dataverse doi:10.7910/DVN/GOQMYV	`Master_DCID_2.0_Release_public_Sep2022.tab` (Access File — original CSV); codebook at ryanmaness.com/cyber-conflict-dataset, not on Dataverse	2000–Jan 2021, 433 incidents	public, CC0	08
15	**EuRepoC Global Dataset of Cyber Incidents v1.3**	European Repository of Cyber Incidents	eurepoc.eu · Zenodo doi:10.5281/zenodo.14965305	`eurepoc_global_dataset_1_3.csv`, `eurepoc_codebook_1_3.pdf`	2000–2024, 3,414 rows in the CSV (the record says 3,416)	free, CC BY-NC 4.0	08
16	**CFR Cyber Operations Tracker**	Council on Foreign Relations	cfr.org/cyber-operations	no official download — browse and hand-code	2005–2024	free to read	08 (reading only)

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
17	**Global Diplomacy Index**	Lowy Institute	globaldiplomacyindex.lowyinstitute.com	direct download `.xlsx` — one sheet per data year (`2023`, `2021`, `2019`, `2017`, `2016`), one row per diplomatic post: `COUNTRY`, `POST CITY`, `POST COUNTRY`, `TYPE OF POST`; **the 2019 sheet is incomplete** (Venezuela only, 40 rows)	2016, 2017, 2019, 2021, 2023 (2024 edition)	free	09
18	**Global Sanctions Data Base, Release 5**	Felbermayr, Kirilakha, Syropoulos, Yalcin & Yotov	globalsanctionsdatabase.com	**request form** → files by e-mail (case-level and dyadic). Case file columns (from a public notebook, not yet checked against the codebook): `case_id`, `sanctioned_state`, `sanctioning_state`, `begin_year`, `end_year`, six 0/1 type columns, `objective`, `success`	1,794 cases; begin years 1949–2025 (the home page says 1950)	free on request; **no redistribution, no commercial use** — never publish raw data to Tableau Public	10

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
19	**TIES — Threat and Imposition of Economic Sanctions v4.0**	Morgan, Bapat & Kobayashi	sanctions.web.unt.edu	`TIESv4-1.xls` + `tiesusers-manu-alv4.pdf`; states as COW codes; threat-only cases have `imposition` = 0	1945–2005	free	10
20	**POLCON — Political Constraint Index**	Witold Henisz, Wharton	POLCON dataset page	`POLCON_2025_FINAL_2021.xlsx` (sheets *Data* and *Key*)	1800–2021	free	12
21	**Worldwide Governance Indicators**	World Bank	worldbank.org — WGI	`wgidataset_with_sourcedata-2025.xlsx`: one sheet per dimension (`pv` = political stability), `Economy (code)`, `Year`, `Governance score (0–100)` — not to be mixed with the estimate column. The 2026 update (1996–2025) is now the landing-page default and recalculates history again: use one edition and log it	1996–2024 (2025 ed.)	free	07, 12

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
22	**SATCAT — satellite catalogue**	CelesTrak (T. S. Kelso), from the US Space Force public catalogue	celestrak.org/satcat/ · format	`pub/satcat.csv` (one row per catalogued object, 17 columns: `OBJECT_TYPE` PAY/R/B/DEB/UNK, `OPS_STATUS_CODE` — active = `+ P B S X`, `OWNER`, `LAUNCH_DATE`, `DECAY_DATE` empty = in space, `ORBIT_CENTER`)	1957 – day of download	free; usage policy: one download per update	11
23	**GCAT — General Catalog of Artificial Space Objects**	Jonathan C. McDowell	planet4589.org/space/gcat	`tsv/cat/launch/launch.tsv` + `tsv/tables/orgs.tsv` (tab-separated; line 2 a "# Updated" comment; orbital = `LaunchCode` starting `0` or `D`; state via `Agency` → `orgs.Code` → `StateCode`)	1942 – last week	free	11 (optional)

B.2 Traps, source by source

Dataset	The trap
GPR	Revised monthly; vintages exist (`data_gpr_export_YYYYMM.xls`). Country indices are the anglophone press's attention to a country, not that country's risk.
WDI	Aggregates ("World", "High income", "G7" is *not* provided — build it) sit in the same file as countries. Current-dollar shares move with exchange rates.

Dataset	The trap
NMC	Uses COW country codes, not ISO; states enter and exit the system (`GMY` absent 1946–1990, when `GFR`/`GDR` cover 1955/1954–1990). Missing components are coded `-9`. Yearly CINC sums run 1.000–1.075, not exactly 1. Ends in 2022.
SIPRI	Several sheets (`Constant (2024) US$`, `Current US$`, `Share of GDP`, `Per capita`, `Share of Govt. spending`, local-currency and `Regional totals`); `...` = data unavailable, `xxx` = the state did not exist — notes atop each sheet, plus a `Footnotes` sheet. Country strings: `United States of America`; `Russia` from 1992, `USSR` before.
UNGA ideal points	Carries both COW `ccode` and ISO `iso3c`; estimated by calendar year since the July 2025 releases (v36/37; current v39, 30 July 2026) — earlier versions by session, so the variable meaning changed between versions. Higher = closer to the US.
IMF IMTS	Exports reported by A ≠ imports reported by B ("mirror" asymmetry). Choose one side and say which.
UNSC members	Two-year terms starting 1 January; election happens the October before — the paper's "election year" matters.
US aid	Obligations vs disbursements; current vs constant dollars; the official site moved in 2025.
ISO Survey	Coverage of certification bodies changed over time: jumps in a country's count can be reporting, not adoption.
Firm-level risk	Listed firms holding earnings calls in English — mostly US. Stops in 2021.
DCID	State-to-state rivals only; `StateA`/`StateB` are the dyad as **names**, not initiator and target — derive the target from `initiator` (COW code) and `Dyadpair`; coded from open sources — attribution is by definition uncertain.
EuRepoC	Much wider than DCID (includes criminal and non-state); filter on `initiator_category` (`State` / `State affiliated actor`) before comparing.
Lowy GDI	66 countries or territories only (including Taiwan and the EU's EEAS); names, not codes, with typos (`Portual`, `Nigaragua`); counts posts, not staff or budget.
GSDB	Licence forbids redistribution: workbooks go to ZoneCours, never Tableau Public with the raw data. "Success" is the coders' judgement of objectives met.
TIES	Includes **threats** of sanctions — a case may never have been imposed.
POLCON / WGI	WGI's 2025 edition changed scale and recalculated history: never mix editions.
SATCAT	Debris carries its parent's `LAUNCH_DATE` (Fengyun-1C debris is dated 1999, not 2007); `OWNER` is not the operator (Starlink `US`, Galileo `ESA`, Russia `CIS`); filter `ORBIT_CENTER` = `EA`; a blank status is not "inactive"; one download per team.
GCAT	The state is the launch provider's, not the customer's (Rocket Lab = New Zealand); vague-date text, padded values.
UCS Satellite Database	Frozen: data current through 1 May 2023. Use SATCAT for anything later.

B.3 Country codes

Join on **ISO 3166 alpha-3** codes, never on names. NMC and DCID (in `initiator` and `Dyadpair`) use **Correlates of War** codes (official list): keep a concordance sheet (COW `ccode` ↔ ISO3) in your workbook. The `countrycode` concordance tables published by Arel-Bundock et al. are the standard reference. The UNGA ideal points carry `iso3c` too. Kosovo is `KOS` at the IMF and `XKX` at the World Bank; the West Bank and Gaza is `WBG` at the IMF.

Appendix C

The Tableau toolkit

Everything you need to install, the one new skill each session adds, the conventions every chart follows, and the traps that cost most marks.

C.1 1. Installation — before Session 01

Tool	How	Used for
Tableau Desktop	Free academic licence through **Tableau for Students** (Tableau Academic Programs). Request it with your HEC e-mail; the key arrives by e-mail.	every practice round
Tableau Prep Builder	Same licence, same key, where the academic programme includes it — optional	cleaning, reshaping and unioning files (from Session 02; essential in Session 07)
Tableau Public	Free account at public.tableau.com	publishing the team dashboard and Risk Briefs
A spreadsheet	Excel, LibreOffice Calc or Numbers	opening files at source, reading codebooks

Test. Open Tableau Desktop → *Sample – Superstore* → `Order Date` to Columns, `Sales` to Rows. A line appears: you are ready.

If the key has not arrived install the 14-day trial and switch to the key when it comes. The academic licence is renewable each year while you are enrolled.

> **If the academic programme no longer covers your case.** Tableau's academic terms have changed more than once (reports in 2025 indicated that free student licences for Desktop and Prep were being withdrawn in some programmes; check tableau.com/academic/students and HEC's own software page at the start of term). The fallback is the free **Tableau Public** desktop edition. Everything in the course works in it — pivots, joins, relationships, LOD, parameters, maps, dashboards — with two differences you must manage:
>
> 1. **It saves only to Tableau Public**, where workbooks are public. For data whose licence forbids redistribution (the Global Sanctions Data Base, possibly the ISO Survey), publish only aggregated views, keep "allow download" off, and hand in screenshots plus your calculated-field definitions instead of a `.twbx`.
> 2. **No Tableau Prep.** Do the Session 07 cleaning in the data source pane (pivot, split, union, calculated fields) — the practice brief gives both routes.

Mac and Windows behave identically for everything in this course, with one exception: on a Mac, right-drag is *Option*-drag.

C.2 2. The skills ladder

One new skill per session, always applied to the session's replication. Nothing is taught twice; everything is assumed afterwards.

Session	Replication	The new skill
01	GPR index	Connect to Excel · fix data types · continuous dates · reference lines · annotations · captions
02	G7 share of world GDP	**Pivot** wide years to long · filter on ISO codes, not names · area chart · **percent of total** (table calculation)
03	Distribution of power	**Calculated fields** · ratios · **parameters** to choose a country · dual axis
04	Trade between blocs	**Joins and relationships** on dyads (exporter × importer × year) · **groups** and **sets** for blocs · indexing to a base year
05	Aid and the Security Council	**LOD expressions** (`FIXED`) to compute "years relative to a term" · event-time charts · reference bands
06	ISO 14001 diffusion	**Maps** · geocoding by ISO code · fixing unmatched countries · the **Pages** shelf for animation · small multiples
07	Firm-level risk and your coded corpus	**Tableau Prep** flows · unions · cleaning text-coded data · **normalisation** (min–max, z-score) · **parameters as weights** in a composite index
08	Cyber incidents	**Highlight tables** · dyad matrices (initiator × target) · **filter and highlight actions**
09	Embassies and exports	**Log scales** · **trend lines** and what their R^2 and p-value do and do not say · clusters
10	Sanctions episodes	**Gantt charts** for episode durations · stacked bars of shares · **quick table calculations**
11	Objects in orbit and the new space race	**Running sum** (table calculation) on pivoted launch/decay events · **Herfindahl–Hirschman index** with LOD · **dashboard actions** across data sources
12	Your team dashboard	**Dashboards** · **stories** · publishing to Tableau Public · writing captions a board can read

C.3 3. Conventions every chart follows

1. **A title that states the finding**, not the variable. *"The G7's share of world GDP fell by a third after 1990"*, not *"GDP share"*.
2. **A source caption**: dataset, producer, version, access date. No caption, no credit.
3. **Units on every axis.** Current US dollars or constant? Index base year? Share or count?
4. **Direct labels** in place of legends wherever there are fewer than six series.
5. **One accent colour** for the series the argument is about; greys for the rest.
6. **Annotate the events** that explain the shape of the line.
7. **Name your decisions** in a footnote: countries dropped, years rebased, codes changed.

C.4 4. The traps that cost marks

Trap	Symptom	Fix
Joining on country names	Côte d'Ivoire, Türkiye, Korea, Rep. silently drop out	Always join on **ISO 3166 alpha-3** codes. Keep a concordance sheet for datasets that use COW or UN codes
Wide years	66 columns called `1960` … `2025`	**Pivot** them in the data source pane (select columns → right-click → *Pivot*)
SUM of an index	A line ten times too high	An index is averaged, not summed. Check the aggregation on every pill
Discrete vs continuous dates	A bar for "May" across all years	Use the *continuous* date (green pill) for time series
Duplicated rows after a join	Totals inflate	Check the row count before and after; use a **relationship** instead of a join when the grain differs
Revised data	Your numbers do not match last year's paper	That is a finding, not an error: log the release you used
Aggregates in country lists	"World", "High income" and "Euro area" counted as countries	Filter on a country list, not on "everything that is not blank"
Percent of total computed the wrong way	Shares that sum to 100% across years instead of within each year	*Edit Table Calculation* → *Compute using* the country dimension, restart every year

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C.5 5. The provenance log

One CSV per practice round, in the session's `data/` folder. Template: `templates/provenance-log.csv`.

Column	What goes in it
`file`	The file name exactly as saved in `raw/`
`source`	The producer: author, agency, organisation
`url`	The landing page you downloaded from — not a mirror
`version_or_release`	Release number, "last updated" date, or edition
`accessed`	Today's date, ISO format: `2026-09-02`

Column	What goes in it
`licence`	Free / registration / request / restricted redistribution
`notes`	Anything odd — and **at least one place where the data disagreed with the paper**

C.6 6. Handing in

- *File → Export Packaged Workbook* (`.twbx`) — it contains the data extract, so the workbook opens on any machine. Do not upload a `.twb`: it does not contain the data.
- For data whose licence forbids redistribution (several of them — see `DATA-SOURCES.md`), upload to **ZoneCours only**, never to Tableau Public.

Appendix D

Readings

The **required reading** of each session is its big work: the article or book whose central figure the practice round rebuilds. It is read and annotated **before** class. **Further readings** are for the Risk Brief, the team memo and the take-home; none is required.

Article–data map: `REPLICATIONS.md`. Datasets: `DATA-SOURCES.md`.

D.1 Core references of the course

These are the principal references listed in the course's official description. Each is the required or further reading of at least one session (in brackets).

- Abbott, K. W., & Snidal, D. (2009). The governance triangle: Regulatory standards institutions and the shadow of the state. In W. Mattli & N. Woods (Eds.), *The politics of global regulation* (pp. 44–88). Princeton University Press. https://doi.org/10.1515/9781400830732.44 *[06]*
- Bjola, C., & Manor, I. (2018). Revisiting Putnam's two-level game theory in the digital age: Domestic digital diplomacy and the Iran nuclear deal. *Cambridge Review of International Affairs*, 31(1), 3–32. https://doi.org/10.1080/09557571.2018.1476836 *[09]*
- Brunsson, N., Rasche, A., & Seidl, D. (2012). The dynamics of standardization: Three perspectives on standards in organization studies. *Organization Studies*, 33(5–6), 613–632. https://doi.org/10.1177/0170840612450120 *[06]*
- Green, J. F. (2013). *Rethinking private authority: Agents and entrepreneurs in global environmental governance*. Princeton University Press. *[06]*
- Kuziemko, I., & Werker, E. (2006). How much is a seat on the Security Council worth? Foreign aid and bribery at the United Nations. *Journal of Political Economy*, 114(5), 905–930. https://doi.org/10.1086/507155 *[05]*
- Le Billon, P., & Shykora, L. (2020). Conflicts, commodities and the environmental geopolitics of supply chains. In S. O'Lear (Ed.), *A research agenda for environmental geopolitics* (pp. 59–73). Edward Elgar. https://doi.org/10.4337/9781788971249.00012 *[04]*
- Mansfield, E. D. (2021). International trade and conflict. In J. C. W. Pevehouse & L. Seabrooke (Eds.), *The Oxford handbook of international political economy* (pp. 660–671). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198793 *[04]*
- Mattli, W., & Büthe, T. (2003). Setting international standards: Technological rationality or primacy of power? *World Politics*, 56(1), 1–42. https://doi.org/10.1353/wp.2004.0006 *[06]*

D.2 Session 01 — What is geopolitical risk?

Required. Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225. https://doi.org/10.1257/aer.20191823

Further. - Kobrin, S. J. (1979). Political risk: A review and reconsideration. *Journal of International Business Studies*, 10(1), 67–80. https://doi.org/10.1057/palgrave.jibs.8490631 - Knight, F. H. (1921). *Risk, uncertainty and profit*. Houghton Mifflin. - Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica*, 77(3), 623–685. https://doi.org/10.3982/ECTA6248 - Bremmer, I., & Keat, P. (2009). *The fat tail: The power of political knowledge for strategic investing*. Oxford University Press.

D.3 Session 02 — The international economic order

Required. Baldwin, R. (2016). *The great convergence: Information technology and the new globalization*. Belknap Press of Harvard University Press. https://doi.org/10.4159/9780674972667

Further. - Ikenberry, G. J. (2011). *Liberal leviathan: The origins, crisis, and transformation of the American world order*. Princeton University Press. - Rodrik, D. (2011). *The globalization paradox: Democracy and the future of the world economy*. W. W. Norton. - Keohane, R. O. (1984). *After hegemony: Cooperation and discord in the world political economy*. Princeton University Press.

D.4 Session 03 — Anarchy and the balance of power

Required. Mearsheimer, J. J. (2001). *The tragedy of great power politics*. W. W. Norton — chapters 1–3. And: Allison, G. (2017). *Destined for war: Can America and China escape Thucydides's trap?* Houghton Mifflin Harcourt — introduction and chapter 1.

Further. - Waltz, K. N. (1979). *Theory of international politics*. Addison-Wesley. - Organski, A. F. K. (1958). *World politics*. Knopf. - Singer, J. D., Bremer, S., & Stuckey, J. (1972). Capability distribution, uncertainty, and major power war, 1820–1965. In B. Russett (Ed.), *Peace, war, and numbers* (pp. 19–48). Sage.

D.5 Session 04 — Trade, conflict and geoeconomic fragmentation

Required. Gopinath, G., Gourinchas, P.-O., Presbitero, A. F., & Topalova, P. (2025). Changing global linkages: A new Cold War? *Journal of International Economics*, 153, 104042. https://doi.org/10.1016/j.jinteco.2024.104042 (IMF Working Paper WP/24/76, free). And: Mansfield (2021), above.

Further. - Bailey, M. A., Strezhnev, A., & Voeten, E. (2017). Estimating dynamic state preferences from United Nations voting data. *Journal of Conflict Resolution*, 61(2), 430–456. https://doi.org/10.1177/0022002715595700 - Martin, P., Mayer, T., & Thoenig, M. (2008). Make trade not war? *Review of Economic Studies*, 75(3), 865–900. https://doi.org/10.1111/j.1467-937X.2008.00492.x - Barbieri, K. (2002). *The liberal illusion: Does trade promote peace?* University of Michigan Press. - Le Billon & Shykora (2020), above — how supply chains carry conflict, through commodities.

D.6 Session 05 — International institutions

Required. Kuziemko & Werker (2006), above. And: Wang, H. (2017). New multilateral development banks: Opportunities and challenges for global governance. *Global Policy*, 8(1), 113–118. https://doi.org/10.1111/1758-5899.12396

Further. - Vreeland, J. R., & Dreher, A. (2014). *The political economy of the United Nations Security Council: Money and influence*. Cambridge University Press. https://doi.org/10.1017/CBO9781139027755 - Morse, J. C., & Keohane, R. O. (2014). Contested multilateralism. *The Review of International Organizations*, 9(4), 385–412. https://doi.org/10.1007/s11558-014-9188-2 - Dreher, A., Sturm, J.-E., & Vreeland, J. R. (2009). Development aid and international politics: Does membership on the UN Security Council influence World Bank decisions? *Journal of Development Economics*, 88(1), 1–18. https://doi.org/10.1016/j.jdeveco.2008.02.003 - Humphrey, C. (2022). *Financing the future: Multilateral development banks in the changing world order of the 21st century*. Oxford University Press. https://doi.org/10.1093/oso/9780192871503.001.0001 - Horn, S., Reinhart, C. M., & Trebesch, C. (2021). China's overseas lending. *Journal of International Economics*, 133, 103539. https://doi.org/10.1016/j.jinteco.2021.103539 - Ikenberry, G. J., & Lim, D. J. (2017). *China's emerging institutional statecraft: The Asian Infrastructure Investment Bank and the prospects for counter-hegemony*. Brookings Institution. - Keohane (1984), above (Session 02). - Abbott & Snidal (2009), above — as the bridge to Session 06.

D.7 Session 06 — Private authority, standards and global regulation

Required. Prakash, A., & Potoski, M. (2006). Racing to the bottom? Trade, environmental governance, and ISO 14001. *American Journal of Political Science*, 50(2), 350–364. https://doi.org/10.1111/j.1540-5907.2006.00188.x. And: Abbott & Snidal (2009), above.

Further. - Mattli & Büthe (2003), above. - Brunsson, Rasche & Seidl (2012), above. - Green (2013), above. - Büthe, T., & Mattli, W. (2011). *The new global rulers: The privatization of regulation in the world economy*. Princeton University Press.

D.8 Session 07 — Measuring risk: content analysis, coding, big data and composite indices

Required. Hassan, T. A., Hollander, S., van Lent, L., & Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4), 2135–2202. https://doi.org/10.1093/qje/qjz021

Further. - Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636. https://doi.org/10.1093/qje/qjw024 - OECD & JRC (2008). *Handbook on constructing composite indicators: Methodology and user guide*. OECD Publishing. https://doi.org/10.1787/9789264043466-en - Krippendorff, K. (2018). *Content analysis: An introduction to its methodology* (4th ed.). SAGE. - Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3), 267–297. https://doi.org/10.1093/pan/mps028

D.9 Session 08 — Cyber geopolitics and digital threats

Required. Valeriano, B., & Maness, R. C. (2015). *Cyber war versus cyber realities: Cyber conflict in the international system*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190204792.001.0001 — chapters 1, 4 and 5.

Further. - Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79. https://doi.org/10.1162/isec_a_00351 - Kello, L. (2017). *The virtual weapon and international order*. Yale University Press. - Rid, T. (2013). *Cyber war will not take place*. Oxford University Press.

D.10 Session 09 — The redefinition of economic diplomacy

Required. Rose, A. K. (2007). The foreign service and foreign trade: Embassies as export promotion. *The World Economy*, 30(1), 22–38. https://doi.org/10.1111/j.1467-9701.2007.00870.x. And: Bjola & Manor (2018), above.

Further. - Putnam, R. D. (1988). Diplomacy and domestic politics: The logic of two-level games. *International Organization*, 42(3), 427–460. https://doi.org/10.1017/S0020818300027697 - Blackwill, R. D., & Harris, J. M. (2016). *War by other means: Geoeconomics and statecraft*. Belknap Press of Harvard University Press.

D.11 Session 10 — Economic sanctions

Required. Felbermayr, G., Kirilakha, A., Syropoulos, C., Yalcin, E., & Yotov, Y. V. (2020). The Global Sanctions Data Base. *European Economic Review*, 129, 103561. https://doi.org/10.1016/j.euroecorev.2020.103561. And: Hufbauer, G. C., Schott, J. J., Elliott, K. A., & Oegg, B. (2007). *Economic sanctions reconsidered* (3rd ed.). Peterson Institute for International Economics — chapters 1–3.

Further. - Pape, R. A. (1997). Why economic sanctions do not work. *International Security*, 22(2), 90–136. https://doi.org/10.1162/isec.22.2.90 - Morgan, T. C., Bapat, N., & Kobayashi, Y. (2014). Threat and imposition of economic sanctions 1945–2005: Updating the TIES dataset. *Conflict Management and Peace Science*, 31(5), 541–558. https://doi.org/10.1177/0738894213520379 - Drezner, D. W. (1999). *The sanctions paradox: Economic statecraft and international relations*. Cambridge University Press.

D.12 Session 11 — Space geopolitics: the orbital commons and the new space race

Required. ESA Space Debris Office. (2026). *ESA's annual space environment report* (Issue 10.0, 1 May 2026; GEN-DB-LOG-00288-OPS-SD). European Space Agency. https://www.sdo.esoc.esa.int/publications/Space_Environment_Report_I10R0_20260501 — executive summary, §1.1–1.2, §2.1, Table 3.1. And: Samson, V., & Brett, K. (Eds.). (2026). *Global counterspace capabilities: An open source assessment*. Secure World Foundation — executive summary. And: Pekkanen, S. M. (2019). Governing the new space race. *AJIL Unbound*, 113, 92–97. https://doi.org/10.1017/aju.2019.16

Further. - Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (1967), 610 U.N.T.S. 205. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.ht - Dolman, E. C. (2002). *Astropolitik: Classical geopolitics in the space age*. Frank Cass. https://doi.org/10.4324/9780203016640 - Bowen, B. E. (2020). *War in space: Strategy, spacepower, geopolitics*. Edinburgh University Press. https://doi.org/10.3366/edinburgh/978 - Kessler, D. J., & Cour-Palais, B. G. (1978). Collision frequency of artificial satellites: The creation of a debris belt. *Journal of Geophysical Research*, 83(A6), 2637–2646. https://doi.org/10.1029/JA083iA06p02637 - Farrell & Newman (2019), above (Session 08).

D.13 Session 12 — Team presentations, and the closing address

Closing readings. - Kobrin (1979), above (Session 01). - Henisz, W. J. (2000). The institutional environment for economic growth. *Economics & Politics*, 12(1), 1–31. https://doi.org/10.1111/1468-0343.00066

Further, for the team memo and the take-home. - Jensen, N. M. (2003). Democratic governance and multinational corporations: Political regimes and inflows of foreign direct investment. *International Organization*, 57(3), 587–616. https://doi.org/10.1017/S0020818303573040 - Kaufmann, D., Kraay, A., & Mastruzzi, M. (2011). The Worldwide Governance Indicators: Methodology and analytical issues. *Hague Journal on the Rule of Law*, 3(2), 220–246.

Appendix E

Assessment

Component	Weight	Individual?	Form	When	Brief
Participation	10%	yes	active engagement, continuous	every session	below
Quizzes	15%	yes	3 × 5%, 20 min, in class, on paper, closed book	Sessions 04, 07, 10	`quizzes/`
Risk Brief	15%	yes	4-minute presentation + one question (a 5-minute slot)	one of Sessions 02–11	`risk-brief/`
Team work	20%	no	risk dashboard for one multinational + memo + presentation	presented Session 12 · memo one week later	`team-project/`
Take-home examination	40%	yes	3 hours of work within a 48-hour window	final examination period	`take-home/`

80% individual · 20% team. In HEC Montréal's terms: A07 Participation (10%), A03 Tests and quizzes (15%), A06 Individual presentation (15%), A09 Team work (20%), A11 Take-home examination (40%).

E.1 How each component maps to the learning outcomes

Outcome	Participation	Quizzes	Risk Brief	Team work	Take-home
Knowledge — define GPR analysis, identify indicators	·	●●	·	·	●
Comprehension — explain the international context of MNEs	●	●●	●	·	●

Outcome	Participation	Quizzes	Risk Brief	Team work	Take-home
Application — economic diplomacy and new risks	●	●	●●	●	●
Analysis — interactions between risks, implications for MNEs	●	·	●	●●	●●
Synthesis — integrate content analysis, coding, big data, statistics	·	·	●	●●	●●
Evaluation — design risk assessment strategies for MNEs	·	·	●	●●	●●

●● principal · ● contributes · · marginal

E.2 Participation — 10%

Marked against a published, absolute bar — never ranked:

Mark	What it looks like across the term
9–10	Regularly moves the discussion forward with reference to the reading; asks real questions after Risk Briefs; fully engaged in every practice round
7–8	Contributes often; engaged in practice rounds; occasional questions
5–6	Present and engaged when asked; rarely volunteers
< 5	Disengaged, or absent from practice rounds without reason

Practice rounds are **formative**: commented on, not marked. What is recorded is that you engaged and handed in the `.twbx` and provenance log.

E.3 Late work and missed assessments

- **Quiz** missed with a documented reason: its weight moves to the take-home. Without: zero.
- **Risk Brief** missed with a documented reason: rescheduled. Without: zero.
- **Team memo**: −5% of its mark per day late.
- **Take-home**: submissions after the window closes are not accepted, per HEC regulations.

E.4 Quizzes

Twenty minutes, at the start of Sessions **04, 07 and 10**, on paper, closed book, individual.

Quiz	Session	Covers
Quiz 1	04	Sessions 01–03: what geopolitical risk is; the international economic order; anarchy and the balance of power
Quiz 2	07	Sessions 04–06: trade, conflict and fragmentation; international institutions; private authority and standards
Quiz 3	10	Sessions 07–09: measuring risk; cyber geopolitics; economic diplomacy

E.4.1 The format — 10 points each

Part	Points	What it asks
A · Concepts	4	Four short definitions or distinctions, one or two sentences each
B · The big work	3	The research question, the data and the central finding of one of the session readings — and one limitation
C · Read the chart	3	An unseen Tableau chart: say what it shows, what decision it was built with, and what it does not license a firm to conclude

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E.4.2 Specimen — Quiz 1

A · Concepts (4 points, 1 each)

1. Distinguish *risk* from *uncertainty* in Knight's sense, and give one geopolitical example of each.
2. What is the difference between Kobrin's *macro* and *micro* political risk?
3. In the GPR index, why are counts of geopolitical articles divided by the total number of articles?
4. What does "anarchy" mean in the realist tradition — and what does it *not* mean?

B · The big work (3 points)

Caldara and Iacoviello (2022) distinguish *GPR Threats* from *GPR Acts*. (a) Which categories enter each? (b) Why does the distinction matter for a firm's investment decision? (c) Name one limitation of the index for a firm whose operations are mainly in sub-Saharan Africa.

C · Read the chart (3 points)

[A stacked area chart: shares of world GDP, current US dollars, 1970–2023, for the G7, China, and the rest of the world.]

(a) State the finding in one sentence. (b) The chart uses current US dollars. How would it change at purchasing-power parity, and why? (c) What does the chart *not* tell a Canadian manufacturer deciding where to build its next plant?

E.4.3 Specimen answers (Part A)

1. *Risk*: probabilities estimable from experience (piracy losses on a shipping route). *Uncertainty*: probabilities not estimable (whether a given strait will be blockaded within five years).
2. *Macro* risk affects all foreign firms in a country (a general expropriation, currency inconvertibility); *micro* risk affects one sector or firm (a mining-code change, a contract cancellation).
3. Because newspapers' total output changes over time; the share removes the effect of size, so the index measures *relative* attention to geopolitical tension.
4. The absence of a government above states — no authority to enforce agreements. It does not mean chaos or constant war.

E.5 The Risk Brief

> **Four minutes. One development. One view. One firm. One recommendation.**

Each student delivers one Risk Brief, at the start of one of Sessions 02–11. Sign-up opens at the end of Session 01; four briefs per session.

E.5.1 The brief

You are the geopolitical risk analyst of a named multinational. Something happened in the last **four weeks**. Brief your chief risk officer.

Minute	What you say
0:00–0:30	**The development.** What happened, where, who decided it. One sentence of context, no history lecture.
0:30–2:00	**The view.** One Tableau chart, built from data **you collected at source**, that puts the development in perspective — is it large, new, accelerating, concentrated?
2:00–3:00	**The exposure.** Through which channel does it reach *your* firm — market access, inputs, finance, assets and people, data and operations? How much of the firm sits in the path?
3:00–4:00	**The recommendation.** One decision, and the indicator and threshold that would make you change it.

One question follows (one minute), from the room or the instructor. Four briefs fill the first twenty minutes of the session.

The firm. Any real multinational with public information on its footprint (annual report, 10-K, sustainability report). It may — but need not — be your team's firm.

The data. Any source in `DATA-SOURCES.md` or any other public source, cited with version and access date. Ideally the concept or dataset of a session you have already had.

E.5.2 What you hand in

Before class on the day: the `.twbx` (or a Tableau Public link) and your provenance log, on ZoneCours.

E.5.3 Rubric (15 points)

Criterion	Points	Full marks look like
The development	2	Precisely dated, sourced, and genuinely geopolitical (between states or political actors)
The view	4	Data collected at source; correct aggregation; title states the finding; source caption; the chart *does* work in the argument
The exposure	4	A specific channel; evidence from the firm's own disclosures; an order of magnitude
The recommendation	3	One decision, an indicator and a threshold that would reverse it
Time and question	2	Four minutes, not six; answers the question asked

Automatic deductions: no source caption (−2); over five minutes (stopped at five); a development older than four weeks (−2).

E.6 Team work

> **Design the geopolitical risk assessment a real multinational should be running — build it, and defend the method behind it to its board.**

Teams of three or four, formed at the start of Session 02. Each team adopts **one named multinational** for the whole term — no two teams the same firm.

E.6.1 What you deliver

Deliverable	Points	When
The dashboard — a Tableau Public workbook	part of the 15 below	one week after Session 12
The memo to the board — 2,500 words, plus appendices	15 (dashboard + memo together)	one week after Session 12
The presentation and defence — 12 minutes + 8 minutes of questions	5	Session 12

E.6.1.1 The dashboard

At least **four datasets** collected in the practice rounds, from at least **three of the six families of indicators** (Session 01): event-based, text-based, expert-based, structural, market-based, firm-level. It must contain:

1. **The firm's footprint** — where its revenue, assets, suppliers and people are (from its own disclosures), as a map or a table.

2. **A composite exposure index** for the firm's main countries, built with the method of Session 07: indicators chosen and justified, normalised, weighted by **parameters** the board can move.
3. **At least two channel views** — e.g. sanctions exposure (S10), dependence on satellite services (S11), bloc alignment of key markets (S04), cyber-incident exposure of its sector (S08).
4. **A watchlist** — three indicators, each with a threshold that triggers a named decision.

E.6.1.2 The memo

Section	Words	Content
Executive summary	250	The three risks that matter, and what to do
The firm and its footprint	300	Where it is exposed, and through which channels
Method	700	Which indicators, why these, how combined, what each cannot see — the Session 07 discipline
Findings	800	What the dashboard shows, risk by risk, interaction by interaction
Recommendations and watchlist	450	Decisions, triggers, owners
Appendices	—	Provenance log for every dataset; data decisions; contribution statement

E.6.2 Rubric (20 points)

Criterion	Points	Full marks look like
Conceptual grounding	3	The risks are framed with the course's concepts — order, anarchy, institutions, diplomacy, sanctions, interdependence — not newspaper vocabulary
Data and provenance	3	Four or more datasets collected at source, every one logged; decisions named
Method of the composite index	4	Indicators justified; normalisation and weighting explained; sensitivity to weights shown
Analysis of interactions	3	Shows how risks compound or offset — e.g. sanctions *and* input concentration *and* bloc alignment
Recommendations	2	Decisions with triggers and owners, not "monitor the situation"
Presentation and defence	5	Twelve minutes, the dashboard driven live, questions answered by any member

Individual accountability. Each memo ends with a contribution statement signed by all members. At the end of term each student submits a confidential peer rating (four dimensions, 1–5). The team mark is multiplied by an individual factor (default 1.00, range 0.80–1.05) only when the peer ratings and the instructor's observation of the practice rounds independently agree.

E.6.3 Milestones

Session	Milestone
02	Team formed, firm chosen and registered on ZoneCours
04	One-paragraph footprint: the firm's five most important countries, with sources
07	Draft composite index — indicators and weights — reviewed in the practice round
10	Dashboard skeleton reviewed in the last twenty minutes of the practice round
12	Presentation and defence
12 + 1 week	Memo and published dashboard

E.7 Take-home examination

Three hours of work, within a 48-hour window, during the final examination period. Individual. Open book, open notes, open internet. No collaboration of any kind.

E.7.1 The format

At the opening of the window you receive, on ZoneCours:

- a **scenario** — a geopolitical development, real or plausible, dated;
- a **firm** — a real multinational, with an extract from its disclosures;
- a **data instruction** — one or two sources from `DATA-SOURCES.md` to collect from, with the variables to use.

You hand in, before the window closes:

Part	Points	What
A · Concepts in context	12	Three short essays (≈ 300 words each) applying the course's concepts to the scenario: e.g. what realism and liberal institutionalism each predict; whether sanctions in this case are likely to work; how the firm's standards exposure changes
B · The evidence	14	One packaged Tableau workbook (`.twbx`) built from data you collected, with its provenance log, and a 400-word reading of it
C · The assessment	14	A 900-word memo to the firm's chief risk officer: the exposure, the interactions between risks, the decision, the watchlist with thresholds

Length limits are strict. Words beyond the limit are not read.

E.7.2 Marking criteria

Criterion	Weight
Correct and precise use of the course's concepts	30%
Data collected at source, correctly prepared, honestly logged	25%
Analysis of interactions between risks and their implications for the firm	25%
A decision with triggers, defensible from the evidence	20%

E.7.3 On language models and integrity

You may use a language model to check syntax or to help you read a codebook. You may not use it to produce any text you submit. Every submission includes a one-paragraph statement of the tools you used and how. Undeclared use, or collaboration with another person, is a violation of HEC Montréal's academic integrity regulations.

E.7.4 Specimen scenario

Dated 15 March. The European Union and the United States announce coordinated export controls on advanced battery-manufacturing equipment to a major Asian economy; that economy responds within a week with an export licensing regime on refined graphite.

Firm: a European automotive manufacturer with an EV assembly plant in Central Europe and a joint venture in the affected Asian economy.

Data instruction: IMF International Trade in Goods by partner country (`XG_FOB_USD`); the Global Sanctions Data Base; UN General Assembly ideal points (Bailey, Strezhnev & Voeten); the firm's last annual report.

Part A (example): Using Farrell and Newman's concept of weaponised interdependence, explain which side holds the stronger *chokepoint* and why. *Part B:* Build a view of goods trade between the EU, the United States and the affected economy since 2000, set against the sanctions cases in force between them and the distance between their UN voting positions. *Part C:* Advise the chief risk officer on the joint venture and on sourcing for the European plant.

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